

Entanglement, Operator Algebras, and the Emergence of Spacetime

A First Introduction for Physicists

Preface

Quantum mechanics is usually taught with the Hilbert space as the starting point: a system has a space of state vectors, observables are operators acting on it, and a composite system is described by a tensor product. This picture works perfectly for a spin, an atom, or any system with finitely many degrees of freedom. It stops working for a quantum field cut into two regions, for a many-body system in the thermodynamic limit, and for quantum gravity in the limit where a classical spacetime appears. In each of these cases the entanglement between a subsystem and its surroundings is infinite, and the tensor-product split that defines “subsystem” in the first place no longer exists.

These notes develop the replacement. A subsystem is described by the algebra of operators an observer can measure, not by a factor of the Hilbert space. The classification of such algebras into types I, II and III turns out to be a classification of how a subsystem is entangled with the rest of the world, and the modular theory of Tomita and Takesaki supplies a notion of time and temperature for any such algebra. In the second half of the notes these tools are applied to holography, where they show that regions of a gravitational spacetime, their causal structure, their horizons and their entropy can all be read off from the algebras of a boundary quantum system.

Who these notes are for

The reader is assumed to know linear algebra (vectors, matrices, eigenvalues) and undergraduate quantum mechanics at the level of Griffiths or Sakurai: kets and bras, operators, the Schrödinger and Heisenberg pictures, spin, the harmonic oscillator, and density matrices. Some familiarity with statistical mechanics (Gibbs states and partition functions) helps in Chapters 3–5. Everything beyond that is built on the page before it is used.

How the notes are organised

Chapters 1–5 develop the mathematics using only finite quantum systems and free fields: why the tensor product fails (Chapter 1), von Neumann algebras and the GNS construction (Chapter 2), algebras with a trace (Chapter 3), type III algebras and modular theory (Chapter 4), and the crossed product that turns a type III algebra into one with a well-defined entropy (Chapter 5). Chapters 6–9 apply these tools to the AdS/CFT correspondence and to simple models of quantum gravity, and Chapter 10 summarises.

Three kinds of boxes appear throughout. *Physics Connection* boxes compute the same quantity twice — once the way you already know from ordinary quantum mechanics, and once with the new construction — and show that the answers agree. *Worked Example* boxes carry out a construction on a small system with explicit numbers. *Key Result* boxes collect theorems and their proofs.

Calculations are written as chains of equalities in which every non-trivial step carries a small number

over its equals sign, for example $A \stackrel{(1)}{=} B \stackrel{(2)}{=} C$. The sentence immediately after the chain explains each numbered step, so no step is left for the reader to reconstruct. Where a result is quoted from the research literature rather than derived, the text says so.

Sources

The logical development follows the lectures of H. Liu [1], which review this subject at research level. The treatment of entanglement in quantum field theory draws on Witten [2]; the mathematical foundations of algebraic quantum mechanics on Strocchi [4] and Haag [10]; the statistical interpretation of states on Slavnov [3]; and the crossed-product treatment of de Sitter entropy on Chandrasekaran, Longo, Penington and Witten [8]. Numerical values quoted in worked examples were checked by direct computation.

Contents

Preface	1
1 Introduction and motivations	6
1.1 What problem are these notes about?	6
1.2 From wavefunctions to density matrices to algebraic states	7
1.3 Why the ordinary quantum-mechanical definition of “subsystem” isn’t good enough	9
1.4 The central example: infinitely many entangled qubit pairs	10
1.5 Two more examples, more briefly	14
1.6 Explicit derivation: why the continuum QFT Hilbert space cannot factorize	15
1.7 The fix: define a subsystem by what you can <i>do</i> , not by how you’d <i>split the Hilbert space</i>	22
1.8 Two organizing claims	23
1.9 Plan of the notes	23
1.10 Conventions and notation	24
2 Introduction to von Neumann algebras	27
2.1 Systems with an infinite amount of entanglement (recap)	27
2.2 Von Neumann algebras as subsystems	27
2.3 Basic properties: C^* vs. von Neumann algebras, and the double commutant theorem	30
2.4 Weights and states	35
2.5 C^* and von Neumann algebras generated by a single operator	37
2.6 Projections	39
2.7 Classification of von Neumann factors	42
2.8 “Emergent” Hilbert space and von Neumann algebra: the GNS construction	48
2.9 Emergent von Neumann algebras of the entangled spin example	58
3 Von Neumann algebras and entanglement: type I and II	60
3.1 Density operators for type I and II algebras	60
3.2 Type I algebras	62
3.3 Type II algebras	65

3.4	Summary	75
4	Von Neumann algebras and entanglement: type III	76
4.1	Emergent times from entanglement — modular flows	76
4.2	Classification of type III factors	85
4.3	The entangled spin example revisited	87
4.4	Local algebras in a relativistic quantum field theory	90
4.5	Emergent times from subalgebras — half-sided modular inclusion	100
4.6	Relative modular flows and relative entropy	103
4.7	A canonical purification: the natural cone	104
5	The crossed product by the modular group	106
5.1	Crossed products in general	106
5.2	Construction of $\widehat{\mathcal{M}}$	107
5.3	$\widehat{\mathcal{M}}$ is always type II	110
5.4	Density operator for $\widehat{\mathcal{M}}$ in a general semiclassical state	115
5.5	Entanglement entropy for $\widehat{\mathcal{M}}$ in a general semiclassical state	115
6	AdS/CFT in the large-N limit: an algebraic formulation	118
6.1	General description	118
6.2	The vacuum sector	124
6.3	The thermofield double state	128
6.4	General black holes	131
6.5	A diagnostic of firewalls in generic highly excited states	132
6.6	Perturbative $1/N$ corrections	132
7	Subregion-subalgebra duality	136
7.1	General formulation	136
7.2	Entanglement wedge reconstruction, algebraically	137
7.3	Quantum informational aspects of entanglement wedge reconstruction	144
7.4	An algebraic formulation of entanglement islands	149
7.5	Boundary description of a bulk causal diamond	153
7.6	Generalized entropy and subregion-subalgebra duality at finite N	155
8	Emergence of bulk geometric concepts	159

8.1	Algebraic characterization of bulk causal structure	159
8.2	Emergence of Kruskal-like time, and resolving the meeting-behind-the-horizon puzzle	162
8.3	Emergent spacetime connectivity: algebraic ER=EPR	165
8.4	Stringy geometry and stringy black holes	167
9	Algebraic approaches to quantum-gravity regimes	168
9.1	A model of gravitational dressing from observers	168
9.2	Application I: quantum volatile black hole spacetime and its entropy	171
9.3	Application II: de Sitter entropy	171
9.4	Application III: generalized entropy for a local spacetime region	174
9.5	A model of dynamical observers	175
9.6	Operator algebras of JT gravity	176
10	Conclusions and outlook	178
10.1	Summary	178
10.2	The mathematical structure of quantum gravity	179

CHAPTER 1

Introduction and motivations

1.1 What problem are these notes about?

Before any equations, it helps to say in plain words what question this subject answers. Once the algebra starts, it is easy to lose sight of the goal.

Here is the question. Ordinary physics, classical and quantum, treats spacetime as a fixed stage. Space and time are given in advance, with a definite geometry (flat, or curved according to some metric), and physics happens *on* that stage. General relativity complicates this a little, because the geometry itself is dynamical: matter tells spacetime how to curve, and spacetime tells matter how to move. Even so, at any moment you can ask “what is the metric right here?” and get a sharp answer.

Quantum gravity is the attempt to quantize this whole picture. Once you do that, the metric itself becomes an operator with quantum fluctuations. The question “what is the geometry right here?” then stops having a sharp answer, in the same way that “where exactly is the electron?” stops having a sharp answer once a particle’s position is quantized. Geometric statements such as “this point lies to the future of that point,” “this region is causally connected to that region,” or “this is a local, self-contained patch of space” are sharply defined only in the classical limit. Physicists write this limit as $G_N \rightarrow 0$, where G_N is Newton’s gravitational constant. Turning G_N off turns off quantum-gravitational effects and leaves an ordinary, fixed classical geometry.

So the research question is this. Suppose you are handed a complete quantum theory of gravity at some finite, nonzero G_N . No classical geometry is assumed, no metric is given in advance, and nothing geometric is built in from the start. As you dial $G_N \rightarrow 0$, familiar geometric structures are supposed to emerge: a well-defined notion of “here,” a causal order, a notion of a local region. **What, physically and mathematically, is doing the emerging?** What feature of the underlying quantum theory produces spacetime geometry as $G_N \rightarrow 0$? And what role, if any, does quantum entanglement play?

The clearest setting in which this question can be asked precisely is the AdS/CFT correspondence. There, a theory of quantum gravity in a $(d+1)$ -dimensional *anti-de Sitter* spacetime is conjectured to be completely equivalent to an ordinary, non-gravitational quantum field theory living on the d -dimensional boundary of that spacetime. Anti-de Sitter space (AdS) is a spacetime of constant negative curvature. It is chosen because it has a well-defined boundary at infinity. The boundary theory is a *conformal field theory* (CFT).

On the CFT side there is a parameter N that counts the number of independent field degrees of freedom. In the gauge theories where the correspondence was first understood, N is the number of colors of an $SU(N)$ gauge group. The AdS/CFT dictionary relates G_N on the gravity side to a

positive power of $1/N$ on the CFT side. So $G_N \rightarrow 0$ (classical, geometric gravity) is the same limit as $N \rightarrow \infty$ on the boundary. The question above then becomes concrete: *how does bulk spacetime, with its geometry, causal structure, and locality, emerge from an ordinary quantum field theory on a lower-dimensional boundary, as the number of degrees of freedom of that theory is sent to infinity?*

The Ryu–Takayanagi proposal [7] gave the first strong hint of the answer. It is a formula that relates the entanglement entropy of a boundary region to the area of a certain minimal surface in the bulk. It is literally an equation between a measure of entanglement and a geometric quantity. If entanglement entropy and bulk area are this closely tied, then entanglement is likely doing real structural work in building the bulk geometry, rather than merely decorating it. That is the motivating intuition.

These notes develop the tools that make this intuition precise. The approach follows a research program developed by Leutheusser, Liu, and collaborators [1]. The key tool is a piece of mathematics from the 1930s: von Neumann algebras. They were originally developed for a different purpose, namely putting ordinary quantum mechanics on a rigorous footing. Only recently have they been recognized as the right language for this question.

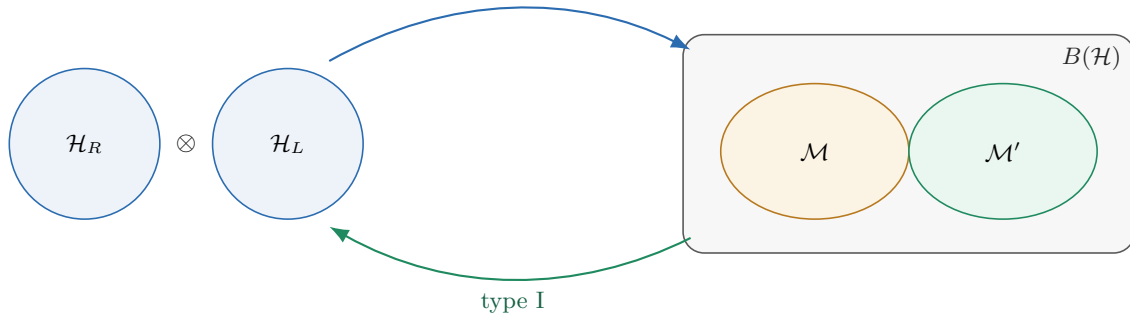


Figure 1.1: From tensor factors to subalgebras. On the left, a subsystem is a tensor factor of $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$, as in standard quantum mechanics. On the right, a subsystem is an algebra of operators $\mathcal{M}$ inside $B(\mathcal{H})$, together with its commutant $\mathcal{M}'$, the operators that commute with all of $\mathcal{M}$. The upper arrow is the generalization; the lower arrow says that type I algebras are the special case in which the tensor-product picture on the left is recovered.

1.2 From wavefunctions to density matrices to algebraic states

To see why the algebraic framework is natural, follow how the idea of a “quantum state” changes as we move from undergraduate quantum mechanics to open systems, quantum field theory, and quantum gravity.

1. **Level 0: the pure wavefunction** $\psi(x) = \langle x | \Psi \rangle$. In textbook quantum mechanics, the basic object is a state vector $|\Psi\rangle \in \mathcal{H}$. It evolves according to the Schrödinger equation. Probabilities come from the Born rule, $P(x) = |\psi(x)|^2$. This description assumes a *closed, isolated system*, in which an experimenter can in principle measure any operator on the whole system.

Symbol / Structure	Mathematical Definition	Physical Interpretation / Role
$\mathcal{H}$	Global Hilbert space of states	Total state space of the system
$B(\mathcal{H})$	Bounded linear operators on $\mathcal{H}$	All bounded operations and observables
$\mathcal{M} \subset B(\mathcal{H})$	von Neumann algebra ($\mathcal{M} = \mathcal{M}''$)	Observables accessible to a local subregion or observer
$\mathcal{M}'$	Commutant: $\{B : [B, A] = 0 \ \forall A \in \mathcal{M}\}$	Observables of the causal complement, or of the complementary subsystem
$\mathcal{Z}(\mathcal{M}) = \mathcal{M} \cap \mathcal{M}'$	Center of the algebra	Classical superselection labels (trivial, $\mathbb{C}\mathbf{1}$, for a factor)
$\omega : \mathcal{M} \rightarrow \mathbb{C}$	State (positive linear functional, $\omega(\mathbf{1}) = 1$)	Expectation values, $\omega(A) = \langle A \rangle$
$(\pi_\omega, \mathcal{H}_\omega, \Omega_\omega\rangle)$	GNS representation	Hilbert space built directly from the state ω
$ \Psi\rangle \in \mathcal{H}$	Cyclic ($\mathcal{M} \Psi\rangle$ dense) and separating (no nonzero $A \in \mathcal{M}$ has $A \Psi\rangle = 0$) vector	Typical entangled state, such as the vacuum of a quantum field theory
S_Ψ	Tomita antilinear operator: $S_\Psi A \Psi\rangle = A^\dagger \Psi\rangle$	State-dependent modular involution
J_Ψ	Modular conjugation ($J_\Psi \mathcal{M} J_\Psi = \mathcal{M}'$)	Antilinear reflection mapping the algebra to its commutant (for a Rindler wedge, a CPT-type reflection)
$\Delta_\Psi = S_\Psi^\dagger S_\Psi$	Modular operator (positive, self-adjoint)	Encodes the entanglement of $ \Psi\rangle$ (equals $\rho_R \otimes \rho_L^{-1}$ in type I)
$h_\Psi \equiv -\log \Delta_\Psi$	Modular Hamiltonian	Generator of the intrinsic time flow of the subsystem (for a Rindler wedge, 2π times the boost generator)
$\sigma_t^\Psi(A) = \Delta_\Psi^{it} A \Delta_\Psi^{-it}$	Modular flow (one-parameter group of automorphisms)	Intrinsic time evolution; the state is thermal (KMS) with respect to it
$\widehat{\mathcal{M}} = \mathcal{M} \rtimes_\sigma \mathbb{R}$	Crossed product algebra, with clock space $L^2(\mathbb{R})$	Gravitationally dressed algebra that includes an observer's energy or clock
τ	Semifinite trace on $\widehat{\mathcal{M}}$ ($\tau(AB) = \tau(BA)$)	Renormalized trace that gives well-defined density matrices
S_{gen}	Generalized entropy: $\frac{\langle \hat{A} \rangle}{4G_N} + S_{\text{bulk}}$	Finite gravitational entropy in semiclassical gravity
$\mathcal{A}_{\text{CFT}}$	Boundary single-trace algebra ($N \rightarrow \infty$)	Generalized free fields on the holographic boundary
$\mathcal{A}_{\text{bulk}}(b_A)$	Bulk algebra in the entanglement wedge b_A	Local semiclassical bulk quantum fields dual to a boundary region

Table 1.1: Summary of the core notation, algebraic structures, and physical interpretations used throughout these notes.

2. Level 1: the density matrix $\rho(x, x')$ and the Wigner function. When a system interacts with an unobserved environment or a thermal bath, pure states are replaced by density operators, $\rho = \sum_k p_k |\psi_k\rangle \langle \psi_k|$. In the position basis the density operator becomes a function of two positions, $\rho(x, x') = \langle x | \rho | x' \rangle$. It is useful to change to the following variables (called Keldysh variables in the study of open systems):

$$x_c \equiv \frac{x + x'}{2} \quad (\text{midpoint}), \quad x_q \equiv x - x' \quad (\text{separation}).$$

The diagonal $x_q = 0$ gives the classical probability distribution, $P(x_c) = \rho(x_c, x_c)$. Nonzero x_q records the off-diagonal elements, which carry quantum interference. Fourier transforming in x_q gives the **Wigner quasi-probability distribution** on phase space (we set $\hbar = 1$):

$$W(x_c, p) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx_q e^{-ipx_q} \rho\left(x_c + \frac{x_q}{2}, x_c - \frac{x_q}{2}\right).$$

Here p is the momentum conjugate to x_q , and the factor $1/(2\pi)$ makes $\int dx_c dp W = 1$. The Wigner function is a bridge between classical statistical physics and quantum mechanics. But the density matrix of a subsystem is obtained by tracing out an environment, so this whole level still presumes that the global Hilbert space factorizes as $\mathcal{H} = \mathcal{H}_{\text{system}} \otimes \mathcal{H}_{\text{environment}}$.

3. **Level 2: observers have limited access.** In realistic situations, no observer can measure everything. An observer has a limited apparatus, or is confined to a spatial subregion R , and can only measure a restricted set of observables $\mathcal{M}$. Describing such an observer’s situation with global pure states gives useless answers. For example, the relative entropy $D(\rho\|\sigma) = \text{Tr}(\rho \log \rho - \rho \log \sigma)$ between two *different* global pure states is always $+\infty$. The reason is that $\log \sigma$ equals $-\infty$ on every direction orthogonal to the state vector of σ , and ρ has weight in such a direction. So the global relative entropy cannot tell two nearly identical pure states apart from two very different ones. What the observer actually needs is a relative entropy of the two states *restricted to* $\mathcal{M}$, which is typically finite and informative. In quantum field theory there is a further problem: as this chapter shows, a region does not even have a reduced density matrix to put into this formula.
4. **Level 3: the algebraic state $\omega : \mathcal{M} \rightarrow \mathbb{C}$.** In algebraic quantum mechanics (developed by von Neumann, Haag, and many others), we drop the assumption that a fixed global Hilbert space is fundamental. The primary physical object is the **algebra of accessible observables**, $\mathcal{M}$. A **state** ω is simply a positive linear functional that assigns expectation values to operators:

$$\omega(A) = \langle A \rangle_\omega, \quad \omega(\mathbf{1}) = 1, \quad \omega(A^\dagger A) \geq 0 \quad \forall A \in \mathcal{M}.$$

The Hilbert space $\mathcal{H}_\omega$ and the state vector $|\Omega_\omega\rangle$ are not postulated in advance. They are **constructed** from the state ω by the Gelfand–Naimark–Segal (GNS) construction of Chapter 2, in such a way that

$$\omega(A) = \langle \Omega_\omega | \pi_\omega(A) | \Omega_\omega \rangle.$$

Key Result / Core Concept

The conceptual takeaway: The wavefunction $\psi(x)$ is not the fundamental container of physical reality. It is one particular GNS representation of an algebraic state ω on a type I algebra. In the thermodynamic limit (infinitely many degrees of freedom), for local subregions in quantum field theory, and for semiclassical black holes, the tensor factorization of the Hilbert space breaks down. The algebraic state ω , however, remains well defined.

1.3 Why the ordinary quantum-mechanical definition of “subsystem” isn’t good enough

To see why a new mathematical tool is needed, we first have to see exactly where the old one breaks. The old tool is the one from every quantum mechanics course. Take a system built from two pieces, R and L . Read them as “right” and “left,” or as “region” and “its complement”; any split into two

parts will do. The Hilbert space of the whole system is then a tensor product,

$$\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L.$$

Concretely, suppose $\mathcal{H}_R$ has orthonormal basis $\{|i\rangle_R\}$ and $\mathcal{H}_L$ has orthonormal basis $\{|a\rangle_L\}$. Then $\mathcal{H}_R \otimes \mathcal{H}_L$ is spanned by all the pairs $|i\rangle_R |a\rangle_L$. A general state of the joint system is a sum $|\Psi\rangle = \sum_{ia} c_{ia} |i\rangle_R |a\rangle_L$ for some array of complex numbers c_{ia} . (The same idea works with continuous or more general index sets.) The state is called **entangled** exactly when the array c_{ia} cannot be written as a product $c_{ia} = \psi_i \chi_a$ of two smaller arrays ψ_i and χ_a . In words: an entangled state does not split into “something about R ” times “something about L .”

Everything you already know about entanglement is built directly on the tensor factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$. This includes the reduced density matrix $\rho_R \equiv \text{Tr}_L |\Psi\rangle\langle\Psi|$ (obtained by tracing out, that is, summing over, the L degrees of freedom), the entanglement entropy $S_R = -\text{Tr}_R(\rho_R \log \rho_R)$, the Rényi entropies, and the relative entropy. If the factorization does not exist, none of these formulas can even be written down. There is no $\mathcal{H}_L$ to trace over, and no well-defined “ R -part” of the state to extract. The familiar machinery of entanglement entropy fails at the first step. It does not give a wrong answer; it gives no answer at all.

The central claim, on which the rest of these notes build, is that this factorization really does fail to exist. This is not an exotic mathematical pathology. It happens in exactly the situations that matter most: gauge theories; many-body systems in a thermodynamic limit (infinite volume or infinite N); quantum field theories cut into two regions by a surface; and, the case these notes care about most, the large- N limit of a holographic boundary theory. Three concrete examples show how and why it fails. The first one, infinitely many entangled qubit pairs, is worked through slowly and completely here, because later chapters keep returning to it.

1.4 The central example: infinitely many entangled qubit pairs

Take a single pair of qubits, one belonging to R and one to L , prepared in the state

$$|\phi_\theta\rangle = \cos \theta |0\rangle_R |0\rangle_L + \sin \theta |1\rangle_R |1\rangle_L, \quad \theta \in (0, \pi/4].$$

This is completely standard. It is a partially entangled pair. It is maximally entangled exactly at $\theta = \pi/4$, where both coefficients equal $\frac{1}{\sqrt{2}}$; this is a Bell pair. Such a state can be prepared in a laboratory with two photons or two trapped ions. Nothing about a single pair is mysterious.

To find the reduced density matrix explicitly, first write out the full 4×4 density operator of the pure pair:

$$\begin{aligned} |\phi_\theta\rangle\langle\phi_\theta| &= \cos^2 \theta |00\rangle\langle 00| + \cos \theta \sin \theta |00\rangle\langle 11| \\ &\quad + \sin \theta \cos \theta |11\rangle\langle 00| + \sin^2 \theta |11\rangle\langle 11|. \end{aligned}$$

Now take the partial trace over subsystem L , summing over the orthonormal basis $\{|0\rangle_L, |1\rangle_L\}$:

$$\rho_R \equiv \text{Tr}_L |\phi_\theta\rangle\langle\phi_\theta| \stackrel{(1)}{=} {}_L\langle 0|\phi_\theta\rangle\langle\phi_\theta|0\rangle_L + {}_L\langle 1|\phi_\theta\rangle\langle\phi_\theta|1\rangle_L$$

$$\stackrel{(2)}{=} \cos^2 \theta |0\rangle_R \langle 0|_R + \sin^2 \theta |1\rangle_R \langle 1|_R \stackrel{(3)}{=} \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.$$

(1) This is the definition of the partial trace: sandwich between the basis vectors of L and sum.

(2) ${}_L \langle 0|\phi_\theta\rangle = \cos \theta |0\rangle_R$ and ${}_L \langle 1|\phi_\theta\rangle = \sin \theta |1\rangle_R$. The cross terms $|00\rangle \langle 11|$ and $|11\rangle \langle 00|$ drop out because $\langle 0|1\rangle_L = 0$. (3) Write the result as a matrix in the basis $|0\rangle_R, |1\rangle_R$.

Because ρ_R is already diagonal, its operator logarithm is found by taking the logarithm of each diagonal entry:

$$\log \rho_R = \begin{pmatrix} \log(\cos^2 \theta) & 0 \\ 0 & \log(\sin^2 \theta) \end{pmatrix}.$$

The von Neumann entanglement entropy of a single pair then follows by taking the trace:

$$s_1(\theta) = -\text{Tr}_R(\rho_R \log \rho_R) = -\left[\cos^2 \theta \log(\cos^2 \theta) + \sin^2 \theta \log(\sin^2 \theta)\right].$$

Two special cases are useful.

- For the maximally entangled Bell pair at $\theta = \pi/4$, we have $\cos^2(\pi/4) = \sin^2(\pi/4) = 1/2$, so

$$s_1(\pi/4) = -\left[\frac{1}{2} \log\left(\frac{1}{2}\right) + \frac{1}{2} \log\left(\frac{1}{2}\right)\right] = \log 2 \approx 0.693147 \text{ nats} \quad (= 1 \text{ bit}).$$

- In the weakly entangled limit $\theta \rightarrow 0$:

$$\begin{aligned} s_1(\theta) &\stackrel{(1)}{=} -(1 - \theta^2) \log(1 - \theta^2) - \theta^2 \log(\theta^2) + O(\theta^4 \log \theta) \\ &\stackrel{(2)}{=} \theta^2 - 2\theta^2 \log \theta + O(\theta^4 \log \theta) = \theta^2(1 - 2 \log \theta) + O(\theta^4 \log \theta) \rightarrow 0. \end{aligned}$$

(1) Substitute $\cos^2 \theta = 1 - \theta^2 + O(\theta^4)$ and $\sin^2 \theta = \theta^2 + O(\theta^4)$. The neglected $O(\theta^4)$ corrections change the result only at order $\theta^4 \log \theta$. (2) Expand $\log(1 - \theta^2) = -\theta^2 + O(\theta^4)$ in the first term, and use $\log \theta^2 = 2 \log \theta$ in the second. The limit is zero because $\theta^2 \log \theta \rightarrow 0$.

Now take N independent copies of this pair. That is, take N separate two-qubit systems, each prepared in the same way, and combine them into one big state:

$$|\Phi_\theta\rangle = |\phi_\theta\rangle_1 \otimes |\phi_\theta\rangle_2 \otimes \cdots \otimes |\phi_\theta\rangle_N.$$

Group all the R -qubits together into “the R system” and all the L -qubits into “the L system.” At any finite N , this is still completely ordinary. The space $\mathcal{H}_R = (\mathbb{C}^2)^{\otimes N}$ is a 2^N -dimensional Hilbert space. The full space $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ factorizes, because it was built as a tensor product. The reduced density operator of R is the N -fold tensor product, $\rho_R^{(N)} = \rho_R^{\otimes N}$. The von Neumann entropy is additive over tensor products, $S(\rho_1 \otimes \rho_2) = S(\rho_1) + S(\rho_2)$. So the total entropy of the N pairs is exactly

$$S_R^{(N)} = N s_1(\theta).$$

At $\theta = \pi/4$, $s_1(\pi/4) = \log 2 \approx 0.693$. So for $N = 1$, $S_R = 0.693$; for $N = 10$, $S_R = 6.93$; for $N = 1000$, $S_R = 693$. Nothing is subtle about any one of these numbers, and each can be checked by direct computation. New physics appears only when we ask what happens as $N \rightarrow \infty$, while staying inside the set of states that a real experiment with finite energy could reach.

Restricting to finite energy

To talk about energy we need a Hamiltonian. Choose one for which $|\Phi_\theta\rangle$ is the ground state:

$$H = f \sum_{i=1}^N (\mathbf{1} - P_i), \quad P_i \equiv |\phi_\theta\rangle \langle \phi_\theta|_i,$$

where P_i is the projector onto the state $|\phi_\theta\rangle$ of the i -th pair and $f > 0$ is a fixed energy scale. Each term $f(\mathbf{1} - P_i)$ has eigenvalues 0 and f . So $H \geq 0$, and $|\Phi_\theta\rangle$, which has $P_i = 1$ for every pair, is the ground state, with energy 0. This Hamiltonian couples each R -qubit to its L -partner. Some coupling of this kind is unavoidable: a Hamiltonian of the form $H_R + H_L$, with no interaction between R and L , always has an unentangled ground state.

Flipping a single spin costs a fixed amount of energy. Apply the spin-flip operator $X = |0\rangle \langle 1| + |1\rangle \langle 0|$ to the R -qubit of pair i . It sends $|\phi_\theta\rangle$ to $\cos \theta |1\rangle_R |0\rangle_L + \sin \theta |0\rangle_R |1\rangle_L$, which is orthogonal to $|\phi_\theta\rangle$. So the flipped pair has $P_i = 0$, and the energy rises by exactly f . Flipping spins in k different pairs costs kf .

A **finite-energy state** built on the reference state $|\Phi_\theta\rangle$ is one obtained by acting on only a finite number k of the pairs, for example by flipping finitely many spins. Such a state differs from $|\Phi_\theta\rangle$ only on those k pairs, and its energy is at most kf . Changing infinitely many pairs costs infinite energy, and no physical experiment supplies infinite energy.

Now comes the key computation. Suppose you want to disentangle R from L completely, that is, to reach some product state $|\psi\rangle_R \otimes |\chi\rangle_L$ with zero entanglement entropy. Here $|\psi\rangle_R$ may be any state of all N of the R -qubits, not necessarily a product over pairs, and similarly for $|\chi\rangle_L$. The energy of any such state is bounded below:

$$\begin{aligned} \langle \psi_R \chi_L | H | \psi_R \chi_L \rangle &\stackrel{(1)}{=} f \sum_{i=1}^N \left(1 - \langle \phi_\theta | \rho_i | \phi_\theta \rangle \right) \\ &\stackrel{(2)}{\geq} f \sum_{i=1}^N (1 - \cos^2 \theta) = Nf \sin^2 \theta. \end{aligned} \tag{1.1}$$

(1) Here ρ_i is the reduced density matrix of pair i in the state $|\psi\rangle_R |\chi\rangle_L$. The expectation value of P_i depends only on ρ_i . **(2)** Because the global state is a product across $R|L$, $\rho_i = \rho_{R_i} \otimes \rho_{L_i}$, which is a mixture of two-qubit product states $|a\rangle |b\rangle$. For each of these, $|\langle \phi_\theta | a b \rangle| = |\cos \theta a_0 b_0 + \sin \theta a_1 b_1| \leq \cos \theta (|a_0| |b_0| + |a_1| |b_1|) \leq \cos \theta$. The first inequality uses $\sin \theta \leq \cos \theta$ for $\theta \leq \pi/4$; the second is the Cauchy–Schwarz inequality for unit vectors. So $\langle \phi_\theta | \rho_i | \phi_\theta \rangle \leq \cos^2 \theta$.

So every unentangled state costs at least $Nf \sin^2 \theta$. There is no shortcut. You cannot remove the entanglement of pair number 573,241 without doing something to pair number 573,241 specifically, and each pair you touch costs a fixed, nonzero amount of energy. As $N \rightarrow \infty$, this cost diverges: *unentangling the R and L subsystems would require changing infinitely many pairs, and so would cost infinite energy*. “Unentangling” has a concrete operational meaning here. It means applying some joint operation to R and L that turns the state into a product of an R -state and an L -state. That is an ordinary (if very large) quantum operation. It simply cannot be done with finite energy in this limit.

Three distinct consequences follow.

1. **Every finite-energy state has infinite entanglement.** Take any state obtained from $|\Phi_\theta\rangle$ by acting on k pairs. The other $N - k$ pairs are untouched. They remain in the state $|\phi_\theta\rangle$, unentangled with the first k pairs. By additivity, the entanglement entropy of such a state is at least $(N - k) s_1(\theta)$, which tends to infinity as $N \rightarrow \infty$. So there is no unentangled, or even weakly entangled, state anywhere in the finite-energy sector. Every physically reachable state is infinitely entangled across the split between R and L .
2. **The Hilbert space itself does not factorize.** Suppose a tensor factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ existed, with the R -operators acting on the first factor and the L -operators on the second. Then product states $|\psi\rangle_R \otimes |\chi\rangle_L$ would exist. They are the simplest vectors you can write down once you have the factorization. But by (1.1), no product state is a finite-energy state. Consider the finite-energy Hilbert space $\mathcal{H}_{\Phi_\theta}$. It is built by taking all finite-energy excitations of $|\Phi_\theta\rangle$ and completing their span. It contains no vectors of the form $|\psi\rangle_R \otimes |\chi\rangle_L$. So $\mathcal{H}_{\Phi_\theta}$ cannot be written as $\mathcal{H}_R \otimes \mathcal{H}_L$ in a way that matches how the R - and L -operators act. This is not a failure to find the right factorization; no such factorization exists. (As an abstract vector space, any infinite-dimensional separable Hilbert space can be written as some tensor product. The point is that no such splitting is compatible with the operators that the R - and L -observers can actually apply.)

There is also a stronger statement, which we quote without proof. Consider the full space obtained from the N -pair system as $N \rightarrow \infty$, before restricting to finite energy. It is not even a *separable* Hilbert space (one with a countable basis). It is a much larger and less tame object, and most of the usual tools of quantum mechanics, which assume separability, do not apply to it directly. Restricting to finite-energy excitations is exactly what picks out a well-behaved separable space $\mathcal{H}_{\Phi_\theta}$ from it.

3. **Different values of θ live in different worlds.** Compare $|\Phi_\theta\rangle$ and $|\Phi_{\theta'}\rangle$ for $\theta \neq \theta'$. They differ on every one of the N pairs, so no operation acting on finitely many pairs turns one into the other. Their overlap is $\langle \Phi_\theta | \Phi_{\theta'} \rangle = \cos^N(\theta - \theta')$, which goes to zero as $N \rightarrow \infty$. Their entanglement entropies differ by $N |s_1(\theta) - s_1(\theta')|$, which diverges. In the strict limit, the finite-energy spaces $\mathcal{H}_{\Phi_\theta}$ and $\mathcal{H}_{\Phi_{\theta'}}$ are best thought of as two separate Hilbert spaces, rather than as two states in one Hilbert space. (In the language of Chapter 2, they carry inequivalent representations of the same algebra of operators.)

To see the size of these numbers directly, here is the $\theta = \pi/4$ case at a few values of N . Each entry follows exactly from additivity of entropy over the N identical pairs, and is shown rounded:

N	1	10	100	10,000
$\dim \mathcal{H}_R^{(N)} = 2^N$	2	1024	$\approx 1.3 \times 10^{30}$	astronomically large
$S_R^{(N)} = N \log 2$	0.693	6.93	69.3	6931

Every entropy entry is just $N \log 2$. Seeing the entropy grow without bound while the dimension of $\mathcal{H}_R^{(N)}$ explodes gives a concrete picture of the abstract statement that no factorized Hilbert space survives the limit.

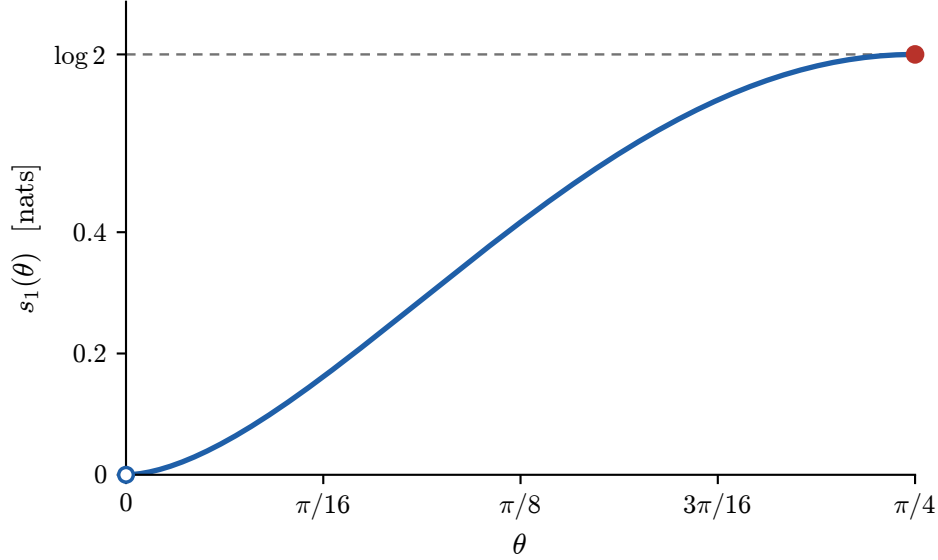


Figure 1.2: The single-pair entanglement entropy $s_1(\theta) = -\cos^2 \theta \log(\cos^2 \theta) - \sin^2 \theta \log(\sin^2 \theta)$ for $\theta \in (0, \pi/4]$. The red dot is the maximally entangled Bell state at $\theta = \pi/4$, where $s_1 = \log 2 \approx 0.693$ (dashed line). The open circle marks the product-state limit $\theta \rightarrow 0$, where the entropy goes smoothly to zero. For N pairs the total entropy is $S_R^{(N)} = N s_1(\theta)$, which tends to infinity as $N \rightarrow \infty$.

1.5 Two more examples, more briefly

Two more examples come back later, so here they are in outline. The second matters most, because it applies to quantum field theory and, ultimately, to AdS/CFT.

Lattice gauge theory. A gauge theory (such as electromagnetism, or the theory of the strong force) has more mathematical variables than physical degrees of freedom. You write down a variable U_{ij} on each link of a lattice. But many different configurations of the U_{ij} describe exactly the same physics. They are related by *gauge transformations*, $U_{ij} \rightarrow V_i U_{ij} V_j^\dagger$. Physical states are required to be invariant under these transformations. This requirement also breaks the naive tensor factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$, by a mechanism quite different from infinite entanglement. The gauge-invariant states do not, in general, lie in a product of an R -Hilbert space and an L -Hilbert space, because gauge invariance ties the two sides together. This example comes back in Chapter 3 as a comparatively mild complication. It adds a classical label (which gauge or superselection sector you are in) on top of an otherwise ordinary tensor-product structure. It is not the deep, qualitative breakdown seen in the spin-pair chain and in field theory.

A quantum field theory cut in half. Take a relativistic quantum field theory, say a scalar field, and split space into two halves, $R = \{x > 0\}$ and $L = \{x < 0\}$. A quantum field has infinitely many degrees of freedom in any interval of space, however small; informally, it has a value (or a mode) at every point. Near the cutting surface $x = 0$, infinitely many field degrees of freedom on the R side sit arbitrarily close to infinitely many on the L side. Each is coupled to its neighbors. This produces infinite entanglement across the cut in *every* finite-energy state, including the vacuum, the lowest-energy state of all.

This is the field-theory version of the mechanism in the spin-pair chain: infinitely many separate contributions to the entanglement, each small, adding up to an infinite total. It shows up as a divergence. Compute the entanglement entropy of the region R with a short-distance cutoff ϵ . (Imagine putting the theory on a lattice with spacing ϵ . This makes everything finite, and we then ask what happens as $\epsilon \rightarrow 0$.) For $d \geq 3$ spacetime dimensions the result is

$$S_R(\epsilon) = b \frac{\text{Area}(\partial R)}{\epsilon^{d-2}} + \cdots, \quad \epsilon \rightarrow 0.$$

Here ∂R is the surface separating R from L , and b is a positive constant that depends on the theory and on how the cutoff is imposed. (In $d = 2$ the divergence is logarithmic instead.) The entropy diverges as the cutoff is removed. It diverges in a specific way: in proportion to the *area* of the cutting surface, not the volume of the region. This “area-law divergence” is one of the most robust facts in quantum field theory. It appears in essentially every explicit calculation, for free and interacting fields alike. It is the field-theory counterpart of the non-factorization seen in the spin-pair example. The next section gives an explicit energy argument for it, and Chapter 4 states it precisely in the language of von Neumann algebra types.

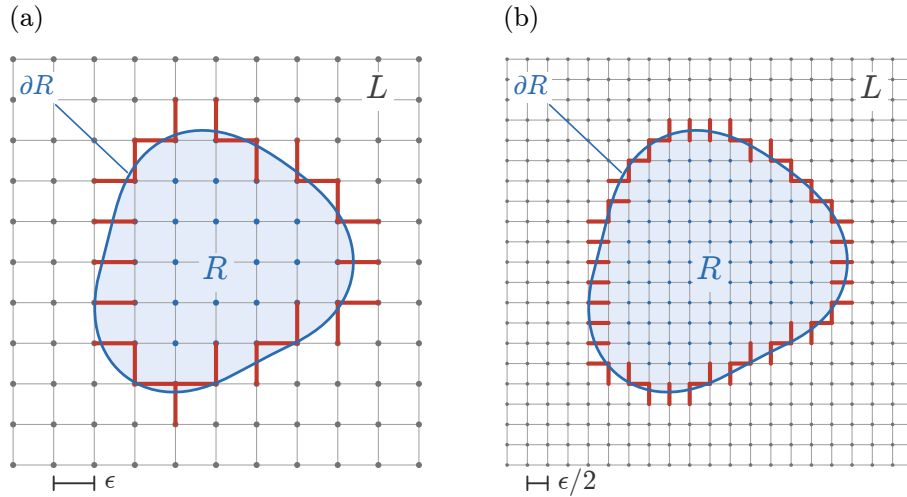


Figure 1.3: Why entanglement across a cut scales with its area. (a) A slice of space is replaced by a lattice of spacing ϵ , and the region R (shaded) is separated from L by the surface ∂R . The red links are the nearest-neighbour couplings that cross ∂R ; each one entangles a short-distance mode of R with one of L . (b) Halving the spacing to $\epsilon/2$ doubles the number of red links, so the entropy grows as $\text{Area}(\partial R)/\epsilon^{d-2}$ and diverges as $\epsilon \rightarrow 0$.

1.6 Explicit derivation: why the continuum QFT Hilbert space cannot factorize

The claim that $\mathcal{H} \neq \mathcal{H}_R \otimes \mathcal{H}_L$ in continuum quantum field theory underlies everything that follows. This section shows where it comes from, using nothing more advanced than coupled harmonic oscillators. The argument below is a physical estimate, not a proof. It shows that an unentangled

state across the cut would cost an energy that diverges as the cutoff is removed. The rigorous version of the statement is quoted at the end of the section.

1. The Hamiltonian and the gradient coupling across the cut

Consider a free, real scalar field $\phi(t, \vec{x})$ of mass m in d spacetime dimensions, that is, in $d - 1$ spatial dimensions. On a slice of constant time, the Hamiltonian is

$$H = \int d^{d-1}x \left[\frac{1}{2} \pi(\vec{x})^2 + \frac{1}{2} (\vec{\nabla} \phi(\vec{x}))^2 + \frac{1}{2} m^2 \phi(\vec{x})^2 \right]. \quad (1.2)$$

Here $\pi(\vec{x})$ is the canonical momentum field, with $[\phi(\vec{x}), \pi(\vec{y})] = i\delta^{(d-1)}(\vec{x} - \vec{y})$.

Divide space into two halves along the plane $x = 0$:

$$R = \{(x, \vec{x}_\perp) : x > 0\}, \quad L = \{(x, \vec{x}_\perp) : x < 0\}.$$

Here $\vec{x}_\perp = (x^2, \dots, x^{d-1})$ are the $d - 2$ spatial coordinates along the dividing surface ∂R . This surface is called the *entangling surface*.

To focus on the physics near the cut, we treat the two kinds of directions differently. We put the perpendicular direction x on a lattice with spacing ϵ . We Fourier transform the transverse coordinates $\vec{x}_\perp$ into transverse momenta $\vec{k}_\perp$:

$$\phi(x, \vec{x}_\perp) = \int \frac{d^{d-2}k_\perp}{(2\pi)^{d-2}} \tilde{\phi}(x, \vec{k}_\perp) e^{i\vec{k}_\perp \cdot \vec{x}_\perp}.$$

The Hamiltonian is quadratic, so different transverse momenta do not mix. For each $\vec{k}_\perp$, the Hamiltonian becomes an independent one-dimensional chain of coupled harmonic oscillators. The transverse gradient $|\vec{k}_\perp|^2$ adds to the mass term, giving the effective mass

$$M^2 \equiv m^2 + |\vec{k}_\perp|^2. \quad (1.3)$$

(Strictly, the Fourier modes are complex, with $\tilde{\phi}(x, -\vec{k}_\perp) = \tilde{\phi}(x, \vec{k}_\perp)^*$. We treat each mode as a real oscillator. This simplification does not affect the energy estimates below.) On the lattice, the field becomes a set of site variables $\phi_j \equiv \tilde{\phi}(j\epsilon, \vec{k}_\perp)$ with conjugate momenta π_j . Sites $j = \dots, -2, -1$ lie in L , and sites $j = 0, 1, 2, \dots$ lie in R .

The gradient in the direction perpendicular to the cut is replaced by a finite difference:

$$\int dx \frac{1}{2} \left(\frac{\partial \phi}{\partial x} \right)^2 \rightarrow \sum_j \frac{\epsilon}{2} \left(\frac{\phi_{j+1} - \phi_j}{\epsilon} \right)^2 = \sum_j \frac{1}{2\epsilon} (\phi_{j+1} - \phi_j)^2. \quad (1.4)$$

The lattice momenta obey $[\phi_j, \pi_k] = i\delta_{jk}/\epsilon$, because the delta function becomes δ_{jk}/ϵ on the lattice. It is convenient to rescale to canonical variables, $\phi_j \rightarrow \sqrt{\epsilon} \phi_j$ and $\pi_j \rightarrow \sqrt{\epsilon} \pi_j$, which obey $[\phi_j, \pi_k] = i\delta_{jk}$. In these variables the chain Hamiltonian for one transverse mode is

$$H_{\vec{k}_\perp} = \sum_j \left[\frac{1}{2} \pi_j^2 + \frac{1}{2} M^2 \phi_j^2 + \frac{1}{2\epsilon^2} (\phi_{j+1} - \phi_j)^2 \right]. \quad (1.5)$$

One link in this sum straddles the cut. It connects site $j = -1$ (the closest site in L) to site $j = 0$ (the closest site in R):

$$H_{\text{cut}} = \frac{1}{2\epsilon^2}(\phi_0 - \phi_{-1})^2 = \frac{1}{2\epsilon^2}(\phi_R - \phi_L)^2, \quad (1.6)$$

where we write $\phi_R \equiv \phi_0$ and $\phi_L \equiv \phi_{-1}$ for short. This single term is the only coupling between R and L for this mode.

2. Two oscillators across the cut

To see the mechanism as simply as possible, keep only the two sites next to the cut, and drop their links to the rest of the chain. This is a crude truncation, but it keeps the essential feature: two oscillators, one on each side, coupled by H_{cut} . (A numerical check of the full chain is given below.) The Hamiltonian of the two oscillators is

$$H_{\text{pair}} = \frac{1}{2}\pi_R^2 + \frac{1}{2}\pi_L^2 + \frac{1}{2}M^2(\phi_R^2 + \phi_L^2) + \frac{1}{2\epsilon^2}(\phi_R - \phi_L)^2. \quad (1.7)$$

This system is quadratic, so it can be solved exactly using normal-mode coordinates:

$$\phi_+ \equiv \frac{\phi_R + \phi_L}{\sqrt{2}}, \quad \phi_- \equiv \frac{\phi_R - \phi_L}{\sqrt{2}}, \quad (1.8)$$

with $\pi_{\pm}$ defined in the same way. This change of variables is a rotation, so $\phi_R^2 + \phi_L^2 = \phi_+^2 + \phi_-^2$ and $\pi_R^2 + \pi_L^2 = \pi_+^2 + \pi_-^2$, while the coupling term becomes $(\phi_R - \phi_L)^2/(2\epsilon^2) = \phi_-^2/\epsilon^2$. The Hamiltonian therefore separates into two independent harmonic oscillators:

$$H_{\text{pair}} = \left(\frac{1}{2}\pi_+^2 + \frac{1}{2}\omega_+^2\phi_+^2 \right) + \left(\frac{1}{2}\pi_-^2 + \frac{1}{2}\omega_-^2\phi_-^2 \right), \quad (1.9)$$

with normal-mode frequencies

$$\omega_+ = M = \sqrt{m^2 + |\vec{k}_{\perp}|^2}, \quad (1.10)$$

$$\begin{aligned} \omega_- &= \sqrt{M^2 + \frac{2}{\epsilon^2}} = \sqrt{m^2 + |\vec{k}_{\perp}|^2 + \frac{2}{\epsilon^2}} \\ &\approx \frac{\sqrt{2}}{\epsilon} \quad \text{as } \epsilon \rightarrow 0. \end{aligned} \quad (1.11)$$

The “sum” mode ϕ_+ feels only the mass. The “difference” mode ϕ_- is stretched by the stiff link across the cut, so its frequency is very high.

3. The vacuum: neighbours across the cut move together

The ground state of (1.9) is a product of two Gaussian ground states in the normal coordinates:

$$\Psi_0(\phi_+, \phi_-) = \left(\frac{\omega_+ \omega_-}{\pi^2} \right)^{1/4} \exp \left[-\frac{1}{2}\omega_+\phi_+^2 - \frac{1}{2}\omega_-\phi_-^2 \right]. \quad (1.12)$$

Its energy is the sum of the two zero-point energies,

$$E_{\text{vac}} = \frac{1}{2}(\omega_+ + \omega_-). \quad (1.13)$$

Transforming back to the local fields ϕ_R, ϕ_L using (1.8) gives

$$\begin{aligned} \Psi_0(\phi_R, \phi_L) = \left(\frac{\omega_+ \omega_-}{\pi^2} \right)^{1/4} \exp \left[-\frac{1}{4}(\omega_+ + \omega_-)(\phi_R^2 + \phi_L^2) \right. \\ \left. - \frac{1}{2}(\omega_+ - \omega_-)\phi_R \phi_L \right]. \end{aligned} \quad (1.14)$$

The exponent contains the cross term $\frac{1}{2}(\omega_- - \omega_+)\phi_R \phi_L$. Because of this term, the wavefunction is not a product of a function of ϕ_R and a function of ϕ_L : the vacuum is entangled across the cut. The entanglement is strong when $\omega_- \approx \sqrt{2}/\epsilon \gg \omega_+$.

A good way to see how strong it is: compute the correlations. Using $\langle \phi_{\pm}^2 \rangle = 1/(2\omega_{\pm})$ and $\phi_{R,L} = (\phi_+ \pm \phi_-)/\sqrt{2}$,

$$\langle \phi_R^2 \rangle_{\Omega} = \frac{1}{4} \left(\frac{1}{\omega_+} + \frac{1}{\omega_-} \right), \quad \langle \phi_R \phi_L \rangle_{\Omega} = \frac{1}{4} \left(\frac{1}{\omega_+} - \frac{1}{\omega_-} \right).$$

The correlation coefficient $\langle \phi_R \phi_L \rangle / \langle \phi_R^2 \rangle = (\omega_- - \omega_+)/(\omega_- + \omega_+)$ tends to 1 as $\epsilon \rightarrow 0$. The two neighbouring field values fluctuate by large amounts, of order $1/\sqrt{M}$, but they fluctuate almost exactly together. In the vacuum $|\Omega\rangle$ this has two consequences.

- The difference between the two sides is small:

$$\langle \Omega | (\phi_R - \phi_L)^2 | \Omega \rangle \stackrel{(1)}{=} 2 \langle \Omega | \phi_-^2 | \Omega \rangle \stackrel{(2)}{=} \frac{2}{2\omega_-} = \frac{1}{\omega_-} \stackrel{(3)}{=} \frac{\epsilon}{\sqrt{2}} + O(\epsilon^3). \quad (1.15)$$

(1) $\phi_R - \phi_L = \sqrt{2}\phi_-$, from the normal-mode definition. **(2)** A harmonic-oscillator ground state of frequency ω has $\langle \phi^2 \rangle = 1/(2\omega)$. **(3)** Expand $1/\omega_- = (\epsilon/\sqrt{2})(1 + M^2\epsilon^2/2)^{-1/2}$ for small ϵ .

- The energy stored in the link across the cut is therefore

$$\langle \Omega | H_{\text{cut}} | \Omega \rangle = \frac{1}{2\epsilon^2} \langle \Omega | (\phi_R - \phi_L)^2 | \Omega \rangle = \frac{1}{2\epsilon^2} \frac{1}{\omega_-} \approx \frac{1}{2\sqrt{2}\epsilon}. \quad (1.16)$$

This is part of the zero-point energy of the vacuum. The vacuum has it anyway, and the energies of other states are measured relative to the vacuum energy (1.13).

4. The energy cost of an unentangled state

Now suppose, for the sake of argument, that the Hilbert space did factorize across the cut:

$$\mathcal{H} \stackrel{?}{=} \mathcal{H}_L \otimes \mathcal{H}_R.$$

Then the Hilbert space would contain unentangled product states,

$$|\Psi_{\text{prod}}\rangle = |\psi_L\rangle \otimes |\chi_R\rangle \in \mathcal{H}, \quad (1.17)$$

with $|\psi_L\rangle \in \mathcal{H}_L$ and $|\chi_R\rangle \in \mathcal{H}_R$. How much energy would such a state have?

In a product state, the expectation value of a product of an L -operator and an R -operator factorizes:

$$\langle \Psi_{\text{prod}} | \phi_R \phi_L | \Psi_{\text{prod}} \rangle = \langle \chi_R | \phi_R | \chi_R \rangle \langle \psi_L | \phi_L | \psi_L \rangle. \quad (1.18)$$

In other words, the fluctuations of ϕ_R and ϕ_L are completely uncorrelated. For the difference of the two fields this gives

$$\begin{aligned} \langle \Psi_{\text{prod}} | (\phi_R - \phi_L)^2 | \Psi_{\text{prod}} \rangle &\stackrel{(1)}{=} \langle \phi_R^2 \rangle_{\chi_R} + \langle \phi_L^2 \rangle_{\psi_L} - 2 \langle \phi_R \rangle_{\chi_R} \langle \phi_L \rangle_{\psi_L} \\ &\stackrel{(2)}{=} (\Delta\phi_R)^2 + (\Delta\phi_L)^2 + (\langle \phi_R \rangle - \langle \phi_L \rangle)^2. \end{aligned} \quad (1.19)$$

(1) Expand the square and factorize the cross term using (1.18). **(2)** Write $\langle \phi^2 \rangle = (\Delta\phi)^2 + \langle \phi \rangle^2$, where $(\Delta\phi)^2$ is the variance, and collect the mean values into a square.

This shows the difficulty. In the vacuum, $\phi_R - \phi_L$ is kept small because the two sides move together. In a product state they cannot move together, so the only way to keep $\phi_R - \phi_L$ small is to make each variance $(\Delta\phi_R)^2$ and $(\Delta\phi_L)^2$ small separately. But by the uncertainty principle, $\Delta\phi \Delta\pi \geq \frac{1}{2}$, so squeezing the field fluctuations increases the momentum fluctuations and hence the kinetic energy. The product state must pay one way or the other.

We can find the exact minimum. Define the frequency

$$\Omega_\epsilon \equiv \sqrt{M^2 + \frac{1}{\epsilon^2}}$$

and the single-site Hamiltonians $h_R = \frac{1}{2}\pi_R^2 + \frac{1}{2}\Omega_\epsilon^2\phi_R^2$ and $h_L = \frac{1}{2}\pi_L^2 + \frac{1}{2}\Omega_\epsilon^2\phi_L^2$. Write $a = \langle \phi_R \rangle$ and $b = \langle \phi_L \rangle$. Then, for any product state,

$$\begin{aligned} \langle H_{\text{pair}} \rangle_{\text{prod}} &\stackrel{(1)}{=} \langle h_R \rangle_{\chi_R} + \langle h_L \rangle_{\psi_L} - \frac{ab}{\epsilon^2} \\ &\stackrel{(2)}{\geq} \left(\frac{1}{2}\Omega_\epsilon + \frac{1}{2}\Omega_\epsilon^2 a^2 \right) + \left(\frac{1}{2}\Omega_\epsilon + \frac{1}{2}\Omega_\epsilon^2 b^2 \right) - \frac{ab}{\epsilon^2} \\ &\stackrel{(3)}{\geq} \Omega_\epsilon + \frac{1}{2} \left(\Omega_\epsilon^2 - \frac{1}{\epsilon^2} \right) (a^2 + b^2) \\ &\stackrel{(4)}{=} \Omega_\epsilon + \frac{1}{2} M^2 (a^2 + b^2) \geq \Omega_\epsilon. \end{aligned} \quad (1.20)$$

(1) Expand $(\phi_R - \phi_L)^2$ in (1.7). The ϕ_R^2 and ϕ_L^2 pieces raise M^2 to Ω_ϵ^2 . The cross term factorizes by (1.18). **(2)** Shift ϕ_R and π_R by their mean values. Then $\langle h_R \rangle$ splits into $\frac{1}{2}\langle \pi_R \rangle^2 + \frac{1}{2}\Omega_\epsilon^2 a^2$ plus the energy of a state with zero mean field and momentum. That last energy is at least the ground-state energy $\frac{1}{2}\Omega_\epsilon$ of an oscillator of frequency Ω_ϵ . Finally drop $\frac{1}{2}\langle \pi_R \rangle^2 \geq 0$. The same holds for h_L . **(3)** Use $ab \leq \frac{1}{2}(a^2 + b^2)$. **(4)** $\Omega_\epsilon^2 - 1/\epsilon^2 = M^2$.

The bound is reached by the product of the ground states of h_R and h_L . It also holds for mixed

product states $\rho_R \otimes \rho_L$, since these are mixtures of pure product states. In that best product state, each field is squeezed to a small spread, of order ϵ , instead of the large vacuum spread of order $1/M$:

$$\langle \phi_R^2 \rangle_{\text{prod, best}} = \frac{1}{2\Omega_\epsilon} \approx \frac{\epsilon}{2}, \quad \text{compared with} \quad \langle \phi_R^2 \rangle_\Omega = \frac{1}{4} \left(\frac{1}{\omega_+} + \frac{1}{\omega_-} \right) \approx \frac{1}{4M}. \quad (1.21)$$

Subtracting the vacuum energy (1.13) gives the minimum extra energy needed to remove the entanglement of this one transverse mode across the cut:

$$\begin{aligned} \Delta E_{\text{cut}}(\vec{k}_\perp) &\equiv \min_{\text{prod}} \langle H_{\text{pair}} \rangle - E_{\text{vac}} = \Omega_\epsilon - \frac{1}{2}(\omega_+ + \omega_-) \\ &= \frac{1}{\epsilon} \left[\sqrt{1+u^2} - \frac{1}{2}(u + \sqrt{u^2+2}) \right], \quad u \equiv M\epsilon. \end{aligned} \quad (1.22)$$

This is strictly positive. Indeed, Ω_ϵ is the square root of the average of $\omega_+^2 = M^2$ and $\omega_-^2 = M^2 + 2/\epsilon^2$, while $\frac{1}{2}(\omega_+ + \omega_-)$ is the average of their square roots. The square root is strictly concave, so the second is always smaller. For $u \rightarrow 0$ (a transverse mode much lighter than the cutoff), the bracket equals $1 - 1/\sqrt{2} \approx 0.29$. At $u = 1$ it equals $\sqrt{2} - \frac{1}{2}(1 + \sqrt{3}) \approx 0.048$, and a quick numerical check shows it decreases steadily in between. So every transverse mode with $M \lesssim 1/\epsilon$ costs an energy of order $1/\epsilon$.

Check on the full chain. The two-site truncation is crude, so we checked the result numerically on the full chain (1.5). Because the Hamiltonian is quadratic, the minimum energy over product states can be computed exactly: it is the sum of the ground-state energies of the two half-chains, with the cut link replaced by the on-site terms $\phi_R^2/(2\epsilon^2)$ and $\phi_L^2/(2\epsilon^2)$. For a chain of 800 sites, the extra energy over the vacuum comes out as about $0.14/\epsilon$ for $M = 0$ and about $0.030/\epsilon$ for $M = 1/\epsilon$. These are roughly half to two-thirds of the two-site values, with the same $1/\epsilon$ scaling. So the truncation changes the numerical constant but not the conclusion.

5. Summing over the entangling surface

To remove the entanglement across the whole surface ∂R , every transverse mode must be disentangled. The energy is a sum over transverse modes, and in a product state across the cut, the reduced state of each mode is itself a (possibly mixed) product state. So the bound (1.22) applies to every mode separately. The number of transverse modes per unit area, with momenta in $d^{d-2}k_\perp$, is $d^{d-2}k_\perp/(2\pi)^{d-2}$. We keep the modes with $|\vec{k}_\perp| \leq 1/\epsilon$, the same cutoff as in the perpendicular direction, and assume the mass is small compared with the cutoff, $m\epsilon \ll 1$. Let $c_{\min} > 0$ be the smallest value of the bracket in (1.22) over this range of u ; it is about 0.048. Then

$$\begin{aligned} \Delta E_{\text{prod}} &\stackrel{(1)}{=} \text{Area}(\partial R) \int_{|\vec{k}_\perp| \leq 1/\epsilon} \frac{d^{d-2}k_\perp}{(2\pi)^{d-2}} \Delta E_{\text{cut}}(\vec{k}_\perp) \\ &\stackrel{(2)}{\geq} \text{Area}(\partial R) \frac{c_{\min}}{\epsilon} \int_{|\vec{k}_\perp| \leq 1/\epsilon} \frac{d^{d-2}k_\perp}{(2\pi)^{d-2}} \\ &\stackrel{(3)}{=} \text{Area}(\partial R) \frac{c_{\min}}{\epsilon} \frac{V_{d-2}}{(2\pi)^{d-2}} \frac{1}{\epsilon^{d-2}}. \end{aligned} \quad (1.23)$$

(1) The transverse modes are independent, so their minimum excitation energies add. (2) Each mode costs at least $c_{\min}/\epsilon$, by (1.22). (3) The integral is the volume of a ball of radius $1/\epsilon$ in $d-2$ dimensions, divided by $(2\pi)^{d-2}$. Here V_{d-2} is the volume of the unit ball in $d-2$ dimensions (with $V_0 = 1$).

Collecting the constants into $C = c_{\min}V_{d-2}/(2\pi)^{d-2}$ gives the main result of this section.

Key Result / Core Concept

$$\Delta E_{\text{prod}} \geq C \cdot \frac{\text{Area}(\partial R)}{\epsilon^{d-1}} \xrightarrow{\epsilon \rightarrow 0} +\infty, \quad (1.24)$$

where $C > 0$ is a positive constant of order one that depends on the dimension (and, through the full-chain correction, on the details of the lattice).

The scaling has a simple interpretation. Disentangling the two sides costs an energy density of order the cutoff scale, $1/\epsilon^d$, in a layer of thickness ϵ around the entangling surface.

Here is what (1.24) shows, and what it does not show.

1. **An infinite energy barrier.** The energy needed to disentangle a region R from its complement L grows like $\epsilon^{-(d-1)}$ and diverges in the continuum limit. A product state would need to break the correlations between neighbouring field values at every point of ∂R , and this costs an energy of order the cutoff scale per mode.
2. **No product state has finite energy.** The physical Hilbert space of a continuum field theory, the vacuum sector, is built from finite-energy excitations of the vacuum. (Mathematically, it is the Fock space, or the GNS space of the vacuum.) Since $\Delta E_{\text{prod}} = +\infty$, no product state across the cut is one of these finite-energy states.
3. **Failure of factorization.** A tensor product $\mathcal{H}_L \otimes \mathcal{H}_R$ is, by definition, spanned by product vectors $|\psi_L\rangle |\chi_R\rangle$. The energy estimate therefore makes the following statement very plausible:

$$\boxed{\mathcal{H} \neq \mathcal{H}_L \otimes \mathcal{H}_R \quad \text{in any continuum relativistic quantum field theory.}} \quad (1.25)$$

Here the factorization is meant in the physical sense: the L -fields act on the first factor and the R -fields on the second. The energy argument alone is not a proof, because it only rules out product states of finite energy. The rigorous statement is a theorem of algebraic quantum field theory, which we quote here and discuss in Chapter 4: the algebra of observables of a region like R is a von Neumann algebra of type III_1 , and an algebra of type III can never be of the form $B(\mathcal{H}_R) \otimes \mathbf{1}$ for any tensor factorization.

6. The algebraic version: type III_1 and the split property

The energy divergence has a precise algebraic counterpart, which we quote without proof.

- Suppose two spatial regions R_1 and R_2 are separated by a **finite buffer** of width $\delta > 0$, so that $\text{dist}(R_1, R_2) = \delta$. Then the gradient energy across the gap is controlled by δ rather than

by the cutoff ϵ . In this case a state that is unentangled between R_1 and R_2 *can* be prepared. Heuristically, the estimate above with ϵ replaced by δ suggests that its energy cost is finite but grows without bound as $\delta \rightarrow 0$. The algebraic statement is the **split property** of quantum field theory (established under mild assumptions by Buchholz, Doplicher, Longo, Wichmann, and others). It says that there is an intermediate type I factor $\mathcal{N}$ with $\mathcal{M}(R_1) \subset \mathcal{N} \subset \mathcal{M}(R_2)'$. Because $\mathcal{N}$ is a type I factor, it provides a tensor factorization $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ with $\mathcal{N} = B(\mathcal{H}_1) \otimes \mathbf{1}$. The operators of R_1 act on the first factor, and those of R_2 act on the second.

- As the buffer shrinks to zero ($\delta \rightarrow 0$, so that the regions touch along a common boundary), this intermediate type I factor ceases to exist, and the estimated energy cost diverges. What remains is the algebra $\mathcal{M}(R)$ of the region itself, which is not type I or type II. It is a **type III₁ factor**, the most strongly entangled type in the classification (Chapter 4).

1.7 The fix: define a subsystem by what you can *do*, not by how you'd *split the Hilbert space*

The way out is simple to state before any formalism is attached to it. Do not define a subsystem R by declaring a factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ in advance. Instead, define it operationally: *R is whatever an observer confined to that subsystem can measure or do*. Concretely, that observer has access to some collection of operators $\mathcal{M} \subset B(\mathcal{H})$, where $B(\mathcal{H})$ is the set of all bounded operators on the full Hilbert space $\mathcal{H}$. The collection $\mathcal{M}$ contains every self-adjoint combination of things the observer's apparatus can measure, and every unitary transformation it can apply. This definition does not require $\mathcal{H}$ to factorize. It only requires that you can say, of any given operator, whether or not it belongs to the observer's toolkit.

This reframing costs nothing in the ordinary case. When $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ does exist, the natural choice is $\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L$, the set of every operator acting only on R . This recovers the familiar picture exactly, as Chapter 2 makes precise. The reframing also works in the broken case, because “the set of operators an R -observer can apply” is a perfectly sensible notion even when there is no factorization for it to come from. You lose nothing, and you gain a definition that survives exactly where the old one fails.

What kind of mathematical object should $\mathcal{M}$ be? It should contain the identity, and be closed under sums, products, and taking the Hermitian conjugate. It should also be closed under some notion of limit. An object with these properties is a **von Neumann algebra**. There are two main candidate notions of “limit,” which turn out to be inequivalent, and the difference between them matters a great deal. Pinning down the right one is the subject of Chapter 2.

Historically, this is not a new invention grafted onto quantum mechanics from outside. John von Neumann was one of the people who built the mathematical foundations of quantum mechanics in the first place. He introduced these algebras in 1929, and in a series of papers from 1936 to 1943 he and Francis Murray classified them into types. For decades this classification lived mostly inside pure mathematics. The modern insight builds on work going back to Rehren around 2000, and has been developed intensively since about 2021 by Leutheusser, Liu, Witten, and others [1]. It is that this same classification, applied to the algebra $\mathcal{M}$ of an observer's accessible operators, tracks how

entangled that observer’s subsystem is with everything the observer cannot access. In this way a piece of abstract functional analysis from the 1930s becomes a working tool for quantum gravity.

1.8 Two organizing claims

Two claims organize everything that follows. The first is:

$$\textit{Classifying von Neumann algebras} \longleftrightarrow \textit{classifying broad types of entanglement}.$$

The second is that finer tools from the same theory give much sharper information about entanglement than the broad type alone. These tools are modular theory (Chapter 4) and the crossed product (Chapter 5).

At this point these claims are *stated*, not yet demonstrated. The double-headed arrow above is not a formal theorem with a name and a citation. It is a dictionary, built up entry by entry over Chapters 3 and 4. Each von Neumann algebra type (I, then II, then the subtypes of III) is matched, on specific worked examples, to a specific qualitative pattern of entanglement between a subsystem and its complement. By the end of Chapter 4 the correspondence will have been shown concretely, on examples you can check by hand.

Similarly, this chapter only motivates the deeper claim that entanglement structure is what *builds* bulk spacetime geometry. The motivation comes from the Ryu–Takayanagi formula and from the broader “entanglement builds geometry” literature (Van Raamsdonk [5], Maldacena–Susskind [6]). The actual mechanism is subregion-subalgebra duality: the statement that a boundary algebra of a specific type is identical to, not merely related to, the algebra of observables of a bulk causal region. That mechanism is developed in Chapter 7. This chapter is a map of where we are going and why we are going there.

1.9 Plan of the notes

The notes split into two halves. Where the split falls says something about how the argument is built.

Chapters 2–5: the mathematics, with no gravity or holography. This half is pure operator-algebra theory. It is developed using only examples from ordinary quantum mechanics, mostly the entangled spin-pair system above, which are simple enough to work out by hand. Chapter 2 explains what a von Neumann algebra is and how such algebras are classified. It also shows how to build a Hilbert space from an algebra and a state (the GNS construction). Chapters 3 and 4 show, type by type, how the classification tracks entanglement structure. Chapter 3 treats the “easy” types, I and II, for which a density matrix and an entropy can still be defined. Chapter 4 treats type III, for which no density matrix exists at all. Saying anything quantitative about type III needs a new tool, Tomita–Takesaki modular theory. Chapter 5 introduces the *crossed product*. This construction takes a type III algebra, attaches an auxiliary quantum clock to it, and produces a new, better-behaved algebra of type II. At that stage it is a purely algebraic device. In Chapter 9 it turns out to describe what a physical observer carrying a physical clock does to gravitational observables.

Chapters 6–10: applying the machinery to quantum gravity. Chapter 6 sets up the large- N algebraic formulation of AdS/CFT. Chapter 7 introduces subregion-subalgebra duality, the central physics claim: a bulk spacetime region and a specific boundary operator algebra are the same object, described in two languages. It uses this duality to restate entanglement-wedge and causal-wedge reconstruction (ideas you may have met in AdS/CFT courses) in algebraic language. Chapter 8 shows how bulk causal structure and horizons can be read off from the type and commutant structure of boundary algebras, with no bulk metric assumed. The same tools even decide whether two disconnected boundary theories are joined by a wormhole. Chapter 9 builds simple, solvable toy models of observers with physical clocks, using the crossed product of Chapter 5. These models reproduce the generalized entropy of black holes and of de Sitter space (up to an additive constant) from the operator-algebraic machinery alone. Chapter 10 summarizes, and discusses what all this suggests about the mathematical structure of quantum gravity.

1.10 Conventions and notation

Each item below is used many times later without further comment. So each one is explained here rather than just listed.

- **Large N versus finite N .** “Large N ” (equivalently $G_N \rightarrow 0$) means working in a perturbative expansion in powers of $1/N$. You may keep more than the leading term, but $1/N$ is still treated as a small expansion parameter, just as in ordinary perturbation theory in quantum mechanics. “Finite N ” (finite G_N) means the opposite. It means treating N (or G_N) as a fixed number, with no expansion at all. This is the nonperturbative regime of full quantum gravity. Most of these notes work at large N . Finite- N questions are the hardest open problems, and they are discussed mainly at the ends of Chapters 7 and 9.
- **$\mathcal{H}$, $B(\mathcal{H})$, and boundedness.** $\mathcal{H}$ is a Hilbert space: a complex vector space with an inner product, complete in the norm defined by that inner product. This is the same notion you know from ordinary quantum mechanics, except that it may be infinite-dimensional. $B(\mathcal{H})$ is the set of *bounded* operators on $\mathcal{H}$. An operator A is bounded if there is some finite number c such that $\|A|\psi\rangle\| \leq c\|\psi\rangle\|$ for every vector $|\psi\rangle \in \mathcal{H}$. Informally, A never stretches the length of a vector by more than a fixed factor, whatever vector you feed it. Finite matrices are automatically bounded.

Some familiar operators are *not* bounded. Position $\hat{x}$ and momentum $\hat{p}$ for a particle on a line are examples. The ratio $\|\hat{x}|\psi\rangle\|/\|\psi\rangle\|$ can be made as large as you like by choosing $|\psi\rangle$ concentrated far from the origin. Similarly, $\|\hat{p}|\psi\rangle\|/\|\psi\rangle\|$ can be made large by choosing $|\psi\rangle$ that oscillates rapidly. We avoid this issue by working with bounded operators throughout. For an unbounded observable, one uses its bounded functions instead. For example, $e^{i\hat{x}}$ is bounded even though $\hat{x}$ is not. In field theory one uses bounded functions of smeared fields in the same way. No physical information is lost, because an unbounded observable is determined by its bounded functions (for instance, by its spectral projections). This convention lets the norm-based definitions of Chapter 2 work cleanly.

- **1** is the identity operator: “do nothing.”

- **$\tilde{J}^\pm(Y)$: causal future and past.** For a region Y of a spacetime, $\tilde{J}^+(Y)$ is the set of all points that can be reached from some point of Y by a future-directed causal curve. A causal curve is a path that never moves faster than light. Physically, $\tilde{J}^+(Y)$ is everything that Y could send a signal to, given unlimited time. $\tilde{J}^-(Y)$ is the mirror image: everything that could have sent a signal to some point of Y .

The tilde means that these sets are computed in the *conformal completion* of the spacetime. This is a technical device, standard in AdS/CFT, that adds a boundary “at infinity,” so that causal curves reaching arbitrarily far away are still tracked properly. For now you can picture the ordinary causal future and past. The tilde starts to matter when the conformal completion appears, in Chapter 6.

- **Causal complement (single prime) and causal completion (double prime) of a spacetime region.** The causal complement Y' of a region Y is the set of all points spacelike separated from every point of Y . These are the points that can neither send a signal to, nor receive a signal from, any point of Y . Loosely, Y' is “everything that Y has no causal contact with, in either direction.”

The causal completion Y'' is the double complement, $(Y')'$. In general Y'' is *larger* than Y . This is a real geometric fact, not a notational curiosity. Y'' is the largest region that has the same causal complement as Y . For instance, the causal completion of a single point is just that point, so nothing is added. But the causal completion of a spatial region R on a Cauchy slice is its entire *domain of dependence* (defined next). This is a bigger set, which extends into the past and the future.

The double-prime notation for spacetime regions deliberately echoes the double-prime notation for commutants of operator algebras $(\mathcal{M}'')$, met in Chapter 2. Once subregion-subalgebra duality is established in Chapter 7, the causal completion of a region and the double commutant of an algebra turn out to be two faces of the same fact.

- **$\bar{R}$, complement of a spatial region; $\hat{R}$, domain of dependence.** Let R be a subregion of a single spatial slice (a snapshot of space at one moment). Then $\bar{R}$ is its ordinary set-complement on that slice: everything on the same slice that is not in R . The domain of dependence $\hat{R}$ is a spacetime region built from R . It is the set of spacetime points p such that *every* causal curve through p , extended as far as it can go in both directions, must cross R . Physics inside $\hat{R}$ is completely determined by initial data given on R alone. That is why it is called the domain of *dependence*. If R is a full spatial slice, $\hat{R}$ is the entire spacetime. If R is a proper subregion, $\hat{R}$ is a diamond-shaped region, called a *causal diamond*. It sits above and below R and is bounded by the light rays that just graze the edge of R . This shape appears constantly from Chapter 6 onward.
- **$\subset$ always means *proper subset*.** Whenever you see $\mathcal{M}_1 \subset \mathcal{M}_2$ in these notes, it means that $\mathcal{M}_1$ is strictly smaller than $\mathcal{M}_2$. Some other texts use the same symbol with the more permissive meaning that allows equality; these notes do not.

With this vocabulary in hand (bounded operators, causal future and past, causal complement and completion, domain of dependence), you can read the geometric statements of later chapters literally, rather than by matching them to what looks familiar. Chapter 2 now builds the algebraic machinery

that this chapter has been pointing toward: C^* and von Neumann algebras, states, projections, the type classification, and the GNS construction that produces a Hilbert space from an algebra and a state.

CHAPTER 2

Introduction to von Neumann algebras

Chapter 1 ended with a proposal. A subsystem should be defined not by splitting the Hilbert space, but by the collection of operators that an observer confined to the subsystem can use. This chapter makes that proposal precise. By the end of it you will know what kind of mathematical object $\mathcal{M}$ must be, how such objects are sorted into types, and how a Hilbert space can be *built* from an algebra and a state instead of being assumed at the start. Most definitions below come with a worked example that uses actual numbers, often small matrices you can multiply out by hand. The definitions are abstract on first reading, and watching them act on something small is the quickest way to make them concrete.

2.1 Systems with an infinite amount of entanglement (recap)

Chapter 1 gave three examples: the chain of N Bell pairs, lattice gauge theory, and a quantum field theory cut in half. They motivate everything in this chapter. The lesson drawn from them was this: a subsystem should be defined by *which operators are available to act with*, not by how the Hilbert space happens to split. This chapter turns that idea into a mathematical definition.

2.2 Von Neumann algebras as subsystems

Setting the stage: what a Hilbert space and an operator are, restated carefully

The new definition is a statement *about* Hilbert spaces and operators. So we first state exactly what those are. Vague pictures of “vectors” and “operators” are not precise enough for what follows.

A Hilbert space $\mathcal{H}$ is a vector space over the complex numbers with two extra properties. First, it has an inner product $\langle \cdot | \cdot \rangle$. This is a rule that assigns a complex number $\langle \xi | \eta \rangle$ to each pair of vectors. It is linear in the second argument and conjugate-linear in the first, and it satisfies $\langle \xi | \xi \rangle \geq 0$, with equality only for the zero vector. Second, it is complete. A *Cauchy sequence* is a sequence of vectors whose terms get arbitrarily close to each other, so that it “ought to” converge. Completeness says that every Cauchy sequence really does converge to some vector in $\mathcal{H}$. For a spin- $\frac{1}{2}$ particle, $\mathcal{H} = \mathbb{C}^2$. Its vectors are pairs of complex numbers (c_1, c_2) , usually written $c_1 |0\rangle + c_2 |1\rangle$. The inner product is the dot product with complex conjugation, $\langle \xi | \eta \rangle = \xi_1^* \eta_1 + \xi_2^* \eta_2$. A von Neumann algebra needs nothing more exotic than this. The infinite-dimensional case (a quantum field, say) uses exactly the same definitions, with infinitely many components instead of two.

An operator A on $\mathcal{H}$ is a rule that turns vectors into vectors, $A : \mathcal{H} \rightarrow \mathcal{H}$, and is linear: $A(c_1 |\xi_1\rangle + c_2 |\xi_2\rangle) = c_1 A |\xi_1\rangle + c_2 A |\xi_2\rangle$. On $\mathbb{C}^2$, every linear operator is a 2×2 matrix, and $A |\xi\rangle$ is ordinary

matrix-vector multiplication. The **Hermitian conjugate** (or adjoint) $A^\dagger$ of A is the unique operator with $\langle \xi | A \eta \rangle = \langle A^\dagger \xi | \eta \rangle$ for all ξ, η . For a finite matrix, this is the familiar rule “transpose and complex-conjugate every entry.” A is **self-adjoint** (or Hermitian) if $A = A^\dagger$. Self-adjoint operators are the ones that represent physical observables, because their eigenvalues are real numbers, the kind of number a measurement can return. $B(\mathcal{H})$ denotes the set of *all* bounded operators on $\mathcal{H}$. Boundedness was defined in Chapter 1. Informally, a bounded operator does not stretch the length of any vector by more than some fixed factor.

The new definition of subsystem

Here is the definition, with every piece spelled out. Suppose you have access to only some subset of operators, $\mathcal{M} \subset B(\mathcal{H})$. This is not everything. It is only what your apparatus, in your location, with your capabilities, can measure or apply. Let $|\Psi\rangle \in \mathcal{H}$ be a state of the full system. Using only $\mathcal{M}$, you can do two things:

- compute expectation values $\langle \Psi | A | \Psi \rangle$ for $A \in \mathcal{M}$, which predict the average reading of a measurement of A over many repetitions;
- act on the state, producing $B |\Psi\rangle$ for $B \in \mathcal{M}$, which changes the system by an operation you have access to.

Whatever $\mathcal{M}$ is, it should be closed under three operations, because an observer with access to $\mathcal{M}$ can already do each of them. If you can measure or apply A and B , you can measure or apply $A + B$ (do both and add the results) and AB (do one after the other). You can also multiply by complex numbers. And if A is available, so is $A^\dagger$. For a self-adjoint A this is automatic, since $A^\dagger = A$. In general it says that your toolkit is closed under taking adjoints. A set closed under sums, products, and adjoints is called a ***-subalgebra** of $B(\mathcal{H})$. The $*$ refers to the adjoint, which mathematics texts often write as A^* instead of $A^\dagger$. In these notes every *-subalgebra is also taken to contain the identity operator $\mathbf{1}$, which is the operation “do nothing.”

This alone is not quite enough. We also want $\mathcal{M}$ to be *complete*: if some operator can be approximated arbitrarily well by elements of $\mathcal{M}$, it should count as an element of $\mathcal{M}$ too. Otherwise the set of things you have access to would have artificial gaps. Here the story becomes subtle. There are two different, inequivalent ways to make “approximated arbitrarily well” precise, and the choice between them matters a great deal. The next section explains both.

With this notion of subsystem in hand, the **complement** of $\mathcal{M}$ is defined to be its commutant,

$$\mathcal{M}' = \{B \in B(\mathcal{H}) : BA = AB \ \forall A \in \mathcal{M}\}.$$

One misreading of this definition is common enough to address right away. $\mathcal{M}'$ is *not* the set of operators that commute *with one another*. It is the set of operators that *each* commute with *every* element of $\mathcal{M}$. Two operators inside $\mathcal{M}'$ can fail to commute with each other. In fact, when $\mathcal{M}'$ is a rich enough algebra (the usual case), it contains non-commuting pairs, just as $\mathcal{M}$ does. What $\mathcal{M}'$ guarantees is only this: acting with anything in $\mathcal{M}'$ produces no interference with any measurement made using $\mathcal{M}$. That is the operational meaning of “complement of a subsystem” here. It is not

“the other half of a tensor factorization you could point to.” It is “everything that, as a matter of algebra, cannot disturb what $\mathcal{M}$ measures.” A fuller treatment, with the functional-analysis machinery spelled out, is given in W. Thirring’s *Quantum Mathematical Physics*. Nothing beyond what is given here is needed for the rest of these notes.

Physics Connection: Spacelike Commutativity

You already know one instance of this idea from relativistic field theory. It is the cleanest bridge from the abstract definition to something you can compute. For a free scalar field ϕ , the commutator of the field at two spacetime points is a fixed function times the identity operator, so it does not depend on the state:

$$[\phi(x), \phi(y)] = i\Delta(x - y).$$

Here Δ , the Pauli–Jordan function, is given by an integral over the on-shell momenta of the field. A standard textbook fact is that $\Delta(x - y) = 0$ whenever $x - y$ is spacelike. The reason uses two properties of Δ . First, Δ is Lorentz invariant: $\Delta(\Lambda z) = \Delta(z)$ for every proper orthochronous Lorentz transformation Λ . Second, Δ is odd, $\Delta(-z) = -\Delta(z)$, because the commutator is antisymmetric, $[\phi(x), \phi(y)] = -[\phi(y), \phi(x)]$. Now let z be a spacelike vector in $3 + 1$ dimensions. Then there is a proper orthochronous Lorentz transformation with $\Lambda z = -z$. (Boost to a frame in which z has no time component, rotate by 180° about an axis perpendicular to z , and boost back.) Hence $\Delta(z) = \Delta(-z) = -\Delta(z)$, so $\Delta(z) = 0$. For a timelike z no such Λ exists, because these transformations never exchange past and future. This is why microcausality, the statement that field operators at spacelike separation commute, comes out of the free-field construction automatically instead of being imposed as an extra axiom.

Now translate this into the language of this chapter. Let $\mathcal{M}(O)$ be the algebra generated by field operators smeared over a spacetime region O . Let O' be its causal complement, the set of points spacelike separated from all of O . The vanishing of Δ at spacelike separation says exactly that $\mathcal{M}(O') \subseteq \mathcal{M}(O)'$. This is the *locality* axiom for the local algebras of a relativistic field theory, stated in general in Chapter 4. So in a relativistic field theory, every operator localized spacelike to O lies in the commutant $\mathcal{M}(O)'$.

The abstract definition of $\mathcal{M}'$ adds one thing to this familiar fact. The commutant is defined by a purely algebraic condition, “commutes with everything in $\mathcal{M}$,” which never mentions spacetime. Nothing in the definition forces every such operator to be localized in O' . In general, $\mathcal{M}(O)'$ can be strictly larger than $\mathcal{M}(O')$. When the two are equal, the theory is said to satisfy *Haag duality* for the region O (Chapter 4). Haag duality holds in many standard cases but not in all, and whether it holds depends on the theory, the region, and the representation. So “commutes with $\mathcal{M}$ ” is a more general notion than “is spacelike separated from O .” Chapter 8 uses exactly this extra room: there, bulk causal structure is read off from boundary commutants that are larger than boundary causality alone would require.

2.3 Basic properties: C^* vs. von Neumann algebras, and the double commutant theorem

Two notions of “the limit of a sequence of operators”

Take a sequence of operators $A_1, A_2, A_3, \dots$ in $\mathcal{M}$, and ask whether it converges to some operator A . For ordinary numbers, “converges” has one obvious meaning. For operators there are several natural meanings, and they are genuinely different.

Norm convergence. Every bounded operator has a norm, $\|A\|$. It is the smallest number such that $\|A|\psi\rangle\| \leq \|A\| \|\psi\rangle\|$ for every vector $|\psi\rangle$: the “maximum stretch factor” of Chapter 1. The norm satisfies the identity

$$\|A^\dagger A\| = \|A\|^2. \quad (2.1)$$

It is easy to check on a simple case. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ acting on $\mathbb{C}^2$. Then $A^\dagger A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, whose largest eigenvalue is 1, so $\|A^\dagger A\| = 1$. The norm $\|A\|$ is the largest singular value of A , which is also 1. So $\|A^\dagger A\| = 1 = 1^2 = \|A\|^2$, as the identity requires. A sequence A_n **converges in norm** to A if $\|A_n - A\| \rightarrow 0$. This says that the largest possible discrepancy between $A_n|\psi\rangle$ and $A|\psi\rangle$, taken over *all* unit vectors $|\psi\rangle$ at once, shrinks to zero. It is a strong, uniform statement.

Strong convergence. A_n **converges strongly** to A if $\|(A_n - A)|\psi\rangle\| \rightarrow 0$ for each fixed vector $|\psi\rangle$. Each vector is tested separately, so the rate of convergence may differ from one vector to another.

Weak convergence. A_n **converges weakly** to A if $\langle \xi | A_n | \eta \rangle \rightarrow \langle \xi | A | \eta \rangle$ for every fixed pair of vectors $|\xi\rangle, |\eta\rangle$. This asks only that individual matrix elements converge, one pair of vectors at a time. It is the weakest of the three demands.

Each notion implies the next one. If $\|A_n - A\| \rightarrow 0$, then $\|(A_n - A)|\psi\rangle\| \leq \|A_n - A\| \|\psi\rangle\| \rightarrow 0$, so norm convergence implies strong convergence. And $|\langle \xi | (A_n - A) | \eta \rangle| \leq \|\xi\rangle\| \|(A_n - A)|\eta\rangle\|$ by the Cauchy–Schwarz inequality, so strong convergence implies weak convergence. In finite dimensions the three notions coincide completely. This is why the distinction never comes up in an ordinary quantum mechanics course. In infinite dimensions they differ, and the gap between them is where infinite-dimensional phenomena (quantum field theory, the $N \rightarrow \infty$ limit) live. The next example shows the difference.

Worked example: weak vs. strong vs. norm convergence in $\ell^2(\mathbb{N})$. Let $\mathcal{H} = \ell^2(\mathbb{N})$ be the infinite-dimensional Hilbert space of square-summable sequences, with standard orthonormal basis $\{|1\rangle, |2\rangle, |3\rangle, \dots\}$. Consider the sequence of rank-one projections

$$P_n \equiv |n\rangle \langle n|.$$

We test the convergence of P_n as $n \rightarrow \infty$ in each of the three senses.

1. **Weak convergence:** $P_n \rightarrow 0$ **weakly**. Take any two vectors $|\xi\rangle = \sum_{k=1}^{\infty} c_k |k\rangle$ and $|\eta\rangle = \sum_{k=1}^{\infty} d_k |k\rangle$, with $\sum |c_k|^2 < \infty$ and $\sum |d_k|^2 < \infty$. The matrix element is

$$\langle \xi | P_n | \eta \rangle = \langle \xi | n \rangle \langle n | \eta \rangle = c_n^* d_n.$$

The Cauchy–Schwarz inequality for series gives $\sum_{n=1}^{\infty} |c_n d_n| \leq \sqrt{\sum_{n=1}^{\infty} |c_n|^2} \sqrt{\sum_{n=1}^{\infty} |d_n|^2} < \infty$. The terms of a convergent series tend to zero, so $c_n^* d_n \rightarrow 0$. Hence $\langle \xi | P_n | \eta \rangle \rightarrow 0$ for *every* pair of vectors, and P_n converges weakly to the zero operator.

2. **Strong convergence:** $P_n \rightarrow 0$ **strongly**. For a fixed vector $|\psi\rangle = \sum c_k |k\rangle$ we compute

$$\|P_n |\psi\rangle\|^2 = \| |n\rangle \langle n | \psi \rangle \|^2 = |c_n|^2 \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

The limit is zero because $|c_n|^2$ are the terms of the convergent series $\sum_k |c_k|^2$. Hence P_n converges strongly to 0.

3. **Norm convergence:** P_n **does not converge in norm**. The operator norm takes the largest stretch over all unit vectors at once. The largest stretch is reached at $|\psi\rangle = |n\rangle$:

$$\|P_n - 0\| = \sup_{\|\psi\|=1} \|P_n |\psi\rangle\| = \|P_n |n\rangle\| = \||n\rangle\| = 1.$$

So $\|P_n - 0\| = 1$ for every n , and this never shrinks. Hence P_n does *not* converge to 0 in norm. It does not converge in norm to anything else either. For $n \neq m$ the operator $P_n - P_m$ has eigenvalues $+1$ and -1 , so $\|P_n - P_m\| = 1$, and the sequence is not even a Cauchy sequence in norm.

This is the standard example to keep in mind. Every matrix element settles down to zero, but the operator as a whole never becomes small in norm.

A $*$ -subalgebra $\mathcal{M}$ that is complete under norm convergence is called a **C^* -algebra**. Here “complete” means that every norm-convergent sequence in $\mathcal{M}$ has its limit back in $\mathcal{M}$. A $*$ -subalgebra that is complete under weak convergence is called a **von Neumann algebra**. (Strictly speaking, the weak closure is defined with nets rather than sequences. Sequences give the right intuition, and nothing in these notes depends on the difference.) Every von Neumann algebra is automatically a C^* -algebra. To see this, suppose $A_n \in \mathcal{M}$ and $A_n \rightarrow A$ in norm. Then $A_n \rightarrow A$ weakly as well, so if $\mathcal{M}$ is weakly closed, $A \in \mathcal{M}$. Weak closure is the stronger requirement, because it asks $\mathcal{M}$ to contain the limits of more sequences. So von Neumann algebras are the more complete, more restrictive notion.

Which notion is the right one for a physical subsystem? Weak convergence, for a simple reason. What a laboratory measures is a matrix element $\langle \xi | A | \eta \rangle$: an expectation value, or more generally a transition amplitude. Weak convergence says exactly that each of these measurable numbers settles down. So von Neumann algebras, and not the less complete C^* -algebras, capture “the set of operators whose physical predictions are under control.” That is why they are the main tool in the rest of these notes.

There is one more difference between the two notions, and it matters for everything that follows. A C^* -algebra can be defined abstractly, with no Hilbert space anywhere. It is a vector space with a product, an adjoint, and a norm that satisfies the identity (2.1), and it is complete in that norm. You could specify one by its multiplication table and its norm, much as you specify a finite group by its multiplication table, without saying what vectors it acts on. The definition of a von Neumann algebra given above cannot be stated this way. Its completeness condition, weak

convergence, is phrased in terms of vectors $|\xi\rangle, |\eta\rangle$ in a specific $\mathcal{H}$. (Mathematicians do have an abstract characterization of von Neumann algebras, due to Sakai, but it is less direct and is not needed here.) This gap is exactly what the GNS construction, worked through later in this chapter, is designed to close. Starting from nothing but an abstract C^* -algebra and a state on it, GNS builds the missing Hilbert space. Only then can you ask whether the algebra, represented on that Hilbert space, is also weakly closed.

The double commutant theorem

Checking weak closure directly means verifying that *every* weakly convergent sequence in $\mathcal{M}$ has its limit in $\mathcal{M}$. That sounds like infinitely many checks. Von Neumann's double commutant theorem replaces this topological chore with an algebraic check. For a $*$ -subalgebra $\mathcal{M} \subset B(\mathcal{H})$ that contains the identity,

$$\mathcal{M} = \mathcal{M}'' \iff \mathcal{M} \text{ is weakly closed (a von Neumann algebra).}$$

Here $\mathcal{M}'' \equiv (\mathcal{M}')'$ is the commutant of the commutant. Read the logic slowly. You take the complement of your subsystem, $\mathcal{M}'$: everything that cannot disturb $\mathcal{M}$. Then you take the complement of *that*, $\mathcal{M}''$: everything that cannot disturb anything in $\mathcal{M}'$. The theorem says that doing this twice brings you back exactly to where you started, provided $\mathcal{M}$ was already a von Neumann algebra. If $\mathcal{M}$ is only a $*$ -subalgebra (closed under sums, products, and adjoints, but not weakly closed), then $\mathcal{M}''$ is in general strictly larger than $\mathcal{M}$. It is the *smallest* von Neumann algebra containing $\mathcal{M}$: $\mathcal{M}$ together with every operator that can be approximated weakly by elements of $\mathcal{M}$. This is not an obvious fact. It is a real theorem, and its proof uses the geometry of Hilbert space. Its value is that a purely algebraic operation (take the commutant twice) carries out a topological completion (weak closure) for you, with no limits in the definition at all.

Worked example, done completely, in four dimensions. Let $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ — two qubits, call them R and L , so $\mathcal{H}$ is 4-dimensional. Let $\mathcal{M}$ be every operator of the form $A \otimes \mathbf{1}_L$ for A an arbitrary 2×2 matrix (every operation you could perform using only apparatus touching qubit R , doing nothing to qubit L). As a vector space, $\mathcal{M}$ is spanned by four matrices: the identity $\mathbf{1}_2$ and the three Pauli matrices $\sigma_x, \sigma_y, \sigma_z$, each tensored with $\mathbf{1}_L$:

$$\mathbf{1}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every 2×2 matrix is some combination $c_0 \mathbf{1}_2 + c_1 \sigma_x + c_2 \sigma_y + c_3 \sigma_z$ (this is just saying the four Pauli matrices, including the identity, form a basis for all 2×2 matrices — a standard fact, and easy to check directly: writing out a general 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and matching coefficients gives $c_0 = \frac{a+d}{2}$, $c_3 = \frac{a-d}{2}$, $c_1 = \frac{b+c}{2}$, $c_2 = \frac{b-c}{2i}$, which always has a solution).

To find $\mathcal{M}'$, look for every 4×4 matrix X that commutes with all the $\sigma_i \otimes \mathbf{1}_L$. Use the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, where the first digit belongs to R and the second to L .

Rather than solving sixteen equations one by one, write X as four 2×2 blocks:

$$X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}, \quad X_{ij} \in M_2(\mathbb{C}).$$

In this basis $\sigma_z \otimes \mathbf{1}_L = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}$ and $\sigma_x \otimes \mathbf{1}_L = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}$. Since σ_x and σ_z generate all 2×2 matrices (their product is $\sigma_x \sigma_z = -i\sigma_y$), it is enough to commute X with these two. First,

$$[\sigma_z \otimes \mathbf{1}_L, X] \stackrel{(1)}{=} \begin{pmatrix} X_{11} & X_{12} \\ -X_{21} & -X_{22} \end{pmatrix} - \begin{pmatrix} X_{11} & -X_{12} \\ X_{21} & -X_{22} \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} 0 & 2X_{12} \\ -2X_{21} & 0 \end{pmatrix}.$$

(1) multiplying by $\sigma_z \otimes \mathbf{1}_L$ on the left flips the sign of the bottom row of blocks; multiplying on the right flips the sign of the right column of blocks. (2) subtract block by block.

Setting this to zero forces $X_{12} = X_{21} = 0$, so X is block-diagonal. Next, for block-diagonal X ,

$$[\sigma_x \otimes \mathbf{1}_L, X] \stackrel{(1)}{=} \begin{pmatrix} 0 & X_{22} \\ X_{11} & 0 \end{pmatrix} - \begin{pmatrix} 0 & X_{11} \\ X_{22} & 0 \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} 0 & X_{22} - X_{11} \\ X_{11} - X_{22} & 0 \end{pmatrix}.$$

(1) multiplying by $\sigma_x \otimes \mathbf{1}_L$ on the left swaps the two rows of blocks; multiplying on the right swaps the two columns of blocks. (2) subtract block by block.

Setting this to zero forces $X_{22} = X_{11}$. Call this common block B . Every X that commutes with $\mathcal{M}$ therefore has the form

$$X = \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix} = \mathbf{1}_R \otimes B, \quad B \in M_2(\mathbb{C}).$$

Since B is an arbitrary 2×2 matrix, $\mathcal{M}' = \mathbf{1}_R \otimes B(\mathcal{H}_L)$. The commutant of “everything R can do” is “everything L can do”.

Now compute the second commutant $\mathcal{M}'' = (\mathcal{M}')'$. An operator $Y = \begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix}$ in $\mathcal{M}''$ must commute with every $\mathbf{1}_R \otimes B = \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix}$. Multiplying out, this says $Y_{ij}B = BY_{ij}$ for each block and for every 2×2 matrix B . The only 2×2 matrices that commute with every 2×2 matrix are multiples of the identity. (Commuting with $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ removes the off-diagonal entries, and commuting with $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ makes the two diagonal entries equal.) So $Y_{ij} = a_{ij}\mathbf{1}_2$, and

$$Y = \begin{pmatrix} a_{11}\mathbf{1}_2 & a_{12}\mathbf{1}_2 \\ a_{21}\mathbf{1}_2 & a_{22}\mathbf{1}_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \otimes \mathbf{1}_L \in \mathcal{M}.$$

Every such Y lies in $\mathcal{M}$, and every element of $\mathcal{M}$ has this form, so $\mathcal{M}'' = \mathcal{M}$. All three algebras are 4-dimensional, and the double commutant theorem holds here by direct calculation.

A second example shows what happens when $\mathcal{M}$ is *not* simply “all operators on one

tensor factor.” It also previews the notion of a *center*, defined just below. Take the same $\mathcal{H} = \mathbb{C}^4$, but now ignore any tensor-product structure. Let $\mathcal{M}$ be spanned by the two matrices $P_1 = \text{diag}(1, 1, 0, 0)$ and $P_2 = \text{diag}(0, 0, 1, 1)$. This is the algebra of an observer who can only tell whether the system is in “block one” or “block two,” and nothing finer. Solving the same kind of linear system shows that $\mathcal{M}'$ is 8-dimensional. It consists of every operator that acts as an arbitrary 2×2 block within each of the two sectors, without mixing them: two independent 2×2 complex matrices, so $4 + 4 = 8$ complex parameters. The second commutant $\mathcal{M}''$ comes out to be exactly $\mathcal{M}$ again, which is 2-dimensional. Something happens here that did not happen in the first example: $\mathcal{M} \subset \mathcal{M}'$. Both P_1 and P_2 commute with *everything* in the block-diagonal algebra $\mathcal{M}'$. So $\mathcal{M}$ is small enough to commute with its own complement. Whenever this happens, $\mathcal{M}$ has a nontrivial *center*, and that has a physical meaning. $\mathcal{M}$ does not describe one indivisible subsystem. It describes a classical label (which block you are in), possibly with further subsystems inside each block. The discussion of the center below comes back to this example.

Where the entangled subsystem lives. Putting the last two subsections together: a subsystem is a von Neumann algebra $\mathcal{M} \subset B(\mathcal{H})$, and its complement is the commutant $\mathcal{M}'$. In this framework, bipartite entanglement is a statement about the pair $(\mathcal{M}, \mathcal{M}')$, not about a tensor factorization. When a factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ does exist, the choice $\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L$ gives back exactly the familiar story. That is the first worked example above. Chapter 3 calls this special case a *type I factor*.

Several further facts about commutants are used freely later, so we collect them here. Let $\mathcal{A}$ be any *-subalgebra, not necessarily weakly closed.

- $\mathcal{A}' = \mathcal{A}'''$. The commutant of $\mathcal{A}$ is already a von Neumann algebra: a short argument shows $\mathcal{A}' = (\mathcal{A}')''$ directly.
- $\mathcal{M} \equiv \mathcal{A}''$ is the smallest von Neumann algebra containing $\mathcal{A}$, and it has the same commutant as $\mathcal{A}$: $\mathcal{M}' = \mathcal{A}''' = \mathcal{A}'$.
- For two von Neumann algebras $\mathcal{M}_1, \mathcal{M}_2$, the **join** $\mathcal{M}_1 \vee \mathcal{M}_2 \equiv (\mathcal{M}_1 \cup \mathcal{M}_2)''$ is the smallest von Neumann algebra containing both. The **meet** $\mathcal{M}_1 \wedge \mathcal{M}_2 \equiv \mathcal{M}_1 \cap \mathcal{M}_2$ is their intersection. The intersection is already a von Neumann algebra, so no double commutant is needed on that side.
- These satisfy a De Morgan-type duality, $(\mathcal{M}_1 \vee \mathcal{M}_2)' = \mathcal{M}_1' \wedge \mathcal{M}_2'$. Joining two algebras and then taking the complement gives the same result as taking each complement first and then intersecting. (The left side is $(\mathcal{M}_1 \cup \mathcal{M}_2)''' = (\mathcal{M}_1 \cup \mathcal{M}_2)'$, and an operator commutes with everything in $\mathcal{M}_1 \cup \mathcal{M}_2$ exactly when it commutes with $\mathcal{M}_1$ and with $\mathcal{M}_2$.)

The **center** of $\mathcal{M}$ is $Z(\mathcal{M}) \equiv \mathcal{M} \cap \mathcal{M}'$. It consists of the operators that commute with everything in $\mathcal{M}$ and with everything in $\mathcal{M}'$. In the second worked example above, $\mathcal{M} \subset \mathcal{M}'$, so $Z(\mathcal{M}) = \mathcal{M}$: the center is as large as it can possibly be. At the other extreme, $\mathcal{M}$ is called a **factor** (or is said to be **primary**) if its center is as small as possible, $Z(\mathcal{M}) = \mathbb{C} \mathbf{1}$, containing only multiples of the

identity. The first worked example, $\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L$, is a factor. Indeed, an operator that has both the form $A \otimes \mathbf{1}_L$ and the form $\mathbf{1}_R \otimes B$ must be a multiple of the identity, as a direct comparison of the two forms shows. A general von Neumann algebra can always be decomposed into a direct sum, or more generally a direct integral, of factors. So the classification of von Neumann algebras reduces to classifying factors, plus bookkeeping for the classical labels carried by the center. For this reason, from here on and for almost all of the notes, we restrict attention to factors.

2.4 Weights and states

The definitions

A **linear functional** ω on $\mathcal{M}$ is a rule that assigns a complex number $\omega(A)$ to every $A \in \mathcal{M}$ and respects addition and multiplication by numbers: $\omega(aA + bB) = a\omega(A) + b\omega(B)$. It is **positive** if $\omega(A^\dagger A) \geq 0$ for every A . This is a natural requirement. The operator $A^\dagger A$ plays the role of $|A|^2$: it is always a non-negative operator, the operator analogue of a squared magnitude, so it should never be assigned a negative number. A positive functional is automatically compatible with the adjoint, $\omega(A^\dagger) = \omega(A)^*$. (This follows from positivity applied to $(\mathbf{1} + A)^\dagger(\mathbf{1} + A)$ and $(\mathbf{1} + iA)^\dagger(\mathbf{1} + iA)$.) We use this property freely. In these notes a positive linear functional is called a **weight**. (In the mathematical literature a weight is also allowed to take the value $+\infty$ on some positive elements, as the ordinary trace does on $B(\mathcal{H})$ when $\mathcal{H}$ is infinite-dimensional. Traces of this kind appear in Chapter 3.) A weight is a **state** if it is also normalized, $\omega(\mathbf{1}) = 1$. It is a **trace** if $\omega(AB) = \omega(BA)$. This is the cyclic property of the ordinary matrix trace, $\text{Tr}(AB) = \text{Tr}(BA)$. Positivity and linearity together give a useful inequality for free. It is the direct analogue of the Cauchy–Schwarz inequality you know from vector inner products:

$$|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \omega(B^\dagger B). \quad (2.2)$$

A state ω is one rule doing one thing: it assigns a single number, $\omega(A)$, to each $A \in \mathcal{M}$. What that number means depends only on which operator you feed it, not on any change to ω itself. Feed it a general observable A , and $\omega(A)$ is the expectation value of A . Feed it a projection P_I instead, and $\omega(P_I)$ is a probability. Here P_I is the spectral projection onto some range I of possible outcomes, defined later in this chapter. Positivity and normalization already force $0 \leq \omega(P_I) \leq 1$, because both $P_I = P_I^\dagger P_I$ and $\mathbf{1} - P_I = (\mathbf{1} - P_I)^\dagger(\mathbf{1} - P_I)$ are of the form $X^\dagger X$. Linearity gives $\omega(P_{I_1}) + \omega(P_{I_2}) = \omega(P_{I_1 \cup I_2})$ for disjoint ranges. It is the same formula and the same ω in both cases. The two readings, expectation value and probability, come entirely from what sits inside the parentheses.

None of these definitions mentions a Hilbert space, a vector, or a density operator. This is the reason for introducing weights and states this way, rather than going straight to a density matrix. ω is defined only as a number assigned to each element of the algebra $\mathcal{M}$. A standard result, stated here without proof, connects this back to density matrices. Call a state *normal* if it is continuous in a suitable weak sense; these are the physically sensible states. Every normal state ω on $\mathcal{M} \subset B(\mathcal{H})$ can be written as $\omega(A) = \text{Tr}(\rho A)$ for some density operator ρ on $\mathcal{H}$. This ρ acts on all of $\mathcal{H}$. It need not belong to $\mathcal{M}$, and in general it is not unique. So the density operator is a consequence of

the definition of ω , not an input to it. ω is **faithful** if $\omega(A^\dagger A) = 0$ forces $A = 0$: nothing nonzero is invisible to it. ω is **pure** if it cannot be written as a nontrivial mixture $\omega = \lambda\omega_1 + (1 - \lambda)\omega_2$ of two different states, with $0 < \lambda < 1$. If it can, ω is said to **dominate** $\lambda\omega_1$ and $(1 - \lambda)\omega_2$, since $\omega - \lambda\omega_1$ and $\omega - (1 - \lambda)\omega_2$ are still positive.

What happens to the wavefunction once the algebra is taken as primary

These definitions look innocent. But they demote the wavefunction from a fundamental object to a derived one, so we unpack them slowly.

An ordinary quantum-mechanical state is a vector $|\psi\rangle$ or, more generally, a density matrix ρ . Both of these live in, or act on, a Hilbert space. You cannot even write down $|\psi\rangle$ without first having $\mathcal{H}$. The definition of ω needs none of that. It is a number assigned to each algebra element, and the algebra $\mathcal{M}$ can be specified abstractly, as a C^* -algebra, with no Hilbert space in the definition. This is more than a cosmetic change. It changes which object is treated as fundamental. In the ordinary picture, you start with $\mathcal{H}$, pick a vector $|\psi\rangle \in \mathcal{H}$, and compute $\langle\psi|A|\psi\rangle$ for whatever operators you like. In the algebraic picture, you start with the algebra $\mathcal{M}$ (what can be measured or done) and a state ω on it (an expected outcome for every element of $\mathcal{M}$). Only afterward, if you want one, do you build a Hilbert space and a vector that reproduce ω . That construction is the GNS construction, covered in full later in this chapter. The vector it produces is not fundamental data. It is a derived bookkeeping device, specific to the state ω you started with. A different state can produce a different Hilbert space that is physically inequivalent to the first.

The linearity of ω , $\omega(A + B) = \omega(A) + \omega(B)$, also tells you something. It says that ω describes an *average* over many repetitions, not a single outcome on one system. A single measurement of $A + B$ on one system does not obviously split into “the result you would have got for A ” plus “the result you would have got for B .” In general you cannot even measure A and B together unless they commute. But the *average* of $A + B$ over a large ensemble of identically prepared systems does split this way, simply because averaging is linear. So ω (equivalently the density matrix ρ , or in the pure case the vector $|\psi\rangle$) is best read as a statement about an ensemble, not about one individual system.

The statistical algebraic approach of Slavnov [3] takes this one level deeper. It also addresses the obvious next question: what, then, describes one individual system? Slavnov starts from something more primitive than ω . It is a functional $\varphi(\hat{A})$ that returns the number a single measurement of $\hat{A}$, on this particular system, would read out. It is not defined linearly on the whole algebra. It is defined on one maximal set of *mutually compatible* observables at a time, that is, observables that commute and so can be measured jointly. There is no averaging in this definition: φ is a record of one concrete measurement outcome. The key point is this. Measuring one observable can disturb your ability to measure another one that does not commute with it. So no experiment can pin down φ completely. An experiment can only tell you that φ belongs to an *equivalence class* of functionals that all agree with your data on the compatible set of observables $\{Q\}$ you chose to measure. Slavnov proposes that this equivalence class, and not φ itself, is what a conventional “quantum state” Ψ_Q is. On this view a quantum state is not a container of hidden, definite properties. It is a precise record of how much you could possibly know, given that measuring one thing costs you the ability to measure something incompatible with it.

On this view, one idea accounts for several facts that otherwise look like separate puzzles:

- **Why quantum probability is not “we just don’t know the hidden variable yet.”** The same individual φ can belong to the class $\Psi_Q = \{\varphi\}_Q$ or to a different class $\Psi_P = \{\varphi\}_P$. Which one depends on which compatible set, $\{Q\}$ or $\{P\}$, the experimenter chooses to measure. So the “quantum state” assigned to the system depends on an experimental choice made after the system was prepared. Slavnov identifies this as the root of the Einstein–Podolsky–Rosen puzzle. On his account, the EPR puzzle is not an additional strange feature of quantum mechanics. It is the same fact seen from another angle.
- **Why ρ , not $|\psi\rangle$, is what enters every physical prediction.** ρ (equivalently ω) is, by construction, one step of ensemble averaging removed from an individual φ . Textbooks such as Sakurai’s replace $|\psi\rangle$ by $\rho = |\psi\rangle\langle\psi|$ as the basic object, so that pure and mixed states are handled in the same way. That is the first half of this demotion. The algebraic framework of these notes completes it: it replaces ρ , which needs a Hilbert space to be written as a matrix, with ω , which does not.
- **Why a Hilbert space appears at all, and in what sense it comes second.** Slavnov states this directly: *Hilbert space vectors (wavefunctions), the operators in the Hilbert space, and other purely mathematical concepts are not primary elements. These concepts only appear at the second stage of the formulation.* The primary elements are the algebra of observables and the state, both tied directly to experiment. The state can be taken in Slavnov’s individual sense φ , or in the ensemble sense ω used throughout these notes. The Hilbert space, its vectors, and the operators acting on them are all built afterward, mechanically, by the GNS construction applied to ω .

To state the conclusion of this viewpoint plainly, in one place: the wavefunction is not a physical entity that stores values the way a hard drive stores bits. It is the name for an equivalence class of individual measurement records. The records in one class cannot be told apart, because no experiment could distinguish them, given what was actually chosen to be measured. The algebra, which says what is measurable and how measurements combine, is the primary, observer-independent structure. The wavefunction depends on a choice of what you decided to look at. This is an interpretation, not a theorem, and the mathematics in the rest of the notes do not depend on adopting it. What the rest of the notes do use is the weaker statement: the algebra and the state come first, and the Hilbert space is built from them.

2.5 C^* and von Neumann algebras generated by a single operator

This section is short. Given the two definitions (C^* -algebra and von Neumann algebra) from earlier in this chapter, it is self-contained. It gives the concrete reason why projections, the subject of the next section, live in von Neumann algebras and generally not in the smaller C^* -algebras.

Take one bounded, self-adjoint operator $\mathcal{O}$. What is the smallest C^* -algebra containing it, $\mathcal{A}(\mathcal{O})$, and the smallest von Neumann algebra containing it, $\mathcal{M}(\mathcal{O})$? Both contain every polynomial $\sum_n c_n \mathcal{O}^n$ in $\mathcal{O}$, since polynomials are made by repeated multiplication and addition. Each must then be completed, in its own sense of limit, to become a full C^* -algebra or von Neumann algebra.

The **spectrum** $\sigma(\mathcal{O})$ of $\mathcal{O}$ is the set of numbers κ for which $\mathcal{O} - \kappa \mathbf{1}$ is not invertible. For a finite Hermitian matrix, the spectrum is its set of eigenvalues: the numbers a measurement of $\mathcal{O}$ can return. A standard fact of functional analysis (stated here without proof) is that $\mathcal{A}(\mathcal{O})$ is isomorphic to the algebra of *continuous* functions on $\sigma(\mathcal{O})$. Every element of $\mathcal{A}(\mathcal{O})$ can be written as $f(\mathcal{O})$ for a continuous function f , and every such $f(\mathcal{O})$ lies in $\mathcal{A}(\mathcal{O})$. A simple example makes this plausible. Let $\mathcal{O}$ be a finite Hermitian matrix with distinct eigenvalues $\kappa_1, \dots, \kappa_n$. Applying a polynomial, or any function, to $\mathcal{O}$ means applying it to each eigenvalue and leaving the eigenvectors alone. So specifying $f(\mathcal{O})$ is the same as specifying the n numbers $f(\kappa_1), \dots, f(\kappa_n)$, that is, a function on the finite set $\{\kappa_1, \dots, \kappa_n\}$. On a finite set every function is continuous, so this example cannot yet show the distinction that matters below. It does show the general shape of the isomorphism.

Physics Connection: Eigenvalues vs. Eigenvectors

The definition of $\sigma(\mathcal{O})$ never mentions an eigenvector or a Hilbert space. A number λ lies in $\sigma(\mathcal{O})$ because $\mathcal{O} - \lambda \mathbf{1}$ has no inverse inside the algebra. So eigenvalues do not need to be demoted, as the vacuum vector $|\Omega\rangle$ will be in the GNS construction later in this chapter. They are already an algebra-level notion, fixed before any state or representation is chosen. What is demoted is the *eigenvector*. An eigenvector lives in whatever Hilbert space GNS builds for a given ω , so it appears only once ω is chosen.

Check this on the number operator $N = a^\dagger a$ of the harmonic oscillator. (Strictly, N is unbounded, so it is not an element of a C^* -algebra. The argument below uses only the commutation relation and positivity, so this does not matter.) We show that the possible eigenvalues of N are fixed by $[a, a^\dagger] = 1$ together with positivity, in every representation at once. Suppose $Nv = \lambda v$ for some nonzero vector v , in any representation of the algebra on a Hilbert space. First, $\lambda \geq 0$:

$$\begin{aligned} \lambda \|v\|^2 &\stackrel{(1)}{=} \langle v | Nv \rangle \\ &\stackrel{(2)}{=} \langle av | av \rangle = \|av\|^2 \geq 0. \end{aligned}$$

(1) $Nv = \lambda v$ by hypothesis. (2) $N = a^\dagger a$ and the definition of the adjoint.

Next, lower with a :

$$\begin{aligned} N(av) &\stackrel{(1)}{=} (aN + [N, a])v \\ &\stackrel{(2)}{=} a(\lambda v) + (-a)v \\ &\stackrel{(3)}{=} (\lambda - 1)(av). \end{aligned}$$

(1) $Na = aN + [N, a]$, the definition of the commutator. (2) $Nv = \lambda v$ by hypothesis, and $[N, a] = [a^\dagger a, a] = [a^\dagger, a]a = -a$ from $[a, a^\dagger] = 1$. (3) collect the two terms, both proportional to av .

So av is either zero or an eigenvector with eigenvalue $\lambda - 1$. Repeating, $a^k v$ is either zero or an eigenvector with eigenvalue $\lambda - k$. Every eigenvalue is ≥ 0 , so the vectors $a^k v$ cannot all be nonzero. Let k be the last index with $a^k v \neq 0$. Then $a(a^k v) = 0$, so $N(a^k v) = a^\dagger a(a^k v) = 0$,

and the eigenvalue $\lambda - k$ must be 0. Hence $\lambda = k \in \{0, 1, 2, \dots\}$. The same computation with $[N, a^\dagger] = a^\dagger$ gives $N(a^\dagger v) = (\lambda + 1)(a^\dagger v)$. Here $a^\dagger v$ is never zero: $a^\dagger v = 0$ would give $aa^\dagger v = (N + 1)v = 0$, so $\lambda = -1$, which contradicts $\lambda \geq 0$. So once one eigenvalue occurs, all of $0, 1, 2, \dots$ occur. These are the quantized levels of the oscillator. They follow from the multiplication table of the algebra and positivity, whichever Hilbert space the operators act on. Choosing a state ω and running GNS only decides which eigenvectors appear, and with what weights. It never changes the spectrum itself.

$\mathcal{M}(\mathcal{O})$, by contrast, is isomorphic to an algebra of *bounded* functions on $\sigma(\mathcal{O})$ that need not be continuous. (Precisely: bounded measurable functions, where two functions count as equal if they differ only on a set that the spectral measure of $\mathcal{O}$ does not see.) Every continuous function on the compact set $\sigma(\mathcal{O})$ is bounded, so $\mathcal{A}(\mathcal{O}) \subset \mathcal{M}(\mathcal{O})$. This fits with von Neumann algebras being the larger, more complete objects. The difference becomes concrete and important when $\sigma(\mathcal{O})$ is a continuum, such as an interval of real numbers, rather than a finite set of isolated points. This happens when $\mathcal{O}$ has continuous spectrum, which is the generic case in field theory and in the $N \rightarrow \infty$ limits these notes are about. On a continuum, the **indicator function** of a subset $I \subset \sigma(\mathcal{O})$,

$$f_I(\kappa) = \begin{cases} 1 & \kappa \in I \\ 0 & \kappa \notin I \end{cases},$$

is bounded, since it only takes the values 0 and 1. But it is discontinuous wherever I has an edge inside the continuum. The corresponding operator $P_I \equiv f_I(\mathcal{O})$ is a **spectral projection**. For a matrix, it projects onto the span of the eigenvectors whose eigenvalues lie in I . In general, it is the operator that answers the yes/no question “does a measurement of $\mathcal{O}$ give a value in the range I ?” Indicator functions are generically discontinuous on a continuum, so spectral projections generically lie in $\mathcal{M}(\mathcal{O})$ but not in $\mathcal{A}(\mathcal{O})$. This is the concrete, checkable reason why von Neumann algebras, and not merely C^* -algebras, are the right setting for the classification in the next two sections. That classification is built entirely out of projections, and a C^* -algebra generally does not contain the sharp yes/no measurement operators it needs.

2.6 Projections

Definitions, and what they mean physically

A **projection** is a self-adjoint operator P with $P^2 = P$. Applying it twice does nothing more than applying it once. This is the algebraic signature of “select a subspace and discard everything orthogonal to it,” which is what a sharp yes/no measurement does. The spectral theorem writes any self-adjoint operator in terms of its spectral projections, as discussed in the previous section. As a result, every element of a von Neumann algebra is a norm limit of finite linear combinations of projections in the algebra. This is why the classification that follows can be phrased entirely in terms of projections, with nothing lost.

Physics Connection: Probabilities from Projections

The section on weights and states noted that $\omega(P_I)$ reads as a probability because P_I is a projection. Here is where that fact comes from. The previous section fixed the possible outcomes of measuring $\mathcal{O}$ as its spectrum $\sigma(\mathcal{O})$. This is an algebra-level fact, decided before any state is chosen. For a range $I \subset \sigma(\mathcal{O})$, the spectral projection P_I , built from $\mathcal{O}$ by the indicator-function construction, is itself an element of the von Neumann algebra. The Born rule of ordinary quantum mechanics gives the probability of an outcome in I as $\langle \psi | P_I | \psi \rangle$. The algebraic version replaces $|\psi\rangle$ by ω , so the probability is $\omega(P_I)$. The probability of an outcome is therefore not a second thing that ω supplies on top of expectation values. It is ω evaluated on one particular algebra element, the spectral projection for that outcome, by the same rule ω uses for everything else.

Worked example: spectral projections and the Born rule on a single qubit. Let $A = \sigma_x + \sigma_z = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, an observable on $\mathbb{C}^2$. We construct its spectral projections explicitly and use them to compute probabilities.

1. **Eigenvalues.** The characteristic equation is $\det(A - \lambda \mathbf{1}) = \lambda^2 - 2 = 0$, so the eigenvalues are $\lambda_{\pm} = \pm\sqrt{2}$.
2. **Spectral projections by Lagrange interpolation.** Let a 2×2 Hermitian matrix A have distinct eigenvalues λ_1, λ_2 . The projection onto the eigenspace of λ_1 is $P_1 = (A - \lambda_2 \mathbf{1})/(\lambda_1 - \lambda_2)$. The reason is that the polynomial $f(x) = (x - \lambda_2)/(\lambda_1 - \lambda_2)$ equals 1 at λ_1 and 0 at λ_2 , so $f(A)$ is the indicator function of $\{\lambda_1\}$ applied to A . For our A this gives

$$P_+ = \frac{A + \sqrt{2} \mathbf{1}}{\sqrt{2} + \sqrt{2}} = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 + \sqrt{2} & 1 \\ 1 & \sqrt{2} - 1 \end{pmatrix},$$

$$P_- = \frac{A - \sqrt{2} \mathbf{1}}{-\sqrt{2} - \sqrt{2}} = \frac{1}{2\sqrt{2}} \begin{pmatrix} \sqrt{2} - 1 & -1 \\ -1 & \sqrt{2} + 1 \end{pmatrix}.$$

3. **Checking the projection properties.**

- $P_+ + P_- = \frac{1}{2\sqrt{2}} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & 2\sqrt{2} \end{pmatrix} = \mathbf{1}_2$.
- $P_+ P_- = \frac{1}{8} \begin{pmatrix} 1 + \sqrt{2} & 1 \\ 1 & \sqrt{2} - 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} - 1 & -1 \\ -1 & \sqrt{2} + 1 \end{pmatrix} = 0$. For example, the top-left entry is $\frac{1}{8} [(1 + \sqrt{2})(\sqrt{2} - 1) - 1] = \frac{1}{8}(1 - 1) = 0$.
- $P_+^2 = P_+$ and $P_-^2 = P_-$. This follows from the two previous lines: $P_+^2 = P_+(\mathbf{1} - P_-) = P_+ - P_+ P_- = P_+$, and in the same way for P_- .
- Spectral decomposition: $\lambda_+ P_+ + \lambda_- P_- = \sqrt{2} P_+ - \sqrt{2} P_- = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = A$.

4. **Born-rule probabilities.** Suppose the system is in the state $|\psi\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. The probabilities of the two outcomes are

$$\text{Prob}(+\sqrt{2}) = \langle 0 | P_+ | 0 \rangle = \frac{1 + \sqrt{2}}{2\sqrt{2}} = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.853553,$$

$$\text{Prob}(-\sqrt{2}) = \langle 0 | P_- | 0 \rangle = \frac{\sqrt{2} - 1}{2\sqrt{2}} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \approx 0.146447.$$

They add up to exactly 1, as they must, since $P_+ + P_- = \mathbf{1}$. The expectation value is

$$\begin{aligned}
 \langle A \rangle &\stackrel{(1)}{=} (+\sqrt{2}) \text{Prob}(+\sqrt{2}) + (-\sqrt{2}) \text{Prob}(-\sqrt{2}) \\
 &\stackrel{(2)}{=} \sqrt{2} \left(\frac{1}{2} + \frac{1}{2\sqrt{2}} \right) - \sqrt{2} \left(\frac{1}{2} - \frac{1}{2\sqrt{2}} \right) \\
 &\stackrel{(3)}{=} 2 \cdot \frac{\sqrt{2}}{2\sqrt{2}} = 1 \\
 &\stackrel{(4)}{=} \langle 0 | \sigma_x + \sigma_z | 0 \rangle.
 \end{aligned}$$

(1) take $\langle 0 | \cdot | 0 \rangle$ of the spectral decomposition $A = \sqrt{2} P_+ - \sqrt{2} P_-$. (2) insert the two probabilities just computed. (3) the two $\sqrt{2} \cdot \frac{1}{2}$ terms cancel, and the two remaining terms are equal. (4) $\langle 0 | \sigma_x | 0 \rangle = 0$ and $\langle 0 | \sigma_z | 0 \rangle = 1$.

Numerically, $\sqrt{2}(0.853553 - 0.146447) = \sqrt{2}(0.707107) = 1.000000$.

The example shows how spectral projections break an observable into exact, mutually orthogonal yes/no questions. Each question has a probability, here $\langle 0 | P | 0 \rangle$, and the expectation value is recovered by weighting each outcome by its probability.

There is an exact correspondence between projections in $B(\mathcal{H})$ and closed subspaces of $\mathcal{H}$. A projection P picks out the subspace $P\mathcal{H}$, which consists of everything you get by applying P to vectors in $\mathcal{H}$. Conversely, every closed subspace has a unique projection onto it. If $P \in \mathcal{M}$, the subspace $P\mathcal{H}$ is said to “belong to $\mathcal{M}$.” Physically, it is a subspace that an observer with access to $\mathcal{M}$ can single out by a measurement. The largest projection is the identity $\mathbf{1}$ (the whole space). Projections are partially ordered: $P \leq Q$ means $PQ = P$, or equivalently $P\mathcal{H} \subseteq Q\mathcal{H}$, so the subspace of P sits entirely inside the subspace of Q .

Two projections $P, Q \in \mathcal{M}$ are called **Murray–von Neumann equivalent**, written $P \sim Q$, if there is a **partial isometry** $V \in \mathcal{M}$ with $P = V^\dagger V$ and $Q = VV^\dagger$. A partial isometry is an operator that maps its input subspace onto its output subspace without stretching or shrinking lengths. So the condition says that V maps the subspace $P\mathcal{H}$ isometrically (preserving lengths) onto the subspace $Q\mathcal{H}$, and that V itself is available inside $\mathcal{M}$. This is the algebra-relative version of “these two subspaces have the same size.” It is not the ordinary linear-algebra notion of dimension, which looks only at the subspace itself. It is tied to what an observer with access to $\mathcal{M}$ can actually do to compare one subspace with another.

Worked example. Return to $\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L$ on $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ from the double-commutant example above. Let $P = |0\rangle_R \langle 0|_R \otimes \mathbf{1}_L$ and $Q = |1\rangle_R \langle 1|_R \otimes \mathbf{1}_L$. Each is a rank-2 projection on the full 4-dimensional $\mathcal{H}$. It picks out a 2-dimensional subspace: the R -qubit is fixed to $|0\rangle$ or to $|1\rangle$, and the L -qubit can be anything. The operator $V = |1\rangle_R \langle 0|_R \otimes \mathbf{1}_L$ is in $\mathcal{M}$, since it has the form $A \otimes \mathbf{1}_L$ with $A = |1\rangle \langle 0|$. Direct matrix multiplication gives $V^\dagger V = |0\rangle_R \langle 0|_R \otimes \mathbf{1}_L = P$ and $VV^\dagger = |1\rangle_R \langle 1|_R \otimes \mathbf{1}_L = Q$. So $P \sim Q$, through a partial isometry that lies entirely inside $\mathcal{M}$. This captures the intuitive fact that “the R -qubit is $|0\rangle$ ” and “the R -qubit is $|1\rangle$ ” are subspaces of the same size, related by a flip that an R -observer can actually perform.

A nonzero projection $P \in \mathcal{M}$ is called **finite** if it is *not* equivalent to any strictly smaller projection $Q < P$ in $\mathcal{M}$. It is called **infinite** if it is equivalent to some strictly smaller projection $Q < P$ in $\mathcal{M}$. This definition takes some getting used to, because it does *not* match the linear-algebra notion of “finite-dimensional.” The following example shows the difference. The useful intuition turns out to be that the smallest projections pick out one copy of an irreducible representation.

Take $\mathcal{M} = B(\mathcal{H}_1) \otimes \mathbf{1}_2$ on $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, where $\mathcal{H}_2$ is *infinite*-dimensional. Let $P = |\psi\rangle\langle\psi| \otimes \mathbf{1}_2$ for a single unit vector $|\psi\rangle \in \mathcal{H}_1$. As a subspace of the full Hilbert space, $P\mathcal{H}$ is infinite-dimensional: it is a whole copy of $\mathcal{H}_2$. Yet P is a *finite* projection of the algebra $\mathcal{M}$. There is no strictly smaller projection in $\mathcal{M}$ equivalent to it, because $\mathcal{M}$ contains no nonzero projection strictly below P at all. Inside $\mathcal{M} \cong B(\mathcal{H}_1)$, the projection P corresponds to a rank-one projection on $\mathcal{H}_1$, the smallest nonzero rank there is. Here is the precise form of the irreducible-representation intuition. As an *abstract algebra*, forgetting the Hilbert space it acts on, $\mathcal{M}$ is isomorphic to $B(\mathcal{H}_1)$. That is a single copy of the algebra of all operators on $\mathcal{H}_1$, and it acts irreducibly on $\mathcal{H}_1$. The factor $\mathcal{H}_2$ in $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ only counts *how many copies* of this irreducible representation sit side by side in the big Hilbert space: one copy for each basis vector of $\mathcal{H}_2$. The projection $P = |\psi\rangle\langle\psi| \otimes \mathbf{1}_2$ picks out one direction *within* the irreducible representation, and it does so in every one of the infinitely many copies at once. That is why P is as small as $\mathcal{M}$ can make anything (finite, and in fact minimal, in the algebra) while being infinite-dimensional as a subspace of $\mathcal{H}$. Finiteness in this algebra-relative sense is a statement about the irreducible representation, not about the dimension of the ambient Hilbert space. Keeping this distinction in mind is what makes the type classification in the next section make sense, rather than look like a strange use of the word “finite.”

A projection P is called **minimal** in $\mathcal{M}$ if it is nonzero and $\mathcal{M}$ contains no nonzero projection strictly smaller than P . This is the algebra-relative notion of “as small as a measurement outcome can be.” Every minimal projection is finite, since there is nothing smaller for it to be equivalent to. The converse fails: a projection can be finite without being minimal. For example, in $\mathcal{M} = B(\mathbb{C}^3)$ every projection is finite, because equivalent projections have equal rank. But $\text{diag}(1, 1, 0)$ is not minimal, since $\text{diag}(1, 0, 0)$ lies strictly below it. More strikingly, an algebra can have no minimal projections at all while still having plenty of finite ones. This possibility is the heart of the classification in the next section.

Now apply these definitions to the largest projection, $P = \mathbf{1}$. For a von Neumann *factor* $\mathcal{M}$, the identity is either finite or infinite, and this one question sorts factors into two coarse classes. If $\mathbf{1}$ is finite, then every projection in $\mathcal{M}$ is also finite (this can be shown from the definitions; we omit the proof), and $\mathcal{M}$ is called a **finite von Neumann factor**. If $\mathbf{1}$ is infinite, $\mathcal{M}$ is called an **infinite von Neumann factor**. In that case, when $\mathcal{H}$ is separable, any two infinite projections in $\mathcal{M}$ are equivalent to each other, so in particular every infinite projection is equivalent to the identity itself (also stated without proof).

2.7 Classification of von Neumann factors

Building a dimension function out of nothing but $\sim$ and $\leq$

Here is the payoff of the last two sections. Using only the equivalence relation $P \sim Q$ and the ordering $P \leq Q$, one can prove the following theorem (a standard result, stated here without proof). Every von Neumann factor $\mathcal{M}$ has a **dimension function** d . It assigns a number $d(P) \in [0, \infty]$ to each projection $P \in \mathcal{M}$, and it has three properties:

- $d(P) = d(Q)$ if and only if $P \sim Q$;
- $d(P + Q) = d(P) + d(Q)$ whenever P and Q are orthogonal, $PQ = 0$;
- $d(P) < \infty$ if and only if P is finite.

These properties fix d up to an overall constant that you are free to rescale. In words: equivalent subspaces get the same size, sizes add, and finite projections get finite sizes. Additivity also makes d respect the ordering. If $P \leq Q$, then $Q - P$ is a projection orthogonal to P , so $d(Q) = d(P) + d(Q - P) \geq d(P)$. The inequality is strict when $P < Q$ and P is finite, because $d(Q - P) > 0$ for the nonzero projection $Q - P$. These are exactly the properties you would want from anything worth calling a “size.” The function d can also be packaged as a **trace**, $\text{tr}(P) \equiv d(P)$, extended from projections to other elements of $\mathcal{M}$ by linearity, using the fact that every self-adjoint operator is built from its spectral projections. When $d(\mathbf{1}) = \infty$, this tr is finite only on part of $\mathcal{M}$, just as the ordinary trace on an infinite-dimensional Hilbert space is finite only for trace-class operators. This tr plays the role of the ordinary matrix trace, generalized to an abstract algebra with no particular Hilbert space singled out.

Why is the ordinary formula “dimension = $\text{Tr } P$ ” not the right notion here? Here Tr is the ordinary Hilbert-space trace, the sum of diagonal entries. The answer shows why an algebra-relative $d(P)$ is needed at all. Take again $P = |\psi\rangle\langle\psi| \otimes \mathbf{1}_2$ on $\mathcal{H}_1 \otimes \mathcal{H}_2$ with $\mathcal{H}_2$ infinite-dimensional, the same example as above. The Hilbert-space trace $\text{Tr } P$ is infinite, because $P\mathcal{H}$ is an infinite-dimensional subspace of $\mathcal{H}$. But from the point of view of the algebra $\mathcal{M} \cong B(\mathcal{H}_1)$, the projection P is the smallest nonzero thing there is: rank one, within $\mathcal{H}_1$. The ordinary trace Tr sees the raw size of the ambient Hilbert space, and for describing what $\mathcal{M}$ can distinguish it gives the wrong answer. The algebra-relative $d(P)$, normalized so that $d(P) = 1$ for this minimal P , correctly reports “this is the smallest measurable unit $\mathcal{M}$ has access to,” however large the corresponding subspace of $\mathcal{H}$ is. This is the concrete sense in which tr is a *renormalized* trace. It strips away the part of $\mathcal{H}$ that $\mathcal{M}$ never had access to, and it reports only the size that $\mathcal{M}$ itself can distinguish.

The three types

The whole classification now follows from one question: what values can $d(P)$ take, as P ranges over all the projections in $\mathcal{M}$?

- **Type I: $\mathcal{M}$ has minimal projections.** Normalize d so that a minimal projection has $d = 1$. You are always free to rescale d by an overall constant, so this is only a choice of units, like choosing what counts as one unit of length. Every projection is then a sum of mutually orthogonal minimal projections, and it can be shown that $d(P)$ is a nonnegative integer or ∞

for every P . This is the familiar notion of dimension, now derived from the algebra instead of assumed. If $n = d(\mathbf{1})$ is finite, $\mathcal{M}$ is called type I_n . If $d(\mathbf{1}) = \infty$, so that there is no upper bound on the number of mutually orthogonal minimal projections, $\mathcal{M}$ is type I_∞ . For an ordinary separable Hilbert space $\mathcal{H}$, the algebra $B(\mathcal{H})$ itself is type I_n if $\mathcal{H}$ is n -dimensional, and type I_∞ if $\mathcal{H}$ is infinite-dimensional with a countable basis. In this case tr is the ordinary matrix trace $\text{Tr}_{\mathcal{H}}$, with no renormalization needed, because there is no part of $\mathcal{H}$ that the algebra fails to see.

The most important fact tying this back to ordinary quantum mechanics is the following. $\mathcal{M}$ is a type I factor *if and only if* there is a Hilbert space factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ with

$$\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L, \quad \text{tr} = \text{Tr}_{\mathcal{H}_R}, \quad \mathcal{M}' = \mathbf{1}_R \otimes B(\mathcal{H}_L).$$

This result closes the type classification, and Chapter 3 uses it constantly. Many books state it without proof. The proof is short enough to give here, and it shows exactly how minimal projections build a tensor product.

Key Result / Core Concept: Derivation of the Type I Factorization Theorem

Theorem: A von Neumann algebra $\mathcal{M} \subseteq B(\mathcal{H})$ is a type I factor if and only if there exists a unitary isomorphism $U : \mathcal{H} \xrightarrow{\sim} \mathcal{H}_R \otimes \mathcal{H}_L$ such that

$$UMU^\dagger = B(\mathcal{H}_R) \otimes \mathbf{1}_L, \quad UM'U^\dagger = \mathbf{1}_R \otimes B(\mathcal{H}_L).$$

Proof ($\implies$):

1. **Minimal projection and orthogonal resolution.** By the definition of type I, $\mathcal{M}$ contains a minimal projection $P \in \mathcal{M}$. Because P is minimal, the compressed algebra $P\mathcal{M}P$ contains only multiples of P :

$$P\mathcal{M}P = \mathbb{C}P.$$

(The reason: $P\mathcal{M}P$ is a von Neumann algebra on $P\mathcal{H}$ whose only projections are 0 and P , and a von Neumann algebra is generated by its projections.) Since $\mathcal{M}$ is a factor, its center is trivial, $\mathcal{Z}(\mathcal{M}) \equiv \mathcal{M} \cap \mathcal{M}' = \mathbb{C}\mathbf{1}$. So the central support of P (the smallest central projection above P) is $c(P) = \mathbf{1}$. By the comparison theorem for projections in a factor, any two minimal projections are Murray–von Neumann equivalent. By Zorn’s lemma, we can choose a maximal family $\{P_i\}_{i \in I}$ of mutually orthogonal minimal projections, each equivalent to P . Maximality together with $c(P) = \mathbf{1}$ implies that their sum is the identity on $\mathcal{H}$:

$$\sum_{i \in I} P_i = \mathbf{1}_{\mathcal{H}}, \quad P_i P_j = \delta_{ij} P_i.$$

2. **Equivalence through partial isometries.** Fix a base index $0 \in I$ with $P_0 \equiv P$.

Since $P_i \sim P$, there are partial isometries $V_i \in \mathcal{M}$ with

$$V_i^\dagger V_i = P, \quad V_i V_i^\dagger = P_i \quad (\text{with } V_0 \equiv P).$$

So V_i maps the base subspace $P\mathcal{H}$ isometrically onto the orthogonal subspace $P_i\mathcal{H}$.

3. Construction of the unitary U . Define two Hilbert spaces:

$$\mathcal{H}_R \equiv \ell^2(I) \quad \text{with orthonormal basis } \{|i\rangle\}_{i \in I}, \quad \mathcal{H}_L \equiv P\mathcal{H}.$$

Define the linear map $U : \mathcal{H} \rightarrow \mathcal{H}_R \otimes \mathcal{H}_L$ by its action on any vector $|\psi\rangle \in \mathcal{H}$:

$$U|\psi\rangle \equiv \sum_{i \in I} |i\rangle \otimes (V_i^\dagger |\psi\rangle).$$

Each $V_i^\dagger |\psi\rangle = PV_i^\dagger |\psi\rangle$ lies in $P\mathcal{H} = \mathcal{H}_L$, so this is well defined. We check that U is an isometry:

$$\begin{aligned} \|U|\psi\rangle\|^2 &= \sum_{i \in I} \|V_i^\dagger |\psi\rangle\|^2 \stackrel{(1)}{=} \sum_{i \in I} \langle \psi | V_i V_i^\dagger | \psi \rangle \\ &\stackrel{(2)}{=} \sum_{i \in I} \langle \psi | P_i | \psi \rangle \stackrel{(3)}{=} \left\langle \psi \left| \sum_{i \in I} P_i \right| \psi \right\rangle \stackrel{(4)}{=} \langle \psi | \psi \rangle. \end{aligned}$$

(1) $\|v\|^2 = \langle v | v \rangle$ applied to $v = V_i^\dagger |\psi\rangle$, using $(V_i^\dagger)^\dagger = V_i$. **(2)** the partial isometry relation $V_i V_i^\dagger = P_i$ from Step 2 above. **(3)** linearity of the inner product, pulling the sum through $\langle \psi | \cdot | \psi \rangle$. **(4)** the resolution of the identity, $\sum_i P_i = \mathbf{1}_{\mathcal{H}}$, established in Step 1.

The adjoint map $U^\dagger : \mathcal{H}_R \otimes \mathcal{H}_L \rightarrow \mathcal{H}$ acts on elementary basis tensors as

$$U^\dagger(|i\rangle \otimes |\phi_L\rangle) = V_i |\phi_L\rangle, \quad |\phi_L\rangle \in P\mathcal{H}.$$

Computing $UU^\dagger$ on basis vectors:

$$\begin{aligned} UU^\dagger(|j\rangle \otimes |\phi_L\rangle) &\stackrel{(1)}{=} U(V_j |\phi_L\rangle) \\ &\stackrel{(2)}{=} \sum_{i \in I} |i\rangle \otimes (V_i^\dagger V_j |\phi_L\rangle) \stackrel{(3)}{=} |j\rangle \otimes (P |\phi_L\rangle) = |j\rangle \otimes |\phi_L\rangle. \end{aligned}$$

(1) the definition of $U^\dagger$ on the basis tensor $|j\rangle \otimes |\phi_L\rangle$. **(2)** the definition of U applied to the vector $V_j |\phi_L\rangle$. **(3)** $V_i^\dagger V_j = V_i^\dagger P_i P_j V_j = \delta_{ij} P$, so only the $i = j$ term in the sum survives, and $P |\phi_L\rangle = |\phi_L\rangle$ since $|\phi_L\rangle \in P\mathcal{H}$.

So U is an isometry with $UU^\dagger = \mathbf{1}$, which means U is unitary.

4. Action on the algebra $\mathcal{M}$. Let $A \in \mathcal{M}$ be any element. Compute $UAU^\dagger$ on

$|j\rangle \otimes |\phi_L\rangle$:

$$\begin{aligned} UAU^\dagger(|j\rangle \otimes |\phi_L\rangle) &\stackrel{(1)}{=} U(AV_j|\phi_L\rangle) \\ &\stackrel{(2)}{=} \sum_{i \in I} |i\rangle \otimes (V_i^\dagger AV_j|\phi_L\rangle). \end{aligned}$$

(1) the definition of $U^\dagger$ on basis tensors, then applying A . (2) the definition of U applied to the vector $AV_j|\phi_L\rangle$.

The key object is the operator $V_i^\dagger AV_j$. It lies in PMP :

$$V_i^\dagger AV_j = (PV_i^\dagger)A(V_jP) = P(V_i^\dagger AV_j)P \in PMP.$$

Because P is minimal, $PMP = \mathbb{C}P$. Therefore $V_i^\dagger AV_j$ is a multiple of P :

$$V_i^\dagger AV_j = a_{ij}P \quad \text{for some number } a_{ij} \in \mathbb{C}.$$

Acting on $|\phi_L\rangle \in P\mathcal{H}$, this gives $V_i^\dagger AV_j|\phi_L\rangle = a_{ij}|\phi_L\rangle$. Thus

$$\begin{aligned} UAU^\dagger(|j\rangle \otimes |\phi_L\rangle) &\stackrel{(1)}{=} \sum_{i \in I} a_{ij} |i\rangle \otimes |\phi_L\rangle \\ &\stackrel{(2)}{=} \left(\sum_{i,k \in I} a_{ik} |i\rangle \langle k| \otimes \mathbf{1}_L \right) (|j\rangle \otimes |\phi_L\rangle). \end{aligned}$$

(1) $V_i^\dagger AV_j|\phi_L\rangle = a_{ij}|\phi_L\rangle$, just shown. (2) $\langle k|j\rangle = \delta_{kj}$ picks out exactly the $k = j$ column of the matrix (a_{ik}) .

So every operator in $\mathcal{M}$ acts trivially on $\mathcal{H}_L$ and as a matrix on $\mathcal{H}_R$. Conversely, every rank-one operator $|i\rangle \langle j| \otimes \mathbf{1}_L$ comes from an element of $\mathcal{M}$: a computation like the one above gives $UV_iV_j^\dagger U^\dagger = |i\rangle \langle j| \otimes \mathbf{1}_L$, and $V_iV_j^\dagger \in \mathcal{M}$. Linear combinations of these operators are weakly dense in $B(\mathcal{H}_R) \otimes \mathbf{1}_L$, and $\mathcal{M}$ is weakly closed. Thus $UMU^\dagger = B(\mathcal{H}_R) \otimes \mathbf{1}_L$.

5. **The commutant and the trace.** Conjugating by the unitary U and using the previous step:

$$\begin{aligned} UM'U^\dagger &\stackrel{(1)}{=} (UMU^\dagger)' \\ &\stackrel{(2)}{=} (B(\mathcal{H}_R) \otimes \mathbf{1}_L)' \stackrel{(3)}{=} \mathbf{1}_R \otimes B(\mathcal{H}_L). \end{aligned}$$

(1) B commutes with every element of $\mathcal{M}$ exactly when $UBU^\dagger$ commutes with every element of $UMU^\dagger$, since $U^\dagger U = \mathbf{1}$. (2) $UMU^\dagger = B(\mathcal{H}_R) \otimes \mathbf{1}_L$ from Step 4. (3) the commutant of “all operators on one tensor factor” is “all operators on the other,” as in the first double-commutant example earlier in this chapter.

The trace tr on $\mathcal{M}$, normalized so that minimal projections have trace 1, corresponds under U to the standard trace $\text{Tr}_{\mathcal{H}_R}$ on $B(\mathcal{H}_R)$.

6. **Proof** ($\Leftarrow$). Suppose $\mathcal{M} \cong B(\mathcal{H}_R) \otimes \mathbf{1}_L$. Its center is $\mathbb{C}\mathbf{1}$ (as in the first double-commutant example), so $\mathcal{M}$ is a factor. Take any one-dimensional projection $p = |i\rangle\langle i|$ on $\mathcal{H}_R$. Then $P = p \otimes \mathbf{1}_L \in \mathcal{M}$. For any $A = a \otimes \mathbf{1}_L \in \mathcal{M}$:

$$\begin{aligned} PAP &\stackrel{(1)}{=} (pap) \otimes \mathbf{1}_L \\ &\stackrel{(2)}{=} \langle i|a|i\rangle (p \otimes \mathbf{1}_L) \stackrel{(3)}{=} \langle i|a|i\rangle P \in \mathbb{C}P. \end{aligned}$$

(1) multiplying tensor products factor by factor, with $\mathbf{1}_L^3 = \mathbf{1}_L$. (2) $pap = |i\rangle\langle i|a|i\rangle\langle i| = \langle i|a|i\rangle p$. (3) the definition $P = p \otimes \mathbf{1}_L$.

So every projection in $\mathcal{M}$ below P is a multiple of P , hence 0 or P . This means P is minimal in $\mathcal{M}$, and $\mathcal{M}$ is a type I factor. ■

So the statement “the algebra describing this subsystem is type I” is the precise, general form of the statement “the ordinary tensor-product picture of textbooks such as Griffiths and Sakurai applies here.” Every worked example given so far in this chapter, including the two-qubit examples above, is type I by construction. Types II and III, described next, are what becomes possible when this is no longer true.

- **Type II: $\mathcal{M}$ has finite projections but no minimal ones.** This is a genuinely new possibility, and it sounds strange at first. The finite/infinite distinction still works, and finite projections can be compared in size using d . But there is no smallest nonzero size: below any nonzero projection there is always a strictly smaller nonzero one. The values of d are then forced to fill out a continuous range. Here is the idea. In a type II factor, any finite projection P can be split into two orthogonal pieces that are equivalent to each other, $P = P_1 + P_2$ with $P_1 \sim P_2$ (stated without proof). Additivity then gives $d(P_1) = d(P_2) = \frac{1}{2}d(P)$. Repeating the split gives projections with $d = 2^{-k}d(P)$ for every k , so d takes arbitrarily small positive values. Adding such pieces gives every dyadic fraction of $d(P)$, and one can show that every value in between is reached as well. So a von Neumann algebra can assign a *real number*, not only an integer, as the “dimension” of a subspace. Nothing like this happens in ordinary finite-dimensional linear algebra, where dimension is a whole number obtained by counting basis vectors. There are two subtypes, depending on whether $d(\mathbf{1})$ is finite. In type II_1 the identity has finite dimension. We normalize $d(\mathbf{1}) = \text{tr}(\mathbf{1}) = 1$, so that $d(P) \in (0, 1]$ for every nonzero projection: the trace is normalized exactly like a probability. In type II_∞ we have $d(\mathbf{1}) = \infty$, so $d(P) \in (0, \infty]$, and there is no natural normalization. Chapter 3 constructs a type II_1 factor by hand, from the $N \rightarrow \infty$ Bell-pair chain of Chapter 1. There, “real-valued dimension” stops being abstract and can be watched in a specific family of projections.
- **Type III: every nonzero projection is infinite.** This is the most extreme case. The finite/infinite distinction has no content, because no projection except zero is finite. Formally, $d(P) = \infty$ for every nonzero P . So there is no useful dimension function, and hence no (normal, semifinite) trace on $\mathcal{M}$ at all. This looks like a mathematical dead end. How could one talk about entanglement in an algebra with no density matrix, when density matrices are defined using a trace? Chapter 4 meets exactly this obstacle and overcomes it with a different

tool, Tomita–Takesaki modular theory; that chapter is the technical heart of the notes. Type III is also the type that is physically generic. The algebras of local regions in relativistic quantum field theory, and the boundary subalgebras of holography in the strict large- N limit, are both type III (in fact type III₁, as later chapters explain). This is not an exotic corner case. It is the everyday situation once you leave the world of finitely many qubits.

Types II and III can sound, on first exposure, like unphysical mathematical curiosities. They are not. The entangled spin chain of Chapter 1 already produces both, explicitly and by direct construction, and Chapter 3 works through that construction completely.

2.8 “Emergent” Hilbert space and von Neumann algebra: the GNS construction

Why this construction is needed

Here is the gap again, now that every piece needed to close it is in hand. The completeness condition of a von Neumann algebra (weak convergence) is stated in terms of matrix elements $\langle \xi | A_n | \eta \rangle$, so it assumes that a Hilbert space already exists. The completeness condition of a C^* -algebra (norm convergence) does not. It needs only an abstract norm with $\|A^\dagger A\| = \|A\|^2$. So you can specify a C^* -algebra $\mathcal{A}$ abstractly, with no Hilbert space anywhere. Suppose you are also given a state ω on $\mathcal{A}$, in the sense defined earlier in this chapter: a positive, normalized linear functional, which needs nothing beyond the algebra itself to define. Can a Hilbert space be *built*, rather than assumed, in which ω is represented by an ordinary vector? The Gelfand–Naimark–Segal (GNS) construction says yes, always. It does so by an explicit, mechanical recipe.

The construction, built up in stages

Stage 1: treat algebra elements as candidate vectors. This is the conceptual leap, so here it is in words before the formulas. An element $C \in \mathcal{A}$ is reinterpreted as standing for “the state you would get by applying the operation C to a fixed reference configuration.” Two elements C_1, C_2 should count as the *same* state if no measurement, computed with ω , could ever tell them apart.

Stage 2: use ω to build an inner product on $\mathcal{A}$ itself. For any $A, B \in \mathcal{A}$, define

$$\langle A | B \rangle \equiv \omega(A^\dagger B).$$

Check that this deserves to be called an inner product. It is positive, $\langle A | A \rangle = \omega(A^\dagger A) \geq 0$, directly from the positivity of ω . And it has the correct conjugate symmetry:

$$\begin{aligned} \langle B | A \rangle &= \omega(B^\dagger A) \stackrel{(1)}{=} \omega((A^\dagger B)^\dagger) \\ &\stackrel{(2)}{=} \omega(A^\dagger B)^* \stackrel{(3)}{=} \langle A | B \rangle^*. \end{aligned}$$

(1) $(A^\dagger B)^\dagger = B^\dagger A$. **(2)** $\omega(X^\dagger) = \omega(X)^*$, which holds for every positive functional (see the definitions of weights and states). **(3)** the definition $\langle A | B \rangle \equiv \omega(A^\dagger B)$.

Both required properties hold, using nothing but the definition of ω .

Stage 3: handle “zero-length” elements. There may be nonzero $X \in \mathcal{A}$ with $\omega(X^\dagger X) = 0$. These are algebra elements that ω cannot see at all. Call the set of all such X the null space $\mathcal{J}$. The Cauchy–Schwarz inequality (2.2) shows that $\mathcal{J}$ is a linear subspace. A short computation shows more: if $X \in \mathcal{J}$ and $A \in \mathcal{A}$ is any element, then $AX \in \mathcal{J}$ too. The proof is again an application of the Cauchy–Schwarz inequality:

$$\begin{aligned} 0 \leq \omega((AX)^\dagger(AX)) &\stackrel{(1)}{=} \omega(X^\dagger A^\dagger AX) \\ &\stackrel{(2)}{\leq} \omega(X^\dagger X)^{1/2} \omega((A^\dagger AX)^\dagger A^\dagger AX)^{1/2} \stackrel{(3)}{=} 0. \end{aligned}$$

(1) regroup $(AX)^\dagger(AX) = X^\dagger A^\dagger AX$. (2) the Cauchy–Schwarz inequality for states, $|\omega(A^\dagger B)|^2 \leq \omega(A^\dagger A) \omega(B^\dagger B)$, with A replaced by X and B replaced by $A^\dagger AX$. This is the only inequality in the chain. (3) $\omega(X^\dagger X) = 0$ because $X \in \mathcal{J}$, so the upper bound is exactly 0 whatever the second factor is.

So $\omega((AX)^\dagger(AX)) = 0$, which means $AX \in \mathcal{J}$. In the language of algebra, $\mathcal{J}$ is a *left ideal*. This computation does real work. It guarantees that quotienting out the null elements in Stage 4 is consistent: multiplying by A never turns a zero-length vector into a vector of nonzero length.

Stage 4: quotient and complete. Declare A and $A + X$ equivalent whenever $X \in \mathcal{J}$: algebra elements that differ only by something ω cannot see count as the same vector. Let $[A]$ denote the equivalence class of A . The inner product of Stage 2 does not depend on the choice of representative, because $|\omega(X^\dagger B)|^2 \leq \omega(X^\dagger X) \omega(B^\dagger B) = 0$ for $X \in \mathcal{J}$. On the quotient space $\mathcal{A}/\mathcal{J}$ the inner product is positive definite: by construction, no nonzero vector has zero length. Completing this space, by filling in the limits of Cauchy sequences just as the rational numbers are completed to the real numbers, gives a Hilbert space, called $\mathcal{H}_\omega$.

Stage 5: define the representation. The algebra acts on $\mathcal{H}_\omega$ by left multiplication:

$$\pi_\omega(A) |[C]\rangle \equiv |[AC]\rangle.$$

This is well defined. If C is replaced by $C + X$ with $X \in \mathcal{J}$, then AC changes by AX , which is also in $\mathcal{J}$ by Stage 3. It respects the multiplication of the algebra, $\pi_\omega(A)\pi_\omega(B) = \pi_\omega(AB)$. Indeed, applying $\pi_\omega(B)$ and then $\pi_\omega(A)$ to $[C]$ gives $[A(BC)] = [(AB)C]$, which is $\pi_\omega(AB)$ applied to $[C]$. For a C^* -algebra, one also has $\|\pi_\omega(A)[C]\| \leq \|A\| \|[C]\|$, so each $\pi_\omega(A)$ is bounded and extends to the completion $\mathcal{H}_\omega$.

Stage 6: identify the special vector. The equivalence class of the identity element, $[1]$, gets its own name, $|\Omega\rangle \equiv |[1]\rangle$. Two facts follow at once from the definitions above:

$$\pi_\omega(A) |\Omega\rangle = |[A \cdot 1]\rangle = |[A]\rangle, \quad \omega(A) = \langle \Omega | \pi_\omega(A) | \Omega \rangle.$$

The second follows from the first together with the definition of the inner product: $\langle \Omega | \pi_\omega(A) | \Omega \rangle = \langle [1] | [A] \rangle = \omega(1^\dagger A) = \omega(A)$. So in the Hilbert space just built, the vector $|\Omega\rangle$ reproduces the abstract state ω exactly, through the ordinary formula $\langle \Omega | A | \Omega \rangle$ from undergraduate quantum mechanics. This is the purpose of the construction. You gave it an abstract algebra and a number-valued rule ω . It gave back a Hilbert space and a vector inside it, and the usual formula for expectation values

holds by construction, not by assumption.

Physics Connection: Origin of the Vacuum $|0\rangle$

$|\Omega\rangle$ is not found somewhere outside this construction. It *is* the identity element of the algebra, $\mathbf{1} \in \mathcal{A}$, renamed once the inner product has been put on $\mathcal{A}$ in Stage 2. A vector and the Hilbert space it lives in are made in the same step, from the same two ingredients, $\mathcal{A}$ and ω . Neither exists before the other.

The construction also explains why a annihilates $|\Omega\rangle$ for the oscillator ground state, without using $|0\rangle$ as an input to define ω . Start from one algebraic condition, $\omega(a^\dagger a) = 0$, together with positivity, $\omega(A^\dagger A) \geq 0$ for every $A \in \mathcal{A}$, and normalization, $\omega(\mathbf{1}) = 1$. For any element Y , the Cauchy–Schwarz inequality (2.2) gives $|\omega(Ya)|^2 \leq \omega(YY^\dagger)\omega(a^\dagger a) = 0$. So $\omega(Ya) = 0$ for every Y , and taking adjoints, $\omega(a^\dagger Y) = \omega(Y^\dagger a)^* = 0$ as well. In particular $\omega(a) = \omega(a^\dagger) = 0$. Using $[a, a^\dagger] = 1$, every monomial in a and $a^\dagger$ can be rewritten as a combination of normal-ordered monomials $(a^\dagger)^m a^n$. Those with $m + n > 0$ either end in a or begin with $a^\dagger$, so ω gives them zero. Only the constant term survives, and $\omega(\mathbf{1}) = 1$ fixes its value. So the one condition $\omega(a^\dagger a) = 0$ determines ω completely. The oscillator box below uses the same bookkeeping to show $\langle 0|a^n (a^\dagger)^n|0\rangle = n!$. Here it is read as *defining* ω from the multiplication table of the algebra, not as evaluating a bra-ket that already existed.

Now apply Stage 3 to this ω . The null ideal is $\mathcal{J} = \{X : \omega(X^\dagger X) = 0\}$, and $a \in \mathcal{J}$, because $\omega(a^\dagger a) = 0$. Stage 4 quotients by exactly this $\mathcal{J}$, so every element of $\mathcal{J}$ becomes the zero vector. So $[a] = 0$, and $\pi_\omega(a)|\Omega\rangle = |[a]\rangle = 0$. The vacuum being annihilated by a is not a separate fact about the world that ω happened to match. It is what $[a] = 0$ means, once $\mathcal{J}$ is fixed by the one number $\omega(a^\dagger a)$.

This is the general answer to where the vacuum and its Hilbert space come from. The vector space is not searched for among candidates that already reproduce ω . It *is* $\mathcal{A}/\mathcal{J}$, completed. Its points are exactly as fine as ω can distinguish, and no finer: two algebra elements become the same vector exactly when ω assigns their difference zero length. Quotienting throws out the redundancy. Completing, by filling in the limits of Cauchy sequences, turns this inner-product space into a Hilbert space in the sense of Chapter 1.

In a field theory, ω is fixed in the same way, only more concretely, by a Euclidean path integral:

$$\omega(O(x_1) \cdots O(x_n)) = \frac{1}{Z} \int \mathcal{D}\phi \, O(x_1) \cdots O(x_n) e^{-S[\phi]}.$$

This formula computes every Euclidean correlator directly (Lorentzian correlators follow by analytic continuation), and no Hilbert space, vector, or bra-ket appears in it. The CFT vacuum of Chapter 6 and the JT-gravity states $|\beta\rangle$ of Chapter 9 are both defined this way. Canonical quantization, with its $|0\rangle$ and its Fock space, is the GNS representation built afterward from these correlators. Chapter 10 applies the same idea to string theory. There, different asymptotic vacua are proposed to be different states ω on one shared, background-independent algebra $\mathcal{A}_{\text{IIB}}$. Each has its own path-integral (or correlator) definition that needs no Hilbert space. Each one’s “ $|0\rangle$ ” is simply $[\mathbf{1}]$ inside the GNS space of *that particular* ω . It is never a single object shared across backgrounds. This is why the resulting Hilbert spaces can be

mutually inequivalent.

The set $\{\pi_\omega(A)|\Omega\rangle : A \in \mathcal{A}\}$ is dense in $\mathcal{H}_\omega$. This is automatic, because $\mathcal{H}_\omega$ was built as the completion of the set of classes $[A] = \pi_\omega(A)|\Omega\rangle$, so every vector in $\mathcal{H}_\omega$ can be approximated arbitrarily well by some $\pi_\omega(A)|\Omega\rangle$. A vector with this property is called **cyclic**: every state can be reached from $|\Omega\rangle$ using only the algebra. This is the same structure as the familiar construction of the harmonic-oscillator Hilbert space, by acting repeatedly on the vacuum $|0\rangle$ with creation operators. The only difference is that here it is a theorem, derived from the algebra and the state, rather than a separate postulate about how the Hilbert space is built.

Physics Connection: GNS and Oscillator Fock Space

The oscillator shows that GNS is the same calculation as the Fock-space construction you already know, run in the opposite direction. We compute the same quantities in two ways and compare the answers.

The ordinary way. Start with a Hilbert space spanned by orthonormal vectors $|0\rangle, |1\rangle, |2\rangle, \dots$, and define two operators by $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$ and $a|n\rangle = \sqrt{n}|n-1\rangle$. Then $[a, a^\dagger] = 1$, $a|0\rangle = 0$, and $(a^\dagger)^m|0\rangle = \sqrt{m!}|m\rangle$. Consequently

$$\langle 0|a^n(a^\dagger)^m|0\rangle = \sqrt{n!m!} \langle n|m\rangle = n! \delta_{nm}.$$

Here the Hilbert space comes first, and the operators are defined on it.

The GNS way. Now forget the Hilbert space. Let $\mathcal{A}$ be the $*$ -algebra generated by $1, a, a^\dagger$ with only the relation $[a, a^\dagger] = 1$. Its elements are polynomials in a and $a^\dagger$. Let ω be the state fixed by the single condition $\omega(a^\dagger a) = 0$, as in the previous box. Since a and $a^\dagger$ are unbounded, $\mathcal{A}$ is a $*$ -algebra but not a C^* -algebra. The GNS steps still go through, with $\pi_\omega(a)$ and $\pi_\omega(a^\dagger)$ defined on the dense subspace of finite combinations of the classes $[(a^\dagger)^n]$. Follow the recipe. The Stage 2 inner product on $\mathcal{A}$ is $\langle (a^\dagger)^n | (a^\dagger)^m \rangle \equiv \omega(a^n(a^\dagger)^m)$. First, it vanishes for $n \neq m$. Write $N = a^\dagger a$. The previous box showed $\omega(a^\dagger Y) = 0$ and $\omega(Ya) = 0$ for every Y , so $\omega(NX) = \omega(XN) = 0$ for every X . Also $[N, a^n(a^\dagger)^m] = (m-n)a^n(a^\dagger)^m$, because $[N, a] = -a$ and $[N, a^\dagger] = a^\dagger$. Hence

$$(m-n)\omega(a^n(a^\dagger)^m) = \omega(Na^n(a^\dagger)^m) - \omega(a^n(a^\dagger)^m N) = 0,$$

so $\omega(a^n(a^\dagger)^m) = 0$ unless $n = m$. For $n = m$, write $\omega(X)$ as $\langle 0|X|0\rangle$, as the ordinary way would; the computation uses only the algebra and $\omega(Ya) = 0$. It is one recursive step:

$$\begin{aligned} \langle 0|a^n(a^\dagger)^n|0\rangle &\stackrel{(1)}{=} \langle 0|a^{n-1}[a, (a^\dagger)^n]|0\rangle + \langle 0|a^{n-1}(a^\dagger)^n a|0\rangle \\ &\stackrel{(2)}{=} n \langle 0|a^{n-1}(a^\dagger)^{n-1}|0\rangle + 0. \end{aligned}$$

(1) split $a(a^\dagger)^n = [a, (a^\dagger)^n] + (a^\dagger)^n a$ and distribute. (2) the oscillator identity $[a, (a^\dagger)^n] = n(a^\dagger)^{n-1}$; the second term has the form $\omega(Ya)$, which is zero.

This is a recursion, $\langle 0|a^n(a^\dagger)^n|0\rangle = n \langle 0|a^{n-1}(a^\dagger)^{n-1}|0\rangle$, with base case $\omega(1) = 1$ at $n = 0$. Its

solution is $n!$. (For example, $n = 1$ gives $1 \cdot 1 = 1 = 1!$ and $n = 2$ gives $2 \cdot 1 = 2 = 2!$.) So the classes $[(a^\dagger)^n]$ are orthogonal, with squared norm $n!$. Normalizing gives $|n\rangle \equiv [(a^\dagger)^n]/\sqrt{n!}$. The representation acts by $\pi_\omega(a^\dagger)[(a^\dagger)^n] = [(a^\dagger)^{n+1}]$, which after normalization is $\pi_\omega(a^\dagger)|n\rangle = \sqrt{n+1}|n+1\rangle$. Also $\pi_\omega(a)[(a^\dagger)^n] = [a(a^\dagger)^n] = n[(a^\dagger)^{n-1}]$ (System 3 of the next box shows this), which after normalization is $\pi_\omega(a)|n\rangle = \sqrt{n}|n-1\rangle$. The cyclic vector $|\Omega\rangle = [1]$ is $|0\rangle$, and $\pi_\omega(a)|\Omega\rangle = [a] = 0$.

Comparison. The two computations give the same answers:

	The ordinary way	The GNS way
Inner products	$\langle 0 a^n(a^\dagger)^m 0\rangle = n!\delta_{nm}$	$\omega(a^n(a^\dagger)^m) = n!\delta_{nm}$
Basis	$ n\rangle = (a^\dagger)^n 0\rangle/\sqrt{n!}$	$ n\rangle = [(a^\dagger)^n]/\sqrt{n!}$
Raising	$a^\dagger n\rangle = \sqrt{n+1} n+1\rangle$	$\pi_\omega(a^\dagger) n\rangle = \sqrt{n+1} n+1\rangle$
Lowering	$a n\rangle = \sqrt{n} n-1\rangle$	$\pi_\omega(a) n\rangle = \sqrt{n} n-1\rangle$
Ground state	$a 0\rangle = 0$	$\pi_\omega(a) \Omega\rangle = [a] = 0$

The map $|n\rangle \mapsto [(a^\dagger)^n]/\sqrt{n!}$ is a unitary between the two Hilbert spaces, and it carries a and $a^\dagger$ of the ordinary way to $\pi_\omega(a)$ and $\pi_\omega(a^\dagger)$. So the two routes give the same Hilbert space, the same basis, the same matrix elements, and the same ground state. These are exact equalities, not approximations.

Nothing here is new physics. It is the Fock-space construction from a first course on the harmonic oscillator, run backward. What differs is only the direction of the logic. In the ordinary way you are handed $\mathcal{H} = L^2(\mathbb{R})$ (or an abstract countable basis) first, and $a, a^\dagger$ are defined as operators on it. In the GNS way the only inputs are the relation $[a, a^\dagger] = 1$ and the state ω (equivalently, “the vacuum is annihilated by a ”). The whole Fock space, every $|n\rangle$ and every matrix element, is an output. This is the sense in which the section on weights and states called the Hilbert space derived rather than fundamental. For the oscillator nothing about the physics changes, because this ω produces the usual Fock space. (For one degree of freedom this is no accident: by the Stone–von Neumann theorem, the usual representation is, up to unitary equivalence, the only well-behaved irreducible one.) The cases that matter later are those in which a different state produces an *inequivalent* Hilbert space from the same algebra. Examples are the N -Bell-pair chain, treated at the end of this chapter and in Chapters 3 and 4, and the different asymptotic vacua of quantum gravity in Chapter 10.

Worked Example: Concrete GNS Construction for Three Physical Systems

To make the six-stage recipe fully mechanical, we now build the GNS Hilbert space, inner product, null ideal, representation, and cyclic vector explicitly for three basic physical systems. In each case we also compare the result with the answer ordinary quantum mechanics gives.

1. System 1: a pure state of a single qubit ($\mathcal{A} = M_2(\mathbb{C})$).

- *Abstract algebra:* $\mathcal{A} = M_2(\mathbb{C})$, the 4-dimensional space of 2×2 complex matrices, spanned by the matrix units $e_{ij} \equiv |i\rangle\langle j|$ for $i, j \in \{0, 1\}$.
- *State:* let $\omega_\uparrow$ be the pure spin-up state, $\omega_\uparrow(A) \equiv \langle 0|A|0\rangle = A_{00}$.

- *Pre-inner product on $\mathcal{A}$* : for any $A, B \in M_2(\mathbb{C})$,

$$\begin{aligned}\langle A|B \rangle_{\mathcal{A}} &= \omega_{\uparrow}(A^{\dagger}B) \stackrel{(1)}{=} (A^{\dagger}B)_{00} \\ &\stackrel{(2)}{=} A_{00}^*B_{00} + A_{10}^*B_{10}.\end{aligned}$$

(1) the definition $\omega_{\uparrow}(X) = X_{00}$. **(2)** matrix multiplication, $(A^{\dagger}B)_{00} = \sum_k (A^{\dagger})_{0k}B_{k0}$, with $(A^{\dagger})_{0k} = A_{k0}^*$.

Only the first column of each matrix enters. The second-column entries $A_{01}, A_{11}, B_{01}, B_{11}$ do not appear at all.

- *Null ideal $\mathcal{J}$* : an element $X \in \mathcal{A}$ is null exactly when

$$\langle X|X \rangle_{\mathcal{A}} = |X_{00}|^2 + |X_{10}|^2 = 0 \iff X_{00} = 0 \quad \text{and} \quad X_{10} = 0.$$

So the null space is the 2-dimensional left ideal of matrices whose first column vanishes:

$$\mathcal{J} = \left\{ \begin{pmatrix} 0 & a \\ 0 & b \end{pmatrix} : a, b \in \mathbb{C} \right\} = \text{span}\{e_{01}, e_{11}\}.$$

- *Quotient space $\mathcal{H}_{\omega} = \mathcal{A}/\mathcal{J}$* : modulo $\mathcal{J}$, every matrix reduces to its first column:

$$\begin{aligned}[A] &\stackrel{(1)}{=} A_{00}[e_{00}] + A_{10}[e_{10}] + A_{01}[e_{01}] + A_{11}[e_{11}] \\ &\stackrel{(2)}{=} A_{00}[e_{00}] + A_{10}[e_{10}].\end{aligned}$$

(1) write $A = \sum_{ij} A_{ij}e_{ij}$ and use linearity of $A \mapsto [A]$. **(2)** $e_{01}, e_{11} \in \mathcal{J}$, so $[e_{01}] = [e_{11}] = 0$.

So the quotient space is 2-dimensional. Define the basis

$$|0\rangle \equiv [e_{00}], \quad |1\rangle \equiv [e_{10}].$$

Its inner products are $\langle 0|0\rangle = 1$, $\langle 1|1\rangle = 1$, $\langle 0|1\rangle = 0$, so it is orthonormal, and $\mathcal{H}_{\omega} \cong \mathbb{C}^2$.

- *Representation π_{ω} and cyclic vector*: left multiplication by A gives

$$\begin{aligned}\pi_{\omega}(A)|0\rangle &= [Ae_{00}] = A_{00}|0\rangle + A_{10}|1\rangle, \\ \pi_{\omega}(A)|1\rangle &= [Ae_{10}] = A_{01}|0\rangle + A_{11}|1\rangle.\end{aligned}$$

This is exactly ordinary matrix multiplication of A on $\mathbb{C}^2$. The cyclic vector is the class of the identity:

$$\begin{aligned}|\Omega\rangle &= [\mathbf{1}_2] \stackrel{(1)}{=} [e_{00} + e_{11}] \\ &\stackrel{(2)}{=} [e_{00}] \stackrel{(3)}{=} |0\rangle.\end{aligned}$$

(1) $\mathbf{1}_2 = e_{00} + e_{11}$. **(2)** $e_{11} \in \mathcal{J}$, so it is zero in the quotient. **(3)** the definition

$|0\rangle \equiv [e_{00}]$.

- *Comparison with ordinary quantum mechanics:* a qubit in the state $|0\rangle$ is described by the Hilbert space $\mathbb{C}^2$, the vector $|0\rangle$, and observables acting as 2×2 matrices. GNS gives exactly this. It reduces the 4-dimensional matrix algebra to the familiar 2-dimensional qubit space $\mathbb{C}^2$. Since $\omega_\uparrow$ is pure, the representation is irreducible: $\pi_\omega(\mathcal{A})$ is all of $B(\mathbb{C}^2)$, as the first structural fact after this box predicts.

2. System 2: a mixed (thermal) state on $M_2(\mathbb{C})$, and the thermofield double.

- *State:* let $\omega_\beta(A) \equiv \text{Tr}(\rho_\beta A)$, with the full-rank density matrix $\rho_\beta = \text{diag}(p_0, p_1)$, where $p_0, p_1 > 0$ and $p_0 + p_1 = 1$.
- *Pre-inner product on $\mathcal{A}$:*

$$\begin{aligned} \langle A|B\rangle_{\mathcal{A}} &= \text{Tr}(\rho_\beta A^\dagger B) = p_0(A_{00}^* B_{00} + A_{10}^* B_{10}) \\ &\quad + p_1(A_{01}^* B_{01} + A_{11}^* B_{11}). \end{aligned}$$

Each column of the matrix now enters, weighted by the corresponding eigenvalue of ρ_β .

- *Null ideal $\mathcal{J}$:* $\langle X|X\rangle_{\mathcal{A}} = 0$ means

$$p_0(|X_{00}|^2 + |X_{10}|^2) + p_1(|X_{01}|^2 + |X_{11}|^2) = 0.$$

Because $p_0 > 0$ and $p_1 > 0$, every matrix entry must vanish, so $X = 0$. The null ideal is trivial, $\mathcal{J} = \{0\}$.

- *The GNS Hilbert space:*

$$\mathcal{H}_\omega = \mathcal{A}/\{0\} \cong M_2(\mathbb{C}) \cong \mathbb{C}^2 \otimes \mathbb{C}^2.$$

The Hilbert space has dimension 4, not 2. Since $\|[e_{ij}]\|^2 = \text{Tr}(\rho_\beta e_{ji} e_{ij}) = p_j$, the normalized basis vectors are

$$\begin{aligned} |00\rangle &\equiv \frac{1}{\sqrt{p_0}}[e_{00}], & |10\rangle &\equiv \frac{1}{\sqrt{p_0}}[e_{10}], \\ |01\rangle &\equiv \frac{1}{\sqrt{p_1}}[e_{01}], & |11\rangle &\equiv \frac{1}{\sqrt{p_1}}[e_{11}]. \end{aligned}$$

The first label is the row index of the matrix and the second label is its column index.

- *The thermofield double vector:* the cyclic vector is

$$|\Omega\rangle \equiv [\mathbf{1}_2] = [e_{00}] + [e_{11}] = \sqrt{p_0}|00\rangle + \sqrt{p_1}|11\rangle.$$

This is an entangled pure state. When $p_j \propto e^{-\beta E_j}$ is a thermal distribution, it is exactly the **thermofield double** (TFD) state. Left multiplication, $[Ae_{ij}] = \sum_k A_{ki}[e_{kj}]$, acts only on the row label, so $\pi_\omega(\mathcal{A}) = M_2(\mathbb{C}) \otimes \mathbf{1}$. Its commutant is $\pi_\omega(\mathcal{A})' = \mathbf{1} \otimes M_2(\mathbb{C})$, acting on the column label.

- *Comparison with ordinary quantum mechanics:* the standard way to describe a mixed state ρ by a pure state is to purify it: add a second copy of the system and write $\sum_j \sqrt{p_j} |j\rangle |j\rangle$. GNS produces exactly this purification. The second tensor factor, which ordinary quantum mechanics introduces by hand, appears here automatically as the column index of the matrix.

3. System 3: the bosonic oscillator (canonical commutation relation) and Fock space.

- *Abstract algebra:* the $*$ -algebra of polynomials in the ladder operators a and $a^\dagger$, with $[a, a^\dagger] = \mathbf{1}$.
- *State:* the vacuum state, defined algebraically by $\omega_0(\mathbf{1}) = 1$ and $\omega_0(a^\dagger a) = 0$. Repeated use of $[a, a^\dagger] = \mathbf{1}$ fixes all the values $\omega_0(a^m (a^\dagger)^n) = n! \delta_{mn}$, as shown in the box above.
- *Null ideal $\mathcal{J}$:* every element that ends with an annihilation operator is null, and the null ideal is exactly $\mathcal{J} = \mathcal{A}a$. Modulo $\mathcal{J}$, every element reduces to a polynomial in $a^\dagger$ alone: $[A] = \sum_n c_n [(a^\dagger)^n]$.
- *Inner product and orthonormal basis:*

$$\left\langle [(a^\dagger)^m] \middle| [(a^\dagger)^n] \right\rangle = \omega_0(a^m (a^\dagger)^n) = n! \delta_{mn}.$$

Normalizing these vectors gives the standard Fock basis:

$$|n\rangle \equiv \frac{1}{\sqrt{n!}} [(a^\dagger)^n], \quad \langle m | n \rangle = \delta_{mn}.$$

- *Action of the representation:*

$$\pi_\omega(a^\dagger) |n\rangle = \sqrt{n+1} |n+1\rangle, \quad \pi_\omega(a) |n\rangle = \sqrt{n} |n-1\rangle.$$

The first follows from $\pi_\omega(a^\dagger)[(a^\dagger)^n] = [(a^\dagger)^{n+1}]$ and the normalization. For the second,

$$\begin{aligned} [a(a^\dagger)^n] &= [[a, (a^\dagger)^n] + (a^\dagger)^n a] \stackrel{(1)}{=} n[(a^\dagger)^{n-1}] + [(a^\dagger)^n a] \\ &\stackrel{(2)}{=} n[(a^\dagger)^{n-1}]. \end{aligned}$$

- (1) the commutator identity $[a, (a^\dagger)^n] = n(a^\dagger)^{n-1}$, the same one used in the box above.
 (2) $(a^\dagger)^n a$ ends in an annihilation operator, so it lies in $\mathcal{J} = \mathcal{A}a$ and vanishes in the quotient.

Dividing by the normalizations gives $\pi_\omega(a) |n\rangle = n\sqrt{(n-1)!}/\sqrt{n!} |n-1\rangle = \sqrt{n} |n-1\rangle$.

- *Comparison with ordinary quantum mechanics:* completing the space gives the infinite-dimensional Fock space $\mathcal{H}_{\text{Fock}} \cong \ell^2(\mathbb{N})$, with exactly the matrix elements of the ordinary oscillator, as in the comparison table in the box above.

Finally, the von Neumann algebra associated with all of this is the double commutant of the

representation,

$$\mathcal{M} \equiv \pi_\omega(\mathcal{A})''. \quad (2.3)$$

This uses the double-commutant machinery from earlier in this chapter to promote the C^* -algebra $\pi_\omega(\mathcal{A})$, which is norm-closed but not necessarily weakly closed, to the von Neumann algebra it generates.

Two structural facts, and what they mean

The GNS triple $(\mathcal{H}_\omega, \pi_\omega, |\Omega\rangle)$ is unique up to unitary equivalence. Any two constructions that start from the same ω give the “same” Hilbert space, possibly described in a different basis. We state this without proof. It is reassuring, because it means the construction does not depend on any arbitrary choice made along the way.

Two further properties of ω translate directly into properties of the representation it produces:

- π_ω is **irreducible** if and only if ω is a **pure** state. Irreducible means $\pi_\omega(\mathcal{A})'' = B(\mathcal{H}_\omega)$: the representation generates *every* bounded operator on $\mathcal{H}_\omega$, leaving nothing outside its reach.
- If ω is **faithful**, then $|\Omega\rangle$ is **separating** for $\pi_\omega(\mathcal{A})$. Separating means that $\pi_\omega(A)|\Omega\rangle = 0$ forces $\pi_\omega(A) = 0$: no nonzero operator in the algebra annihilates the reference vector. (Indeed, $\pi_\omega(A)|\Omega\rangle = [A] = 0$ means $\omega(A^\dagger A) = 0$, so $A = 0$.) Conversely, if $|\Omega\rangle$ is separating, then ω is faithful on the represented algebra $\pi_\omega(\mathcal{A})$.

The combination *cyclic and separating* deserves attention now, although its full importance appears only in Chapter 4. Cyclic says that the algebra reaches every vector, starting from $|\Omega\rangle$. Separating says that different elements of the algebra act differently on $|\Omega\rangle$, so nothing nonzero is wasted. The two notions are linked: a vector is cyclic for $\mathcal{M}$ if and only if it is separating for $\mathcal{M}'$ (a standard fact). So a vector that is cyclic and separating for $\mathcal{M}$ is also cyclic and separating for $\mathcal{M}'$. In the finite-dimensional examples of this chapter, this happens exactly when $|\Omega\rangle$ is entangled between the two tensor factors with full Schmidt rank on both sides. This property is the foundation of Tomita–Takesaki modular theory, the subject of Chapter 4.

The worked example, done completely with numbers

We now work one standard example all the way through with numbers, rather than leaving it symbolic. It is the most important computation in this chapter to see in full.

Let $\mathcal{A} = M_3(\mathbb{C})$, the algebra of all 3×3 complex matrices, and take

$$\rho = \begin{pmatrix} 0.6 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \omega(A) \equiv \text{Tr}(\rho A).$$

First check that ω is a state in the sense defined earlier. It is linear, because the trace and matrix multiplication are linear. It is positive: since ρ is diagonal, $\text{Tr}(\rho A^\dagger A) = \sum_j \rho_{jj} (A^\dagger A)_{jj} =$

$\sum_j \rho_{jj} \sum_k |A_{kj}|^2 \geq 0$. It is normalized: $\text{Tr}(\rho \mathbf{1}_3) = \text{Tr} \rho = 0.6 + 0.4 + 0 = 1$. The matrix ρ has rank 2, not 3. It ignores the third basis direction completely, so ω is *not* faithful. For example, $e_{33} \neq 0$ but $\omega(e_{33}^\dagger e_{33}) = \rho_{33} = 0$. This property will show up directly in the construction.

Now follow the recipe, exactly as for System 2 above. The GNS inner product is

$$\begin{aligned} \langle A|B \rangle &= \text{Tr}(\rho A^\dagger B) \stackrel{(1)}{=} \sum_{j=1}^3 \rho_{jj} (A^\dagger B)_{jj} \\ &\stackrel{(2)}{=} 0.6 \sum_{k=1}^3 A_{k1}^* B_{k1} + 0.4 \sum_{k=1}^3 A_{k2}^* B_{k2}. \end{aligned}$$

(1) ρ is diagonal. (2) $(A^\dagger B)_{jj} = \sum_k A_{kj}^* B_{kj}$, and $\rho_{33} = 0$ removes the $j = 3$ term.

So only the first two columns of a matrix matter. The null space $\mathcal{J}$ consists of the matrices whose first two columns vanish, with the third column arbitrary. It is 3-dimensional, so the quotient $\mathcal{A}/\mathcal{J}$ has dimension $9 - 3 = 6$. An orthonormal basis is $|k\rangle_R |j\rangle_L \equiv [e_{kj}]/\sqrt{\rho_{jj}}$, for $k = 1, 2, 3$ and $j = 1, 2$. Left multiplication, $[Ae_{kj}] = \sum_l A_{lk}[e_{lj}]$, acts only on the row label k , so $\pi_\omega(A) = A \otimes \mathbf{1}_2$. The cyclic vector is $|\mathbf{1}_3\rangle = [e_{11}] + [e_{22}] + [e_{33}] = [e_{11}] + [e_{22}]$, since $e_{33} \in \mathcal{J}$. Altogether, $\mathcal{H}_\omega \cong \mathbb{C}^3 \otimes \mathbb{C}^2$ is 6-dimensional: 3 (the size of the matrices) times 2 (the rank of ρ). The cyclic vector and the representation are

$$|\Omega\rangle = \sqrt{0.6} |1\rangle_R |1\rangle_L + \sqrt{0.4} |2\rangle_R |2\rangle_L, \quad \pi_\omega(A) = A \otimes \mathbf{1}_2.$$

We also checked this numerically. For several random complex 3×3 matrices A, B , we compared $\omega(A) = \text{Tr}(\rho A)$ with $\langle \Omega | \pi_\omega(A) | \Omega \rangle$, and the GNS inner product $\langle A|B \rangle = \text{Tr}(\rho A^\dagger B)$ with $\langle \Omega | \pi_\omega(A)^\dagger \pi_\omega(B) | \Omega \rangle$. In every trial the two sides agreed to about one part in 10^{16} , the rounding error of ordinary double-precision arithmetic. The failure of faithfulness is also visible: $\pi_\omega(e_{13}) |\Omega\rangle = 0$ although $e_{13} \neq 0$. So $|\Omega\rangle$ is cyclic but not separating, as the second structural fact leads us to expect.

Recognize what this is. $|\Omega\rangle$ is the **purification** of ρ , familiar from quantum information. Purification is the standard way to write a mixed state ρ on a small Hilbert space as a pure state on a larger one, by introducing an auxiliary system (here the L factor) entangled with the original. The GNS construction produces this purification automatically, with no separate step of “now introduce an auxiliary system.” The auxiliary factor $\mathcal{H}_L$ appears inside $\mathcal{A}/\mathcal{J}$ as the column label of the matrix $[A]$. Its dimension is the rank of ρ , because $\mathcal{J}$ removes exactly the columns on which ρ vanishes.

Two variations of the same example show the two structural facts at work, on numbers you can check directly:

- Take $\rho = \text{diag}(1, 0, 0)$ instead, which has rank 1 and is a pure state. Then $\mathcal{H}_\omega$ reduces to a single copy of $\mathbb{C}^3$: the L factor becomes 1-dimensional and drops out. The representation π_ω is irreducible, consistent with the first structural fact, since ω is now pure.
- Take $\rho = \frac{1}{3} \mathbf{1}_3$ instead, which has full rank 3 and is maximally mixed. Then $\mathcal{H}_\omega = \mathbb{C}^3 \otimes \mathbb{C}^3$, and $|\Omega\rangle = \frac{1}{\sqrt{3}} \sum_{a=1}^3 |a\rangle_R |a\rangle_L$ is both cyclic *and* separating. This is consistent with the second structural fact, since this ω is faithful: $\omega(A^\dagger A) = \frac{1}{3} \text{Tr}(A^\dagger A) > 0$ for every nonzero A . This

maximally mixed state, with its maximally entangled reference vector, is the 3-dimensional analogue of the Bell pair at $\theta = \pi/4$ from Chapter 1. It is the same structure, one size larger.

2.9 Emergent von Neumann algebras of the entangled spin example

This closing section does one thing. It takes the GNS machinery just built and applies it explicitly to the $N \rightarrow \infty$ Bell-pair chain of Chapter 1. This turns the informal description given there (“the finite-energy excitations form some Hilbert space $\mathcal{H}_{\Phi_\theta}$ ”) into an actual construction, with an actual algebra and an actual state.

Define the algebra of **finite-energy operations**, $\mathcal{A}$. It is generated by operators of the form

$$O = \alpha_1 \otimes \alpha_2 \otimes \cdots \otimes \alpha_n \otimes \cdots, \quad \text{all but finitely many of the } \alpha_i \text{ equal to } \mathbf{1}_2 \otimes \mathbf{1}_2.$$

So O acts on only finitely many of the infinitely many spin pairs, and does nothing (the identity) to every pair beyond some finite point. This restriction is not arbitrary. It is the algebraic form of the statement that O corresponds to a process reachable with finite energy. Recall from Chapter 1 that flipping infinitely many spins costs infinite energy. An operator that acts nontrivially on infinitely many pairs at once is exactly what a finite-energy process could never do. In the $N \rightarrow \infty$ limit, O inherits a well-defined norm from the finite- N operators it is built from. Completing $\mathcal{A}$ in that norm makes it a C^* -algebra. It is defined abstractly, with no particular Hilbert space singled out yet.

The reference state $|\Phi_\theta\rangle$, the infinite chain of Bell-like pairs from Chapter 1, defines a state ω_θ on this algebra in the obvious way:

$$\omega_\theta(O) = \langle \Phi_\theta | O | \Phi_\theta \rangle.$$

This makes sense because O touches only finitely many pairs, so the expectation value uses only finitely much of the state. No infinite sum or ill-defined limit is hidden in this definition.

Now apply the GNS construction of the previous section, exactly as given, to $(\mathcal{A}, \omega_\theta)$. It produces a Hilbert space $\mathcal{H}_{\Phi_\theta}$. This *is* the space of finite-energy excitations described informally in Chapter 1, now obtained as a mathematical construction rather than a descriptive placeholder. Heuristically, and this is Stage 1 of the GNS recipe applied to this case, $\mathcal{H}_{\Phi_\theta}$ is the completion of the set of states reachable from $|\Phi_\theta\rangle$ by flipping a finite number of spins. That was exactly the informal description in Chapter 1. In this representation, the von Neumann algebra you get (the double commutant (2.3)) is all of $B(\mathcal{H}_{\Phi_\theta})$, every bounded operator on this emergent Hilbert space. The reason is the first structural fact: $|\Phi_\theta\rangle$ is a pure state of the whole chain, so the representation is irreducible.

Now suppose, as in Chapter 1, that you have access only to the R -half of the spin system. Let $\mathcal{M}_R$ be the subalgebra of $B(\mathcal{H}_{\Phi_\theta})$ consisting of operators that act only on the R -spins, completed under weak convergence *within this particular* $\mathcal{H}_{\Phi_\theta}$. It is a von Neumann algebra. Its *type* (I, II, or III, depending on θ) is the subject of Chapters 3 and 4. Define $\mathcal{M}_L$ in the same way for the L -spins. Operations on the L -spins commute with operations on the R -spins, because they act on physically distinct degrees of freedom. This shows directly that $\mathcal{M}_L \subseteq \mathcal{M}'_R$. In fact the two are equal, $\mathcal{M}_L = \mathcal{M}'_R$. This equality is a theorem about infinite tensor products of this kind, which we state without proof. So the L -algebra is exactly the commutant of the R -algebra. This is the “complement of a subsystem” picture from the start of this chapter, now realized concretely.

One further point removes an apparent asymmetry. This example is completely symmetric between R and L : nothing distinguishes them, and the construction could equally well have started from $\mathcal{M}_L$. It turns out that the same Hilbert space $\mathcal{H}_{\Phi_\theta}$ can be rebuilt using *only* the R -algebra, with no reference to L at all. Concretely, redo the GNS construction using $\mathcal{A}_R$ (finite-energy operations on the R -spins only) and the state $\omega_\theta(A) = \langle \Phi_\theta | A | \Phi_\theta \rangle$ restricted to $A \in \mathcal{A}_R$. This restricted ω_θ is a *faithful* state of $\mathcal{A}_R$, because each pair's reduced density matrix, $\text{diag}(\cos^2 \theta, \sin^2 \theta)$, has full rank for $\theta \in (0, \pi/4]$. On the full algebra $\mathcal{A}$, by contrast, ω_θ is not faithful. For example, the projection $\mathbf{1} - |\phi_\theta\rangle\langle\phi_\theta|$ on a single pair is nonzero, but ω_θ gives it zero. So faithfulness is a property of a state *relative to* a specific algebra, not a property of the state alone. The GNS Hilbert space built from $(\mathcal{A}_R, \omega_\theta)$ is the same $\mathcal{H}_{\Phi_\theta}$ as before, because $|\Phi_\theta\rangle$ is cyclic for $\mathcal{M}_R$ (stated here without proof for the infinite chain).

The intuition is clearest for a single pair. Acting on $|\phi_\theta\rangle$ with operators on the right spin alone already produces every vector in the pair's four-dimensional space $\mathbb{C}^2 \otimes \mathbb{C}^2$. Indeed,

$$(A \otimes \mathbf{1}) |\phi_\theta\rangle = \cos \theta (A |0\rangle) |0\rangle + \sin \theta (A |1\rangle) |1\rangle,$$

and as A ranges over all 2×2 matrices, this covers all of $\mathbb{C}^2 \otimes \mathbb{C}^2$ when $\cos \theta$ and $\sin \theta$ are both nonzero. (We confirmed numerically that the four vectors $(e_{ij} \otimes \mathbf{1}) |\phi_\theta\rangle$ are linearly independent.) So operators on the right spin reach everything that operators on the left spin, or on both spins together, can reach. For the whole chain, the same statement is that $|\Phi_\theta\rangle$ is cyclic for $\mathcal{M}_R$. This mirrors, on a much larger scale, the point made earlier about minimal projections and irreducible representations. $\mathcal{M}_R$, as an abstract algebra, already contains everything needed to rebuild the Hilbert space; $\mathcal{H}_L$ only ever tracked multiplicity. In this symmetric example, R alone generates the same Hilbert space that R and L together do.

With $\mathcal{H}_{\Phi_\theta}$, $\mathcal{M}_R$, and $\mathcal{M}_L = \mathcal{M}'_R$ now built explicitly rather than only described, the stage is set for the question that occupies the next two chapters. For different values of θ , what *type*, in the precise sense of the classification above, is $\mathcal{M}_R$? Chapter 3 answers this by explicit computation for $\theta = \pi/4$, where the answer is type II_1 . It also sets up the general case $\theta \neq \pi/4$, which is type III and is completed in Chapter 4.

CHAPTER 3

Von Neumann algebras and entanglement: type I and II

Chapter 2 built the basic tools: algebras, states, projections, the type classification, and the GNS construction. There the focus was on the algebra $\mathcal{M}$ itself, sitting inside some larger $B(\mathcal{H})$. This chapter adds a specific physical state $|\Psi\rangle$ and asks the question the whole framework was built to answer. Given Ψ , what does the *type* of $\mathcal{M}$ tell us about how Ψ entangles the subsystem described by $\mathcal{M}$ with everything outside it?

We treat type I and type II here. Both types have a trace. Because of that, we can still define a density operator and an entropy by a direct generalization of the formulas you already know. Type III has no trace at all. It needs a different tool, and it is the subject of Chapter 4.

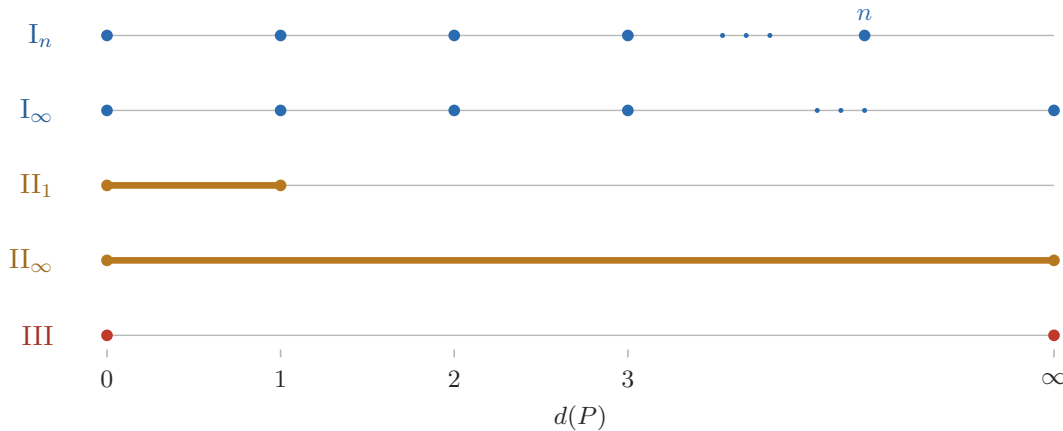


Figure 3.1: The types of von Neumann factors, told apart by the values that the dimension function $d(P)$ takes on projections (one row per type, dots and bars marking the allowed values). Types I_n and I_∞ take only whole-number values, because they have minimal projections (projections onto single rays). Types II_1 and II_∞ have no minimal projections but have a trace, finite or infinite, so $d(P)$ fills the interval $[0, 1]$ or $[0, \infty]$. Type III has no trace, and every nonzero projection is infinite, so only 0 and ∞ occur.

3.1 Density operators for type I and II algebras

We start with the central new formula of this chapter. Suppose the full system is in a state $|\Psi\rangle \in \mathcal{H}$. Let $\mathcal{M}$ be a von Neumann algebra with a trace tr . By the type classification of Chapter 2, this

Factor	Values of $d(P)$	Trace	Density operator	Example	Entropy
I_n	$\{0, 1, \dots, n\}$	yes, finite	$\rho_R = \text{Tr}_L \Psi\rangle\langle\Psi $	n -level system; qubits	ordinary, $S \geq 0$
I_∞	$\{0, 1, 2, \dots, \infty\}$	yes, semifinite	$\rho_R = \text{Tr}_L \Psi\rangle\langle\Psi $	harmonic oscillator; Fock space	ordinary, $S \geq 0$
II_1	$[0, 1]$	yes, finite, $\text{tr}(\mathbf{1}) = 1$	$\rho_{\mathcal{M}}$ from eq. (3.1)	infinite chain of Bell pairs at $\theta = \pi/4$	$S_{\mathcal{M}} \leq 0$; minus a relative entropy
II_∞	$[0, \infty]$	yes, semifinite	exists (Chapter 5)	crossed product $\mathcal{M} \rtimes \mathbb{R}$	defined up to a constant; S_{gen} (Chapter 5)
III_0	$\{0, \infty\}$	none	none	few physical examples	relative entropy only
III_λ	$\{0, \infty\}$	none	none	spin chain with $\tan^2 \theta = \lambda$	relative entropy only
III_1	$\{0, \infty\}$	none	none	QFT subregion; Rindler wedge	relative entropy only; crossed product gives II_∞

Table 3.1: The factor types and how each one handles density operators and entropy.

means $\mathcal{M}$ is type I or type II. Define $\rho_{\mathcal{M}}$ to be the operator in $\mathcal{M}$ that satisfies

$$\text{tr}(A\rho_{\mathcal{M}}) = \langle\Psi|A|\Psi\rangle \quad \text{for every } A \in \mathcal{M}. \quad (3.1)$$

This equation has exactly one solution $\rho_{\mathcal{M}}$, and that solution is positive. We quote this result without proof. It is a noncommutative version of the Radon–Nikodym theorem, and it needs two conditions: the trace must be faithful and normal, and the state must be normal. Both hold in every example of this chapter. The normalization is automatic: setting $A = \mathbf{1}$ gives $\text{tr} \rho_{\mathcal{M}} = \langle\Psi|\Psi\rangle = 1$. So $\rho_{\mathcal{M}}$ deserves the name “density operator.”

One point is essential. The operator $\rho_{\mathcal{M}}$ is required to lie in $\mathcal{M}$ itself, not merely in $B(\mathcal{H})$. We solve eq. (3.1) *inside* the algebra. (If $\rho_{\mathcal{M}}$ is unbounded, the precise statement is that $\rho_{\mathcal{M}}$ is *affiliated* with $\mathcal{M}$: all of its spectral projections lie in $\mathcal{M}$.)

Once we have $\rho_{\mathcal{M}}$, we define the entanglement entropy in the expected way:

$$S_{\mathcal{M}} \equiv -\text{tr}(\rho_{\mathcal{M}} \log \rho_{\mathcal{M}}). \quad (3.2)$$

This has the same form as the ordinary formula $S_R = -\text{Tr}(\rho_R \log \rho_R)$. Two things have been replaced. The algebra’s own trace tr from Chapter 2, which may be renormalized, stands in for the ordinary Hilbert-space trace. And $\rho_{\mathcal{M}}$ stands in for the ordinary reduced density matrix.

Compare this with the formula you already know, the partial trace $\rho_R = \text{Tr}_L |\Psi\rangle\langle\Psi|$. That formula needs the complement L to be given explicitly, as a separate tensor factor that you can trace over. In other words, to build an object that describes the part of the world you *can* access, you must know about the part you *cannot* access. Equation (3.1) needs nothing of the sort. It uses only the trace of $\mathcal{M}$ and the expectation values of operators in $\mathcal{M}$.

This raises an obvious question. If only $\mathcal{M}$ enters the definition, in what sense does $\rho_{\mathcal{M}}$ capture entanglement with the outside world? The answer depends on the *type* of $\mathcal{M}$. Working it out type by type is the content of the rest of this chapter and of Chapter 4.

3.2 Type I algebras

The type I factor case: nothing is lost

When $\mathcal{M}$ is a type I *factor*, Chapter 2 showed that

$$\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L, \quad \mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L, \quad \mathcal{M}' = \mathbf{1}_R \otimes B(\mathcal{H}_L),$$

and the trace is $\text{tr}(a \otimes \mathbf{1}_L) = \text{Tr}_{\mathcal{H}_R}(a)$. Let $\rho_R = \text{Tr}_{\mathcal{H}_L} |\Psi\rangle \langle \Psi|$ be the ordinary reduced density matrix. For any $A = a \otimes \mathbf{1}_L$ in $\mathcal{M}$,

$$\begin{aligned} \langle \Psi | a \otimes \mathbf{1}_L | \Psi \rangle &\stackrel{(1)}{=} \text{Tr}_{\mathcal{H}} ((a \otimes \mathbf{1}_L) |\Psi\rangle \langle \Psi|) \\ &\stackrel{(2)}{=} \text{Tr}_{\mathcal{H}_R} (a \text{Tr}_{\mathcal{H}_L} |\Psi\rangle \langle \Psi|) \stackrel{(3)}{=} \text{Tr}_{\mathcal{H}_R} (a \rho_R) \\ &\stackrel{(4)}{=} \text{tr} ((a \otimes \mathbf{1}_L)(\rho_R \otimes \mathbf{1}_L)). \end{aligned}$$

(1) an expectation value is the trace of the operator times the projector $|\Psi\rangle \langle \Psi|$. (2) do the trace over $\mathcal{H}_L$ first; the operator a does not act on $\mathcal{H}_L$, so it comes out of that partial trace. (3) definition of ρ_R . (4) $(a \otimes \mathbf{1}_L)(\rho_R \otimes \mathbf{1}_L) = a \rho_R \otimes \mathbf{1}_L$, and tr of this is $\text{Tr}_{\mathcal{H}_R}(a \rho_R)$.

Comparing with the defining equation (3.1), we read off $\rho_{\mathcal{M}} = \rho_R \otimes \mathbf{1}_L$. This is the ordinary reduced density matrix, written as an element of $\mathcal{M}$. The new definition, which uses only the algebra, and the old one, which uses a partial trace, give *exactly* the same operator whenever a type I factorization is available. Nothing is lost by switching definitions. It follows that $S_{\mathcal{M}} = S_R$.

Two remarks follow. They are the seeds of everything that happens once type I is no longer available.

1. The two recipes are conceptually different, even though they agree here. The partial-trace recipe needs the full global state $|\Psi\rangle$, including the part in $\mathcal{H}_L$ that an observer on R cannot access. Equation (3.1) needs only expectation values of operators in $\mathcal{M}$. An observer on R could, in principle, collect those data. So in the type I case, all of the entanglement information was already contained in the internal data of $\mathcal{M}$. The tensor-product recipe was a roundabout way of getting the same answer, and it is the recipe that breaks once no factorization exists.
2. Here tr equals the ordinary $\text{Tr}_{\mathcal{H}_R}$, which counts basis states of an honest Hilbert space $\mathcal{H}_R$. So $S_{\mathcal{M}} = S_R$ keeps its usual statistical meaning: roughly, the logarithm of the number of effectively occupied states. This meaning, too, becomes much more delicate once we leave type I.

General type I: a nontrivial center, and where lattice gauge theory fits

Now suppose $\mathcal{M}$ is type I but *not* a factor. Then it has a nontrivial center. (Recall the block-diagonal example with projections P_1, P_2 in Chapter 2.) The Hilbert space splits into a direct sum of sectors, one for each value α of the central label. This label is classical: every operator in $\mathcal{M}$ commutes

with it. We have

$$\mathcal{H} = \bigoplus_{\alpha} \mathcal{H}_{\alpha}, \quad \mathcal{H}_{\alpha} = \mathcal{H}_{R\alpha} \otimes \mathcal{H}_{L\alpha}, \quad \mathcal{M} = \bigoplus_{\alpha} (B(\mathcal{H}_{R\alpha}) \otimes \mathbf{1}_{L\alpha}).$$

Within each sector, the tensor-product story of the factor case holds exactly. The sectors sit side by side and never mix. Every operator in $\mathcal{M}$ is block diagonal, so it never sends a state in sector α to a state in a different sector α' . A general element of $\mathcal{M}$ has the form $A = \bigoplus_{\alpha} (a_{\alpha} \otimes \mathbf{1}_{L\alpha})$, and the trace is built sector by sector:

$$\text{tr } A = \sum_{\alpha} \text{Tr}_{\mathcal{H}_{R\alpha}} a_{\alpha}.$$

A general density matrix ρ on $\mathcal{H}$ can have blocks that connect different sectors. Operators in $\mathcal{M}$ cannot detect those off-diagonal blocks, because they are block diagonal. So only the diagonal blocks matter, and we write them as $p_{\alpha}\rho_{\alpha}$. Here p_{α} is the classical probability of being in sector α , and ρ_{α} is a normalized density matrix within that sector. Let $\rho_{R\alpha} = \text{Tr}_{\mathcal{H}_{L\alpha}} \rho_{\alpha}$ be the ordinary reduced density matrix within sector α . For $A \in \mathcal{M}$,

$$\begin{aligned} \text{Tr}_{\mathcal{H}}(\rho A) &\stackrel{(1)}{=} \sum_{\alpha} p_{\alpha} \text{Tr}_{\mathcal{H}_{\alpha}} (\rho_{\alpha} (a_{\alpha} \otimes \mathbf{1}_{L\alpha})) \stackrel{(2)}{=} \sum_{\alpha} p_{\alpha} \text{Tr}_{\mathcal{H}_{R\alpha}} (\rho_{R\alpha} a_{\alpha}) \\ &\stackrel{(3)}{=} \text{tr} \left(A \bigoplus_{\alpha} p_{\alpha} (\rho_{R\alpha} \otimes \mathbf{1}_{L\alpha}) \right). \end{aligned}$$

(1) A is block diagonal, so only the diagonal blocks $p_{\alpha}\rho_{\alpha}$ of ρ contribute. **(2)** the type I factor computation above, done inside each sector. **(3)** the sector-by-sector formula for tr .

Comparing with eq. (3.1) gives

$$\rho_{\mathcal{M}} = \bigoplus_{\alpha} p_{\alpha} (\rho_{R\alpha} \otimes \mathbf{1}_{L\alpha}).$$

Now put this into eq. (3.2). Let $S_{\alpha} = -\text{Tr}_{\mathcal{H}_{R\alpha}} (\rho_{R\alpha} \log \rho_{R\alpha})$ be the ordinary entanglement entropy within sector α . Then

$$\begin{aligned} S_{\mathcal{M}} &\stackrel{(1)}{=} - \sum_{\alpha} \text{Tr}_{\mathcal{H}_{R\alpha}} (p_{\alpha} \rho_{R\alpha} \log(p_{\alpha} \rho_{R\alpha})) \\ &\stackrel{(2)}{=} - \sum_{\alpha} p_{\alpha} \text{Tr}_{\mathcal{H}_{R\alpha}} (\rho_{R\alpha} (\log p_{\alpha} + \log \rho_{R\alpha})) \\ &\stackrel{(3)}{=} - \sum_{\alpha} p_{\alpha} \log p_{\alpha} + \sum_{\alpha} p_{\alpha} S_{\alpha}. \end{aligned} \tag{3.3}$$

(1) $\rho_{\mathcal{M}}$ is block diagonal, so $\log \rho_{\mathcal{M}}$ is computed block by block, and tr is the sum of the traces of the R parts. **(2)** $\log(cX) = \log c \mathbf{1} + \log X$ for a number $c > 0$. **(3)** $\text{Tr}_{\mathcal{H}_{R\alpha}} \rho_{R\alpha} = 1$, and the definition of S_{α} .

The entropy splits into two recognizable pieces. The first is the classical (Shannon) entropy of *which sector you are in*. The second is the probability-weighted average of the quantum entanglement entropy *within* each sector. Nothing here is conceptually new. It is what you would compute by hand if someone told you: “the system is in configuration A with probability p_A , entangled by an amount S_A , or in configuration B with probability p_B , entangled by an amount S_B , and you don’t know which.” The new point is that this result comes directly out of the single formula (3.1). We

do not have to reason it out from scratch each time.

Worked Example: Superselection sectors in a two-site lattice gauge model

Consider a toy lattice with two sites, x_1 and x_2 , joined by one gauge link that crosses the cut between them. Let the electric flux on the link take the values $q \in \{0, 1\}$. By Gauss's law, this flux can be measured from either side of the cut, so it commutes with every gauge-invariant operator on each side. It is therefore central, and it labels the sectors. The physical Hilbert space splits into two sectors:

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_{q=0} \oplus \mathcal{H}_{q=1} = (\mathcal{H}_{R,0} \otimes \mathcal{H}_{L,0}) \oplus (\mathcal{H}_{R,1} \otimes \mathcal{H}_{L,1}).$$

Suppose the state is a mixture of the two sectors,

$$\rho = p_0 \rho_0 \oplus p_1 \rho_1, \quad p_0 = 0.8, \quad p_1 = 0.2.$$

In sector $q = 0$, take the two sites to be in a Bell state, so $\rho_{R,0} = \text{diag}(1/2, 1/2)$ and $S_0 = \log 2 \approx 0.6931$. In sector $q = 1$, take them to be in a product state, so $S_1 = 0$. We evaluate eq. (3.3) one piece at a time (all logarithms are natural, so entropies are in nats).

1. Classical Shannon entropy of the flux label:

$$\begin{aligned} H(p) &= -p_0 \log p_0 - p_1 \log p_1 = -0.8 \log 0.8 - 0.2 \log 0.2 \\ &\approx 0.1785 + 0.3219 = 0.5004. \end{aligned}$$

2. Average entanglement within the sectors:

$$\sum_{\alpha} p_{\alpha} S_{\alpha} = 0.8 \times \log 2 + 0.2 \times 0 \approx 0.5545.$$

3. Total entropy of the algebra:

$$S_{\mathcal{M}} = H(p) + \sum_{\alpha} p_{\alpha} S_{\alpha} \approx 0.5004 + 0.5545 = 1.0549.$$

One formula captures both the classical uncertainty in the flux and the genuine quantum entanglement. It does not need the Hilbert space to factorize across the cut.

This is exactly the situation of the lattice gauge theory example of Chapter 1. On a finite lattice, the algebra of gauge-invariant local operators is type I. It still has minimal projections, and it still counts states in the ordinary way, sector by sector. But it is not a factor. Gauge invariance imposes a classical superselection label, such as the electric flux through the boundary. The single formula (3.1) handles all of this at once. A type I factor is the special case with a trivial center: only one sector, with $p_{\alpha} = 1$. So even at this early and mostly familiar stage, the algebraic language is already unifying cases that otherwise need separate treatment.

3.3 Type II algebras

This is where something genuinely new happens. The clearest way to see it is to build a type II₁ factor by hand, out of the infinite Bell-pair chain of Chapter 2. We do every step of the construction below. It is the first place in these notes where we can watch a new kind of mathematical object being built. The only raw material is an infinite chain of ordinary qubits.

Building a trace at $\theta = \pi/4$

Recall the Bell-pair chain from Chapter 2. The algebra $\mathcal{A}_R$ consists of finite-energy operations on the right-hand spins, meaning operators that act on only finitely many of them. The state is $\omega_\theta(A) = \langle \Phi_\theta | A | \Phi_\theta \rangle$, where $|\Phi_\theta\rangle$ is the infinite product of pairs $\cos \theta |00\rangle + \sin \theta |11\rangle$. The von Neumann algebra $\mathcal{M} \equiv \mathcal{M}_R$ is built from $\mathcal{A}_R$ and ω_θ by the GNS construction. Now fix $\theta = \pi/4$, so that every pair is maximally entangled, and *define*

$$\text{tr } A \equiv \langle \Phi_{\pi/4} | A | \Phi_{\pi/4} \rangle, \quad A \in \mathcal{M}. \quad (3.4)$$

In notation, this is just the expectation value $\omega_{\pi/4}(A)$ from Chapter 2. The claim is that at $\theta = \pi/4$, and only there, this expectation value *also* has the cyclic property $\text{tr}(AB) = \text{tr}(BA)$ that defines a trace (Chapter 2). The rest of this section rests on this single fact, so we verify it.

Step 1: a single pair. Take one spin pair at $\theta = \pi/4$, in the state $|\phi_{\pi/4}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Let $a = \begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix}$ be any operator on the right spin alone, acting on the pair as $a \otimes \mathbf{1}_L$. We compute its expectation value by expanding the state:

$$\begin{aligned} \langle \phi_{\pi/4} | a \otimes \mathbf{1}_L | \phi_{\pi/4} \rangle &\stackrel{(1)}{=} \frac{1}{2} (\langle 00 | + \langle 11 |) (a \otimes \mathbf{1}_L) (|00\rangle + |11\rangle) \\ &\stackrel{(2)}{=} \frac{1}{2} \left(\langle 00 | a \otimes \mathbf{1} | 00 \rangle + \langle 00 | a \otimes \mathbf{1} | 11 \rangle \right. \\ &\quad \left. + \langle 11 | a \otimes \mathbf{1} | 00 \rangle + \langle 11 | a \otimes \mathbf{1} | 11 \rangle \right) \\ &\stackrel{(3)}{=} \frac{1}{2} (a_{00} \langle 0|0 \rangle_L + a_{01} \langle 0|1 \rangle_L + a_{10} \langle 1|0 \rangle_L + a_{11} \langle 1|1 \rangle_L) \\ &\stackrel{(4)}{=} \frac{1}{2} (a_{00} + a_{11}) \stackrel{(5)}{=} \frac{1}{2} \text{Tr}_2(a). \end{aligned} \quad (3.5)$$

(1) substitute $|\phi_{\pi/4}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ on both sides. **(2)** expand the product of two two-term sums into four terms. **(3)** $a \otimes \mathbf{1}_L$ acts only on the first (right) qubit, so each term factorizes into a matrix element of a times an inner product on L . **(4)** $\langle 0|0 \rangle_L = \langle 1|1 \rangle_L = 1$ and $\langle 0|1 \rangle_L = \langle 1|0 \rangle_L = 0$, so the cross terms vanish. **(5)** $a_{00} + a_{11}$ is the ordinary matrix trace $\text{Tr}_2(a)$.

Here Tr_2 is the ordinary 2×2 matrix trace. So in a single maximally entangled pair, the expectation value of any operator on one side is exactly one half of its ordinary trace. The coefficient $\frac{1}{2}$ is $1/\dim(\mathcal{H}_2)$ for a single qubit. It will come back when we compute the dimensions of projections below.

Step 2: many pairs. A typical element of $\mathcal{A}_R$ is a product $A = a_1 \otimes a_2 \otimes \cdots$, where only finitely many of the a_i differ from the identity. Say $a_{i_1}, \dots, a_{i_k}$ are the nontrivial ones. General elements of

$\mathcal{A}_R$ are finite sums of such products, and general elements of $\mathcal{M}_R$ are limits of these sums. Since $|\Phi_{\pi/4}\rangle$ is a product over pairs,

$$\begin{aligned} \langle \Phi_{\pi/4} | A | \Phi_{\pi/4} \rangle &\stackrel{(1)}{=} \prod_i \langle \phi_{\pi/4} | a_i \otimes \mathbf{1}_L | \phi_{\pi/4} \rangle \\ &\stackrel{(2)}{=} \frac{1}{2^k} \text{Tr}_2(a_{i_1}) \cdots \text{Tr}_2(a_{i_k}). \end{aligned} \quad (3.6)$$

(1) both A and $|\Phi_{\pi/4}\rangle$ are products over pairs, so the expectation value is a product of single-pair expectation values. (2) each touched pair gives $\frac{1}{2} \text{Tr}_2(a_{i_s})$ by eq. (3.5); each untouched pair gives $\frac{1}{2} \text{Tr}_2(\mathbf{1}_2) = 1$.

So the expectation value in the infinite chain reduces to a product of ordinary 2×2 traces, with one factor of $\frac{1}{2}$ for each of the k pairs that A actually touches.

Step 3: cyclicity. Take two product operators $A = \bigotimes_i a_i$ and $B = \bigotimes_i b_i$. Choose n large enough that both act trivially beyond pair n . Then $AB = \bigotimes_i (a_i b_i)$, and

$$\text{tr}(AB) \stackrel{(1)}{=} \prod_{i=1}^n \frac{1}{2} \text{Tr}_2(a_i b_i) \stackrel{(2)}{=} \prod_{i=1}^n \frac{1}{2} \text{Tr}_2(b_i a_i) \stackrel{(3)}{=} \text{tr}(BA).$$

(1) eq. (3.6) applied to the product operator AB ; pairs where $a_i b_i = \mathbf{1}_2$ contribute 1. (2) the ordinary matrix trace is cyclic, $\text{Tr}_2(a_i b_i) = \text{Tr}_2(b_i a_i)$, pair by pair. (3) step (1) read backwards, for the product operator BA .

Both sides are linear in A and in B , so cyclicity extends to finite sums of products, that is, to all of $\mathcal{A}_R$. It then extends to all of $\mathcal{M}_R$ by continuity. For fixed A , both $B \mapsto \text{tr}(AB)$ and $B \mapsto \text{tr}(BA)$ are continuous under the weak limits that build $\mathcal{M}_R$ out of $\mathcal{A}_R$, because a vector expectation value is weakly continuous. So we may first take limits in B and then, by the same argument, in A . This establishes eq. (3.4) as a genuine trace on $\mathcal{M}$.

This is exactly where $\theta = \pi/4$ earns its special status. For a general angle, Step 1 gives $\langle \phi_\theta | a \otimes \mathbf{1}_L | \phi_\theta \rangle = \cos^2 \theta a_{00} + \sin^2 \theta a_{11}$. This is a weighted sum, not a multiple of the trace, and cyclicity fails. A concrete check: take $a = |0\rangle\langle 1|$ and $b = |1\rangle\langle 0|$, so that $ab = |0\rangle\langle 0|$ and $ba = |1\rangle\langle 1|$. Then

$$\omega_\theta(ab) = \cos^2 \theta, \quad \omega_\theta(ba) = \sin^2 \theta,$$

and these are equal only at $\theta = \pi/4$. So the state ω_θ is not a trace when $\theta \neq \pi/4$.

More is true: for $\theta \neq \pi/4$ the algebra $\mathcal{M}_R$ admits no normalized trace of *any* kind. This can be proved directly, using the average magnetization of many spins and the law of large numbers.

Key Result / Core Concept: No normalized trace away from $\theta = \pi/4$

Theorem. Let $\mathcal{A}_R = \bigotimes_{k=1}^\infty M_2(\mathbb{C})$ be the algebra of the right-hand spins, generated by operators that act on finitely many sites. Let $|\Phi_\theta\rangle = \bigotimes_{k=1}^\infty (\cos \theta |00\rangle + \sin \theta |11\rangle)_k$ with $\theta \in (0, \pi/2)$, and let $\mathcal{M}_R \equiv \pi_\theta(\mathcal{A}_R)'' \subseteq B(\mathcal{H}_\theta)$ be the von Neumann algebra given by the GNS construction. If $\theta \neq \pi/4$, then **there is no normal tracial state on $\mathcal{M}_R$.**

Proof.

1. **A trace vanishes on each σ_z .** Suppose, to reach a contradiction, that there is a normal tracial state $\tau : \mathcal{M}_R \rightarrow \mathbb{C}$ with $\tau(\mathbf{1}) = 1$. Being a trace means $\tau(AB) = \tau(BA)$ for all $A, B \in \mathcal{M}_R$. Equivalently, $\tau([A, B]) = 0$ for every commutator. At any single site k , the Pauli operators $\sigma_x^{(k)}, \sigma_y^{(k)}, \sigma_z^{(k)}$ lie in $\mathcal{A}_R \subset \mathcal{M}_R$. They obey $[\sigma_x, \sigma_y] = 2i\sigma_z$, so

$$\sigma_z^{(k)} = \frac{1}{2i} [\sigma_x^{(k)}, \sigma_y^{(k)}].$$

Applying τ gives

$$\tau(\sigma_z^{(k)}) \stackrel{(1)}{=} \frac{1}{2i} \tau([\sigma_x^{(k)}, \sigma_y^{(k)}]) \stackrel{(2)}{=} 0 \quad \text{for all } k \geq 1.$$

- (1) substitute the commutator identity and pull the constant $\frac{1}{2i}$ out by linearity of τ .
 (2) a trace vanishes on every commutator.

2. **A trace vanishes on the average magnetization.** Define the average magnetization of the first N sites,

$$M_N \equiv \frac{1}{N} \sum_{k=1}^N \sigma_z^{(k)} \in \mathcal{M}_R.$$

Then

$$\tau(M_N) \stackrel{(1)}{=} \frac{1}{N} \sum_{k=1}^N \tau(\sigma_z^{(k)}) \stackrel{(2)}{=} 0 \quad \text{for all } N \geq 1.$$

- (1) linearity of τ . (2) every term vanishes by item 1.

3. **In the GNS state, the magnetization settles to a fixed value.** Now look at M_N in the GNS reference vector $|\Omega_\theta\rangle \equiv |\Phi_\theta\rangle$. For a single pair,

$$\langle \phi_\theta | \sigma_z \otimes \mathbf{1}_L | \phi_\theta \rangle \stackrel{(1)}{=} \cos^2 \theta \langle 0 | \sigma_z | 0 \rangle + \sin^2 \theta \langle 1 | \sigma_z | 1 \rangle \stackrel{(2)}{=} \cos^2 \theta - \sin^2 \theta \stackrel{(3)}{=} \cos(2\theta).$$

- (1) σ_z is diagonal, so the cross terms between $|00\rangle$ and $|11\rangle$ vanish. What remains are the two diagonal terms, weighted by $\cos^2 \theta$ and $\sin^2 \theta$. (2) $\langle 0 | \sigma_z | 0 \rangle = +1$ and $\langle 1 | \sigma_z | 1 \rangle = -1$. (3) the double-angle identity.

So the average magnetization has expectation value

$$\langle \Omega_\theta | M_N | \Omega_\theta \rangle = \frac{1}{N} \sum_{k=1}^N \cos(2\theta) = \cos(2\theta).$$

Write $c \equiv \cos(2\theta)$. The state $|\Phi_\theta\rangle$ is a product over pairs, so the fluctuations at different sites $k \neq j$ are uncorrelated. At the same site, $(\sigma_z^{(k)})^2 = \mathbf{1}$. Together these give

$$\langle \Omega_\theta | (\sigma_z^{(k)} - c)(\sigma_z^{(j)} - c) | \Omega_\theta \rangle = \delta_{kj}(1 - c^2) = \delta_{kj} \sin^2(2\theta).$$

The variance of M_N in the GNS vector is then

$$\|(M_N - c\mathbf{1})|\Omega_\theta\rangle\|^2 \stackrel{(1)}{=} \frac{1}{N^2} \sum_{k,j=1}^N \delta_{kj} \sin^2(2\theta) \stackrel{(2)}{=} \frac{\sin^2(2\theta)}{N} \xrightarrow{N \rightarrow \infty} 0.$$

(1) write $M_N - c\mathbf{1} = \frac{1}{N} \sum_k (\sigma_z^{(k)} - c)$, expand the squared norm into a double sum, and use the correlation formula just derived. (2) the Kronecker delta leaves N equal terms.

This is the law of large numbers for N independent spins. The average of many independent measurements of σ_z settles to its mean c , with fluctuations of order $1/\sqrt{N}$.

4. **From one vector to all vectors.** We now show that $M_N \rightarrow c\mathbf{1}$ on *every* vector, not only on $|\Omega_\theta\rangle$. Take any A' in the commutant $\mathcal{M}'_R$. Since A' commutes with $M_N \in \mathcal{M}_R$,

$$\|(M_N - c\mathbf{1})A'|\Omega_\theta\rangle\| \stackrel{(1)}{=} \|A'(M_N - c\mathbf{1})|\Omega_\theta\rangle\| \stackrel{(2)}{\leq} \|A'\| \|(M_N - c\mathbf{1})|\Omega_\theta\rangle\| \xrightarrow{N \rightarrow \infty} 0.$$

(1) A' commutes with M_N . (2) the operator-norm bound $\|A'v\| \leq \|A'\| \|v\|$.

For $\theta \in (0, \pi/2)$ every pair has both Schmidt coefficients nonzero, so $|\Omega_\theta\rangle$ is separating for $\mathcal{M}_R$ (Chapter 2). A vector is separating for $\mathcal{M}_R$ exactly when it is cyclic for $\mathcal{M}'_R$, so the vectors $A'|\Omega_\theta\rangle$ are dense in $\mathcal{H}_\theta$. The operators $M_N - c\mathbf{1}$ are uniformly bounded, $\|M_N - c\mathbf{1}\| \leq 2$, so convergence on a dense set implies convergence on every vector. Hence M_N converges in the strong operator topology to a multiple of the identity:

$$\text{s-}\lim_{N \rightarrow \infty} M_N = \cos(2\theta)\mathbf{1}.$$

5. **The contradiction.** A normal state is continuous under strong limits of uniformly bounded sequences. (On bounded sets, strong convergence implies weak convergence, and a normal state is weakly continuous there.) Combining this with the result $\tau(M_N) = 0$ of item 2 gives two evaluations of the same limit:

$$\begin{aligned} 0 &= \lim_{N \rightarrow \infty} \tau(M_N) \stackrel{(1)}{=} \tau\left(\text{s-}\lim_{N \rightarrow \infty} M_N\right) \\ &\stackrel{(2)}{=} \tau(\cos(2\theta)\mathbf{1}) \stackrel{(3)}{=} \cos(2\theta)\tau(\mathbf{1}) \stackrel{(4)}{=} \cos(2\theta). \end{aligned}$$

(1) normality of τ , applied to the bounded sequence M_N and its strong limit from item 4. (2) substitute the limit $\text{s-}\lim M_N = \cos(2\theta)\mathbf{1}$. (3) linearity of τ . (4) normalization, $\tau(\mathbf{1}) = 1$.

So $\cos(2\theta) = 0$. But for $\theta \in (0, \pi/2)$ with $\theta \neq \pi/4$ we have $\cos(2\theta) \neq 0$. This is a contradiction, so no normal tracial state τ exists on $\mathcal{M}_R$. ■

Two comments on what this theorem does and does not show. First, at $\theta = \pi/4$ the argument gives no contradiction: $\cos(2\theta) = 0$, so $M_N \rightarrow 0$, which agrees with $\text{tr}(M_N) = 0$. Second, the theorem rules out a *normalized* trace, that is, types II_1 and I_n with finite n . It does not by itself rule out an infinite (semifinite) trace, as in types I_∞ and II_∞ . Excluding those as well takes more work. The

full result, due to Powers (1967), is that for $\theta \neq \pi/4$ the algebra $\mathcal{M}_R$ is a type III_λ factor with $\lambda = \tan^2 \theta$ (for $\theta < \pi/4$). Chapter 4 derives this from the spectrum of the modular operator.

No minimal projection, and dimensions that are real numbers

We now have a genuine trace, so $\mathcal{M}_R$ at $\theta = \pi/4$ is type I or type II. Which one? We use one fact that we quote without proof: $\mathcal{M}_R$ is a factor, meaning that its center contains only multiples of the identity. Now consider projections built by forcing some finite set of spins to be spin-up and leaving the rest alone:

$$\mathbf{1} \equiv \mathbf{1}_2 \otimes \mathbf{1}_2 \otimes \cdots, \quad P_1 = \mathbf{1}_2 \otimes P_\uparrow \otimes \mathbf{1}_2 \otimes \cdots, \quad P_2 = P_\uparrow \otimes \mathbf{1}_2 \otimes P_\uparrow \otimes \mathbf{1}_2 \otimes \cdots.$$

Here $P_\uparrow = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is the projector onto spin-up for a single qubit. In general, we can choose *any* finite set of spins and replace their identity factors with $P_\uparrow$.

Dimensions. We use the trace normalized so that $d(\mathbf{1}) = \text{tr}(\mathbf{1}) = 1$. This holds automatically, since $\text{tr}(\mathbf{1}) = \langle \Phi_{\pi/4} | \Phi_{\pi/4} \rangle = 1$. By eq. (3.6), and using $\text{Tr}_2(P_\uparrow) = 1$,

$$d(P_1) = \text{tr}(P_1) = \frac{1}{2} \text{Tr}_2(P_\uparrow) = \frac{1}{2}, \quad d(P_2) = \text{tr}(P_2) = \frac{1}{2^2} \text{Tr}_2(P_\uparrow) \text{Tr}_2(P_\uparrow) = \frac{1}{4}.$$

In general, a projection that fixes k spins has dimension exactly 2^{-k} . This is positive for every finite k , so every such projection is nonzero. Fixing one more spin halves the dimension, and this never reaches zero at any finite step.

No minimal projection. Take any nonzero projection P of this fixed-spin form. It fixes only finitely many spins, so we can pick one more spin that P leaves alone and fix it too. This gives a new projection $\tilde{P}$ with $\tilde{P} < P$, and $\tilde{P}$ is still nonzero, since $\text{tr}(\tilde{P}) = \frac{1}{2} \text{tr}(P) > 0$. So no projection of this form is minimal.

That argument covers only the fixed-spin projections. To rule out minimal projections of every kind, we use the trace. Suppose $\mathcal{M}_R$ had a minimal projection E . Then $\mathcal{M}_R$ would be a type I factor, and two standard facts about type I factors apply. All minimal projections are equivalent, so they all have the same trace t . Every nonzero projection contains a minimal one. The trace is faithful, because $|\Phi_{\pi/4}\rangle$ is separating, so $t = \text{tr}(E) > 0$. Now pick k so large that $2^{-k} < t$, and take any projection P that fixes k spins. It contains some minimal projection $E' \leq P$, so $t = \text{tr}(E') \leq \text{tr}(P) = 2^{-k} < t$. This is a contradiction. So $\mathcal{M}_R$ has no minimal projections, and by the type classification of Chapter 2 it is not type I.

Every dimension in $[0, 1]$. Here something appears that has no counterpart in finite-dimensional linear algebra. In ordinary quantum mechanics, the dimension of a subspace is a whole number, found by counting basis vectors. There is no subspace of dimension $1/\pi = 0.3183\dots$. In $\mathcal{M}_R$ there is. Let $P_\downarrow = \mathbf{1}_2 - P_\uparrow$, and define projections that act on the first n spins as

$$Q_n = \underbrace{P_\downarrow \otimes \cdots \otimes P_\downarrow}_{n-1} \otimes P_\uparrow \otimes \mathbf{1}_2 \otimes \cdots, \quad \text{tr}(Q_n) = 2^{-n}.$$

Any two of them are orthogonal: for $n < m$, the spin at site n is up in Q_n and down in Q_m . Now take any number $x \in [0, 1]$ with binary digits $x = \sum_n b_n 2^{-n}$, $b_n \in \{0, 1\}$. The partial sums of

$\sum_n b_n Q_n$ are projections in $\mathcal{M}_R$, and they converge strongly. Since $\mathcal{M}_R$ is closed under such limits, the limit $P_x = \sum_n b_n Q_n$ is a projection in $\mathcal{M}_R$. The trace is a vector expectation value, so it adds up over orthogonal projections, and

$$d(P_x) = \sum_n b_n \operatorname{tr}(Q_n) = \sum_n b_n 2^{-n} = x.$$

So $d(P)$ takes every real value in $[0, 1]$.

This continuous range of dimensions, together with the absence of minimal projections, is the defining signature of type II_1 (Chapter 2). Finite projections exist and can be compared in size, but there is no smallest unit of dimension, only a continuum that can be refined forever. Since $\mathcal{M}_R$ is a factor, we conclude: at $\theta = \pi/4$, $\mathcal{M} = \mathcal{M}_R$ is a type II_1 factor. It is the first genuinely new object in these notes, and we built it from nothing more than an infinite chain of ordinary spin- $\frac{1}{2}$ qubits.

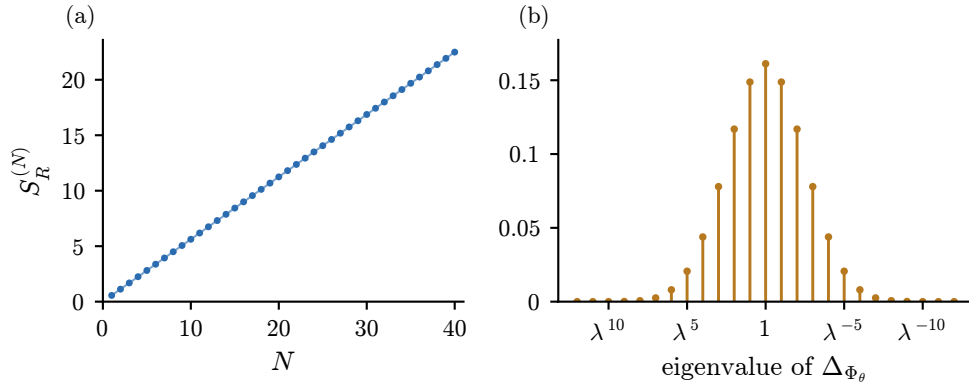


Figure 3.2: The Bell-pair chain at a generic angle, here $\theta = \pi/6$. (a) The entanglement entropy of N pairs, in nats, grows linearly in N , so it diverges as $N \rightarrow \infty$ and no factorized $\mathcal{H}_R$ survives the limit. (b) At $N = 12$, the eigenvalues of the modular operator Δ_{Φ_θ} are the powers λ^n with $\lambda = \tan^2 \theta$, evenly spaced on a logarithmic axis; bar heights give the fraction of all 4^N eigenvalues equal to each power. As $N \rightarrow \infty$ this becomes the geometric ladder of a type III_λ factor (Chapter 4). At $\theta = \pi/4$, $\lambda = 1$ and the whole ladder collapses to the single eigenvalue 1 of the type II_1 factor of this chapter.

Interpreting $\rho_{\mathcal{M}}$ and $S_{\mathcal{M}}$: an entropy that can be negative

With the type II_1 trace (3.4), we can use eq. (3.1) to compute $\rho_{\mathcal{M}}$, and eq. (3.2) to compute $S_{\mathcal{M}}$, for many states. The results have a surprising feature that looks at first like a mistake. So we work it out with actual numbers.

Take a state $|\Psi\rangle$ that equals $|\Phi_{\pi/4}\rangle$ everywhere except on a finite number k of pairs, labelled $i_1, \dots, i_k$. Those pairs are instead prepared in other pure two-qubit states $|\eta_{i_s}\rangle$. Let ρ_i be the ordinary reduced density matrix of pair i on its right qubit. For the unchanged pairs, $\rho_i = \frac{1}{2}\mathbf{1}_2$. We claim that

$$\rho_{\mathcal{M}}(\Psi) = 2\rho_1 \otimes 2\rho_2 \otimes \cdots \otimes 2\rho_n \otimes \cdots, \quad (3.7)$$

where $2\rho_i = \mathbf{1}_2$ for every pair not among $i_1, \dots, i_k$. To check the claim, take any product operator

$A = \bigotimes_i a_i$. Then $A\rho_{\mathcal{M}} = \bigotimes_i (a_i 2\rho_i)$, and

$$\begin{aligned} \text{tr}(A\rho_{\mathcal{M}}) &\stackrel{(1)}{=} \prod_i \frac{1}{2} \text{Tr}_2(a_i 2\rho_i) \stackrel{(2)}{=} \prod_i \text{Tr}_2(a_i \rho_i) \\ &\stackrel{(3)}{=} \prod_i \langle \eta_i | a_i \otimes \mathbf{1}_L | \eta_i \rangle \stackrel{(4)}{=} \langle \Psi | A | \Psi \rangle. \end{aligned}$$

(1) eq. (3.6); only finitely many factors differ from $\mathbf{1}_2$. (2) the $\frac{1}{2}$ cancels the 2. (3) the ordinary partial-trace formula in each pair, with $|\eta_i\rangle = |\phi_{\pi/4}\rangle$ for the unchanged pairs. (4) $|\Psi\rangle$ is a product over pairs.

By linearity and continuity, the same holds for every $A \in \mathcal{M}$, so eq. (3.7) solves eq. (3.1). The factor of 2 comes directly from the $\frac{1}{2}$ in eq. (3.5). As a check, set $\Psi = \Phi_{\pi/4}$. Then every factor is $\mathbf{1}_2$ and $\rho_{\mathcal{M}}(\Phi_{\pi/4}) = \mathbf{1}$. This must be so, because $\text{tr}(A \cdot \mathbf{1}) = \text{tr}(A) = \langle \Phi_{\pi/4} | A | \Phi_{\pi/4} \rangle$ by the definition of tr . The state that defines the trace has the identity as its density operator.

Now put eq. (3.7) into the entropy formula (3.2). The logarithm of a tensor product is the sum of the logarithms of the factors, and $\log(2\rho_i) = \log \mathbf{1}_2 = 0$ for every unchanged pair. Therefore

$$\begin{aligned} S_{\mathcal{M}}(\Psi) = -\text{tr}(\rho_{\mathcal{M}} \log \rho_{\mathcal{M}}) &\stackrel{(1)}{=} -\sum_{s=1}^k \frac{1}{2} \text{Tr}_2(2\rho_{i_s} \log(2\rho_{i_s})) \\ &\stackrel{(2)}{=} -\sum_{s=1}^k \left[\text{Tr}_2(\rho_{i_s} \log \rho_{i_s}) + \log 2 \text{Tr}_2 \rho_{i_s} \right] \\ &\stackrel{(3)}{=} \sum_{s=1}^k S_2(\rho_{i_s}) - k \log 2. \end{aligned} \tag{3.8}$$

(1) only the k changed pairs contribute to $\log \rho_{\mathcal{M}}$. For each one, the trace factorizes pair by pair as in eq. (3.6). The pair itself gives $\frac{1}{2} \text{Tr}_2$, and every other factor gives $\frac{1}{2} \text{Tr}_2(2\rho_j) = \text{Tr}_2 \rho_j = 1$. (2) $\log(2\rho) = \log 2 \mathbf{1}_2 + \log \rho$, and the $\frac{1}{2}$ cancels the 2. (3) $\text{Tr}_2 \rho_{i_s} = 1$, and $S_2(\rho) \equiv -\text{Tr}_2(\rho \log \rho)$ is the ordinary single-qubit entropy.

Here $S_2(\rho_{i_s})$ is the ordinary entanglement entropy of the single changed pair i_s . It lies between 0 and $\log 2$.

Worked Example: One changed pair at angle $\phi = 0.3$

Take $k = 1$. Let the changed pair be prepared at angle $\phi = 0.3$ instead of $\pi/4 \approx 0.785$, so $|\eta_{i_1}\rangle = \cos(0.3)|00\rangle + \sin(0.3)|11\rangle$. This pair is less than maximally entangled. Its reduced density matrix and the corresponding factor of $\rho_{\mathcal{M}}$ are

$$\rho_{i_1} = \begin{pmatrix} \cos^2 0.3 & 0 \\ 0 & \sin^2 0.3 \end{pmatrix} \approx \begin{pmatrix} 0.9127 & 0 \\ 0 & 0.0873 \end{pmatrix}, \quad 2\rho_{i_1} \approx \begin{pmatrix} 1.8253 & 0 \\ 0 & 0.1747 \end{pmatrix}.$$

Directly from $\rho_{\mathcal{M}}$. Only this one factor of $\rho_{\mathcal{M}}$ differs from $\mathbf{1}_2$, so

$$S_{\mathcal{M}}(\Psi) = -\frac{1}{2} \text{Tr}_2(2\rho_{i_1} \log(2\rho_{i_1})) = -\frac{1}{2}(1.8253 \log 1.8253 + 0.1747 \log 0.1747)$$

$$\approx -0.5492 + 0.1524 = -0.3968.$$

From eq. (3.8). The ordinary entropy of the pair is $S_2(\rho_{i_1}) = -0.9127 \log 0.9127 - 0.0873 \log 0.0873 \approx 0.2963$, and $\log 2 \approx 0.6931$, so

$$S_{\mathcal{M}}(\Psi) \approx 0.2963 - 0.6931 = -0.3968.$$

From ordinary quantum mechanics. The ordinary relative entropy of the qubit state ρ_{i_1} with respect to the maximally mixed state $\frac{1}{2}\mathbf{1}_2$ is

$$S(\rho_{i_1} \parallel \tfrac{1}{2}\mathbf{1}_2) = \text{Tr}_2 \rho_{i_1} (\log \rho_{i_1} - \log \tfrac{1}{2}\mathbf{1}_2) = \log 2 - S_2(\rho_{i_1}) \approx 0.3968.$$

All three numbers agree to four decimals (we checked them numerically). So $S_{\mathcal{M}}(\Psi)$ is a genuinely *negative* number, and it equals minus an ordinary relative entropy of one qubit.

The negative sign is not an accident of the angle chosen. The ordinary two-level entropy is at most $\log 2$, with equality only for a maximally entangled pair. So every term in the sum in eq. (3.8) obeys $S_2(\rho_{i_s}) - \log 2 \leq 0$. Hence $S_{\mathcal{M}}(\Psi) \leq 0$ for *every* state built this way. Equality holds only when every changed pair is still maximally entangled.

At first sight this is alarming. The operator $\rho_{\mathcal{M}}$ is a genuine density operator: it is positive and $\text{tr} \rho_{\mathcal{M}} = 1$. Yet $-\text{tr}(\rho_{\mathcal{M}} \log \rho_{\mathcal{M}})$ is negative. For an ordinary finite-dimensional density matrix this can never happen, since $-\text{Tr}(\rho \log \rho) \geq 0$, with equality only for a pure state. There is no contradiction, and the resolution is simple. Here tr is *not* the ordinary matrix trace that counts basis states one at a time. We built it from expectation values in the maximally entangled reference state $|\Phi_{\pi/4}\rangle$. In a type II algebra there is no minimal projection to serve as “one state.” So tr never had the counting interpretation that makes the ordinary entropy non-negative.

What eq. (3.8) actually computes is minus a *relative* entropy:

$$S_{\mathcal{M}}(\Psi) = -S_{\mathcal{M}}(\Psi \parallel \Phi_{\pi/4}).$$

Here $S(\rho \parallel \sigma) = \text{tr} \rho (\log \rho - \log \sigma)$ is the relative entropy of quantum information theory (mentioned in Chapter 1), applied with $\rho = \rho_{\mathcal{M}}(\Psi)$ and $\sigma = \rho_{\mathcal{M}}(\Phi_{\pi/4}) = \mathbf{1}$. Since $\log \mathbf{1} = 0$, the identity follows at once from eq. (3.2). It holds for every state that has a density operator, not only for the simple states used above. That the right-hand side agrees with the general definition of relative entropy in a von Neumann algebra is a standard result, which we quote without proof.

Relative entropy measures how distinguishable a state is from a chosen reference state. It always obeys $S(\rho \parallel \sigma) \geq 0$, so $-S(\rho \parallel \sigma) \leq 0$. This matches what we just computed. So in a type II algebra, $S_{\mathcal{M}}$ should not be read as “how many microstates the state occupies.” That reading only makes sense when there is a minimal projection to count from. Instead, $S_{\mathcal{M}}$ is **a signed distance from the maximally entangled reference state that defines the trace**. It is a relative quantity, not an absolute one.

This is the first place in the notes where relative entropy, rather than entropy itself, turns out to be the more fundamental and more robust object. A suitable generalization of relative entropy (due to

Araki) survives through type III (Chapter 4) and into the holographic applications that begin in Chapter 6. Ordinary entropy, as we have just seen, already changes its meaning at type II.

Physics Connection: Negative relative entropy in qubits

Is a negative “entropy” a sensible thing for a theory to produce? Ordinary finite-dimensional quantum mechanics already contains the same phenomenon in disguise. Take a d -dimensional quantum system, and compute the relative entropy of any state ρ with respect to the maximally mixed state $\sigma = \mathbf{1}/d$:

$$S(\rho\|\sigma) \stackrel{(1)}{=} \text{Tr } \rho(\log \rho - \log(\mathbf{1}/d)) \stackrel{(2)}{=} -S(\rho) + \log d.$$

(1) the definition of relative entropy, with $\sigma = \mathbf{1}/d$. (2) $\log(\mathbf{1}/d) = -\log d \mathbf{1}$ and $\text{Tr } \rho = 1$; here $S(\rho) \equiv -\text{Tr}(\rho \log \rho)$ is the ordinary von Neumann entropy.

The entropy never exceeds its maximally mixed value, $S(\rho) \leq \log d$, so $S(\rho\|\sigma) \geq 0$, as a relative entropy must be. Now flip the sign: $-S(\rho\|\sigma) = S(\rho) - \log d$. For a qubit ($d = 2$) in the state $\rho = \text{diag}(0.9, 0.1)$, we get $S(\rho) \approx 0.325$ and $\log 2 \approx 0.693$, so $-S(\rho\|\sigma) \approx -0.368$. This is a negative number, obtained from an ordinary qubit entropy minus a constant.

This is exactly the mechanism behind eq. (3.8), with the constant written out. In a type II algebra, $S_{\mathcal{M}}$ is built the same way: an ordinary-looking entropy, shifted down by the baseline set by the maximally entangled reference state. In finite dimensions the baseline is the finite number $\log d$. In the type II₁ Bell-pair algebra the same baseline is present, but it has been absorbed into the definition of tr , which was built from the maximally entangled state $|\Phi_{\pi/4}\rangle$. It is no longer a separate $\log d$ that we subtract by hand.

The arithmetic of entropy has not changed. Subtracting a constant from an ordinary entropy could always give a negative number. What is new in the type II case is that the algebra offers no other baseline. With no minimal projection there is no “ d ” to normalize against. Moreover, a type II₁ factor has exactly one normalized trace. So the reference state is not a free choice: it is fixed by the algebra.

Changing infinitely many pairs: the law of large numbers

Equation (3.8) says that each changed pair lowers $S_{\mathcal{M}}$ by $\log 2 - S_2(\rho_i) \geq 0$. What happens if we keep changing more and more pairs? Let Ψ_N be the state in which the first N pairs are at angle ϕ and all the others are at $\pi/4$. Then

$$S_{\mathcal{M}}(\Psi_N) = -N[\log 2 - S_2(\phi)].$$

For $\phi = 0.3$ the deficit per pair is 0.3968, so after $N = 100$ pairs $S_{\mathcal{M}} \approx -39.68$. As $N \rightarrow \infty$, $S_{\mathcal{M}}(\Psi_N) \rightarrow -\infty$. The limiting state is the chain Φ_ϕ with every pair at angle ϕ , and it has no density operator at all with respect to this trace. The law of large numbers shows why.

Where the state lives. By eq. (3.7), $\rho_{\mathcal{M}}(\Psi_N) = (2\rho_\phi)^{\otimes N} \otimes \mathbf{1} \otimes \cdots$, where $\rho_\phi = \text{diag}(\cos^2 \phi, \sin^2 \phi)$.

On a configuration in which m of the first N right spins are up, this operator has the eigenvalue

$$2^N \cos^{2m} \phi \sin^{2(N-m)} \phi.$$

In the state Ψ_N , each of the first N right spins is up with probability $\cos^2 \phi$, independently. By the law of large numbers, $m/N \rightarrow \cos^2 \phi$. So the eigenvalue that the state typically sees is

$$\begin{aligned} 2^N \cos^{2m} \phi \sin^{2(N-m)} \phi &\stackrel{(1)}{=} \exp \left\{ N [\log 2 + \cos^2 \phi \log \cos^2 \phi + \sin^2 \phi \log \sin^2 \phi] \right\} \\ &\stackrel{(2)}{=} e^{N[\log 2 - S_2(\phi)]} \xrightarrow{N \rightarrow \infty} \infty. \end{aligned}$$

(1) take the logarithm and set $m = N \cos^2 \phi$. (2) $S_2(\phi) = -\cos^2 \phi \log \cos^2 \phi - \sin^2 \phi \log \sin^2 \phi$.

Where the trace lives. Under the trace, each spin is up with probability $\frac{1}{2}$, so the trace concentrates on configurations with $m/N \rightarrow \frac{1}{2}$. The configurations where Ψ_N lives, with $m \approx N \cos^2 \phi$, number roughly $e^{NS_2(\phi)}$, and each has trace weight 2^{-N} . Their total trace is therefore about $e^{-N[\log 2 - S_2(\phi)]} \rightarrow 0$. As N grows, the state piles up on a set of vanishing trace, with a density that grows without bound. In the limit, the density would have to be infinite on a set of trace zero, and no operator can do that.

The sharp version. This is the same law of large numbers that drives the theorem above. Apply item 3 of its proof at $\theta = \pi/4$. The variance is $1/N$, so $M_N \rightarrow 0$ strongly on the $\pi/4$ Hilbert space. Suppose Φ_ϕ had a density operator ρ in $\mathcal{M}_R$. The state $A \mapsto \text{tr}(A\rho)$ would then be normal, and so $\text{tr}(\rho M_N) \rightarrow 0$. But $\langle \Phi_\phi | M_N | \Phi_\phi \rangle = \cos(2\phi)$ for every N , which is nonzero unless $\phi = \pi/4$. So Φ_ϕ is not a normal state of the $\pi/4$ algebra. It is not described by any vector or density matrix in the Hilbert space $\mathcal{H}_{\pi/4}$. The density operator of eq. (3.1) exists for states that differ from $\Phi_{\pi/4}$ on finitely many pairs, or by changes that die off fast enough along the chain. It does not exist for a change of fixed size repeated on infinitely many pairs.

Type II_∞

For a type II_∞ factor, the same interpretation of $\rho_{\mathcal{M}}$ and $S_{\mathcal{M}}$ carries over, with one difference. In the II_1 case the condition $d(\mathbf{1}) = 1$ fixes the normalization of the trace. In the II_∞ case $\text{tr}(\mathbf{1}) = \infty$, so there is no such condition. The trace is unique only up to a positive constant, $\text{tr}' = c \text{tr}$. Rescaling the trace changes the density operator to $\rho' = \rho/c$, since $\text{tr}'(A\rho/c) = \text{tr}(A\rho)$. The entropy then shifts by a constant:

$$\begin{aligned} S' &= -\text{tr}'(\rho' \log \rho') \stackrel{(1)}{=} -c \text{tr} \left(\frac{\rho}{c} (\log \rho - \log c \mathbf{1}) \right) \\ &\stackrel{(2)}{=} -\text{tr}(\rho \log \rho) + \log c \text{tr} \rho \stackrel{(3)}{=} S + \log c. \end{aligned}$$

(1) substitute $\text{tr}' = c \text{tr}$ and $\rho' = \rho/c$, and use $\log(\rho/c) = \log \rho - \log c \mathbf{1}$. (2) the factors c and $1/c$ cancel. (3) $\text{tr} \rho = 1$.

So in type II_∞ , the entropy is defined only up to an additive constant that is the same for every state. Entropy differences between states are unambiguous. This is exactly the situation we will meet in the crossed product of Chapter 5.

3.4 Summary

We can now combine the type I and type II results into one statement, before Chapter 4 introduces type III. The *type* of the algebra $\mathcal{M}$ that describes a subsystem determines, independently of the state, what *kind* of entanglement every state of the full system can have.

- **Type I factor.** A genuine tensor factorization $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ exists. Unentangled product states exist, and every state is built from them by superposition or mixing. This is the ordinary textbook picture of quantum mechanics, and $\rho_{\mathcal{M}}$ is the ordinary reduced density matrix.
- **Type I with a center.** The Hilbert space splits into superselection sectors. The entropy is the classical Shannon entropy of the sector label plus the average entanglement within the sectors, eq. (3.3). Lattice gauge theory is an example.
- **Type II factor.** There are no minimal projections. A standard result says that a normal pure state on $\mathcal{M}$ exists only if $\mathcal{M}$ has a minimal projection. So no state of the full system restricts to a pure state on $\mathcal{M}$. There is no unentangled reference state anywhere. In this sense *every* state of the full system is entangled across $(\mathcal{M}, \mathcal{M}')$. There is no “ground floor” of zero entanglement to compare against. There is only the maximally entangled reference state that defines the trace, and the entropy (3.2) is measured relative to it. In type II_1 this entropy is minus a relative entropy and is never positive. In type II_∞ it is defined up to a state-independent constant.

In type III, not even a trace exists, so eqs. (3.1) and (3.2) cannot be written down at all. Type III is the subject of the next chapter. It is not an exotic corner case. It is the type that governs local regions of relativistic quantum field theory, and holographic boundary subalgebras in the strict large- N limit. These are the central objects of these notes.

CHAPTER 4

Von Neumann algebras and entanglement: type III

This chapter is the technical heart of the notes. A type III algebra has no trace. So the density operator $\rho_{\mathcal{M}}$ and the entropy $S_{\mathcal{M}}$ of Chapter 3 cannot even be defined. Type III is not a rare special case, though. It is the type of the algebra of a local region in a relativistic quantum field theory. As Chapter 7 will show, it is also the type of the holographic boundary algebras in the strict large- N limit. To say anything about entanglement in these settings, we need a different tool. That tool is **Tomita–Takesaki modular theory**. Its core physical idea fits in one sentence, before any formalism: *when a subsystem is highly entangled with its complement, and we know nothing about the complement, the subsystem looks thermal*. Seen from inside one half, entanglement looks like heat. This chapter makes that one idea precise.

4.1 Emergent times from entanglement — modular flows

Starting from something you already understand: the type I case, retold

To motivate the general construction, go back to an ordinary type I bipartite pure state $|\Psi\rangle$. Assume that its reduced density matrix ρ_R is full-rank. “Full-rank” means invertible: every eigenvalue is strictly positive, so ρ_R^{-1} exists. This will matter in a moment. Instead of studying the entropy S_R directly, we package the same information in a different way. Write

$$\rho_R = e^{-K_R}, \quad K_R \equiv -\log \rho_R.$$

The operator K_R is called the **entanglement Hamiltonian**. It satisfies $K_R \geq 0$, because the eigenvalues of ρ_R lie in $(0, 1]$, so their negative logarithms are all ≥ 0 . The full set of eigenvalues of K_R is called the *entanglement spectrum*. It carries more information than the entropy S_R , and more than any finite list of Rényi entropies. Each of those numbers is one particular weighted sum over the spectrum, while the spectrum itself is the unsummed data. This is why condensed-matter physicists study entanglement spectra directly, and not only the single number S_R . It has proved to be a very useful diagnostic.

Now use K_R to generate a flow:

$$A(s) = e^{iK_R s} A e^{-iK_R s} \in B(\mathcal{H}_R), \quad A \in B(\mathcal{H}_R). \quad (4.1)$$

This has the same form as ordinary Heisenberg time evolution, $A(t) = e^{iHt} A e^{-iHt}$. Here K_R plays the role of the Hamiltonian and s plays the role of time. The flow is called **modular flow**, and s is

called **modular time**. The key physical claim is the following. An observer confined to R , who uses s as their time, should see their system in thermal equilibrium. The reason is simple. By construction, $\rho_R = e^{-K_R}$ is an ordinary Gibbs density matrix for the “Hamiltonian” K_R , at the dimensionless inverse temperature $\beta = 1$. In precise terms, correlation functions of modular-flowed operators satisfy the **Kubo–Martin–Schwinger (KMS) condition**. This condition is the mathematical signature of thermal equilibrium. It is spelled out precisely below.

The same construction works on the L side, with $K_L = -\log \rho_L$. To treat both sides at once, define

$$\Delta_\Psi \equiv \rho_R \otimes \rho_L^{-1}, \quad -\log \Delta_\Psi = K_R - K_L, \quad (4.2)$$

and let the modular flow act on operators of *either* side:

$$\begin{aligned} \sigma_s(A) &\equiv \Delta_\Psi^{-is} A \Delta_\Psi^{is} \in B(\mathcal{H}_R), & A &\in B(\mathcal{H}_R), \\ \sigma_s(A') &\equiv \Delta_\Psi^{-is} A' \Delta_\Psi^{is} \in B(\mathcal{H}_L), & A' &\in B(\mathcal{H}_L). \end{aligned}$$

Here, as usual, an operator A on $\mathcal{H}_R$ stands for $A \otimes \mathbf{1}_L$, and an operator A' on $\mathcal{H}_L$ stands for $\mathbf{1}_R \otimes A'$. Let us check that this reduces to the one-sided flow (4.1) for $A \in B(\mathcal{H}_R)$:

$$\begin{aligned} \Delta_\Psi^{-is}(A \otimes \mathbf{1}_L)\Delta_\Psi^{is} &\stackrel{(1)}{=} (\rho_R^{-is} \otimes \rho_L^{is})(A \otimes \mathbf{1}_L)(\rho_R^{is} \otimes \rho_L^{-is}) \\ &\stackrel{(2)}{=} \rho_R^{-is} A \rho_R^{is} \otimes \rho_L^{is} \rho_L^{-is} \stackrel{(3)}{=} \rho_R^{-is} A \rho_R^{is} \otimes \mathbf{1}_L. \end{aligned}$$

(1) a power of a tensor product of positive operators is the tensor product of the powers. (2) operators on different tensor factors multiply factor by factor. (3) $\rho_L^{is} \rho_L^{-is} = \mathbf{1}_L$. Since $\rho_R^{-is} = e^{iK_R s}$, this is exactly (4.1). So the single joint object Δ_Ψ reproduces the one-sided flow on each side. The operator Δ_Ψ is called the **modular operator**.

Worked example, with actual numbers. Take the two-qubit state $|\phi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$ from Chapter 1, with $0 < \theta < \pi/2$ so that both coefficients are nonzero. In each basis vector $|ab\rangle$, the first label refers to the R qubit and the second to the L qubit. Then $\rho_R = \rho_L = \text{diag}(\cos^2\theta, \sin^2\theta)$. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, the modular operator $\Delta_\Psi = \rho_R \otimes \rho_L^{-1}$ is the diagonal matrix

$$\Delta_\Psi = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cot^2\theta & 0 & 0 \\ 0 & 0 & \tan^2\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \text{diag}(1, \cot^2\theta, \tan^2\theta, 1).$$

For example, the $|01\rangle$ entry is $\cos^2\theta/\sin^2\theta = \cot^2\theta$. The eigenvalues are 1 (twice, on $|00\rangle$ and $|11\rangle$), $\lambda \equiv \tan^2\theta$ (on $|10\rangle$), and $\lambda^{-1} = \cot^2\theta$ (on $|01\rangle$).

Now let us build three more objects from first principles, using only the algebra and the state. They are the Tomita operator S_Ψ , its adjoint $F_\Psi \equiv S_\Psi^\dagger$, and the modular conjugation J_Ψ . (Remark (c) after the Tomita–Takesaki theorem below defines them in general.) We will see that Δ_Ψ comes out of them.

1. **Action of the algebra on $|\phi_\theta\rangle$.** Any operator $A \in B(\mathcal{H}_R)$ is a 2×2 matrix

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Acting on $|\phi_\theta\rangle$ it gives

$$(A \otimes \mathbf{1}_L) |\phi_\theta\rangle = a \cos \theta |00\rangle + b \sin \theta |01\rangle + c \cos \theta |10\rangle + d \sin \theta |11\rangle.$$

Now take an arbitrary vector $|x\rangle = x_{00} |00\rangle + x_{01} |01\rangle + x_{10} |10\rangle + x_{11} |11\rangle$. Matching coefficients shows that $|x\rangle = (A \otimes \mathbf{1}_L) |\phi_\theta\rangle$ for exactly one A , namely $a = x_{00}/\cos \theta$, $b = x_{01}/\sin \theta$, $c = x_{10}/\cos \theta$, $d = x_{11}/\sin \theta$.

2. **The Tomita operator S_Ψ .** It is defined by $S_\Psi(A \otimes \mathbf{1}_L) |\phi_\theta\rangle = (A^\dagger \otimes \mathbf{1}_L) |\phi_\theta\rangle$. It is antilinear, which means $S_\Psi(z|x\rangle) = z^* S_\Psi|x\rangle$ for a complex number z . Since $A^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$, we have

$$(A^\dagger \otimes \mathbf{1}_L) |\phi_\theta\rangle = a^* \cos \theta |00\rangle + c^* \sin \theta |01\rangle + b^* \cos \theta |10\rangle + d^* \sin \theta |11\rangle.$$

Now substitute the expressions for a, b, c, d in terms of the x_{ij} :

$$S_\Psi \begin{pmatrix} x_{00} \\ x_{01} \\ x_{10} \\ x_{11} \end{pmatrix} \stackrel{(1)}{=} \begin{pmatrix} x_{00}^* \\ \frac{\sin \theta}{\cos \theta} x_{10}^* \\ \frac{\cos \theta}{\sin \theta} x_{01}^* \\ x_{11}^* \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} x_{00}^* \\ \tan \theta x_{10}^* \\ \cot \theta x_{01}^* \\ x_{11}^* \end{pmatrix}.$$

(1) insert $a^* = x_{00}^*/\cos \theta$, $c^* = x_{10}^*/\cos \theta$, $b^* = x_{01}^*/\sin \theta$ and $d^* = x_{11}^*/\sin \theta$ into the four coefficients $a^* \cos \theta$, $c^* \sin \theta$, $b^* \cos \theta$ and $d^* \sin \theta$. The angles are real, so conjugation does not affect them. (2) $\sin \theta / \cos \theta = \tan \theta$ and $\cos \theta / \sin \theta = \cot \theta$. Applying S_Ψ twice gives back the original vector. For example, the second slot becomes $\tan \theta (\cot \theta x_{01}^*)^* = x_{01}$. So $S_\Psi^2 = \mathbf{1}$. Also $S_\Psi |\phi_\theta\rangle = |\phi_\theta\rangle$, because $|\phi_\theta\rangle$ is $(A \otimes \mathbf{1}_L) |\phi_\theta\rangle$ with $A = \mathbf{1}$, and $\mathbf{1}^\dagger = \mathbf{1}$.

3. **The adjoint operator $F_\Psi \equiv S_\Psi^\dagger$.** For an antilinear operator, the adjoint is defined by $\langle y | S_\Psi x \rangle = \langle x | F_\Psi y \rangle$ for all vectors $|x\rangle, |y\rangle$. Writing out both sides in components and comparing gives

$$F_\Psi \begin{pmatrix} y_{00} \\ y_{01} \\ y_{10} \\ y_{11} \end{pmatrix} = \begin{pmatrix} y_{00}^* \\ \cot \theta y_{10}^* \\ \tan \theta y_{01}^* \\ y_{11}^* \end{pmatrix}.$$

4. **The modular operator $\Delta_\Psi = F_\Psi S_\Psi$.** Multiply the two operators:

$$\Delta_\Psi \begin{pmatrix} x_{00} \\ x_{01} \\ x_{10} \\ x_{11} \end{pmatrix} \stackrel{(1)}{=} F_\Psi \begin{pmatrix} x_{00}^* \\ \tan \theta x_{10}^* \\ \cot \theta x_{01}^* \\ x_{11}^* \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} x_{00} \\ \cot^2 \theta x_{01} \\ \tan^2 \theta x_{10} \\ x_{11} \end{pmatrix}.$$

(1) apply S_Ψ first, using the formula from item 2. (2) apply F_Ψ from item 3 to the vector $(y_{00}, y_{01}, y_{10}, y_{11}) = (x_{00}^*, \tan \theta x_{10}^*, \cot \theta x_{01}^*, x_{11}^*)$. The second slot becomes $\cot \theta y_{10}^* = \cot \theta \cdot \cot \theta x_{01}$. The third becomes $\tan \theta y_{01}^* = \tan \theta \cdot \tan \theta x_{10}$. The two

complex conjugations cancel.

This is exactly the diagonal matrix found at the start,

$$\Delta_\Psi = \text{diag}(1, \cot^2 \theta, \tan^2 \theta, 1).$$

So the modular operator can be obtained from the Tomita operator alone, as $\Delta_\Psi = S_\Psi^\dagger S_\Psi$.

5. **The modular conjugation** $J_\Psi = S_\Psi \Delta_\Psi^{-1/2}$. The positive square root of Δ_Ψ is

$$\Delta_\Psi^{1/2} = \text{diag}(1, \cot \theta, \tan \theta, 1).$$

Apply S_Ψ to $\Delta_\Psi^{-1/2} |x\rangle$:

$$J_\Psi \begin{pmatrix} x_{00} \\ x_{01} \\ x_{10} \\ x_{11} \end{pmatrix} \stackrel{(1)}{=} S_\Psi \begin{pmatrix} x_{00} \\ \tan \theta x_{01} \\ \cot \theta x_{10} \\ x_{11} \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} x_{00}^* \\ x_{10}^* \\ x_{01}^* \\ x_{11}^* \end{pmatrix}.$$

(1) $\Delta_\Psi^{-1/2} = \text{diag}(1, \tan \theta, \cot \theta, 1)$, the inverse of the square root just written.

(2) apply S_Ψ . The second slot becomes $\tan \theta \cdot (\cot \theta x_{10})^* = x_{10}^*$. The third becomes $\cot \theta \cdot (\tan \theta x_{01})^* = x_{01}^*$. So J_Ψ conjugates every component and then exchanges $|01\rangle \leftrightarrow |10\rangle$. That exchange is the SWAP of the two qubits. J_Ψ is antiunitary and satisfies $J_\Psi^2 = \mathbf{1}$. It also satisfies $J_\Psi \Delta_\Psi J_\Psi = \Delta_\Psi^{-1}$, because the swap exchanges the two diagonal entries $\cot^2 \theta$ and $\tan^2 \theta$.

6. **Exact modular flow on Pauli matrices.** The modular flow of an observable $A \in B(\mathcal{H}_R)$ is $\sigma_s(A) = \Delta_\Psi^{-is}(A \otimes \mathbf{1}_L)\Delta_\Psi^{is}$.

- Take $A = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. The operator $\sigma_z \otimes \mathbf{1}_L = \text{diag}(1, 1, -1, -1)$ is diagonal, so it commutes with Δ_Ψ . Hence

$$\sigma_s(\sigma_z) = \sigma_z.$$

- Take the raising operator $\sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $\sigma_+ \otimes \mathbf{1}_L = |00\rangle\langle 10| + |01\rangle\langle 11|$, and

$$\begin{aligned} \sigma_s(\sigma_+) &= \Delta_\Psi^{-is}(\sigma_+ \otimes \mathbf{1}_L)\Delta_\Psi^{is} \stackrel{(1)}{=} (\tan^2 \theta)^{is} \sigma_+ \otimes \mathbf{1}_L \\ &\stackrel{(2)}{=} e^{2is \log \tan \theta} \sigma_+ \otimes \mathbf{1}_L. \end{aligned}$$

(1) Δ_Ψ is diagonal. So conjugating $|00\rangle\langle 10|$ multiplies it by $(\Delta_{00})^{-is}(\Delta_{10})^{is} = 1 \cdot (\tan^2 \theta)^{is}$. Conjugating $|01\rangle\langle 11|$ multiplies it by $(\cot^2 \theta)^{-is} \cdot 1 = (\tan^2 \theta)^{is}$. Both terms get the same factor. (2) $x^{is} = e^{is \log x}$ and $\log \tan^2 \theta = 2 \log \tan \theta$. In the same way, $\sigma_s(\sigma_-) = e^{-2is \log \tan \theta} \sigma_-$.

- Now use $\sigma_x = \sigma_+ + \sigma_-$ and $\sigma_y = -i(\sigma_+ - \sigma_-)$. Write $\alpha \equiv 2s \log \tan \theta$ for

brevity. Then

$$\begin{aligned}
\sigma_s(\sigma_x) &= \sigma_s(\sigma_+) + \sigma_s(\sigma_-) \stackrel{(1)}{=} e^{i\alpha}\sigma_+ + e^{-i\alpha}\sigma_- \\
&\stackrel{(2)}{=} \cos \alpha (\sigma_+ + \sigma_-) + i \sin \alpha (\sigma_+ - \sigma_-) \\
&\stackrel{(3)}{=} \cos \alpha \sigma_x - \sin \alpha \sigma_y.
\end{aligned}$$

(1) substitute the two results just derived, $\sigma_s(\sigma_{\pm}) = e^{\pm i\alpha}\sigma_{\pm}$. (2) Euler's formula, $e^{\pm i\alpha} = \cos \alpha \pm i \sin \alpha$, then collect the terms multiplying $\cos \alpha$ and those multiplying $\sin \alpha$. (3) $\sigma_+ + \sigma_- = \sigma_x$ by definition. Inverting $\sigma_y = -i(\sigma_+ - \sigma_-)$ gives $\sigma_+ - \sigma_- = i\sigma_y$, so $i \sin \alpha (\sigma_+ - \sigma_-) = i \sin \alpha \cdot i\sigma_y = -\sin \alpha \sigma_y$. The same computation for σ_y (same substitution, terms collected the other way) gives

$$\sigma_s(\sigma_y) = \cos \alpha \sigma_y + \sin \alpha \sigma_x, \quad \alpha = 2s \log \tan \theta.$$

So on the qubit R , modular flow is an exact rotation of the Bloch sphere about the z -axis. The rotation angle is $-\alpha = 2s \log \cot \theta$, so the angular frequency is the constant $2 \log \cot \theta$. It is positive for $0 < \theta < \pi/4$.

Two facts about Δ_Ψ deserve to be stated plainly. They are the two facts that carry over, unchanged in spirit, to the fully general (type III) theory below.

1. $\Delta_\Psi |\Psi\rangle = |\Psi\rangle$, and also $\Delta_\Psi^{-1} |\Psi\rangle = |\Psi\rangle$. Equivalently, $(K_R - K_L) |\Psi\rangle = 0$. To see this in general, write the Schmidt decomposition $|\Psi\rangle = \sum_n \sqrt{\lambda_n} |n\rangle_R |n\rangle_L$. Each term $|n\rangle_R |n\rangle_L$ is an eigenvector of K_R with eigenvalue $-\log \lambda_n$, and of K_L with the same eigenvalue, so $K_R - K_L$ annihilates it. In the worked example, $|\Psi\rangle = |\phi_\theta\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle$ is built entirely out of the two eigenvectors of Δ_Ψ with eigenvalue 1, so Δ_Ψ fixes it. Physically, modular flow is an emergent, internal time evolution under which the state itself never changes. It still acts nontrivially on operators in R and in L . This fact is also the operator version of the elementary statement $S_R = S_L$ for a pure global state (Chapter 1). By the Schmidt decomposition, ρ_R and ρ_L have the same nonzero eigenvalues, so K_R and K_L have the same spectrum. The equation $(K_R - K_L) |\Psi\rangle = 0$ expresses this symmetry as an operator statement.
2. Correlators of modular-flowed operators satisfy the KMS condition, which is made precise just below. This is the mathematical signature of thermal equilibrium at $\beta = 1$. An observer with access only to R (or only to L), who uses modular time as their clock, sees a thermal state.

The swap operator J_Ψ

A third object will appear again and again, so it pays to build it explicitly now. Requiring both ρ_R and ρ_L to be full-rank forces $\dim \mathcal{H}_R = \dim \mathcal{H}_L$. The reason is short. The number of nonzero Schmidt coefficients of $|\Psi\rangle$ equals the rank of ρ_R , and it also equals the rank of ρ_L . If ρ_R is full-rank, this number is $\dim \mathcal{H}_R$. If ρ_L is full-rank, it is $\dim \mathcal{H}_L$. So the two dimensions must be equal. Because the dimensions match, we can build an explicit **swap operator**. Write the

Schmidt decomposition $|\Psi\rangle = \sum_n \sqrt{\lambda_n} |n\rangle_R |n\rangle_L$. Expand a general vector in the same Schmidt basis, $|\phi\rangle = \sum_{mn} \phi_{mn} |m\rangle_R |n\rangle_L$. Define

$$J_\Psi |\phi\rangle = \sum_{m,n} \phi_{mn}^* |n\rangle_R |m\rangle_L.$$

This operator swaps the labels $R \leftrightarrow L$ and complex-conjugates every coefficient. For the two-qubit example it is exactly the J_Ψ found in item 5 above. J_Ψ is **antiunitary**: because of the complex conjugation it is a different kind of map from an ordinary unitary, and it needs its own name. It satisfies $J_\Psi^2 = \mathbf{1}$, since swapping twice and conjugating twice gives back the starting vector. A direct check shows $J_\Psi(A \otimes \mathbf{1}_L)J_\Psi = \mathbf{1}_R \otimes \bar{A}$, where $\bar{A}$ is the matrix of A in the Schmidt basis with every entry complex-conjugated. So conjugating an operator in $\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L$ by J_Ψ produces an operator in $\mathcal{M}' = \mathbf{1}_R \otimes B(\mathcal{H}_L)$. In other words, J_Ψ exchanges the subsystem with its complement at the level of operators, not just of vectors. It is called the **modular conjugation operator**.

The key limitation, and the reformulation that removes it

Everything above required ρ_R and ρ_L to be full-rank. This is a real restriction. Among other things, it forces $\dim \mathcal{H}_R = \dim \mathcal{H}_L$, which already rules out many ordinary type I examples. It also says nothing about type II, where there is no factorization $\mathcal{H}_R \otimes \mathcal{H}_L$ and the density operators of Chapter 3 are of a different kind, or about type III, where there is no density operator at all. The strategy now is to restate “ ρ_R and ρ_L are full-rank” in a way that never mentions ρ_R or ρ_L . Then the condition can survive in a setting where those objects do not exist.

The restatement uses the two properties singled out at the end of Chapter 2’s discussion of the GNS construction: **cyclic** and **separating**. Recall the definitions. The vector $|\Psi\rangle$ is cyclic with respect to $\mathcal{M}$ if $\{A|\Psi\rangle : A \in \mathcal{M}\}$ is dense in $\mathcal{H}$. In words, the algebra can reach essentially every vector, starting from $|\Psi\rangle$. The vector $|\Psi\rangle$ is separating with respect to $\mathcal{M}$ if $A|\Psi\rangle = 0$ forces $A = 0$. In words, no nonzero operator in $\mathcal{M}$ annihilates $|\Psi\rangle$. A short exercise with the Schmidt decomposition shows two things. First, $|\Psi\rangle$ is cyclic for $\mathcal{M} = B(\mathcal{H}_R) \otimes \mathbf{1}_L$ exactly when ρ_L is full-rank. Second, $|\Psi\rangle$ is cyclic for the commutant $\mathcal{M}' = \mathbf{1}_R \otimes B(\mathcal{H}_L)$ exactly when ρ_R is full-rank. So “ ρ_R and ρ_L are both full-rank” is *exactly equivalent* to “ $|\Psi\rangle$ is cyclic with respect to both $\mathcal{M}$ and $\mathcal{M}'$.”

A further simplification is available. For any von Neumann algebra, $|\Psi\rangle$ is cyclic with respect to $\mathcal{M}'$ if and only if it is separating with respect to $\mathcal{M}$. (Cyclic for the complement is the same as separating for the algebra itself.) So we never need to look at the two algebras separately. The whole condition becomes a single statement about $\mathcal{M}$ alone: **$|\Psi\rangle$ is cyclic and separating with respect to $\mathcal{M}$** . This is an important step in the logic. It shows once more that the relevant facts about bipartite entanglement can be stated using the algebra of one side only, with no reference to the complement.

The Tomita–Takesaki theorem, stated in full

Here is the theorem. It holds for *any* von Neumann algebra $\mathcal{M}$ with a cyclic and separating vector $|\Psi\rangle$. Its statement never mentions a trace, a density matrix, or a tensor factorization. This is exactly why it survives into type III.

1. There is a positive operator Δ_Ψ , still called the *modular operator*, that leaves $|\Psi\rangle$ invariant: $\Delta_\Psi |\Psi\rangle = |\Psi\rangle$. The operator $K_\Psi \equiv -\log \Delta_\Psi$ generates an automorphism of $\mathcal{M}$ (a map from $\mathcal{M}$ to itself that respects all of the algebra structure), and separately an automorphism of $\mathcal{M}'$:

$$\sigma_s(A) \equiv \Delta_\Psi^{-is} A \Delta_\Psi^{is} \in \mathcal{M} \quad \forall A \in \mathcal{M}, \quad \sigma_s(A') \equiv \Delta_\Psi^{-is} A' \Delta_\Psi^{is} \in \mathcal{M}' \quad \forall A' \in \mathcal{M}'. \quad (4.3)$$

So at every modular time s , the flow stays inside the algebra it started in. This holds for $\mathcal{M}$ and for $\mathcal{M}'$ separately.

2. There is an antiunitary **modular conjugation** J_Ψ . It satisfies $J_\Psi |\Psi\rangle = |\Psi\rangle$, $J_\Psi = J_\Psi^{-1} = J_\Psi^\dagger$, $J_\Psi \Delta_\Psi J_\Psi = \Delta_\Psi^{-1}$, and $J_\Psi \mathcal{M} J_\Psi = \mathcal{M}'$, $J_\Psi \mathcal{M}' J_\Psi = \mathcal{M}$. So J_Ψ swaps $\mathcal{M}$ and $\mathcal{M}'$. This is the role the explicit swap operator played in the type I example above, now established as a general fact.
3. There is a technical analyticity statement. It is included for completeness and is not needed for the physics that follows. As a function of s , the vector $\Delta_\Psi^{-is} A |\Psi\rangle$ extends to complex values of s in the strip $0 < \text{Im } s < \frac{1}{2}$. On the upper edge of that strip it satisfies $\Delta_\Psi^{-i(t+i/2)} A |\Psi\rangle = \Delta_\Psi^{-it} J_\Psi A^\dagger |\Psi\rangle$.
4. Consider correlation functions of modular-flowed operators, $f_{AB}(s) \equiv \langle \Psi | \sigma_s(A) B | \Psi \rangle$. They extend to analytic functions in the strip $-1 < \text{Im } s < 0$, of width 1, and satisfy the **KMS relation**

$$f_{AB}(s) = f_{BA}(-s - i). \quad (4.4)$$

This is the precise mathematical statement of “looks thermal at $\beta = 1$.” It is the same periodicity-in-imaginary-time relation that an ordinary thermal correlator $\text{Tr}(e^{-\beta H} A(t) B)$ satisfies, here with $\beta = 1$. For an ordinary thermal state, the relation comes from the cyclic property of the trace $\text{Tr}(e^{-\beta H} \dots)$. The KMS relation encodes that property without needing $e^{-\beta H}$ to exist as a trace-class operator. That is exactly what is needed once there is no trace at all, as in type III.

Physics Connection: KMS Condition in Thermal QFT

You have almost certainly met the KMS relation already, perhaps not by that name, in a course on finite-temperature quantum mechanics or statistical field theory. Before trusting it for a type III algebra with no Hamiltonian at all, let us check the same relation on an ordinary example. Take a two-level system, $H = \omega |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & \omega \end{pmatrix}$, in the thermal (Gibbs) state

$$\rho = \frac{e^{-\beta H}}{Z} = \frac{1}{1 + e^{-\beta\omega}} \begin{pmatrix} 1 & 0 \\ 0 & e^{-\beta\omega} \end{pmatrix}.$$

The computation below in fact works for any Hamiltonian on a finite-dimensional space. For two operators A, B , define $f_{AB}(t) \equiv \text{Tr}(\rho A(t) B)$, where $A(t) = e^{iHt} A e^{-iHt}$. Write $B(s) \equiv e^{iHs} B e^{-iHs}$ for the same flow applied to B . Substituting the imaginary time $s = -i\beta$

gives $B(-i\beta) = e^{\beta H} B e^{-\beta H}$, an analytic continuation of the flow to imaginary time. Then:

$$\begin{aligned}
 f_{AB}(t) &= \text{Tr} \left(\frac{e^{-\beta H}}{Z} e^{iHt} A e^{-iHt} B \right) \\
 &\stackrel{(1)}{=} \text{Tr} \left(B \frac{e^{-\beta H}}{Z} e^{iHt} A e^{-iHt} \right) \\
 &\stackrel{(2)}{=} \text{Tr} \left(\frac{e^{-\beta H}}{Z} [e^{\beta H} B e^{-\beta H}] e^{iHt} A e^{-iHt} \right) \\
 &\stackrel{(3)}{=} \text{Tr} (\rho B(-i\beta) A(t)) \\
 &\stackrel{(4)}{=} \text{Tr} (\rho e^{-iHt} B(-i\beta) e^{iHt} A) \\
 &\stackrel{(5)}{=} \text{Tr} (\rho B(-t - i\beta) A) = f_{BA}(-t - i\beta).
 \end{aligned}$$

(1) cyclicity of the ordinary matrix trace, moving B to the front. (2) insert $\mathbf{1} = e^{-\beta H} e^{\beta H}$ in front of B . This changes nothing, since it is the identity, but it sets up the next step. (3) $e^{-\beta H}/Z = \rho$, the bracket is $B(-i\beta)$, and $e^{iHt} A e^{-iHt} = A(t)$ by definition. (4) cyclicity again: move the factor e^{-iHt} at the right end of $A(t)$ around to the front, then use that ρ commutes with e^{-iHt} (both are functions of H). (5) the group property of the flow, $e^{-iHt} B(-i\beta) e^{iHt} = B(-t - i\beta)$: flowing by $-i\beta$ and then by $-t$ is the same as flowing by $-t - i\beta$. The last equality is the definition of f_{BA} .

This algebraic proof uses only the cyclicity of the trace and the group property of the flow. It shows why thermal correlators are periodic in imaginary time, with period β . Tomita–Takesaki theory takes this exact property as the definition of a thermal state at $\beta = 1$. It needs no trace and no Hamiltonian.

So what changed between this ordinary example and the general Tomita–Takesaki statement? Only this. Here the flow generating $A(t)$ came from an honest Hamiltonian H , and ρ was a normalizable density matrix that you could write down as a finite matrix. In the general theorem, neither needs to exist. The flow σ_s can be generated by no Hamiltonian that lives in $\mathcal{M}$. This is exactly the defining property of type III, given by the inner-automorphism criterion (4.5) below. There may also be no ρ from which to build a trace. The KMS relation (4.4) survives all of this unchanged. It was never really a statement about H or ρ separately. It is a statement about the correlator f_{AB} alone. That is why the general theory takes the KMS relation, rather than the Hamiltonian or the density matrix, as the basic object.

Read physically, the theorem says the following. *Any* operator algebra with a cyclic and separating reference vector comes with a canonical, built-in notion of time flow. Under this flow, the reference state looks exactly thermal at inverse temperature 1. This holds whether or not $\mathcal{M}$ has a trace. So the type I story above is not being generalized by analogy. It is included as the special case in which the canonical flow is generated by an honest Hamiltonian K_R that lives inside $\mathcal{M}$ itself.

Several remarks complete the theorem:

- (a) Physically: given $\mathcal{M}$ and a cyclic and separating $|\Psi\rangle$, there is an emergent time evolution,

internal to $\mathcal{M}$ (or to $\mathcal{M}'$), that leaves $|\Psi\rangle$ fixed. Relative to this time, an observer confined to $\mathcal{M}$ (or to $\mathcal{M}'$) feels a temperature $1/\beta = 1$.

- (b) For type I, the cyclic and separating condition was a real restriction, since it forced $\dim \mathcal{H}_R = \dim \mathcal{H}_L$. For the type III situations that matter most in these notes, namely quantum field theory and statistical mechanics in the thermodynamic limit, the condition holds very widely. You will see this explicitly later in this chapter, in the $N \rightarrow \infty$ entangled-spin construction and in the Reeh–Schlieder theorem.
- (c) The theorem is usually *proved*, not just stated, by first defining an antilinear **Tomita operator** S_Ψ directly from $\mathcal{M}$ and $|\Psi\rangle$:

$$S_\Psi A |\Psi\rangle = A^\dagger |\Psi\rangle \quad (A \in \mathcal{M}), \quad S_\Psi^2 = \mathbf{1}, \quad S_\Psi |\Psi\rangle = |\Psi\rangle.$$

Its adjoint $F_\Psi = S_\Psi^\dagger$ does the same job for the commutant:

$$F_\Psi A' |\Psi\rangle = A'^\dagger |\Psi\rangle \quad (A' \in \mathcal{M}').$$

(In the two-qubit worked example one can check directly that it is F_Ψ , not S_Ψ , that acts this way on vectors $A' |\Psi\rangle$; the two operators differ unless $\theta = \pi/4$.) The operator S_Ψ is well defined precisely because $|\Psi\rangle$ is separating. If $A |\Psi\rangle = B |\Psi\rangle$, then $(A - B) |\Psi\rangle = 0$, so $A = B$, and therefore $A^\dagger |\Psi\rangle = B^\dagger |\Psi\rangle$. This is the same role the separating property played in the GNS construction of Chapter 2. One then takes the **polar decomposition** $S_\Psi = J_\Psi \Delta_\Psi^{1/2}$, with $\Delta_\Psi = S_\Psi^\dagger S_\Psi$. This is the operator analogue of writing a complex number as $z = e^{i\phi} |z|$: an antiunitary “phase” J_Ψ times a positive “modulus” $\Delta_\Psi^{1/2}$. Every statement of the theorem is then derived from this one definition. The derivation is a technical piece of functional analysis and is not reproduced here. What matters is that the whole construction starts from this one simple map.

- (d) There is a converse, which is used to *identify* the modular operator in physical examples, including the Rindler-wedge computation later in this chapter. Suppose you find a one-parameter group of unitaries that leaves $|\Psi\rangle$ invariant, maps $\mathcal{M}$ to itself in the sense of (4.3), and whose flow satisfies the KMS relation (4.4). Then it *must be* the modular flow Δ_Ψ^{-is} for $|\Psi\rangle$. Only one flow has these properties. So finding one by any means is enough, whether by a physical argument, a symmetry, or a guess that is later checked.
- (e) If $\mathcal{M}$ is type II, the density operators $\rho_{\mathcal{M}} \in \mathcal{M}$ and $\rho_{\mathcal{M}'} \in \mathcal{M}'$ of Chapter 3 exist. Then Δ_Ψ can still be built as in (4.2), with $\rho_{\mathcal{M}}, \rho_{\mathcal{M}'}$ in place of ρ_R, ρ_L . The tensor product is replaced by an ordinary product of these two commuting operators, $\Delta_\Psi = \rho_{\mathcal{M}} \rho_{\mathcal{M}'}^{-1}$. If $|\Psi\rangle$ is the tracial state itself (so $\rho_{\mathcal{M}} = \rho_{\mathcal{M}'} = \mathbf{1}$), then $\Delta_\Psi = \mathbf{1}$. This matches the worked example above at $\theta = \pi/4$.
- (f) If $\mathcal{M}$ is type III, Δ_Ψ cannot be split into a piece from $\mathcal{M}$ and a piece from $\mathcal{M}'$ at all, because there is no $\rho_{\mathcal{M}}$. Δ_Ψ still exists, by the theorem above, but it is now a single joint object with no factorized description.

Finally, one lemma does real work starting in Chapter 7. It is the mechanism that lets entanglement-wedge reconstruction go strictly beyond causal-wedge reconstruction.

Lemma (an “ergodic” property of modular flow). Let $\mathcal{N} \subset \mathcal{X}$ be two von Neumann algebras, and let $|\Psi\rangle$ be cyclic and separating for both. Apply the modular flow σ_s of the *larger* algebra $\mathcal{X}$ to the *smaller* algebra $\mathcal{N}$. The result regenerates all of $\mathcal{X}$: $\{\sigma_s(A) : A \in \mathcal{N}, s \in \mathbb{R}\}'' = \mathcal{X}$.

In words: flow a small piece of an algebra in modular time, using the larger algebra’s own modular clock, and you sweep out the entire larger algebra. This may sound too strong to be true. It follows from a theorem of Takesaki. That theorem says that a subalgebra of $\mathcal{X}$ that is mapped into itself by the modular flow of $\mathcal{X}$, and for which $|\Psi\rangle$ is still cyclic, must be all of $\mathcal{X}$. The algebra generated by all the $\sigma_s(A)$ is such a subalgebra. It contains $\mathcal{N}$, so $|\Psi\rangle$ is cyclic for it.

4.2 Classification of type III factors

Telling type III apart from type I and II, using modular flow alone

Here is a clean, checkable criterion. It was met briefly in Chapter 2’s discussion of inner automorphisms, and it is now established as the actual dividing line between the types:

$$\sigma_s \text{ is an inner automorphism of } \mathcal{M} \text{ for every } s \in \mathbb{R} \quad \Longleftrightarrow \quad \mathcal{M} \text{ is type I or type II.} \quad (4.5)$$

Here σ_s is an **inner automorphism** if there is a unitary U_s that belongs to $\mathcal{M}$ itself (not just some unitary on $\mathcal{H}$) and implements the flow: $\sigma_s(A) = U_s A U_s^\dagger$. For type I, this holds with $U_s = \rho_R^{-is} = e^{iK_R s}$. For type II it holds with the analogous operator built from $\rho_{\mathcal{M}}$. This is the ordinary type I story from earlier in this chapter, now described as “the flow is inner.” Now read (4.5) in the other direction. **For a type III algebra, there must be *some* value of s at which the modular flow cannot be implemented by any unitary inside $\mathcal{M}$.** This gives a practical definition of type III: modular time flows, but no clock that generates the flow can be found inside the algebra being flowed.

State-independence: relating the flows of different reference vectors

The modular operator Δ_Ψ , and hence the flow σ_s^Ψ , depends on the choice of the cyclic and separating reference vector $|\Psi\rangle$. A different choice $|\Omega\rangle$ gives a different Δ_Ω and a different flow σ_s^Ω . A reassuring theorem, due to Connes, says that these flows are never wildly unrelated. There is a family of unitaries $u_{\Psi\Omega}(s) \in \mathcal{M}$, one for each s , relating the two:

$$\sigma_s^\Psi(A) = u_{\Psi\Omega}(s) \sigma_s^\Omega(A) u_{\Psi\Omega}(s)^\dagger, \quad \forall A \in \mathcal{M}. \quad (4.6)$$

So modular flows for different reference states differ only by an inner automorphism, never by anything more drastic. These unitaries also obey a chain rule through a third vector Φ , $u_{\Psi\Omega}(t) u_{\Omega\Phi}(t) = u_{\Psi\Phi}(t)$. This is what one wants if “relating two flows” is to be a consistent notion.

This result lets us define two **state-independent** invariants of the algebra $\mathcal{M}$ itself. They are sets of numbers that depend only on $\mathcal{M}$, not on the reference vector used to compute them. That is

exactly why they are useful for *classifying* algebras rather than states. The first is

$$T(\mathcal{M}) \equiv \{t \in \mathbb{R} : \sigma_t^\Psi \text{ is inner on } \mathcal{M}\}.$$

It does not depend on Ψ , by (4.6): composing with an inner automorphism does not change whether a map is inner. The second is

$$S(\mathcal{M}) \equiv \bigcap_{\Psi} \sigma(\Delta_\Psi) \subset \mathbb{R}_{\geq 0}, \quad (4.7)$$

where $\sigma(\Delta_\Psi)$ denotes the spectrum of Δ_Ψ . It is the intersection, over *every* cyclic and separating reference vector, of the spectrum of the corresponding modular operator. Whatever survives this intersection is a property of $\mathcal{M}$ alone. These are **Connes' two fundamental invariants**. By (4.5), $T(\mathcal{M}) = \mathbb{R}$ for type I and type II, since every modular time is inner. For type III, $T(\mathcal{M})$ is a proper subgroup of $\mathbb{R}$. (For the type III examples below it is a discrete set.) For type I or type II, $S(\mathcal{M}) = \{1\}$. The reason is that there is a tracial reference for which $\Delta = \mathbf{1}$, whose spectrum is the single point $\{1\}$. For the I_∞ and II_∞ cases the trace is not a normalizable state. The precise definition of $S(\mathcal{M})$ therefore takes the intersection over a slightly larger class of reference functionals (called weights), which includes the trace.

For a type III factor, Connes proved that $S(\mathcal{M})$ with the point 0 removed is a closed multiplicative subgroup of $\mathbb{R}_{>0}$. This is a hard theorem and is not re-derived here. The closed subgroups of $\mathbb{R}_{>0}$ are $\{1\}$, the powers $\lambda^{\mathbb{Z}}$ of a single number, and all of $\mathbb{R}_{>0}$. So there are exactly three possibilities, and no others:

$$\begin{aligned} \text{Type III}_0 : S(\mathcal{M}) &= \{0, 1\}, & \text{Type III}_\lambda : S(\mathcal{M}) &= \{0\} \cup \{\lambda^n : n \in \mathbb{Z}\}, \lambda \in (0, 1), \\ \text{Type III}_1 : S(\mathcal{M}) &= \mathbb{R}_{\geq 0}. \end{aligned} \quad (4.8)$$

This is as fine as the classification gets. Every physically relevant type III algebra in the rest of these notes is one of these three.

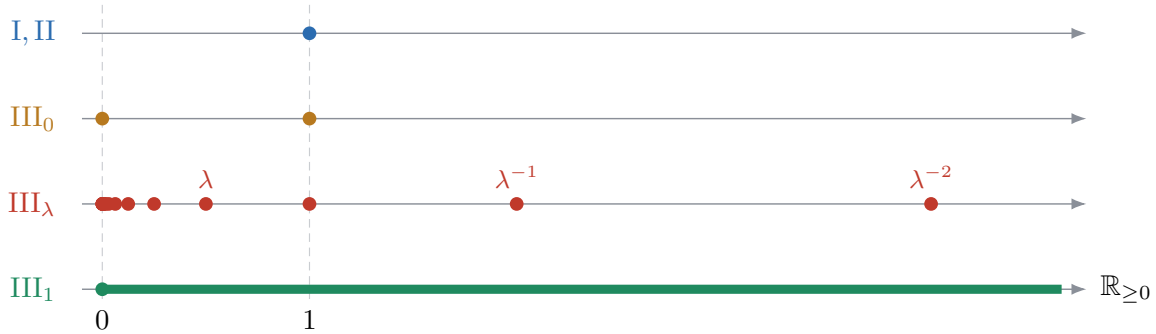


Figure 4.1: Connes' invariant $S(\mathcal{M})$ for each kind of factor, drawn as a set of points on the half-line $\mathbb{R}_{\geq 0}$ (dashed guides mark 0 and 1). Types I and II give the single point 1, and type III_0 gives the two points 0 and 1. Type III_λ gives 0 together with every integer power of λ (drawn for $\lambda = 1/2$), and these powers pile up at 0. Type III_1 fills the whole half-line.

The unitaries $u_{\Psi\Omega}(s)$ have further technical properties. They satisfy a so-called cocycle identity. Conversely, any family of unitaries satisfying that identity comes from some other reference state

(more precisely, from a weight). These facts serve later as machinery rather than as physical content, so they are only mentioned here.

4.3 The entangled spin example revisited

We can now compute $S(\mathcal{M}_\theta)$ and $T(\mathcal{M}_\theta)$ explicitly, by hand, for the algebra $\mathcal{M}_\theta \equiv \mathcal{M}_R(\theta)$ of the infinite chain of entangled spin pairs from Chapter 2. We will find that different values of θ give different type III $_\lambda$ subtypes.

Setting up the finite- N computation

At finite N , the algebra of the R system is an ordinary type I algebra. The reduced density matrices are N -fold tensor products of the single-pair result from Chapter 1:

$$\rho_R(\Phi_\theta) = \rho_r(\phi_\theta)^{\otimes N}, \quad \rho_r(\phi_\theta) = \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.$$

The same holds for ρ_L , since each pair is symmetric between R and L . For $\theta \in (0, \pi/4)$ both are strictly positive (full rank, since every diagonal entry is nonzero). So $|\Phi_\theta\rangle$ is cyclic and separating with respect to the R -algebra at every finite N , and the modular operator can be built directly from (4.2):

$$\Delta_{\Phi_\theta} = \rho_R(\Phi_\theta) \otimes \rho_L(\Phi_\theta)^{-1} \stackrel{(1)}{=} \delta_\theta^{\otimes N}, \quad \delta_\theta \equiv \rho_r(\phi_\theta) \otimes \rho_l(\phi_\theta)^{-1}.$$

(1) ρ_R and ρ_L^{-1} are both N -fold tensor products. Regroup the factors pair by pair, with the i -th r factor next to the i -th l factor. The whole operator is then the N -fold tensor power of the single-pair operator δ_θ .

The single-pair operator δ_θ is exactly the modular operator of the two-qubit worked example earlier in this chapter. Its eigenvalues were computed there: 1 (twice), λ and λ^{-1} , with $\lambda = \tan^2 \theta$. The N pairs are independent, identical copies. So the eigenvalues of the full Δ_{Φ_θ} are all possible products,

$$\bigotimes_{i=1}^N (1, \lambda, \lambda^{-1}) = \prod_{i=1}^N \lambda^{\alpha_i}, \quad \alpha_i \in \{0, 1, -1\}. \quad (4.9)$$

Each of the N pairs independently contributes a factor 1, λ or λ^{-1} to the product, and every combination of choices appears as an eigenvalue.

Taking $N \rightarrow \infty$, and reading off the type

The eigenvalue (4.9) depends only on the exponent $n = \sum_i \alpha_i$, where each term is 0 or ± 1 . As $N \rightarrow \infty$, every integer n becomes achievable, each with a growing multiplicity. So the eigenvalues are the numbers λ^n with $n \in \mathbb{Z}$, and the spectrum is

$$\sigma(\Delta_{\Phi_\theta}) = \{0\} \cup \{\lambda^n : n \in \mathbb{Z}\}. \quad (4.10)$$

The point $\{0\}$ appears because the spectrum is a closed set. Since $0 < \lambda < 1$, the eigenvalues λ^n come arbitrarily close to zero as $n \rightarrow +\infty$ (choose more and more $\alpha_i = +1$). The eigenvalue 0 itself is never reached. It is a limit point, and the spectrum includes all its limit points. The keyresult below makes this infinite- N statement precise.

Key Result / Core Concept: Spectral Derivation of the Connes Invariant for the Powers Factor

Goal: Show that for the infinite entangled spin chain $\mathcal{M}_\theta$ with $\theta \in (0, \pi/4)$ and $\lambda = \tan^2 \theta$,

$$\begin{aligned} \text{Spec}(\Delta_{\Phi_\theta}) &= \{0\} \cup \{\lambda^n : n \in \mathbb{Z}\}, \\ S(\mathcal{M}_\theta) &\equiv \bigcap_{\Psi} \text{Spec}(\Delta_\Psi) = \{0\} \cup \lambda^{\mathbb{Z}}. \end{aligned}$$

Steps 1 and 2 are a complete computation of the first line. Step 3 is a sketch of the second line, which rests on a theorem quoted from the literature.

Derivation:

1. **Action on the local GNS basis.** The GNS Hilbert space $\mathcal{H}_\theta$ is the completion of the span of local excitations of $|\Phi_\theta\rangle$:

$$|\Psi_{\{a_k\}}\rangle \equiv (a_1 \otimes a_2 \otimes \cdots \otimes a_m \otimes \mathbf{1} \otimes \cdots) |\Phi_\theta\rangle.$$

Expand each single-site 2×2 matrix in the standard basis $\{e_{00}, e_{01}, e_{10}, e_{11}\}$, where $e_{ij} \equiv |i\rangle\langle j|$. Use the Tomita operator $S_\phi(a|\phi\rangle) = a^\dagger|\phi\rangle$ on a single pair $|\phi\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$:

$$\begin{aligned} S_\phi(e_{00}|\phi\rangle) &= S_\phi(\cos\theta|00\rangle) \stackrel{(1)}{=} e_{00}^\dagger|\phi\rangle \stackrel{(2)}{=} e_{00}|\phi\rangle = \cos\theta|00\rangle, \\ S_\phi(e_{11}|\phi\rangle) &= S_\phi(\sin\theta|11\rangle) \stackrel{(1)}{=} e_{11}^\dagger|\phi\rangle \stackrel{(2)}{=} e_{11}|\phi\rangle = \sin\theta|11\rangle, \\ S_\phi(e_{01}|\phi\rangle) &= S_\phi(\sin\theta|01\rangle) \stackrel{(1)}{=} e_{01}^\dagger|\phi\rangle \stackrel{(2)}{=} e_{10}|\phi\rangle = \cos\theta|10\rangle, \\ S_\phi(e_{10}|\phi\rangle) &= S_\phi(\cos\theta|10\rangle) \stackrel{(1)}{=} e_{10}^\dagger|\phi\rangle \stackrel{(2)}{=} e_{01}|\phi\rangle = \sin\theta|01\rangle. \end{aligned}$$

(1) the defining property of the Tomita operator, $S_\phi(a|\phi\rangle) = a^\dagger|\phi\rangle$. (2) e_{00} and e_{11} are Hermitian, while $e_{01}^\dagger = e_{10}$ and $e_{10}^\dagger = e_{01}$.

S_ϕ is antilinear and the angles are real. So the last two lines say $S_\phi|01\rangle = \cot\theta|10\rangle$ and $S_\phi|10\rangle = \tan\theta|01\rangle$. This is exactly the 4×4 Tomita operator found in the worked example earlier in this chapter. Computing the adjoint $S_\phi^\dagger$ and the modular operator $\delta_\theta = S_\phi^\dagger S_\phi = \rho_r \otimes \rho_l^{-1}$ on this basis gives four exact eigenvectors:

$$\begin{aligned} \delta_\theta(e_{00}|\phi\rangle) &= 1 \cdot (e_{00}|\phi\rangle), & \delta_\theta(e_{11}|\phi\rangle) &= 1 \cdot (e_{11}|\phi\rangle), \\ \delta_\theta(e_{10}|\phi\rangle) &= \tan^2\theta \cdot (e_{10}|\phi\rangle) = \lambda \cdot (e_{10}|\phi\rangle), \\ \delta_\theta(e_{01}|\phi\rangle) &= \cot^2\theta \cdot (e_{01}|\phi\rangle) = \lambda^{-1} \cdot (e_{01}|\phi\rangle). \end{aligned}$$

2. **Eigenvalue spectrum on the infinite chain.** Consider a product excitation $|\Psi\rangle$ in which n_+ sites carry the factor e_{10} , n_- sites carry e_{01} , and all other sites carry e_{00} , e_{11} or

1. The modular operator acts site by site, so

$$\Delta_{\Phi_\theta} |\Psi\rangle = \lambda^{n_+ - n_-} |\Psi\rangle = \lambda^n |\Psi\rangle, \quad n = n_+ - n_- \in \mathbb{Z}.$$

The numbers n_+ and n_- can be any non-negative integers. So the set of eigenvalues is exactly the geometric progression $\{\lambda^n : n \in \mathbb{Z}\}$. These eigenvectors span a dense subspace of $\mathcal{H}_\theta$, so the spectrum is the closure of this set. Since $\lambda \in (0, 1)$, we have $\lambda^n \rightarrow 0$ as $n \rightarrow +\infty$. So 0 is an accumulation point and belongs to the spectrum:

$$\text{Spec}(\Delta_{\Phi_\theta}) = \{0\} \cup \{\lambda^n : n \in \mathbb{Z}\}.$$

3. **Independence of the reference state (sketch).** Why does the intersection over all states not shrink this set? Let $|\Psi\rangle$ be any other cyclic and separating vector. By Connes' theorem (4.6), the two modular flows are related by unitaries $u_t \in \mathcal{M}_\theta$:

$$\sigma_t^\Psi(A) = u_t \sigma_t^{\Phi_\theta}(A) u_t^\dagger.$$

The algebra $\mathcal{M}_\theta$ is an infinite tensor product of finite type I factors (an “ITPFI” factor; this particular one is called a Powers factor). Any normal state on it can be approximated in norm by states that differ from Φ_θ only on finitely many sites $1, \dots, K$. On the infinite tail $k > K$, such a state is identical to Φ_θ . Heuristically, its modular operator then factorizes as

$$\Delta_\Psi \approx \Delta_{\Psi, \text{local}} \otimes \bigotimes_{k=K+1}^{\infty} \delta_\theta^{(k)},$$

and the infinite tail still produces all the eigenvalues λ^n . The precise version is a theorem. Araki and Woods defined an invariant of such infinite tensor products, the asymptotic ratio set, that is built only from the tail of the chain. Connes later showed that it equals $S(\mathcal{M})$ for these algebras. For the Powers factor it equals $\{0\} \cup \lambda^\mathbb{Z}$. Hence

$$S(\mathcal{M}_\theta) \equiv \bigcap_{\Psi} \text{Spec}(\Delta_\Psi) = \{0\} \cup \{\lambda^n : n \in \mathbb{Z}\} = \{0\} \cup \lambda^\mathbb{Z}.$$

Comparing with Connes' classification, $\mathcal{M}_\theta$ is a **type III $_\lambda$** factor with $\lambda = \tan^2 \theta$. ■

The spectrum (4.10) belongs to *one specific* reference vector, $|\Phi_\theta\rangle$. The Connes invariant $S(\mathcal{M}_\theta)$ needs the intersection over *every* cyclic and separating reference vector, as in (4.7). The Araki–Woods–Connes result quoted in step 3 says that for this example the intersection is the same set (4.10). Now compare with the three-way classification (4.8). The set $\{0\} \cup \{\lambda^n\}$ with $\lambda = \tan^2 \theta \in (0, 1)$ is exactly the signature of **type III $_\lambda$** . **So $\mathcal{M}_\theta$ is type III $_{\tan^2 \theta}$, for every $\theta \in (0, \pi/4)$.** Different entangling angles produce different, inequivalent von Neumann algebras.

The other Connes invariant, $T(\mathcal{M}_\theta)$, can also be read off from (4.9). The operator $\Delta_{\Phi_\theta}^{it}$ equals 1 exactly when $\lambda^{int} = 1$ for every eigenvalue at once, that is, when $t \log \lambda \in 2\pi\mathbb{Z}$. At those times the flow does nothing, so it is trivially inner, implemented by the identity. This holds at every finite N , and hence also in the $N \rightarrow \infty$ limit. One can show that no other values of t give an inner flow; this

part is quoted, not proved here. The result is

$$T(\mathcal{M}_\theta) = \left\{ \frac{2\pi n}{\log \lambda} : n \in \mathbb{Z} \right\}, \quad \lambda = \tan^2 \theta.$$

This is a discrete proper subgroup of $\mathbb{R}$. It is not all of $\mathbb{R}$, as the type III criterion (4.5) requires.

The endpoints, and how to reach III_0 and III_1 too

Two special values of θ need to be reconciled with all this. At $\theta = \pi/4$, Chapter 3 already showed that the algebra is type II_1 , not type III. Indeed $\lambda = \tan^2(\pi/4) = 1$ here. This sits just outside the range $\lambda \in (0, 1)$ of type III_λ , consistent with this being a different type, one that has a trace and is easier to handle (Chapter 3). At $\theta = 0$, the two spins in each pair are not entangled at all. The $N \rightarrow \infty$ construction then never leaves ordinary type I, because there is no entanglement to produce anything new.

Can type III_0 and type III_1 be reached from a variant of the same family? Yes. Seeing how shows that the classification is sensitive to fine details of *how* the infinite limit is taken, not just to a single overall parameter. Instead of using the same θ for every pair, give the i -th pair its own angle θ_i . Then $\lambda_i = \tan^2 \theta_i$ can vary from pair to pair. The eigenvalue formula (4.9) becomes $\prod_i \lambda_i^{\alpha_i}$. What happens in the $N \rightarrow \infty$ limit now depends on the *asymptotic behavior* of the whole sequence $\{\lambda_i\}$, not on any single number. Araki and Woods worked out the answer. Stated roughly, it is as follows. If every λ_i equals the same $\lambda \in (0, 1)$, one gets type III_λ ; this is the case just worked out. If $\lambda_i \rightarrow 0$ fast enough that $\sum_i \lambda_i$ is finite, one gets an ordinary type I_∞ algebra. The entanglement dies off so quickly that nothing new happens in the limit. If $\lambda_i \rightarrow 0$ more slowly, one gets a type III algebra, and type III_0 can arise this way. Finally, sequences that do not settle down to a single value typically give type III_1 , the “most chaotic” member of the family. A simple example is a sequence that keeps returning to two values whose logarithms have an irrational ratio.

There is a second way to build type III_1 directly. It will return, in a quite different physical form, when the Rindler wedge is discussed in the next section. Instead of qubits, use pairs of *qutrits* (three-level systems, such as spin-1) in an entangled state. The single-pair reduced density matrix is then a 3×3 diagonal matrix, $\rho_r = \frac{1}{1+\lambda+\tilde{\lambda}} \text{diag}(1, \lambda, \tilde{\lambda})$, with two independent parameters $\lambda, \tilde{\lambda} > 0$. The single-pair modular eigenvalues are the ratios of these entries. So in the $N \rightarrow \infty$ limit the modular eigenvalues are all products $\lambda^n \tilde{\lambda}^m$ with integers n, m . Suppose the ratio $\log \lambda / \log \tilde{\lambda}$ is irrational, which is the generic case. Then these products are dense in all of $\mathbb{R}_{\geq 0}$, so $\sigma(\Delta_\Psi) = \mathbb{R}_{\geq 0}$. By the same Araki–Woods–Connes argument, $S(\mathcal{M}) = \mathbb{R}_{\geq 0}$, which is type III_1 again. This time the result comes not from varying θ_i from pair to pair, but from two incommensurate “frequencies,” $\log \lambda$ and $\log \tilde{\lambda}$, built into a single repeated building block.

4.4 Local algebras in a relativistic quantum field theory

The Rindler wedge, and why the modular flow is a boost

Now return to the third example of Chapter 1. There, a relativistic quantum field theory in 1+1-dimensional Minkowski space, with coordinates (t, x) , was cut in half at the surface $x = 0$. Chapter 1 showed that the vacuum state $|\Omega\rangle$ has infinite entanglement across this cut, so no tensor factorization exists. In the language now available, we describe this entanglement through the algebra $\mathcal{M}_R$ of operators localized in the right region $R = \{x > 0\}$.

One geometric fact is used constantly from here on, so we state it explicitly. In a relativistic theory, time evolution is causal: nothing propagates faster than light. So $\mathcal{M}_R$, built from operators localized at $x > 0$ at a single instant, is the same as the algebra of operators localized anywhere in the *domain of dependence* $\hat{R}$ of that region (the notion from Chapter 1). For the half-line $x > 0$, the domain of dependence is the **right Rindler wedge**, $\{(t, x) : x > |t|\}$. It is everything that data on the initial slice $x > 0$ determines completely, both to the future and to the past. Causality alone gives $\mathcal{M}_L \subseteq \mathcal{M}'_R$: operators in the left wedge commute with those in the right wedge. For wedges the stronger statement $\mathcal{M}'_R = \mathcal{M}_L = \mathcal{M}_{\hat{L}}$ also holds. The commutant is exactly the algebra of the causally complementary (left) wedge, with no gap between them. This equality is known as wedge duality, and it follows from the Bisognano–Wichmann theorem discussed below.

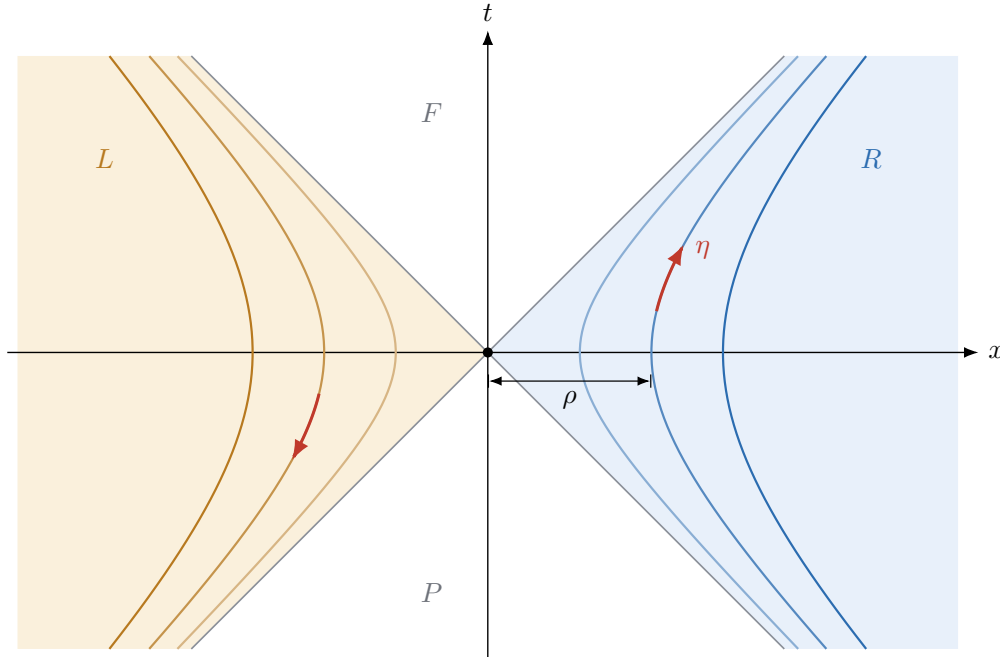


Figure 4.2: The Rindler wedges. The light cone through the origin (dot) splits Minkowski spacetime into the right wedge R ($x > |t|$, blue), the left wedge L ($x < -|t|$, gold), the future F and the past P . The curves are the hyperbolae $x^2 - t^2 = \rho^2$ (the bar marks ρ for one of them), worldlines of uniformly accelerated observers with proper acceleration $a = 1/\rho$ and proper time $\tau = \rho\eta$, where η is the boost parameter. The red arrows show which way a boost that increases η moves points: toward the future in R and toward the past in L .

The **Reeh–Schlieder theorem**, stated here and sketched below, says the following. In a relativistic quantum field theory, act on the vacuum $|\Omega\rangle$ with operators localized in *any* open spacetime region,

however small. The resulting set of states is dense in the entire Hilbert space.

Key Result / Core Concept: Derivation of the Reeh–Schlieder Theorem

Theorem: Let $\mathcal{O} \subset \mathbb{R}^{1,d-1}$ be a nonempty open region in Minkowski spacetime. Let $\mathcal{M}(\mathcal{O})$ be the local von Neumann algebra generated by fields smeared with test functions supported in $\mathcal{O}$. In any relativistic QFT satisfying the Wightman axioms:

1. $|\Omega\rangle$ is **cyclic** for $\mathcal{M}(\mathcal{O})$: $\overline{\mathcal{M}(\mathcal{O})|\Omega\rangle} = \mathcal{H}$.
2. If the causal complement of $\mathcal{O}$ contains a nonempty open region (as it does for any bounded region, and for a wedge), then $|\Omega\rangle$ is also **separating** for $\mathcal{M}(\mathcal{O})$: $A|\Omega\rangle = 0 \implies A = 0$ for all $A \in \mathcal{M}(\mathcal{O})$.

Proof sketch (treating the fields as if they were defined at points, a standard shortcut):

1. **Orthogonality hypothesis.** To prove cyclicity, suppose some state $|\chi\rangle \in \mathcal{H}$ is orthogonal to $\mathcal{M}(\mathcal{O})|\Omega\rangle$:

$$\langle \chi | A | \Omega \rangle = 0 \quad \text{for all } A \in \mathcal{M}(\mathcal{O}).$$

Consider field states $\phi(x_1) \cdots \phi(x_n) |\Omega\rangle$ with all points x_j in $\mathcal{O}$. Use translation covariance, $\phi(x) = e^{iP_\mu x^\mu} \phi(0) e^{-iP_\mu x^\mu}$, and translation invariance of the vacuum, $P_\mu |\Omega\rangle = 0$. Define the function

$$F(x_1, \dots, x_n) \equiv \langle \chi | \phi(x_1) \phi(x_2) \cdots \phi(x_n) | \Omega \rangle.$$

2. **Relativistic spectrum condition and analyticity.** Change to the difference coordinates $\xi_j \equiv x_j - x_{j+1}$ ($j = 1, \dots, n-1$). Inserting the translation formula and using $e^{-iP \cdot x_n} |\Omega\rangle = |\Omega\rangle$ gives

$$F = \left\langle \chi \left| e^{iP \cdot x_1} \phi(0) e^{-iP \cdot \xi_1} \phi(0) e^{-iP \cdot \xi_2} \cdots e^{-iP \cdot \xi_{n-1}} \phi(0) \right| \Omega \right\rangle.$$

By the relativistic spectrum condition, the joint spectrum of the energy-momentum operator $P^\mu = (H, \vec{P})$ lies in the closed forward lightcone:

$$\text{Spec}(P^\mu) \subseteq \bar{V}^+ = \{p^\mu : p^0 \geq |\vec{p}|\}.$$

Now continue the differences into the complex domain:

$$\xi_j \longrightarrow \zeta_j = \xi_j - i\eta_j, \quad \eta_j \in V^+ \quad (\eta_j^0 > |\vec{\eta}_j|).$$

On a state with four-momentum $p \in \bar{V}^+$, the exponent becomes

$$-ip \cdot (\xi_j - i\eta_j) = -ip \cdot \xi_j - p \cdot \eta_j.$$

For $p \in \bar{V}^+$ and $\eta_j \in V^+$, the Lorentzian inner product $p \cdot \eta_j = p^0 \eta_j^0 - \vec{p} \cdot \vec{\eta}_j$ is non-negative (and strictly positive unless $p = 0$). So the factor $e^{-p \cdot \eta_j}$ is at most 1. It damps high momenta, which keeps the operator product bounded and makes it depend holomorphically on the ζ_j . The same argument applies to x_1 , continued to $x_1 + i\eta_0$ with $\eta_0 \in V^+$. Therefore

F is holomorphic in a tube domain $\mathcal{T}$: the set of complex arguments whose imaginary parts lie in these forward cones.

3. **Edge-of-the-wedge theorem and global vanishing.** By assumption, the boundary value $F(x_1, \dots, x_n)$ vanishes when all $x_j \in \mathcal{O}$. The set of such real points is a nonempty open set. The edge-of-the-wedge theorem is a many-variable version of the Schwarz reflection principle. Together with the identity theorem for holomorphic functions, it implies the following: a function holomorphic in a tube domain whose boundary values vanish on a real open set must vanish *identically* on the whole tube:

$$F \equiv 0 \quad \text{on } \mathcal{T}.$$

Taking the boundary limit back to real arguments gives

$$\langle \chi | \phi(x_1) \phi(x_2) \cdots \phi(x_n) | \Omega \rangle = 0 \quad \text{for all } x_1, \dots, x_n \in \mathbb{R}^{1,d-1}.$$

4. **Conclusion of cyclicity.** By the Wightman axioms, polynomials in fields smeared over all of spacetime, acting on $|\Omega\rangle$, give a dense subspace of $\mathcal{H}$. The vector $|\chi\rangle$ is orthogonal to this dense subspace, so $|\chi\rangle = 0$. Thus $\overline{\mathcal{M}(\mathcal{O})|\Omega\rangle} = \mathcal{H}$, and $|\Omega\rangle$ is **cyclic**.
5. **Separating property.** Suppose $A \in \mathcal{M}(\mathcal{O})$ satisfies $A|\Omega\rangle = 0$. Choose a nonempty open region $\mathcal{O}'$ in the causal (spacelike) complement of $\mathcal{O}$. By microcausality, $[A, B'] = 0$ for all $B' \in \mathcal{M}(\mathcal{O}')$. Therefore

$$A(B'|\Omega\rangle) = B'(A|\Omega\rangle) = B'(0) = 0.$$

By the cyclicity of step 4, applied to $\mathcal{O}'$, vectors of the form $B'|\Omega\rangle$ are dense in $\mathcal{H}$. A bounded operator that vanishes on a dense subspace is zero, so $A = 0$. Hence $|\Omega\rangle$ is **separating** for $\mathcal{M}(\mathcal{O})$. ■

Apply this to the two wedges. It shows that $|\Omega\rangle$ is cyclic with respect to both $\mathcal{M}_R$ and $\mathcal{M}_L$. Since $\mathcal{M}_L \subseteq \mathcal{M}'_R$, it is also cyclic for $\mathcal{M}'_R$. By the cyclic–separating duality established earlier in this chapter, $|\Omega\rangle$ is therefore cyclic *and* separating with respect to $\mathcal{M}_R$ alone. So Tomita–Takesaki theory applies directly, with no further assumption.

Now comes a striking physical fact. It is found using the uniqueness remark (d) after the Tomita–Takesaki theorem: find *any* flow with the right properties that satisfies the KMS condition, and it must be *the* modular flow. The result is that the modular operator for $\mathcal{M}_R$ in the vacuum state is

$$K_\Omega \equiv -\log \Delta_\Omega = 2\pi K, \tag{4.11}$$

where K is the ordinary **boost generator**. This is the operator that generates Lorentz boosts in special relativity. It is a geometric symmetry of Minkowski space, and nothing about its definition is abstract or algebraic. This result is the **Bisognano–Wichmann theorem**. To justify (4.11), two things must be checked. (i) The flow generated by K maps $\mathcal{M}_R$ to itself and leaves $|\Omega\rangle$ invariant. This is immediate: a boost maps the Rindler wedge $\widehat{R}$ to itself, and the vacuum is Lorentz invariant.

- (ii) Correlators of boosted operators satisfy the KMS relation with $\beta = 2\pi$.

Worked Example: Verification of the Bisognano–Wichmann KMS Condition

Goal: Show, by a Euclidean path-integral argument, that the boost flow $\alpha_\eta(A) = e^{iK\eta} A e^{-iK\eta}$ satisfies the KMS condition at $\beta = 2\pi$ for any operators $A, B \in \mathcal{M}_R$:

$$\langle \Omega | A \alpha_{i2\pi}(B) | \Omega \rangle = \langle \Omega | B A | \Omega \rangle.$$

Here $\alpha_{i2\pi}(B) = e^{-2\pi K} B e^{2\pi K}$ is the flow continued to the imaginary parameter $\eta = 2\pi i$.

Calculation:

1. **Rindler coordinates and Euclidean rotation.** The right Rindler wedge $\hat{R} = \{(t, x) : x > |t|\}$ is described by a proper distance $\rho > 0$ and a boost parameter $\eta \in \mathbb{R}$:

$$t = \rho \sinh \eta, \quad x = \rho \cosh \eta.$$

A boost by parameter s shifts $\eta \rightarrow \eta + s$. Now rotate to Euclidean signature by setting $t_E \equiv -it$ and $\eta_E \equiv -i\eta$, so that $t = it_E$ and $\eta = i\eta_E$. Since $\sinh(i\eta_E) = i \sin \eta_E$ and $\cosh(i\eta_E) = \cos \eta_E$, this gives

$$t_E = \rho \sin \eta_E, \quad x = \rho \cos \eta_E.$$

The Euclidean version of the Minkowski metric is

$$ds_E^2 = dt_E^2 + dx^2 + dx_\perp^2 = d\rho^2 + \rho^2 d\eta_E^2 + dx_\perp^2,$$

where $dx_\perp^2$ collects the transverse directions, if there are any. In the (ρ, η_E) plane these are standard polar coordinates: ρ is the radius and η_E is the polar angle.

2. **Regularity and Euclidean 2π -periodicity.** The point $\rho = 0$ is the Euclidean image of the horizon. For the plane to be smooth there, with no conical singularity, the Euclidean angle must have period 2π :

$$\eta_E \sim \eta_E + 2\pi.$$

In terms of the Lorentzian boost parameter $\eta = i\eta_E$, the full turn $\eta_E \rightarrow \eta_E + 2\pi$ is an imaginary shift of the rapidity:

$$\eta \longrightarrow \eta + 2\pi i.$$

Geometrically, rotating η_E by π maps $(t_E, x) \rightarrow (-t_E, -x)$. This sends an operator in the right wedge R to the left wedge L . Rotating by 2π goes once around the point $\rho = 0$ and returns to the right wedge.

3. **Path integral and operator ordering.** In the Euclidean path integral that prepares the vacuum $|\Omega\rangle$, flowing an operator by the imaginary boost parameter $i\vartheta$ moves its insertion to the Euclidean angle $\eta_E = \vartheta$. So for $0 < \vartheta < 2\pi$, the correlator is a path integral with A

inserted at angle 0 and B at angle ϑ . The path integral orders operators by their angle:

$$\begin{aligned} \langle \Omega | A e^{-\vartheta K} B e^{\vartheta K} | \Omega \rangle \\ = \langle \Omega | \mathcal{T}_{\eta_E} [A(\eta_E=0) B(\eta_E=\vartheta)] | \Omega \rangle, \quad 0 < \vartheta < 2\pi. \end{aligned}$$

The right side is a single smooth function of the angle ϑ , because in the Euclidean plane no angle is special. As $\vartheta \rightarrow 0$, the insertion of B approaches A from one side, and the ordered product becomes AB . As $\vartheta \rightarrow 2\pi$, it approaches the same point from the other side, and the ordering puts B to the left of A :

$$\lim_{\vartheta \rightarrow 2\pi} \mathcal{T}_{\eta_E} [A(0) B(\vartheta)] = B A.$$

Therefore

$$\langle \Omega | A(e^{-2\pi K} B e^{2\pi K}) | \Omega \rangle = \langle \Omega | B A | \Omega \rangle.$$

This is the KMS relation with inverse temperature $\beta = 2\pi$ with respect to the boost parameter η . The argument treats the path integral formally, so it is a heuristic. The rigorous statement, for any QFT satisfying the Wightman axioms, is the Bisognano–Wichmann theorem. Given the KMS property, and property (i) above, the uniqueness remark (d) identifies the modular operator as $\Delta_\Omega = e^{-2\pi K}$. ■

The physical meaning of condition (ii) is the **Unruh effect**. The geometry deserves to be spelled out, because “a Rindler observer” is not just a figure of speech. Write the Minkowski metric as $ds^2 = -dt^2 + dx^2 = -\rho^2 d\eta^2 + d\rho^2$. In these Rindler coordinates, ρ plays the role of a radius and η the role of a boost angle. A trajectory of constant ρ is a hyperbola in the (t, x) plane. It is the worldline of an observer with constant proper acceleration $a = 1/\rho$. Along it, the boost parameter η is proportional to the observer’s own proper time: $d\tau = \rho d\eta$. Condition (ii) says that observers who use η as their clock, which means exactly these uniformly accelerated observers, see the Minkowski vacuum as a thermal bath. For an inertial observer the vacuum contains no particles at all. The accelerated observers see the temperature

$$T_\rho = \frac{1}{2\pi\rho} = \frac{a}{2\pi}.$$

This is Unruh’s 1976 result. Here it appears as a direct consequence of Tomita–Takesaki theory applied to the vacuum of a relativistic field. The factor 2π in (4.11) converts the universal, dimensionless modular temperature $\beta = 1$ into this physical temperature, once η is converted to the accelerated observer’s own proper time τ .

Physics Connection: a finite-matrix version of the Unruh effect, from quantum optics

The field-theory argument above is rigorous but abstract. It involves every mode of the field, an honest boost, and a full QFT vacuum. The same thermality already appears in a finite, checkable calculation, using a toy model that every quantum-optics course covers: a two-mode squeezed state. Take two copies of a single harmonic-oscillator mode, labelled R and L , and

build

$$|\text{TFD}_x\rangle = \sqrt{1-x^2} \sum_{n=0}^{\infty} x^n |n\rangle_R |n\rangle_L, \quad x \in (0, 1).$$

This is the standard two-mode squeezed vacuum. It is already in Schmidt form, with probabilities $p_n = (1-x^2)x^{2n}$. We truncated at 40 Fock levels and computed $\rho_R = \text{Tr}_L |\text{TFD}_x\rangle \langle \text{TFD}_x|$ numerically at $x = 0.6$. The off-diagonal entries vanish, as they must, since the state is already diagonal in the Fock basis. The trace $\text{Tr} \rho_R$ equals 1 to machine precision (the truncation error is $x^{80} \approx 2 \times 10^{-18}$). The diagonal entries match $p_n = (1-x^2)x^{2n}$ to machine precision for every n . Now compare this with an ordinary Gibbs state:

$$p_n = (1-x^2)x^{2n} \stackrel{(1)}{=} (1-e^{-\beta\omega})e^{-n\beta\omega}.$$

(1) set $x^2 \equiv e^{-\beta\omega}$. The right side is the Boltzmann distribution of a single mode of frequency ω at inverse temperature β .

At $x = 0.6$ this gives $\beta\omega = -\log(x^2) \approx 1.022$. **The observer with access only to mode R looks at a pure, zero-entropy global state and measures an exactly thermal distribution.** No approximation or large- N limit is involved. It is just the ordinary partial trace, applied to a state that happens to be diagonal. This is more than an analogy to the Rindler-wedge argument above. It is the same calculation with the geometry stripped away. In free field theory, rewriting the Minkowski vacuum in Rindler modes produces exactly this two-mode squeezed form, mode by mode. The squeezing parameter is tied to the proper acceleration a by $x = e^{-\pi\omega/a}$, so $\beta = 2\pi/a$. Summing this thermal spectrum over every field mode is the full field-theoretic Unruh calculation.

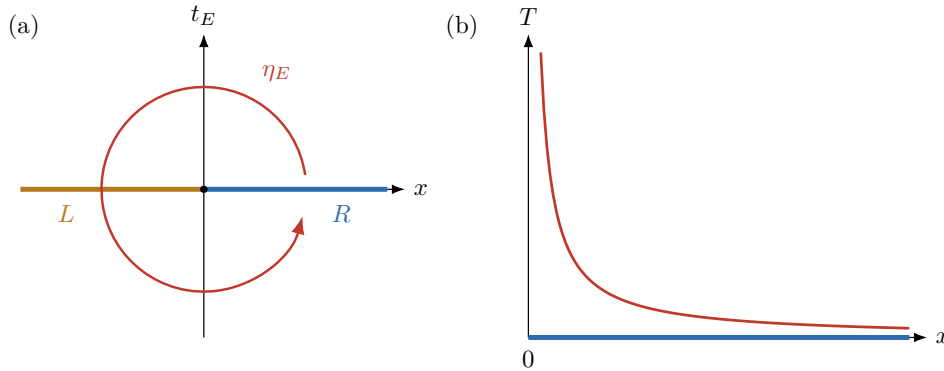


Figure 4.3: The Bisognano–Wichmann theorem and the Unruh effect. (a) The Euclidean plane (x, t_E) : the $t = 0$ slice is the horizontal axis, split at the horizon point $\rho = 0$ (dot) into the half-lines R (blue) and L (gold), and the Euclidean boost angle η_E (red) turns about that point, carrying R onto L after a half turn and back after a full turn of 2π , the periodicity behind the KMS condition at $\beta = 2\pi$. (b) The resulting Unruh temperature $T = 1/(2\pi x)$ seen by the accelerated observer passing through the point x of R at $t = 0$; it diverges at the horizon and falls off far from it.

The boost generator K has continuous spectrum covering the whole real line, $(-\infty, \infty)$. (The only eigenvector is the vacuum, with eigenvalue 0.) Boosts form a noncompact group: there are boosts

of arbitrarily large rapidity, and no smallest nonzero one. So $\Delta_\Omega = e^{-2\pi K}$ has spectrum $\mathbb{R}_{\geq 0}$. One can show that $S(\mathcal{M}_R)$ equals this spectrum. By the classification (4.8), $\mathcal{M}_R$ is **type III₁**. The modular conjugation J_Ω also has a geometric meaning. It is *CRT*: charge conjugation, combined with a reflection of the x direction and time reversal. So the CPT symmetry of relativistic quantum field theory shows up as part of the modular structure of the vacuum, built from nothing but the algebra and the state.

This whole story extends at once to higher spacetime dimensions. Any transverse directions just come along unchanged, since the boost acts only in the t - x plane. More importantly, it extends in part to a *general* open region O on a Cauchy slice, not just to a half-space. Reeh–Schlieder guarantees that $|\Omega\rangle$ is still cyclic and separating for $\mathcal{M}_O$, so Tomita–Takesaki theory applies. In general the modular operator can no longer be written in closed form. It depends on the precise theory and on the precise shape of O . Still, a scale-invariance argument pins down its type. The argument needs the theory to have a scale-invariant fixed point at short distances, which is believed to be true of essentially every interacting quantum field theory. Very close to the boundary Σ_O of the region O , the boundary looks like a flat plane. So locally the geometry cannot be told apart from the Rindler wedge just worked out, and

$$-\log \Delta_\Omega \approx 2\pi K, \quad \text{very close to } \Sigma_O,$$

with K now the boost that locally leaves Σ_O fixed. This forces the same continuous spectrum $\mathbb{R}_{\geq 0}$, and hence type III₁ again. As stated, this is a heuristic. It can be made rigorous under a technical assumption about the short-distance (scaling) limit of the theory. **So the type III₁ nature of local algebras in a relativistic quantum field theory can be traced to the local Rindler structure near any entangling surface, and that structure comes from relativistic causality.** This is the precise sense in which “the causal structure of a relativistic QFT requires type III₁.” Correspondingly, a *non*-relativistic field theory, with no light-cone structure to force this local Rindler behavior, need not have type III₁ local algebras at all.

The split property

Chapter 1 gave a heuristic picture: non-factorization and type III structure come from infinite entanglement among short-distance degrees of freedom right at the boundary of a region. This picture can be made precise. The precise version also resolves what could look like a contradiction. How can two regions be infinitely entangled (type III, no factorization) and yet, intuitively, “mostly independent” away from their shared boundary?

Separate R and L by a small but nonzero buffer distance ϵ_b . There is then a thin strip I_ϵ of “no man’s land” between them. The **split property** states that there *is* a genuine tensor factorization $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, with

$$\begin{aligned} \mathcal{M}_R &\subset B(\mathcal{H}_1) \otimes \mathbf{1}_{\mathcal{H}_2} \subset \mathcal{M}'_L = \mathcal{M}_{R_\epsilon}, \\ \mathcal{M}_L &\subset \mathbf{1}_{\mathcal{H}_1} \otimes B(\mathcal{H}_2) \subset \mathcal{M}'_R = \mathcal{M}_{L_\epsilon}, \end{aligned}$$

where $R_\epsilon \equiv R \cup I_\epsilon$, and similarly for L_ϵ . So a type I factor, $B(\mathcal{H}_1) \otimes \mathbf{1}_{\mathcal{H}_2}$, sits between the two type III₁ algebras $\mathcal{M}_R$ and $\mathcal{M}_{R_\epsilon}$. This has real physical content. Once R and L are separated by *any* nonzero distance, however small, they *can* be completely disentangled. There are honest

product states with respect to the factors $\mathcal{H}_1, \mathcal{H}_2$, and on them $\mathcal{M}_R$ and $\mathcal{M}_L$ act independently. In other words, the entanglement obstruction of Chapter 1 is a short-distance effect localized at the boundary. It disappears as soon as the two regions are separated by any finite gap. More generally, take two regions $O_1 \subset O_2$ whose boundaries do not touch (the closure of O_1 lies inside the interior of O_2). The split property then guarantees a type I factor $\mathcal{N}$ with $\mathcal{M}_{O_1} \subset \mathcal{N} \subset \mathcal{M}_{O_2}$. A type I “buffer” can always be inserted, as long as there is some geometric gap to put it in.

The split property is not an extra assumption pulled from nowhere. It can be shown to follow from a technical condition called the *nuclear property*. This is a statement about how quickly the number of available states of the theory grows with energy. It is believed to hold for any “reasonable” relativistic QFT. Given the split property, one can further show that $\mathcal{M}_O$, for an open region O , is not only type III₁ but also **hyperfinite**. This means it can be built as the weak closure of an increasing sequence of ordinary finite-dimensional matrix algebras. That is exactly the kind of construction used throughout these notes: bigger and bigger, but always finite, matrices, taken to a limit. A deep uniqueness theorem then applies: up to isomorphism, there is only *one* hyperfinite type III₁ factor. The consequence is striking. **The local algebra of any region in any “reasonable” relativistic QFT (free or interacting, weakly or strongly coupled, in any spacetime dimension) is abstractly isomorphic to the local algebra of any other region in any other such theory.** Different theories are not distinguished by what their local algebras *are*, since abstractly these are all the same object. They are distinguished by how the algebras sit relative to one another: which operators are shared between overlapping regions, how correlators between distant regions behave, and so on. This statement is surprising, and it is given here at full strength for a reason. It shows clearly how much structure the type classification captures. It also shows how much it deliberately does *not* capture: the dynamics, which lives entirely in the relations between algebras, not in the abstract type of any single algebra.

Two further consequences follow. The first comes with a short proof. The second is one of the more startling facts in the whole subject.

Strong local preparability. Take the split setup just described, with regions R and L separated by a buffer I_ϵ . In a general state ω , operators in R and in L are correlated: $\omega(AB) \neq \omega(A)\omega(B)$ for $A \in \mathcal{M}_R$, $B \in \mathcal{M}_L$. Nothing is surprising there. The startling claim is the following. There is an operation W , supported only in the slightly larger region R_ϵ (R plus the thin buffer strip), that turns ω into a *new* state ω_W with three properties. (i) There is *no* correlation at all between $\mathcal{M}_R$ and $\mathcal{M}_L$. (ii) On $\mathcal{M}_R$, the new state equals an arbitrarily chosen target state ϕ . (iii) On $\mathcal{M}_L$, the new state is the same as the original ω . This sounds nearly impossible. It says you can locally erase all correlation with a distant system and, at the same time, re-prepare your own region in any state you like, without touching the distant system. Yet it follows in a few lines from the split property and the type III structure.

Here is the argument. By the split property, $\mathcal{M}_R \subset B(\mathcal{H}_1) \otimes \mathbf{1}_{\mathcal{H}_2}$. So there is a vector $|\xi\rangle \in \mathcal{H}_1$ that represents the target state, $\phi(A) = \langle \xi | A | \xi \rangle$. (This step is justified more carefully in the section on the natural cone at the end of this chapter.) The projection $P_\xi = |\xi\rangle \langle \xi| \otimes \mathbf{1}_{\mathcal{H}_2}$ lies in $\mathcal{M}_{R_\epsilon}$. Since $\mathcal{M}_{R_\epsilon}$ is type III, *every* nonzero projection in it is equivalent to the identity. This is Chapter 2’s discussion of finite and infinite projections pushed to its extreme: in type III nothing is finite, so every projection is as “big” as the whole algebra. So there is an isometry $W \in \mathcal{M}_{R_\epsilon}$ with $WW^\dagger = P_\xi$ and $W^\dagger W = \mathbf{1}$. Define $\omega_W(X) \equiv \omega(W^\dagger X W)$. Because $W \in \mathcal{M}_{R_\epsilon} \subset \mathcal{M}'_L$, it commutes

with everything in $\mathcal{M}_L$. So for $B \in \mathcal{M}_L$,

$$\omega_W(B) \equiv \omega(W^\dagger B W) \stackrel{(1)}{=} \omega(W^\dagger W B) \stackrel{(2)}{=} \omega(B).$$

(1) W commutes with B , because $W \in \mathcal{M}'_L$. (2) $W^\dagger W = \mathbf{1}$.

So the L side is untouched, as claimed. For the R side, we use the relation $P_\xi A P_\xi = \phi(A) P_\xi$, which follows at once from $|\xi\rangle$ representing ϕ . For $A \in \mathcal{M}_R$ it gives

$$\begin{aligned} W^\dagger A W &\stackrel{(1)}{=} W^\dagger (W W^\dagger) A (W W^\dagger) W \\ &\stackrel{(2)}{=} W^\dagger P_\xi A P_\xi W \stackrel{(3)}{=} \phi(A) W^\dagger P_\xi W \stackrel{(4)}{=} \phi(A) \mathbf{1}. \end{aligned}$$

(1) $W^\dagger W = \mathbf{1}$ implies $W^\dagger = W^\dagger W W^\dagger$ and $W = W W^\dagger W$. (2) $W W^\dagger = P_\xi$. (3) $P_\xi A P_\xi = \phi(A) P_\xi$. (4) $W^\dagger P_\xi W = W^\dagger W W^\dagger W = \mathbf{1}$.

Combining the two results gives the factorized, re-prepared state. For $A \in \mathcal{M}_R$ and $B \in \mathcal{M}_L$,

$$\omega_W(AB) \equiv \omega(W^\dagger A B W) \stackrel{(1)}{=} \omega(W^\dagger A W B) \stackrel{(2)}{=} \phi(A) \omega(B).$$

(1) W commutes with $B \in \mathcal{M}_L$, so $BW = WB$. (2) $W^\dagger A W = \phi(A) \mathbf{1}$, from the chain just above.

The Connes–Størmer transitivity theorem. For a type III₁ factor, *any* two states ϕ and ω can be connected to arbitrary precision $\epsilon > 0$ by a unitary W *inside the algebra itself*: $\|\phi - \omega_W\| < \epsilon$. In words, every state can be prepared locally, to arbitrary accuracy, starting from any other state. This is such a strong form of ergodicity that it is fair to call the state space *homogeneous*. No state is structurally special or hard to reach from another. This is in sharp contrast with a type I factor $B(\mathcal{H})$. There, unitary conjugation $\rho \rightarrow U \rho U^\dagger$ preserves the eigenvalues of the density matrix, so a pure state can never be brought close to a mixed one. With a nontrivial center it is worse still: different superselection sectors are, by definition, unreachable from one another by anything in the algebra.

The collection of algebras $\{\mathcal{M}(O)\}$, and Haag duality

One more structural point returns directly in Chapter 7. Consider the full collection of local algebras $\{\mathcal{M}(O)\}$, one for every open spacetime region O . It is expected to satisfy several basic consistency relations. Each one is an algebraic translation of an ordinary physical principle.

- **Isotony**, $\mathcal{M}(O_1) \subseteq \mathcal{M}(O_2)$ for $O_1 \subseteq O_2$. A bigger region gives access to at least as many operations as a smaller region inside it. This is almost a definition.
- **The time-slice axiom**, $\mathcal{M}(O) = \mathcal{M}(\widehat{O})$. The equations of motion are causal. So operators anywhere in the domain of dependence $\widehat{O}$ can be rewritten, using time evolution, in terms of operators in O itself. Knowing the algebra on a single Cauchy slice therefore determines it everywhere.
- **Locality (commutativity)**, $\mathcal{M}(O') \subseteq \mathcal{M}(O)'$. Operators in the causal complement O'

(spacelike separated from all of O) commute with everything in $\mathcal{M}(O)$. This is the operator-algebra form of ordinary microcausality.

When the last relation holds with *equality*, $\mathcal{M}(O') = \mathcal{M}(O)'$, it is called **Haag duality**. This is strictly stronger. It says that *every* operator commuting with $\mathcal{M}(O)$ already comes from the causal complement, with nothing extra hiding elsewhere. Haag duality is not automatic. It can fail for topologically nontrivial regions. It is expected to hold quite generally for the vacuum sector of an ordinary relativistic QFT on topologically simple regions. Two further relations complete the package. The first is **additivity**, $\mathcal{M}(O_1 \cup O_2) = \mathcal{M}(O_1) \vee \mathcal{M}(O_2)$: no “extra,” genuinely nonlocal operators hide in a union beyond those built from the pieces. The second is the **intersection property**, $\mathcal{M}(O_1 \cap O_2) = \mathcal{M}(O_1) \cap \mathcal{M}(O_2)$. It can be derived from Haag duality and additivity by taking commutants, in cases where the causal complement of $O_1 \cap O_2$ is the union of O_1' and O_2' . A theory that satisfies all of these relations for every region (not just topologically trivial ones) is called “complete.” It helps to know these relations in outline, not for their own sake, but because Chapter 7 builds subregion-subalgebra duality on exactly this dictionary, translated to the boundary theory of AdS/CFT.

4.5 Emergent times from subalgebras — half-sided modular inclusion

This section introduces a second, distinct notion of emergent time. It does not come from the modular flow of a single algebra. It comes from the relationship between an algebra and a carefully chosen subalgebra of it. It is special to type III_1 algebras. It is also the mechanism, taken up again in Chapters 7 and 8, that lets a single band of boundary time generate the entire interior of an emergent black-hole horizon.

A new, positive “Hamiltonian” G

Let $\mathcal{M}$ be a von Neumann algebra with a cyclic and separating vector $|\Omega\rangle$. Its modular data are $\Delta_{\mathcal{M}} = e^{-K_{\mathcal{M}}}$ and $J_{\mathcal{M}}$, as in the Tomita–Takesaki theorem. Now suppose $\mathcal{N} \subset \mathcal{M}$ is a subalgebra, and $|\Omega\rangle$ is *also* cyclic for $\mathcal{N}$. It is then automatically separating for $\mathcal{N}$ too: it is separating for the bigger algebra $\mathcal{M}$, and $\mathcal{N} \subset \mathcal{M}$. So $\mathcal{N}$ has its own modular data, $\Delta_{\mathcal{N}} = e^{-K_{\mathcal{N}}}$ and $J_{\mathcal{N}}$, with respect to the same $|\Omega\rangle$.

Here is the first new structural fact, with the reason it holds. Because $\mathcal{N} \subset \mathcal{M}$, the Tomita operator $S_{\mathcal{M}}$ (built from $\mathcal{M}$ and $|\Omega\rangle$) is an extension of $S_{\mathcal{N}}$ (built from $\mathcal{N}$ and the same $|\Omega\rangle$). That is, $S_{\mathcal{M}}$ agrees with $S_{\mathcal{N}}$ wherever $S_{\mathcal{N}}$ is defined, but it is also defined on the larger set $\mathcal{M}|\Omega\rangle \supseteq \mathcal{N}|\Omega\rangle$. There is a general fact about unbounded operators: if X extends Y , then $X^\dagger X \leq Y^\dagger Y$ as quadratic forms. The two forms agree wherever both are defined. The form of $X^\dagger X$ is defined on more vectors, and in this ordering that makes it the smaller one. Applied to $\Delta_{\mathcal{M}} = S_{\mathcal{M}}^\dagger S_{\mathcal{M}}$ and $\Delta_{\mathcal{N}} = S_{\mathcal{N}}^\dagger S_{\mathcal{N}}$, this gives $\Delta_{\mathcal{M}} \leq \Delta_{\mathcal{N}}$. The logarithm preserves inequalities between positive operators, so $\log \Delta_{\mathcal{M}} \leq \log \Delta_{\mathcal{N}}$, which means $K_{\mathcal{M}} \geq K_{\mathcal{N}}$. Define

$$G \equiv \frac{1}{2\pi}(K_{\mathcal{M}} - K_{\mathcal{N}}) \geq 0, \quad G|\Omega\rangle = 0. \quad (4.12)$$

The factor 2π is a normalization chosen for convenience later. The equation $G|\Omega\rangle = 0$ holds because $K_{\mathcal{M}}$ and $K_{\mathcal{N}}$ each annihilate $|\Omega\rangle$. This is the invariance $\Delta|\Omega\rangle = |\Omega\rangle$ from the Tomita–Takesaki theorem, applied to each algebra. G is a positive operator, so it deserves to be called a Hamiltonian. The flow e^{iGs} that it generates is a legitimate new notion of “time.” It is distinct from the modular flows of both $\mathcal{M}$ and $\mathcal{N}$, and it also leaves the reference vector $|\Omega\rangle$ fixed.

The half-sided modular inclusion condition, and what it buys you

Everything so far works for *any* subalgebra $\mathcal{N} \subset \mathcal{M}$ that shares a cyclic and separating vector with $\mathcal{M}$. Much more follows if $\mathcal{N}$ satisfies one additional, geometric-looking condition, called **half-sided modular inclusion**:

$$\mathcal{N}_t \equiv \Delta_{\mathcal{M}}^{-it} \mathcal{N} \Delta_{\mathcal{M}}^{it} \subset \mathcal{N}, \quad \text{for every } t \leq 0. \quad (4.13)$$

In words: flowing $\mathcal{N}$ by $\mathcal{M}$ ’s *own* modular flow, for negative modular time, always shrinks $\mathcal{N}$ or leaves it the same. It never grows it. When this holds, a theorem of Wiesbrock, building on earlier work of Borchers, gives three consequences at once. We quote it without proof. None of the three is assumed; all are derived from (4.13) alone.

1. $K_{\mathcal{M}}$, $K_{\mathcal{N}}$ and G satisfy the commutation relations

$$[K_{\mathcal{M}}, K_{\mathcal{N}}] = -4\pi^2 i G, \quad [K_{\mathcal{M}}, G] = 2\pi i G,$$

together with a companion relation for the modular conjugations. Since $K_{\mathcal{N}} = K_{\mathcal{M}} - 2\pi G$, these are the commutation relations of the dilations and translations of a line. They form a two-dimensional subalgebra of the Möbius algebra, which is the global conformal symmetry of a line. So a recognizable piece of conformal symmetry is produced from nothing but one algebraic inclusion condition on two von Neumann algebras.

2. The flow generated by G maps $\mathcal{M}$ into itself for one whole half of the time axis: $e^{iGs} \mathcal{M} e^{-iGs} \subset \mathcal{M}$ for $s \leq 0$. In particular, $\mathcal{N} = e^{-iG} \mathcal{M} e^{iG}$. So $\mathcal{N}$ is exactly the image of $\mathcal{M}$ under one unit of this new time translation generated by G . Running G continuously gives a nested, continuously parametrized family of algebras $e^{-iGu} \mathcal{M} e^{iGu}$ for $u \geq 0$. For $0 \leq u_1 < u_2$ they satisfy

$$e^{-iGu_2} \mathcal{M} e^{iGu_2} \subset e^{-iGu_1} \mathcal{M} e^{iGu_1} \subset \mathcal{M},$$

with $u = 0$ giving $\mathcal{M}$ itself and $u = 1$ giving $\mathcal{N}$. This is a new time translation, distinct from $\mathcal{M}$ ’s own modular flow. It acts by relating the algebra to smaller and smaller copies of itself.

3. $\mathcal{M}$ **must be type III₁** (as long as $\mathcal{N} \neq \mathcal{M}$). Half-sided modular inclusion is not just a convenient technical condition. Its existence forces the ambient algebra to be the “most chaotic” type identified earlier in this chapter.

The proof of this theorem builds several further objects along the way. They include the unitaries $D(t) = \Delta_{\mathcal{M}}^{-it} \Delta_{\mathcal{N}}^{it}$ and $V = J_{\mathcal{M}} J_{\mathcal{N}}$, and a chain of nested algebras built by repeatedly conjugating with V . These are used to *prove* the theorem and to show that this half-sided inclusion structure, once it exists, is unique. They are careful pieces of functional analysis. They are named here so

that the notation is not a surprise in the research literature. The three numbered consequences above are the physical content that matters for everything that follows.

The concrete example: light-cone translations in the Rindler wedge

Here is a worked example that turns the abstract theorem into something you can picture. Use light-cone coordinates $x^\pm = x^0 \pm x^1$, so that the right Rindler wedge $\hat{R}$ from earlier in this chapter is $\{x^+ > 0, x^- < 0\}$. Take $\mathcal{M}$ to be the algebra of $\hat{R}$. Let $\mathcal{N}$ be the algebra of the smaller region $\{x^+ > 0, x^- < -1\}$. This is a wedge nested strictly inside $\hat{R}$, with its tip moved by one unit along the x^- direction, to the point $(x^+, x^-) = (0, -1)$. Now flow $\mathcal{N}$ using $K_{\mathcal{M}} = 2\pi K$ (the boost generator, (4.11)). A boost acts on light-cone coordinates by a simple rescaling. With the sign convention in which the modular flow moves the right wedge forward in time, the flow $e^{iK_{\mathcal{M}}t} = e^{2\pi i K t}$ is a boost by $\eta = 2\pi t$ that acts as $x^\pm \rightarrow e^{\pm 2\pi t} x^\pm$. It turns the defining condition $x^- < -1$ into $x^- < -e^{-2\pi t}$. So

$$\mathcal{N}_t \equiv e^{iK_{\mathcal{M}}t} \mathcal{N} e^{-iK_{\mathcal{M}}t} = \text{algebra of the region } \{x^+ > 0, x^- < -e^{-2\pi t}\}.$$

For $t < 0$ we have $e^{-2\pi t} > 1$. So this region is *smaller* than the original one, since it requires x^- to be even more negative. This is exactly $\mathcal{N}_t \subset \mathcal{N}$ for $t < 0$, the half-sided modular inclusion condition. Here it is checked directly on an explicit geometric example, not just asserted.

Identifying G here takes a short computation with the ordinary Poincaré algebra of special relativity. Write $P^\pm = \frac{1}{2}(P^0 \pm P^1)$ for the light-cone components of the energy-momentum operator. The boost generator and the momentum satisfy $[K, P^\pm] = \pm i P^\pm$. This is the statement that a boost rescales energy and momentum in the same way that it rescales light-cone coordinates. The region of $\mathcal{N}$ is the region of $\mathcal{M}$ translated so that its tip moves from the origin to $(x^+, x^-) = (0, -1)$. So $K_{\mathcal{N}}$ is $K_{\mathcal{M}}$ conjugated by this translation, which is the unitary e^{-iP^+} :

$$K_{\mathcal{N}} = e^{-iP^+} K_{\mathcal{M}} e^{iP^+} \stackrel{(1)}{=} K_{\mathcal{M}} - i[P^+, K_{\mathcal{M}}] \stackrel{(2)}{=} K_{\mathcal{M}} - 2\pi P^+.$$

(1) expand $e^{-iP^+} K_{\mathcal{M}} e^{iP^+} = K_{\mathcal{M}} - i[P^+, K_{\mathcal{M}}] + \dots$; the higher terms vanish, because $[P^+, K_{\mathcal{M}}]$ is proportional to P^+ , which commutes with P^+ . (2) $[P^+, K_{\mathcal{M}}] = -2\pi[K, P^+] = -2\pi i P^+$.

Then the definition (4.12), $G = \frac{1}{2\pi}(K_{\mathcal{M}} - K_{\mathcal{N}})$, gives

$$G = P^+.$$

So in this example, the abstract “positive Hamiltonian” G is just the ordinary light-cone momentum operator P^+ . The new “time flow” it generates is just ordinary translation in the x^- direction. It is positive, as it must be: $P^+ \geq 0$ by the spectrum condition. The general statements 1 and 2 above (the commutation relations and the half-sided translation structure) can be checked directly on this example with ordinary special-relativistic kinematics. For instance, $[K_{\mathcal{M}}, G] = 2\pi[K, P^+] = 2\pi i G$. Statement 3 agrees with the type III₁ result for the Rindler wedge found earlier. This is the mechanism that Chapter 8 reuses to show that a single band of *boundary* time in AdS/CFT can generate the entire interior of an emergent black-hole horizon. There, the interior is built by exactly this kind of half-sided modular inclusion light-cone translation, with G

playing the role of a positive bulk momentum.

One further variant is the mirror image. Take $\mathcal{N}$ to be the region $\{x^+ > 1, x^- < 0\}$, shifted along x^+ instead of x^- . This gives a half-sided inclusion on the *positive* half of the time axis, $\mathcal{N}_t \subset \mathcal{N}$ for $t \geq 0$. The commutation relations have the opposite sign, and the translation generator is $G = P^-$ instead of P^+ .

4.6 Relative modular flows and relative entropy

This section extends two familiar objects, the relative entropy $S(\rho||\sigma)$ of Chapter 1 and the Tomita operator S_Ψ of this chapter, to a setting with *two* states $|\Psi\rangle$ and $|\Omega\rangle$, both cyclic and separating for the same $\mathcal{M}$. The aim is a relative entropy that still makes sense when $\mathcal{M}$ is type III, where the ordinary entropy $S_{\mathcal{M}}$ cannot be defined at all (Chapter 3).

The **relative Tomita operator** is defined like the ordinary one, except that it maps between the two vectors:

$$S_{\Psi\Omega} A |\Omega\rangle = A^\dagger |\Psi\rangle \quad (A \in \mathcal{M}), \quad F_{\Psi\Omega} A' |\Omega\rangle = A'^\dagger |\Psi\rangle \quad (A' \in \mathcal{M}'),$$

with $F_{\Psi\Omega} = S_{\Psi\Omega}^\dagger$, exactly as S_Ψ and F_Ψ act on $\mathcal{M}$ and $\mathcal{M}'$ in the single-state case. Its polar decomposition $S_{\Psi\Omega} = J_{\Psi\Omega} \Delta_{\Psi\Omega}^{1/2}$ defines the **relative modular operator** $\Delta_{\Psi\Omega} \geq 0$ and the **relative modular conjugation** $J_{\Psi\Omega}$. When $\Psi = \Omega$ these are the ordinary Δ_Ψ and J_Ψ . For the type I case $\mathcal{M} = B(\mathcal{H}_1) \otimes \mathbf{1}$, the same calculation that gave $\Delta_\Psi = \rho_R \otimes \rho_L^{-1}$ at the start of this chapter now gives

$$\Delta_{\Psi\Omega} = \rho_\Psi \otimes \rho'_\Omega{}^{-1},$$

the density matrix of Ψ on the $\mathcal{M}$ side and the inverse density matrix of Ω on the commutant side. (Here ρ_Ψ and ρ'_Ψ are the reduced density matrices of Ψ on $\mathcal{H}_1$ and $\mathcal{H}_2$, the ρ_R and ρ_L of the opening example.) The relative modular operator lets one convert correlation functions in one state into correlation functions in the other. For $A, B \in \mathcal{M}$,

$$\begin{aligned} \langle \Psi | AB | \Psi \rangle &\stackrel{(1)}{=} \langle A^\dagger \Psi | B \Psi \rangle \stackrel{(2)}{=} \langle S_{\Psi\Omega} A \Omega | S_{\Psi\Omega} B^\dagger \Omega \rangle \\ &\stackrel{(3)}{=} \langle \Delta_{\Psi\Omega}^{1/2} B^\dagger \Omega | \Delta_{\Psi\Omega}^{1/2} A \Omega \rangle \stackrel{(4)}{=} \langle \Omega | B \Delta_{\Psi\Omega} A | \Omega \rangle. \end{aligned} \quad (4.14)$$

(1) move A to the left as $A^\dagger$. (2) the definition of $S_{\Psi\Omega}$, used twice: $A^\dagger \Psi = S_{\Psi\Omega} A \Omega$ and $B \Psi = S_{\Psi\Omega} B^\dagger \Omega$. (3) $S_{\Psi\Omega} = J_{\Psi\Omega} \Delta_{\Psi\Omega}^{1/2}$, and the antiunitary J satisfies $\langle J\xi | J\eta \rangle = \langle \eta | \xi \rangle$, which swaps the two slots. (4) $\Delta_{\Psi\Omega}^{1/2}$ is self-adjoint, and $(B^\dagger)^\dagger = B$.

This is the two-state version of the KMS relation. For two random states of a pair of qutrits it was checked numerically, with $\Delta_{\Psi\Omega} = \rho_\Psi \otimes \rho'_\Omega{}^{-1}$ and random A, B : both sides agree to fifteen digits.

Two further facts are quoted without proof. On $\mathcal{M}$, the flow generated by $\Delta_{\Psi\Omega}$ is the same as the flow generated by the ordinary Δ_Ψ , and on $\mathcal{M}'$ it is the same as the flow generated by Δ_Ω . And the unitary $u_{\Omega\Psi}(s) \equiv \Delta_{\Omega\Phi}^{-is} \Delta_{\Psi\Phi}^{is}$, built with the help of a third state $|\Phi\rangle$ but independent of which Φ is used, is the explicit unitary in $\mathcal{M}$ that relates σ_s^Ψ to σ_s^Ω in (4.6).

Relative entropy for type III, and a proof that it is positive

For a type III algebra no entropy can be assigned to a single state. A **relative** entropy between two states can be, using the relative modular operator:

$$S_{\mathcal{M}}(\Psi||\Omega) \equiv -\langle \Psi | \log \Delta_{\Omega\Psi} | \Psi \rangle. \quad (4.15)$$

In the type I case this reduces to the familiar formula:

$$\begin{aligned} -\langle \Psi | \log \Delta_{\Omega\Psi} | \Psi \rangle &\stackrel{(1)}{=} -\langle \Psi | \log \rho_{\Omega} \otimes \mathbf{1} | \Psi \rangle + \langle \Psi | \mathbf{1} \otimes \log \rho'_{\Psi} | \Psi \rangle \\ &\stackrel{(2)}{=} -\text{tr}(\rho_{\Psi} \log \rho_{\Omega}) + \text{tr}(\rho'_{\Psi} \log \rho'_{\Psi}) \stackrel{(3)}{=} \text{tr}(\rho_{\Psi} \log \rho_{\Psi}) - \text{tr}(\rho_{\Psi} \log \rho_{\Omega}). \end{aligned}$$

(1) $\Delta_{\Omega\Psi} = \rho_{\Omega} \otimes \rho'_{\Psi}{}^{-1}$, and the logarithm of a tensor product of commuting positive factors is the sum of the logarithms. (2) the expectation value of an operator acting on one factor is its trace against that factor's reduced density matrix. (3) for a pure state $|\Psi\rangle$, ρ_{Ψ} and ρ'_{Ψ} have the same nonzero eigenvalues (Schmidt decomposition), so $\text{tr}(\rho'_{\Psi} \log \rho'_{\Psi}) = \text{tr}(\rho_{\Psi} \log \rho_{\Psi})$.

This is the relative entropy of Chapter 1. For the same random qutrit states, the left side of the chain and the familiar formula both give 2.10961, agreeing to thirteen digits. The definition (4.15) never uses a trace, however, so it carries over unchanged to type III.

Positivity takes only a few lines. The inequality $\log x \leq x - 1$ holds for every positive real x , with equality only at $x = 1$. (The function $f(x) = x - 1 - \log x$ has $f(1) = 0$ and $f'(x) = 1 - 1/x$, which is negative for $x < 1$ and positive for $x > 1$, so $x = 1$ is its unique minimum.) Then

$$\begin{aligned} -\langle \Psi | \log \Delta_{\Omega\Psi} | \Psi \rangle &\stackrel{(1)}{\geq} \langle \Psi | \mathbf{1} - \Delta_{\Omega\Psi} | \Psi \rangle \\ &\stackrel{(2)}{=} \langle \Psi | \Psi \rangle - \langle \Omega | \Omega \rangle \stackrel{(3)}{=} 0. \end{aligned}$$

(1) $\log x \leq x - 1$, applied to the spectrum of the positive operator $\Delta_{\Omega\Psi}$ and then averaged in the state $|\Psi\rangle$. (2) the two-state KMS relation (4.14) with Ψ and Ω exchanged and $A = B = \mathbf{1}$ gives $\langle \Psi | \Delta_{\Omega\Psi} | \Psi \rangle = \langle \Omega | \Omega \rangle$ (numerically 0.99999... for the qutrit example). (3) both states are normalized.

So $S_{\mathcal{M}}(\Psi||\Omega) \geq 0$ for every type of algebra. Chapter 3 found that the ordinary entropy $S_{\mathcal{M}}$ of a type II algebra can be negative. Relative entropy does not have this problem, because it measures how distinguishable Ψ is from a fixed reference state Ω . It never needs an absolute zero of entropy, which type II and type III algebras do not have.

4.7 A canonical purification: the natural cone

This last section answers a natural question about purifications. Given a state ω on $\mathcal{M}$, is there a *canonical* vector $|\xi\rangle \in \mathcal{H}$ with $\omega(A) = \langle \xi | A | \xi \rangle$?

In general many vectors do the job. For $\mathcal{M} = B(\mathcal{H}_1) \otimes \mathbf{1}_{\mathcal{H}_2}$, any vector in $\mathcal{H}_1 \otimes \mathcal{H}_2$ whose reduced density matrix on $\mathcal{H}_1$ is the right one works, and ordinary quantum mechanics gives no reason to prefer one of them. Suppose, however, that $\mathcal{H}$ contains *some* cyclic and separating vector for $\mathcal{M}$.

The pair $(\mathcal{M}, \mathcal{H})$ is then said to be in **standard form**. This holds, for instance, for $\mathcal{M} = \mathcal{M}_O$ with O an open region in the vacuum sector of a relativistic QFT (by Reeh–Schlieder), and for any GNS representation built from a faithful state (Chapter 2). In standard form there is a canonical choice.

Fix a cyclic and separating reference vector $|\Omega\rangle$. The **natural cone** P_Ω is the closure of the set $\{A j_\Omega(A) |\Omega\rangle : A \in \mathcal{M}\}$, where $j_\Omega(A) \equiv J_\Omega A J_\Omega$. An equivalent description is the closure of $\{\Delta_\Omega^{1/4} A^\dagger A |\Omega\rangle : A \in \mathcal{M}\}$. Every normal state ω on $\mathcal{M}$ has exactly one representative vector in P_Ω , and that vector is the canonical purification. It depends on the choice of Ω : a different reference vector gives a different cone.

For two qubits the cone is easy to see. Take $\mathcal{M} = B(\mathbb{C}^2) \otimes \mathbf{1}$ and the reference vector $|\Omega\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Write a vector as $|\xi\rangle = \sum_{ia} \xi_{ia} |i\rangle |a\rangle$, that is, as a 2×2 matrix ξ . For this Ω , J_Ω acts by $\xi \mapsto \xi^\dagger$, and $A j_\Omega(A) |\Omega\rangle$ corresponds to the matrix $AA^\dagger/\sqrt{2}$, which is positive semidefinite. The natural cone is therefore the set of positive semidefinite 2×2 matrices. A state with density matrix ρ on the first qubit is represented by every ξ with $\xi \xi^\dagger = \rho$, but only one of these is positive, namely $\xi = \sqrt{\rho}$. The canonical purification is $|\xi_\rho\rangle = \sum_{ia} (\sqrt{\rho})_{ia} |i\rangle |a\rangle$, the familiar “square-root” purification of ordinary quantum information. The inner product of two such vectors is $\text{tr}(\sqrt{\rho}\sqrt{\sigma})$, which is real and non-negative.

Vectors in the natural cone have several further properties, quoted here without proof. The inner product of any two vectors in P_Ω is real and non-negative, as in the two-qubit example. Vectors fixed by J_Ω can be decomposed into cone vectors. The distance between two cone vectors is bounded by the distance between the states they represent. Two different cyclic and separating vectors in the same natural cone have the same J and the same cone. A cyclic and separating vector outside the cone can be moved into it by an explicit unitary. Every automorphism of $\mathcal{M}$ can be implemented by a unitary that preserves the cone. The physics in the rest of these notes uses only the existence and uniqueness of the canonical purification and the positivity of inner products.

With this, the tools for entanglement are in place for every type of von Neumann algebra: I, II, and III. Chapter 5 takes up the problem noted in remark (f) after the Tomita–Takesaki theorem. A type III algebra has no trace and so no entropy of its own, which is a serious problem if the goal is to compute a black-hole entropy. The **crossed product** fixes this. It attaches an extra quantum system, a clock, to a type III algebra and turns it into a type II algebra, which does have a trace.

CHAPTER 5

The crossed product by the modular group

Chapter 4 ended with a problem. A type III algebra has no trace. Without a trace there is no density operator, and without a density operator there is no entropy. This is true even for the weaker, “relative to a reference” kind of entropy that worked for type II in Chapter 3. Suppose the goal is to compute something like a black-hole entropy using only operator algebras. Then this is a real obstruction, not a technicality.

The **crossed product** is the construction that removes the obstruction. It does not get around the no-trace theorem. Instead, it builds a *different, larger* algebra out of the original type III algebra. This larger algebra always turns out to be type II, and a type II algebra does have a trace. The construction is mechanical and completely general. It comes back, almost unchanged, in Chapter 9, where it is the physical mechanism behind gravitational entropy.

One limitation should be stated at the start. It explains why this chapter has no qubit model of the whole construction. Every finite-dimensional von Neumann algebra is type I. The reason is simple. In finite dimensions every projection has finite rank. So a nonzero projection of smallest rank in the algebra cannot be split any further, and an algebra with such minimal projections is type I (Chapter 2). As a result, no finite matrix example can show the passage from type III to type II itself. That passage happens only in infinite dimensions. What small, explicit examples *can* check is each individual step: the dressing condition, the commutator identities, the trace formula and its cyclicity, and the shape of the entropy formula. These steps are checked explicitly below. (The cyclicity of a type II trace was already checked on Bell-pair matrices in Chapter 3.)

5.1 Crossed products in general

The general construction does not mention modular theory at all. The modular case used in these notes is one special instance. Suppose a group G acts on a von Neumann algebra $\mathcal{M}$ by unitaries, $\alpha_g(A) = U_g A U_g^\dagger$. Think of α as a family of symmetries of $\mathcal{M}$, labelled by the group elements g . Given the triple $(\mathcal{M}, G, \alpha)$, there is a standard way to build a *new* von Neumann algebra, called the **crossed product** and written $\widehat{\mathcal{M}} \equiv \mathcal{M} \rtimes_\alpha G$. It acts on the enlarged Hilbert space $\widehat{\mathcal{H}} = \mathcal{H} \otimes L^2(G)$. The second factor $L^2(G)$ consists of square-integrable functions on G . One can think of it as an extra quantum system whose configuration space is the group G itself.

In these notes the group is $G = \mathbb{R}$, with generator K and action

$$\alpha_t(A) = e^{iKt} A e^{-iKt}. \quad (5.1)$$

Then $\widehat{\mathcal{H}} = \mathcal{H} \otimes L^2(\mathbb{R})$. The extra system is simply a quantum particle on a line.

From now on, take $K = -\log \Delta_\Psi$. This is the *modular* generator of $\mathcal{M}$ itself, built from some cyclic

and separating vector $|\Psi\rangle$. With this choice, (5.1) is exactly the modular flow σ_t built in Chapter 4. The main result of this chapter is proved step by step below. If $\mathcal{M}$ is type III, the crossed product $\widehat{\mathcal{M}}$ is **always type II**, whichever type III subtype $\mathcal{M}$ started as. This makes the tools of Chapter 3 (density operators and entropy) available at once, applied to $\widehat{\mathcal{M}}$ instead of $\mathcal{M}$. A second fact is justified at the end of the next section. The algebra $\widehat{\mathcal{M}}$ depends only on $\mathcal{M}$, not on which reference state $|\Psi\rangle$ was used to build the modular flow. So the construction is *intrinsic* to $\mathcal{M}$. It is not an artifact of an arbitrary choice. In Chapter 9 the same construction explains black-hole and de Sitter entropy. There, the clock position $\hat{q}$ introduced below becomes the energy of a physical observer.

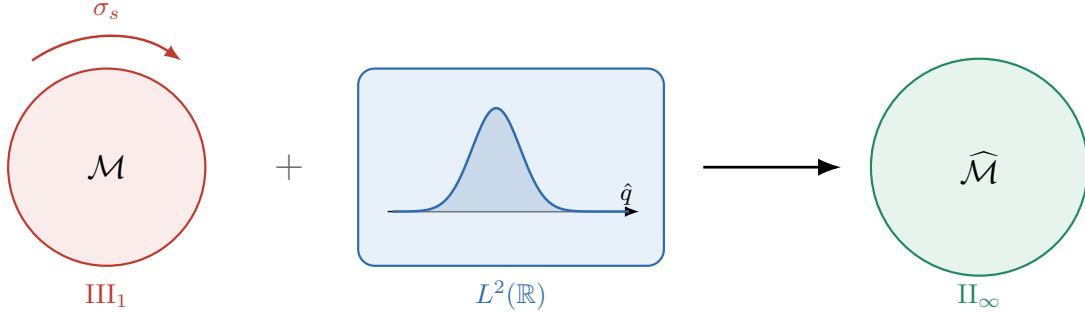


Figure 5.1: The crossed product. A type III₁ algebra $\mathcal{M}$, which has no trace, comes with its modular flow σ_s (red arrow). Adding a clock, a particle on a line with position $\hat{q}$ and Hilbert space $L^2(\mathbb{R})$ (a wavepacket is sketched), and combining the two produces the larger algebra $\widehat{\mathcal{M}}$. It is type II_∞ and does have a trace.

5.2 Construction of $\widehat{\mathcal{M}}$

Attach a genuine one-dimensional quantum system to the original system. This is a particle on a line, with position $\hat{q}$ and momentum $\hat{p}$ obeying the ordinary canonical commutation relation $[\hat{q}, \hat{p}] = i$. We call it the **clock**. The enlarged Hilbert space is $\mathcal{H} \otimes L^2(\mathbb{R})$. Define the single operator

$$C \equiv K + \hat{q}, \quad (5.2)$$

and let $\widehat{\mathcal{M}}$ be the set of all operators in $\mathcal{M} \otimes B(L^2(\mathbb{R}))$ that commute with C . In symbols, $a \in \widehat{\mathcal{M}}$ if and only if $[a, C] = 0$.

Why is this the right condition? It is not obvious at first sight. The operator C generates two translations at once. The piece K generates the modular flow. The piece $\hat{q}$ generates translations of the clock's momentum. So e^{iCs} shifts both by the same amount. Asking for invariance under C therefore says that only the *relative* reading, “clock minus modular time,” matters, and neither one matters on its own. Mathematically this is the same as the familiar idea that only relative positions matter, never the choice of an origin for the coordinates. In the applications of Chapter 9, $\hat{q}$ is an observer's own energy and $\hat{p}$ is the observer's proper time. There the invariance condition becomes the statement that physical observables must be diffeomorphism-invariant: they cannot depend on an arbitrary choice of where to put the zero of a clock.

Physics Connection: this is the same trick as separating center-of-mass motion

You have already solved a version of this problem in a first mechanics course, in the two-body problem. Two particles have positions x_1, x_2 and momenta p_1, p_2 . Their total momentum $P \equiv p_1 + p_2$ generates a *simultaneous* translation of both particles, e^{iPs} . This is the same job that $C = K + \hat{q}$ does here: it generates modular-time flow and clock translation together. Suppose the Hamiltonian depends only on the relative coordinate $x_{\text{rel}} \equiv x_1 - x_2$. This is translation invariance: nothing picks out an absolute position. Then the useful, physically meaningful variables are the ones that are unchanged by the combined shift. These are the relative coordinate itself and the total momentum P . Check the elementary commutator directly:

$$[x_{\text{rel}}, P] = [x_1 - x_2, p_1 + p_2] \stackrel{(1)}{=} [x_1, p_1] - [x_2, p_2] \stackrel{(2)}{=} i - i = 0.$$

(1) expand the commutator term by term. The cross terms $[x_1, p_2]$ and $[x_2, p_1]$ vanish because the two particles are independent degrees of freedom. (2) the canonical commutator $[x_j, p_j] = i$ for each particle separately.

So x_{rel} commutes with the generator of the shift that we want the physics to ignore. Below, the dressed operator $\hat{A} = e^{iK\hat{p}} A e^{-iK\hat{p}}$ plays exactly this role for C . The phrase “translate both particles” is replaced by “translate modular time and the clock together.”

The center-of-mass coordinate X_{cm} behaves differently. It does *not* commute with P . Instead $[X_{\text{cm}}, P] = i$, the ordinary canonical commutator. It plays the role that $\hat{q}$ would play if you forgot to dress it: an absolute reference that the physics was never supposed to depend on.

None of the mathematics here is new. The only new features are these. The “coordinate” being made relative is modular time, not ordinary position. And the “other particle” is the modular generator K , not a second physical body.

Two families of operators obviously satisfy $[a, C] = 0$, and together they generate all of $\widehat{\mathcal{M}}$.

The first family is built from $\hat{q}$ itself. It commutes with $C = K + \hat{q}$, because $[\hat{q}, \hat{q}] = 0$ and $\hat{q}$ acts on a different tensor factor than K . So every bounded function of $\hat{q}$ belongs to $\widehat{\mathcal{M}}$. An example is the unitary $e^{-i\hat{q}s}$.

The second family is the nontrivial one. An ordinary element $A \in \mathcal{M}$ does *not* commute with C by itself. The reason is that $[A, K] \neq 0$ in general; this is exactly what it means for K to generate a nontrivial flow on A . But a specific *dressed* version of A does commute with C :

$$[e^{iK\hat{p}} A e^{-iK\hat{p}}, C] = 0, \quad A \in \mathcal{M}. \quad (5.3)$$

This identity follows from the canonical commutation relations alone. Here is the calculation. Write $\hat{A} \equiv e^{iK\hat{p}}(A \otimes \mathbf{1})e^{-iK\hat{p}}$. Because $[\hat{q}, \hat{p}] = i$, the position operator acts on functions of $\hat{p}$ like a derivative: $[\hat{q}, f(\hat{p})] = i f'(\hat{p})$. Apply this to the two exponentials. Since K acts on $\mathcal{H}$ and $\hat{p}$ acts on the clock, K can be treated as a constant when differentiating with respect to p :

$$[\hat{q}, e^{iK\hat{p}}] \stackrel{(a)}{=} i(iK) e^{iK\hat{p}} = -K e^{iK\hat{p}},$$

$$[\hat{q}, e^{-iK\hat{p}}] \stackrel{(b)}{=} i(-iK)e^{-iK\hat{p}} = Ke^{-iK\hat{p}}.$$

(a) $[\hat{q}, f(\hat{p})] = if'(\hat{p})$ with $f(p) = e^{iKp}$, so $f'(p) = iKe^{iKp}$. **(b)** the same rule with $f(p) = e^{-iKp}$, so $f'(p) = -iKe^{-iKp}$.

Now compute the commutator $[\hat{q}, \hat{A}]$ with the product rule for commutators:

$$\begin{aligned} [\hat{q}, \hat{A}] &\stackrel{(1)}{=} [\hat{q}, e^{iK\hat{p}}](A \otimes \mathbf{1})e^{-iK\hat{p}} + e^{iK\hat{p}}(A \otimes \mathbf{1})[\hat{q}, e^{-iK\hat{p}}] \\ &\stackrel{(2)}{=} -Ke^{iK\hat{p}}(A \otimes \mathbf{1})e^{-iK\hat{p}} + e^{iK\hat{p}}(A \otimes \mathbf{1})e^{-iK\hat{p}}K \\ &\stackrel{(3)}{=} -K\hat{A} + \hat{A}K = -[K, \hat{A}]. \end{aligned}$$

(1) the product rule $[\hat{q}, XYZ] = [\hat{q}, X]YZ + X[\hat{q}, Y]Z + XY[\hat{q}, Z]$, with the middle term zero because $[\hat{q}, A \otimes \mathbf{1}] = 0$. **(2)** substitute (a) and (b). In the second term, $Ke^{-iK\hat{p}} = e^{-iK\hat{p}}K$, because K commutes with any function of K and $\hat{p}$. **(3)** both terms contain $e^{iK\hat{p}}(A \otimes \mathbf{1})e^{-iK\hat{p}} = \hat{A}$, by definition.

Now take the commutator with the full constraint $C = K + \hat{q}$:

$$[C, \hat{A}] = [K + \hat{q}, \hat{A}] \stackrel{(4)}{=} [K, \hat{A}] + [\hat{q}, \hat{A}] \stackrel{(5)}{=} [K, \hat{A}] - [K, \hat{A}] = 0.$$

(4) the commutator is linear in each slot, so it splits over the sum $K + \hat{q}$. **(5)** substitute $[\hat{q}, \hat{A}] = -[K, \hat{A}]$ from step (3).

The dressing by the clock momentum $\hat{p}$ exactly cancels the failure of A to commute with K . So $\hat{A}$ is exactly invariant under C . The crossed product is therefore

$$\widehat{\mathcal{M}} = \{e^{iK\hat{p}}Ae^{-iK\hat{p}}, e^{-i\hat{q}s} \mid A \in \mathcal{M}, s \in \mathbb{R}\}'', \quad (5.4)$$

the von Neumann algebra generated by the two families. A general element has the schematic form $\hat{A} = \int ds A(\hat{p}; s) e^{-i\hat{q}s}$. Here $A(\hat{p}; s) \equiv e^{iK\hat{p}}A(s)e^{-iK\hat{p}}$, built from an operator-valued function $A(s) \in \mathcal{M}$. Since $\hat{p}$ is now an operator, $A(\hat{p}; s)$ is an operator that depends on another operator, not on a fixed number. In the applications of Chapter 9, where $\hat{p}$ is an observer's time, this means the operator is evaluated at a time that is itself quantum.

There is a second, equivalent description of the same algebra. It comes from the unitary change of frame $U = e^{-iK\hat{p}}$, which maps $\widehat{\mathcal{M}} \rightarrow U\widehat{\mathcal{M}}U^\dagger$ and $C \rightarrow UC U^\dagger$. The same derivative rule as in (a) gives $U\hat{q}U^\dagger = \hat{q} - K$, so $C \rightarrow \hat{q}$ and $e^{-i\hat{q}s} \rightarrow e^{i(K-\hat{q})s}$. The dressed operator $\hat{A}$ goes back to the undressed A . This second form is more convenient for almost everything that follows:

$$\begin{aligned} \widehat{\mathcal{M}} &= \{A, e^{i(K-\hat{q})s} \mid A \in \mathcal{M}, s \in \mathbb{R}\}'', \\ \hat{A} &= \int ds A(s) e^{is(K-\hat{q})}, \quad A(s) \in \mathcal{M}. \end{aligned} \quad (5.5)$$

In this frame, ordinary (undressed) elements of $\mathcal{M}$ sit directly inside $\widehat{\mathcal{M}}$. All the dressing has been moved into the clock-dependent generator $K - \hat{q}$. In this frame, $e^{i\hat{q}s}$ commutes with every $A \in \mathcal{M}$, because they act on different factors. It also commutes with every $e^{i(K-\hat{q})s'}$, because $[K, \hat{q}] = 0$. So $e^{i\hat{q}s}$ belongs to the commutant $\widehat{\mathcal{M}}'$. (This matches the constraint: in this frame C has become $\hat{q}$,

and everything in $\widehat{\mathcal{M}}$ commutes with it.)

5.3 $\widehat{\mathcal{M}}$ is always type II

Finding the modular operator of $\widehat{\mathcal{M}}$

Now set $K = -\log \Delta_\Psi$, the modular generator of $\mathcal{M}$. Work in the frame (5.5) and take the reference vector

$$|\widehat{\Psi}\rangle = |\Psi\rangle \otimes |p=0\rangle. \quad (5.6)$$

The clock is in a state of definite momentum, so it is completely spread out in position. This vector cannot be normalized: it is a plane wave. Strictly speaking it defines a *weight* rather than a state. This technical point does not change any conclusion below. The vector $|\widehat{\Psi}\rangle$ is cyclic and separating for $\widehat{\mathcal{M}}$ (a standard fact that we quote), so Tomita–Takesaki theory applies. The modular operator $\widehat{\Delta}$ can then be found from the KMS relation that characterizes it,

$$\langle \widehat{\Psi} | \widehat{A} \widehat{B} | \widehat{\Psi} \rangle = \langle \widehat{\Psi} | \widehat{B} \widehat{\Delta} \widehat{A} | \widehat{\Psi} \rangle, \quad (5.7)$$

or, equivalently, from the Tomita operator $\widehat{S}$. The calculation below shows a clean cancellation.

Key Result / Core Concept: Derivation of the Crossed Product Modular Operator $\widehat{\Delta}$

Goal: On $\widehat{\mathcal{H}} = \mathcal{H} \otimes L^2(\mathbb{R})$ with reference vector $|\widehat{\Psi}\rangle = |\Psi\rangle \otimes |p=0\rangle$, find the Tomita operator $\widehat{S}$ and show that the modular operator is

$$\widehat{\Delta} \equiv \widehat{S}^\dagger \widehat{S} = \Delta_\Psi \otimes \mathbf{1} = \Delta_\Psi.$$

Derivation:

1. **A general operator acting on the reference vector.** In the frame (5.5), a general operator $\widehat{A} \in \widehat{\mathcal{M}}$ has the form

$$\widehat{A} = \int_{-\infty}^{\infty} ds A(s) e^{is(K-\hat{q})}, \quad A(s) \in \mathcal{M}.$$

Act with $\widehat{A}$ on $|\widehat{\Psi}\rangle = |\Psi\rangle \otimes |p=0\rangle$. Two facts are needed. First, $K = -\log \Delta_\Psi$ and $\Delta_\Psi |\Psi\rangle = |\Psi\rangle$, so the modular Hamiltonian annihilates the state: $K |\Psi\rangle = 0$, and therefore $e^{isK} |\Psi\rangle = |\Psi\rangle$. Second, on the clock $\hat{q} = i\partial_p$ in the momentum representation, so $e^{-is\hat{q}}$ shifts the clock momentum:

$$e^{-is\hat{q}} |p=0\rangle = |p=-s\rangle.$$

Since K and $\hat{q}$ act on different factors, they commute, and

$$e^{is(K-\hat{q})} |\widehat{\Psi}\rangle \stackrel{(1)}{=} (e^{isK} |\Psi\rangle) \otimes (e^{-is\hat{q}} |p=0\rangle) \stackrel{(2)}{=} |\Psi\rangle \otimes |p=-s\rangle.$$

- (1) $[K, \hat{q}] = 0$, so the exponential factorizes into a piece acting on $\mathcal{H}$ and a piece acting on

the clock. **(2)** $e^{isK} |\Psi\rangle = |\Psi\rangle$ and $e^{-is\hat{q}} |p=0\rangle = |p=-s\rangle$, both shown just above.

Now multiply by $A(s) \in \mathcal{M}$ and integrate:

$$\hat{A} |\hat{\Psi}\rangle \stackrel{(3)}{=} \int_{-\infty}^{\infty} ds A(s) |\Psi\rangle \otimes |p=-s\rangle \stackrel{(4)}{=} \int_{-\infty}^{\infty} dp A(-p) |\Psi\rangle \otimes |p\rangle.$$

(3) substitute the result of steps (1)–(2) inside the s integral. **(4)** change the integration variable to $p = -s$. The sign from $dp = -ds$ cancels the sign from swapping the limits.

So the clock momentum p labels the different pieces $A(s)$ of the operator: the component at clock momentum p is $A(-p) |\Psi\rangle$.

2. The adjoint $\hat{A}^\dagger$ acting on the reference vector. The adjoint is

$$\hat{A}^\dagger = \int_{-\infty}^{\infty} ds e^{-is(K-\hat{q})} A(s)^\dagger.$$

Insert $\mathbf{1} = e^{is(K-\hat{q})} e^{-is(K-\hat{q})}$ to the right of $A(s)^\dagger$:

$$\hat{A}^\dagger = \int_{-\infty}^{\infty} ds \left(e^{-is(K-\hat{q})} A(s)^\dagger e^{is(K-\hat{q})} \right) e^{-is(K-\hat{q})}.$$

Since $[\hat{q}, A(s)^\dagger] = 0$, the conjugation in brackets is just the modular flow of the original algebra $\mathcal{M}$:

$$e^{-is(K-\hat{q})} A(s)^\dagger e^{is(K-\hat{q})} \stackrel{(5)}{=} e^{-isK} A(s)^\dagger e^{isK} \stackrel{(6)}{=} \Delta_\Psi^{is} A(s)^\dagger \Delta_\Psi^{-is} \equiv \alpha_{-s}(A(s)^\dagger).$$

(5) $\hat{q}$ commutes with $A(s)^\dagger$ and with K , so the $\hat{q}$ exponentials cancel against each other.

(6) $K = -\log \Delta_\Psi$, so $e^{-isK} = \Delta_\Psi^{is}$.

The last factor $e^{-is(K-\hat{q})}$ acts on $|\hat{\Psi}\rangle$ as in step (2), with $s \rightarrow -s$, and gives $|\Psi\rangle \otimes |p=s\rangle$. Hence

$$\hat{A}^\dagger |\hat{\Psi}\rangle = \int_{-\infty}^{\infty} ds \alpha_{-s}(A(s)^\dagger) |\Psi\rangle \otimes |p=s\rangle.$$

Now use $\Delta_\Psi^{-is} |\Psi\rangle = |\Psi\rangle$ and the definition of the Tomita operator, $S_\Psi A |\Psi\rangle = A^\dagger |\Psi\rangle$:

$$\alpha_{-s}(A(s)^\dagger) |\Psi\rangle = \Delta_\Psi^{is} A(s)^\dagger |\Psi\rangle = \Delta_\Psi^{is} S_\Psi A(s) |\Psi\rangle.$$

3. The Tomita operator $\hat{S}$. The Tomita operator of $\hat{\mathcal{M}}$ is defined by $\hat{S}(\hat{A} |\hat{\Psi}\rangle) = \hat{A}^\dagger |\hat{\Psi}\rangle$.

Compare the two results. The input has the vector $A(s) |\Psi\rangle$ at clock momentum $p = -s$. The output has the vector $\Delta_\Psi^{is} S_\Psi A(s) |\Psi\rangle$ at clock momentum $p = +s$. Three things happen. The clock momentum is reflected, $p \rightarrow -p$; call this reflection $\mathcal{P}_p |p\rangle \equiv |-p\rangle$. The $\mathcal{H}$ part is acted on by S_Ψ . Finally, the component at clock momentum p is multiplied by Δ_Ψ^{ip} . So

$$\hat{S} = \Delta_\Psi^{ip} (S_\Psi \otimes \mathcal{P}_p),$$

where $\Delta_\Psi^{ip} = e^{-iK\hat{p}}$ is a unitary operator. (One can check that S_Ψ commutes with Δ_Ψ^{is} for real s , so the order of S_Ψ and Δ_Ψ^{ip} at fixed p does not matter.)

4. **Cancellation in the modular operator.** Now evaluate the modular operator $\widehat{\Delta} \equiv \widehat{S}^\dagger \widehat{S}$:

$$\begin{aligned} \widehat{\Delta} &\stackrel{(7)}{=} (S_\Psi^\dagger \otimes \mathcal{P}_p^\dagger) \Delta_\Psi^{-i\hat{p}} \Delta_\Psi^{i\hat{p}} (S_\Psi \otimes \mathcal{P}_p) \\ &\stackrel{(8)}{=} (S_\Psi^\dagger S_\Psi) \otimes (\mathcal{P}_p^\dagger \mathcal{P}_p) \stackrel{(9)}{=} \Delta_\Psi \otimes \mathbf{1}_{L^2(\mathbb{R})} = \Delta_\Psi. \quad \blacksquare \end{aligned}$$

(7) the adjoint of a product reverses the order, and $(\Delta_\Psi^{i\hat{p}})^\dagger = \Delta_\Psi^{-i\hat{p}}$. (8) the unitary factor cancels, $\Delta_\Psi^{-i\hat{p}} \Delta_\Psi^{i\hat{p}} = \mathbf{1}$; then products of tensor products multiply factor by factor. (9) $S_\Psi^\dagger S_\Psi = \Delta_\Psi$ is the definition of the modular operator, and the reflection $\mathcal{P}_p$ is unitary, $\mathcal{P}_p^\dagger \mathcal{P}_p = \mathbf{1}_{L^2(\mathbb{R})}$.

The answer is very simple, given how much machinery went into building $\widehat{\mathcal{M}}$:

$$\widehat{\Delta} = \Delta_\Psi. \quad (5.8)$$

In this reference vector, the modular operator of the new algebra is the same operator as the modular operator of the original algebra. Nothing new had to be invented. The crossed product inherits its modular structure directly from $\mathcal{M}$.

Why this forces type II

The key algebraic step is short. Start from $\widehat{\Delta} = e^{-K}$ and split the exponential:

$$\widehat{\Delta} = e^{-K} = e^{-(K-\hat{q})} e^{-\hat{q}} \equiv \rho \rho'^{-1}, \quad \rho \equiv e^{-(K-\hat{q})}, \quad \rho' \equiv e^{\hat{q}}. \quad (5.9)$$

Splitting the exponential is allowed because K and $\hat{q}$ commute. Hence

$$\widehat{\Delta}^{-is} = e^{i(K-\hat{q})s} e^{i\hat{q}s}.$$

Now compare the two factors on the right with the descriptions of $\widehat{\mathcal{M}}$ and its commutant. The factor $e^{i(K-\hat{q})s}$ is a unitary inside $\widehat{\mathcal{M}}$ itself; it is one of the generators in (5.5). The factor $e^{i\hat{q}s}$ is a unitary inside the commutant $\widehat{\mathcal{M}}'$. **So $\widehat{\Delta}^{-is}$ is a unitary in $\widehat{\mathcal{M}}$ times a unitary in $\widehat{\mathcal{M}}'$.**

When we conjugate an element of $\widehat{\mathcal{M}}$ by $\widehat{\Delta}^{-is}$, the factor in $\widehat{\mathcal{M}}'$ commutes with it and drops out. So the modular flow of $\widehat{\mathcal{M}}$ is implemented by a unitary that lies inside $\widehat{\mathcal{M}}$. This is the definition of an *inner* automorphism. By the criterion (4.5) of Chapter 4, the modular flow is inner for *all* s if and only if the algebra is type I or type II, never type III. **So $\widehat{\mathcal{M}}$ cannot be type III**, whatever $\mathcal{M}$ was. No explicit trace was needed for this step. It followed only from identifying the modular operator and checking the inner-automorphism criterion.

It remains to rule out type I. Here is an intuitive reason, which is not a proof. Attaching a clock and imposing an invariance condition does not create a tensor factorization of $\widehat{\mathcal{H}} = \mathcal{H} \otimes L^2(\mathbb{R})$ adapted to $\widehat{\mathcal{M}}$. The obstruction that made $\mathcal{M}$ type III is still present.

A sharper argument works when $\widehat{\mathcal{M}}$ is a factor, which is the case when $\mathcal{M}$ is type III₁. Shift the clock by conjugating with $e^{i\hat{p}s}$, which sends $\hat{q} \rightarrow \hat{q} + s$. This maps $\widehat{\mathcal{M}}$ to itself: each $A \in \mathcal{M}$ is

unchanged, and $e^{i(K-\hat{q})t}$ only picks up a phase e^{-ist} . Using the trace (5.10) constructed in the next subsection, together with $e^{-i\hat{p}s}|p=0\rangle = |p=0\rangle$, one finds that this shift multiplies the trace by e^s . In a type I factor $B(\mathcal{K})$ every automorphism has the form $X \rightarrow VXV^\dagger$ with V unitary, and such a map never changes the trace. So a type I factor has no automorphism that rescales its trace, and $\widehat{\mathcal{M}}$ cannot be type I. **By elimination, $\widehat{\mathcal{M}}$ is type II.** For a type III₁ factor $\mathcal{M}$, the crossed product is a type II_∞ factor. For the other type III subtypes, $\widehat{\mathcal{M}}$ is still type II but has a nontrivial center. That last statement is a standard theorem of Takesaki that we quote without proof.

Building the trace explicitly, and checking cyclicity by hand

We now know *that* a trace exists. It is useful to write one down and check that it works. Define

$$\mathrm{tr} \hat{A} \equiv \langle \widehat{\Psi} | \hat{A} \rho^{-1} | \widehat{\Psi} \rangle, \quad (5.10)$$

with $\rho = e^{-(K-\hat{q})}$ from (5.9). The motivation is as follows. If a trace exists, expectation values in the reference vector should have the form $\langle A \rangle = \mathrm{tr}(A\rho)$ for some density operator ρ . So to recover the trace from an expectation value, one removes ρ by multiplying with ρ^{-1} .

Cyclicity means $\mathrm{tr}(\hat{A}\hat{B}) = \mathrm{tr}(\hat{B}\hat{A})$. It follows from nothing more than the KMS relation (5.7). In the chain below, $\langle \cdots \rangle$ is shorthand for $\langle \widehat{\Psi} | \cdots | \widehat{\Psi} \rangle$:

$$\begin{aligned} \mathrm{tr}(\hat{A}\hat{B}) &= \langle \hat{A}\hat{B}\rho^{-1} \rangle \stackrel{(1)}{=} \langle \hat{B}\rho^{-1}\hat{A} \rangle \\ &\stackrel{(2)}{=} \langle \hat{B}\rho'^{-1}\hat{A} \rangle \stackrel{(3)}{=} \langle \hat{B}\hat{A}\rho'^{-1} \rangle \stackrel{(4)}{=} \mathrm{tr}(\hat{B}\hat{A}). \end{aligned}$$

(1) the KMS relation (5.7), with $\hat{A}$ as the first operator and $\hat{B}\rho^{-1}$ as the second. This is allowed because $\rho^{-1} = e^{K-\hat{q}}$ is built from a generator of $\widehat{\mathcal{M}}$. (2) $\rho^{-1}\hat{A} = \rho^{-1}\rho\rho'^{-1} = \rho'^{-1}$, directly from the factorization $\hat{A} = \rho\rho'^{-1}$. (3) $\rho'^{-1} = e^{-\hat{q}}$ is built from the commutant $\widehat{\mathcal{M}}'$. So it commutes with $\hat{A} \in \widehat{\mathcal{M}}$ and can be moved to the far right. (4) since $K|\Psi\rangle = 0$, we have $\rho^{-1}|\Psi\rangle = e^{K-\hat{q}}|\Psi\rangle = e^{-\hat{q}}|\Psi\rangle = \rho'^{-1}|\Psi\rangle$. So $\langle \hat{B}\hat{A}\rho'^{-1} \rangle = \langle \hat{B}\hat{A}\rho^{-1} \rangle$, which is the definition (5.10) of $\mathrm{tr}(\hat{B}\hat{A})$.

Every step is either a definition or something already established. No new assumption enters. (The manipulations are formal, since ρ^{-1} is unbounded and $|\widehat{\Psi}\rangle$ is not normalizable, but they can be made precise.) The same mechanism appeared in a much more concrete setting in Chapter 3. There the trace on the infinite chain of maximally entangled Bell pairs was checked to be cyclic by explicit 2×2 matrix computation. In both cases the consistency conditions of a specific reference state force cyclicity. Here the argument is written in the general language of modular theory, which also works for type III, rather than worked out on small matrices.

Type II_∞, or type II₁ if the clock's energy is bounded below

Evaluate the trace (5.10) on the identity. Since $K|\Psi\rangle = 0$, only the clock factor $e^{-\hat{q}}$ acts nontrivially. A plane wave has $|\langle q|p=0\rangle|^2 = 1/2\pi$ at every q , so

$$\mathrm{tr}(\mathbf{1}) = \langle p=0 | e^{-\hat{q}} | p=0 \rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} dq e^{-q} = \infty.$$

The trace of the identity diverges. So, by the classification of Chapter 2, $\widehat{\mathcal{M}}$ is type II_∞ .

The divergence comes from the range of the clock variable. In the construction above, the clock's "energy" $\hat{q}$ is like the position x , or the momentum p , of a particle on a line. Its spectrum is the whole real line, and the weight e^{-q} blows up as $q \rightarrow -\infty$. That is what gives type II_∞ .

There is a simple, physically motivated fix, used again in Chapter 9. Require the clock's energy to be bounded below, $q \geq 0$. This says that a real observer's energy cannot be negative, which is a reasonable requirement for any physical clock. For example, an observer whose energy is a kinetic energy $p^2/2m \geq 0$ has possible energies filling the half-line $[0, \infty)$, not the whole line.

To impose this, insert a projector. In the frame (5.4), the operator $\hat{q}$ itself belongs to $\widehat{\mathcal{M}}$, and the projector is $\Pi = \theta(\hat{q})$. Here θ is the step function, equal to 1 for $q \geq 0$ and 0 for $q < 0$. Define $\widehat{\mathcal{M}}_+ \equiv \Pi \widehat{\mathcal{M}} \Pi$. In that frame it acts on the smaller Hilbert space $\mathcal{H} \otimes L^2(\mathbb{R}_{\geq 0})$. In the frame (5.5), where the trace formula is written, the same projector reads $\Pi = \theta(\hat{q} - K)$. This is a function of the generator $K - \hat{q}$, so it lies in $\widehat{\mathcal{M}}$, and it acts on $|\hat{\Psi}\rangle$ as $\theta(\hat{q})$ because $K|\Psi\rangle = 0$. The identity element of $\widehat{\mathcal{M}}_+$ is Π , and

$$\text{tr}_{\widehat{\mathcal{M}}_+}(\mathbf{1}) \stackrel{(1)}{=} \text{tr}_{\widehat{\mathcal{M}}}(\Pi) \stackrel{(2)}{=} \frac{1}{2\pi} \int_{-\infty}^{\infty} dq e^{-q} \theta(q) \stackrel{(3)}{=} \frac{1}{2\pi} \int_0^{\infty} dq e^{-q} = \frac{1}{2\pi}.$$

(1) the identity of $\widehat{\mathcal{M}}_+$ is the projector Π , viewed as an element of $\widehat{\mathcal{M}}$. (2) the trace formula (5.10); acting on $|\hat{\Psi}\rangle$, the projector inserts the factor $\theta(q)$ into the clock integral. (3) $\theta(q)$ cuts the integration range down to $q \geq 0$.

The result is finite. A trace is only defined up to an overall positive constant, so multiplying it by 2π normalizes it to $\text{tr}(\mathbf{1}) = 1$. **So restricting the observer's clock to positive energy is exactly the algebraic step that turns type II_∞ into type II_1 .**

The same condition decides whether an ordinary Gibbs state exists. A partition function $Z = \text{Tr} e^{-\beta H}$ can converge only if H is bounded below. For the harmonic oscillator, with energies $\omega(n + \frac{1}{2})$ for $n \geq 0$, the geometric sum converges. For a Hamiltonian whose spectrum runs down to $-\infty$, the weights $e^{-\beta E}$ grow without bound and $Z = \infty$. An example is a particle in a linear potential $V = -Fx$ on the whole line: its spectrum is all of $\mathbb{R}$. The two integrals $\int_0^\infty e^{-q} dq$ (finite) and $\int_{-\infty}^\infty e^{-q} dq$ (infinite) express the same fact for the clock, whose "energy" is $\hat{q}$.

One caution about the analogy. Being bounded below is necessary for a finite Z , but it is not sufficient. A free particle on the whole line has $p^2/2m \geq 0$, yet its thermal Z is infinite, because the infinite volume gives infinitely many states in any energy window. The clock trace has no such volume factor: it weights each value of q with the flat measure $dq/2\pi$. So for the clock, the lower bound on the spectrum is the only thing that matters.

This fork has real physical consequences. Black-hole entropy (type II_∞) and de Sitter entropy (type II_1) sit on opposite sides of it, as Chapter 9 shows.

The crossed product does not depend on the reference state

The construction used a specific cyclic and separating vector $|\Psi\rangle$ to define the modular generator K . Choosing a different vector $|\Phi\rangle$ gives only a *unitarily equivalent* algebra,

$$\widehat{\mathcal{M}}_\Phi = u'_{\Phi\Psi}(\hat{p}) \widehat{\mathcal{M}}_\Psi u_{\Phi\Psi}^{\dagger}(\hat{p}).$$

Here $u'_{\Phi\Psi}$ is the intertwining unitary of relative modular theory (Chapter 4). It relates the modular flows of the two states. In the formula above, its flow parameter is replaced by the clock momentum operator $\hat{p}$. The computation that proves this is a direct but somewhat long manipulation of the relative modular identities of Chapter 4, and we do not reproduce it. What matters is the conclusion: up to unitary equivalence, $\widehat{\mathcal{M}}$ is an *intrinsic* invariant of $\mathcal{M}$ alone.

5.4 Density operator for $\widehat{\mathcal{M}}$ in a general semiclassical state

With a trace in hand, the tools of Chapter 3 become available for $\widehat{\mathcal{M}}$. In particular, a density operator $\rho_{\widehat{\mathcal{M}}}$ for a state is defined by $\text{tr}(A\rho_{\widehat{\mathcal{M}}}) = \langle \text{state} | A | \text{state} \rangle$ for all $A \in \widehat{\mathcal{M}}$. Consider the physically motivated class of states

$$|\widehat{\Phi}\rangle = |\Phi\rangle \otimes |g\rangle, \quad (5.11)$$

where $|g\rangle$ is a fixed clock wavefunction. It is normalized, $\int dq |g(q)|^2 = 1$, and nonzero everywhere.

Solving the defining equation for the density operator is a longer computation. It uses the two-state KMS relation of relative modular theory (Chapter 4) twice. It also uses the explicit form $\langle q | e^{-iK\hat{p}} | g \rangle = g(q - K)$, which describes how the clock wavefunction changes between the two frames (5.4) and (5.5). Once the KMS relation is accepted, the rest is mostly bookkeeping, and we do not reproduce it. The result is clean [8]:

$$\rho_{\widehat{\Phi}} = 2\pi g(\hat{q} - K) e^{\hat{q}} \Delta_{\Phi\Psi} g^*(\hat{q} - K). \quad (5.12)$$

This expression is written in the frame (5.5), the same frame as the trace (5.10). (There, $\hat{q} - K$ is minus the generator $K - \hat{q}$, so functions of it lie in $\widehat{\mathcal{M}}$.) The expression in the other frame follows by conjugating with $U^\dagger$. It is an explicit operator with three ingredients: the clock wavefunction g ; the factor $e^{\hat{q}}$, which comes from the normalization of the trace; and the relative modular operator $\Delta_{\Phi\Psi}$ of Chapter 4, which compares the state of interest Φ with the reference Ψ used to build the crossed product. The factor 2π matches the $1/2\pi$ in the plane-wave normalization used for the trace.

5.5 Entanglement entropy for $\widehat{\mathcal{M}}$ in a general semiclassical state

The final formula, and why it has exactly the shape it does

Insert $\rho_{\widehat{\Phi}}$ into the ordinary entropy formula, $S_{\widehat{\mathcal{M}}} = -\text{tr}(\rho_{\widehat{\Phi}} \log \rho_{\widehat{\Phi}})$. Then specialize to a **semiclassical** clock wavefunction. This means one that varies slowly, with $g'(q) \propto \epsilon$ for a small parameter ϵ , and we work to leading order in ϵ . The expansion has three steps. First, expand $-\log \rho_{\widehat{\Phi}}$ using

the relative-modular identity $K_{\Phi\Psi} = K_{\Phi} + K - K_{\Psi\Phi}$ of Chapter 4. Second, evaluate the resulting expectation value term by term, using $K_{\Phi}|\Phi\rangle = 0$. Third, recognize the surviving piece as the relative entropy of Chapter 4. The result, due to Chandrasekaran, Longo, Penington and Witten [8], is

$$S_{\widehat{\mathcal{M}}}(\widehat{\Phi}) = -S_{\mathcal{M}}(\Phi\|\Psi) - \bar{q} + S_o, \quad (5.13)$$

$$\bar{q} = \int dq q |g(q)|^2, \quad S_o = - \int dq |g(q)|^2 \log |g(q)|^2.$$

Here $\bar{q}$ is the mean clock reading in the probability distribution $|g(q)|^2$. The quantity S_o is the ordinary (differential) Shannon entropy of that same distribution. Both depend only on the shape of the clock wavefunction g , not on the state Φ . (The formula holds up to an additive constant, which depends on how the trace is normalized.)

Up to terms fixed entirely by the clock, the entropy of the type II crossed-product algebra is minus the type III relative entropy of the original algebra $\mathcal{M}$. This closes the loop opened in Chapter 4. There, relative entropy was introduced as the one entropy-like quantity that survives in type III, where ordinary entropy does not. Now it turns out to be an ordinary entropy, of a different and larger algebra built by attaching a clock.

Worked Example: the clock terms of the entropy formula, from a Gaussian clock

The clock terms $-\bar{q} + S_o$ in (5.13) can be checked on a fully explicit example. Take $\Phi = \Psi$, so the relative-entropy term is zero. Then look only at the operators in $\widehat{\mathcal{M}}$ that are functions of $X \equiv \hat{q} - K$ (in the frame (5.5)). These form a commutative subalgebra, so this is a toy calculation on a subalgebra, not the full entropy of $\widehat{\mathcal{M}}$. It does check where the clock terms come from.

The trace on this subalgebra. Since $K|\Psi\rangle = 0$, a function $f(X)$ acts on $|\widehat{\Psi}\rangle$ as $f(\hat{q})$. With $|\langle q|p=0\rangle|^2 = 1/2\pi$, the trace (5.10) becomes

$$\text{tr } f(X) = \frac{1}{2\pi} \int dq e^{-q} f(q).$$

The density operator. For $\Phi = \Psi$ we have $\Delta_{\Phi\Psi} = \Delta_{\Psi} = e^{-K}$, and (5.12) reduces to $\rho(X) = 2\pi |g(X)|^2 e^X$. Check that it reproduces the state:

$$\text{tr } (\rho(X) f(X)) \stackrel{(1)}{=} \frac{1}{2\pi} \int dq e^{-q} 2\pi |g(q)|^2 e^q f(q) \stackrel{(2)}{=} \int dq |g(q)|^2 f(q).$$

(1) the trace formula just derived, applied to the function ρf . (2) the factors 2π and $e^{\pm q}$ cancel.

The right side is the expectation value of $f(X)$ in $|\Psi\rangle \otimes |g\rangle$, because X acts on that vector as $\hat{q}$ and the clock distribution is $|g(q)|^2$. With $f = 1$ this also gives $\text{tr } \rho = 1$.

The entropy.

$$\begin{aligned} -\text{tr}(\rho \log \rho) &\stackrel{(3)}{=} - \int dq |g(q)|^2 (\log 2\pi + \log |g(q)|^2 + q) \\ &\stackrel{(4)}{=} -\bar{q} + S_o - \log 2\pi. \end{aligned}$$

(3) apply step (2) with $f = \log \rho = \log 2\pi + \log |g|^2 + X$. (4) $\int |g|^2 = 1$, and the definitions of $\bar{q}$ and S_o in (5.13).

The constant $-\log 2\pi$ disappears if the trace is rescaled by 2π . So the clock terms have exactly the shape $-\bar{q} + S_o$.

A Gaussian example. Take a Gaussian clock distribution of width σ centered at $\bar{q}_0$, $|g(q)|^2 = \frac{1}{\sqrt{2\pi}\sigma} \exp(-(q - \bar{q}_0)^2/2\sigma^2)$. Then $\bar{q} = \bar{q}_0$ and $S_o = \frac{1}{2} \log(2\pi e\sigma^2)$, so

$$-\text{tr}(\rho \log \rho) = -\bar{q}_0 + \frac{1}{2} \log(2\pi e\sigma^2) - \log 2\pi.$$

We checked this numerically by doing the q integrals directly (for $\bar{q}_0 = 0.4$, $\sigma = 0.8$), and also checked $\text{tr} \rho = 1$. This example tests the clock terms only. The relative-entropy term, which needs $\Phi \neq \Psi$ and the full non-commutative algebra, is not tested by it.

This completes the operator-algebra toolkit of Chapters 2–5. A type III algebra has no trace and no entropy of its own. But it can always be crossed with its own modular group to produce a type II algebra that has both. And the entropy of that type II algebra is the relative entropy of the original algebra, in disguise. Chapter 6 applies all of these tools (the type classification, modular theory, and the crossed product) to the large- N limit of the AdS/CFT correspondence.

CHAPTER 6

AdS/CFT in the large- N limit: an algebraic formulation

Chapters 2–5 developed their tools on their own terms. They used nothing more exotic than qubits, matrix algebras, and (in Chapter 4) an ordinary free quantum field. This chapter applies all of those tools (the type classification, the GNS construction, modular theory, and the crossed product) to the physical system these notes are aiming at. That system is the AdS/CFT correspondence, in the limit $N \rightarrow \infty$, or equivalently $G_N \rightarrow 0$, where bulk spacetime is supposed to emerge.

6.1 General description

The dictionary, restated precisely

AdS/CFT is a conjecture. It says that quantum gravity on $(d+1)$ -dimensional anti-de Sitter space (AdS) is completely equivalent to an ordinary conformal field theory (CFT), without gravity, living on the d -dimensional boundary of that space. The standard example is type IIB string theory on $\text{AdS}_5 \times S^5$, which is dual to $\mathcal{N} = 4$ super-Yang-Mills theory with gauge group $SU(N)$.

The core entries of the dictionary are these.

- The two theories share the same Hilbert space, $\mathcal{H}_{\text{bulk}} = \mathcal{H}_{\text{CFT}} \equiv \mathcal{H}$.
- The classical-gravity limit $G_N \rightarrow 0$ is the same as the large- N limit of the boundary gauge theory.
- The limit $\alpha' \rightarrow 0$ is the same as the limit of infinite 't Hooft coupling, $\lambda \rightarrow \infty$, in the boundary theory. Here α' sets the string length, so $\alpha' \rightarrow 0$ switches off stringy corrections.
- Elementary bulk fields correspond to **single-trace operators**. These are boundary operators built as a single trace over color indices, $\text{Tr}(\cdots)$.
- A classical bulk geometry ϕ_c corresponds to a specific boundary state $|\Psi\rangle$, called a **semiclassical state**.

The precise map from bulk fields to boundary operators is the **extrapolate dictionary**:

$$O(x) = \lim_{r \rightarrow \infty} r^\Delta \phi(r, x). \quad (6.1)$$

In words: take a bulk field ϕ and evaluate it at radial coordinate r and boundary point x . Multiply by r^Δ , where Δ is the conformal dimension of the operator, a number fixed by the mass of the field.

Then send $r \rightarrow \infty$, which moves the point out to the AdS boundary. What survives this limit is the corresponding boundary operator $O(x)$.

This equation is not just a convenient definition. In some holographic theories an independent definition of $O(x)$ already exists, such as $\text{Tr}(\dots)$ in super-Yang-Mills. There, (6.1) is a derived fact. In more general holographic systems no such independent definition is available. There, (6.1) *is* taken as the definition of what counts as a single-trace operator.

A semiclassical state $|\Psi\rangle$ is one with a well-defined $N \rightarrow \infty$ limit. This means it can be built as the limit of a sequence of states $\{|\Psi_N\rangle\}$, one for each finite- N theory. This chapter works out two examples. The first is the vacuum $|\Omega\rangle$, dual to empty AdS. The second is the thermofield double $|\Psi_\beta\rangle$, which at high enough temperature is dual to an eternal black hole. A third example is a single-sided black hole formed by collapse. It is generally harder to write down explicitly.

Building the boundary operator algebra with GNS, and why it can be bigger than expected

Here the tools of Chapters 2 and 4 are used directly. Say that an operator A has a well-defined large- N limit in the state $|\Psi\rangle$ if two things hold. First, A is the limit of some sequence of finite- N operators $\{A_N\}$, with corresponding states $\{|\Psi_N\rangle\}$. Second, the expectation values converge to something finite:

$$\lim_{N \rightarrow \infty} \langle \Psi_N | A_N | \Psi_N \rangle < \infty.$$

This is the same kind of finite-energy restriction met for the infinite chain of Bell pairs in Chapters 1 and 2, now applied to a real gauge theory. Call the collection of all such operators $\mathcal{A}_\Psi$. One subset is present for every semiclassical state:

$$\mathcal{S} \equiv \text{the } * \text{-algebra generated by single-trace operators} \subseteq \mathcal{A}_\Psi.$$

The inclusion need not be an equality. The set $\mathcal{A}_\Psi$ may contain *more* operators than those built from single-trace operators, and which extra operators survive can depend on the state Ψ . This point is easy to skip on a first reading. But the fact that the inclusion can be strict turns out to be exactly what allows a description of a black-hole *interior*. This is discussed in the section on general black holes below.

Assume that $\mathcal{A}_\Psi$ is closed under products, so it is a $*$ -algebra. Assume also that it is completed in the norm it inherits, so it is a C^* -algebra, exactly the kind of object studied in Chapter 2. Then expectation values in $|\Psi\rangle$ define a state ω_Ψ on $\mathcal{A}_\Psi$. The GNS construction of Chapter 2 applies directly. It gives a GNS Hilbert space $\mathcal{H}_\Psi^{\text{GNS}}$ built from $(\mathcal{A}_\Psi, \omega_\Psi)$, together with a representation $\pi_\Psi(\mathcal{A}_\Psi)$. This Hilbert space is, quite literally, “the space of small excitations around $|\Psi\rangle$.” It is the same construction that Chapter 2 carried out by hand on qubits.

Suppose the bulk geometry dual to Ψ is smooth and Ψ is pure. Then ω_Ψ is expected to be a pure state on $\mathcal{A}_\Psi$. Chapter 2 showed that a pure state gives an irreducible GNS representation. So

$$B(\mathcal{H}_\Psi^{\text{GNS}}) = \pi_\Psi(\mathcal{A}_\Psi)'' . \quad (6.2)$$

This can fail if the dual geometry has a singularity that can be reached on some Cauchy slice. We

return to this in the discussion of firewalls later in this chapter. Write

$$Y \equiv (\pi_\Psi(\mathcal{S}))'', \quad Y_O \equiv (\pi_\Psi(\mathcal{S}_O))''$$

for the von Neumann algebra generated by single-trace operators alone, and for its restriction to a boundary region O . The algebra Y may be strictly smaller than $\pi_\Psi(\mathcal{A}_\Psi)''$.

Matching this to an ordinary bulk Fock space

On the gravity side, expand every bulk field around its classical background: $\phi = \phi_c + \kappa \delta\phi$, with $\kappa = \sqrt{8\pi G_N}$. This gives an action $S[\phi_c] + S_2[\delta\phi] + S_{\text{int}}[\delta\phi]$, with $S_{\text{int}} = \kappa S_3 + \kappa^2 S_4 + \dots$. At leading order, $\kappa \rightarrow 0$, only the quadratic piece S_2 survives. It describes an ordinary free quantum field $\delta\phi$ on the fixed background ϕ_c . Quantizing it in the standard way, with some vacuum $|0\rangle_{\phi_c}$, gives a Fock space $\mathcal{H}_\Psi^{\text{Fock}}$.

We now have two Hilbert spaces. One is built from the boundary algebra by GNS. The other is built from the bulk field theory by ordinary quantization. For the AdS/CFT duality to hold, they must agree:

$$\mathcal{H}_\Psi^{\text{Fock}} = \mathcal{H}_\Psi^{\text{GNS}}. \quad (6.3)$$

This requires the GNS representation to have exactly the structure of a Fock space. In other words, the boundary theory must be Gaussian at large N : all correlators must factorize into products of two-point functions. This property is called “large- N factorization.” It follows from ’t Hooft’s planar scaling, as the following sketch shows.

Key Result / Core Concept: Sketch of the large- N generalized free field and its CCR algebra

Goal: Explain why, in a large- N $SU(N)$ gauge theory with fixed ’t Hooft coupling $\lambda = g_{YM}^2 N$:

1. Connected correlators of n single-trace operators scale as $\langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle_{\text{conn}} \sim N^{2-n}$.
2. Commutators of single-trace operators become c -numbers: $[\mathcal{O}_A, \mathcal{O}_B] = c_{AB} \mathbf{1} + O(1/N)$.
3. The resulting boundary algebra is a CCR algebra, whose GNS space is a free Fock space $\mathcal{H}_\Psi^{\text{Fock}}$.

This is a standard diagrammatic argument, not a rigorous proof.

Derivation sketch:

1. **The ’t Hooft topological expansion.** Consider an adjoint matrix field theory with action $S = \frac{N}{\lambda} \int d^d x \text{Tr} [\frac{1}{2}(\partial\Phi)^2 + V(\Phi)]$. So the propagator carries a factor $g_{YM}^2 = \lambda/N$ and each interaction vertex a factor $1/g_{YM}^2 = N/\lambda$. In ’t Hooft’s double-line notation, a vacuum Feynman graph with V vertices, E edges (propagators), and F closed index loops scales as

$$(g_{YM}^2)^{E-V} N^F \stackrel{(1)}{=} \lambda^{E-V} N^{V-E+F} \stackrel{(2)}{=} \lambda^{E-V} N^\chi \stackrel{(3)}{=} \lambda^{E-V} N^{2-2g}.$$

(1) substitute $g_{YM}^2 = \lambda/N$, so $(g_{YM}^2)^{E-V} = \lambda^{E-V} N^{-(E-V)}$. (2) $\chi \equiv V - E + F$ is the Euler characteristic of the surface on which the double-line graph can be drawn. (3) for a closed orientable surface of genus g , $\chi = 2 - 2g$.

The leading diagrams are planar, with the topology of a sphere ($g = 0$, $\chi = 2$). They contribute at order N^2 .

2. **Scaling of connected correlators.** Define the single-trace operators, with their vacuum value subtracted,

$$\mathcal{O}_i(x) \equiv \text{Tr}(\Phi^{k_i}(x)) - \langle \text{Tr}(\Phi^{k_i}(x)) \rangle.$$

Now consider a connected correlator $\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_{\text{conn}}$. Treat each operator insertion as an extra vertex of the graph, so the graph has V_{int} interaction vertices and n insertions. An interaction vertex carries a factor N/λ from the action. An insertion carries no such factor. Therefore

$$\begin{aligned} \left(\frac{N}{\lambda}\right)^{V_{\text{int}}} \left(\frac{\lambda}{N}\right)^E N^F &\stackrel{(4)}{=} \lambda^{E-V_{\text{int}}} N^{(V_{\text{int}}+n)-E+F} N^{-n} \\ &\stackrel{(5)}{=} \lambda^{E-V_{\text{int}}} N^{\chi-n} \stackrel{(6)}{=} \lambda^{E-V_{\text{int}}} N^{2-n}. \end{aligned}$$

(4) collect the powers of N , and add and subtract n in the exponent. (5) counting the insertions as vertices, the Euler characteristic is $\chi = (V_{\text{int}} + n) - E + F$. (6) a connected planar graph has $\chi = 2$.

So

$$\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_{\text{conn}} \sim N^{2-n}.$$

In terms of canonically normalized fields $\varphi = \Phi/g_{YM}$, the same operators are $\mathcal{O}_i = (\lambda/N)^{k_i/2} \text{Tr} \varphi^{k_i}$ minus its mean. For $k_i = 2$ this is $\lambda N^{-1} \text{Tr} \varphi^2$, the normalization used in the worked example later in this chapter. The leading powers of N are:

- $n = 2$: $\langle \mathcal{O}_1 \mathcal{O}_2 \rangle_{\text{conn}} \sim N^0 = O(1)$. The two-point function is finite and nonzero.
- $n = 3$: $\langle \mathcal{O}_1 \mathcal{O}_2 \mathcal{O}_3 \rangle_{\text{conn}} \sim N^{-1} \xrightarrow{N \rightarrow \infty} 0$.
- $n \geq 3$ in general: $\langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle_{\text{conn}} \sim N^{2-n} \xrightarrow{N \rightarrow \infty} 0$.

3. **Wick's theorem and Gaussian factorization.** All connected correlators with $n \geq 3$ vanish as $N \rightarrow \infty$. By the cumulant expansion, every higher correlation function then splits into a sum of products of two-point functions:

$$\lim_{N \rightarrow \infty} \langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_{2m}(x_{2m}) \rangle = \sum_{\text{pairings } \pi} \prod_{(i,j) \in \pi} \langle \mathcal{O}_i(x_i) \mathcal{O}_j(x_j) \rangle.$$

Correlators with an odd number of operators vanish. A field with this property is called a **generalized free field** (GFF).

4. **Commutators become c -numbers.** Now look at the commutator $[\mathcal{O}_A(x), \mathcal{O}_B(y)]$. Its

expectation value is a number,

$$c_{AB}(x, y) \equiv \langle [\mathcal{O}_A(x), \mathcal{O}_B(y)] \rangle \sim O(1).$$

The quantum fluctuation of the commutator around this number is controlled by connected four-point functions:

$$\begin{aligned} \left\| ([\mathcal{O}_A(x), \mathcal{O}_B(y)] - c_{AB}(x, y)\mathbf{1}) |\Omega\rangle \right\|^2 &\sim \langle \mathcal{O}_A \mathcal{O}_B \mathcal{O}_A \mathcal{O}_B \rangle_{\text{conn}} \\ &\sim N^{2-4} = \frac{1}{N^2} \xrightarrow{N \rightarrow \infty} 0. \end{aligned}$$

The fluctuations vanish as $N \rightarrow \infty$, so in the limit the commutator is a c -number:

$$[\mathcal{O}_A(x), \mathcal{O}_B(y)] = c_{AB}(x, y) \mathbf{1}.$$

5. **A bulk Fock space appears.** A commutator that is a c -number is the canonical commutation relation (CCR) of a free quantum field. The GNS representation of this CCR algebra, built on the cyclic vector $|1\rangle_\Psi$, is a multi-particle Fock space:

$$\mathcal{H}_\Psi^{\text{GNS}} = \overline{\text{span}\{\mathcal{O}_{i_1} \cdots \mathcal{O}_{i_k} |1\rangle_\Psi\}} \cong \mathcal{H}_\Psi^{\text{Fock}}.$$

The boundary single-trace operators act as creation and annihilation operators for the free bulk fluctuations $\delta\phi$. This is the content of (6.3).

It is then natural to identify the two vacua, $|0\rangle_{\phi_c} = |1\rangle_\Psi$. Here $|1\rangle_\Psi$ is the GNS vector that corresponds to the identity operator. It is the same object that was called $|\Omega\rangle$ throughout Chapter 2.

Disjoint sectors: no single Hilbert space survives $N \rightarrow \infty$

This consequence is the large- N version of the obstruction met in Chapter 1. Different semiclassical states correspond to different classical bulk geometries. Their energies typically differ by an amount of order $1/G_N$. This is an ordinary classical-gravity energy difference, and it diverges as $G_N \rightarrow 0$. By contrast, states *within* a single $\mathcal{H}_\Psi^{\text{GNS}}$ differ from Ψ only by energies of order G_N^0 , which is the size of an ordinary quantum fluctuation and stays finite. So the GNS Hilbert spaces $\mathcal{H}_{\Psi_1}$ and $\mathcal{H}_{\Psi_2}$, built around two different classical geometries, cannot overlap at any finite order in perturbation theory in G_N .

There is a further, more subtle point. Two semiclassical states with the *same* energy can still lie in different GNS sectors. This happens if they are separated by an infinite entanglement barrier, exactly the mechanism of Chapter 1.

So in the large- N limit there is no longer one single Hilbert space for the theory. The space of states splits into disjoint sectors, one for each semiclassical background. Each sector has its own GNS Hilbert space and its own emergent operator algebra. This mirrors the $N \rightarrow \infty$ behavior of the entangled spin chain with different angles θ in Chapter 2. There, different

values of θ gave disjoint sectors. Here, different classical geometries give disjoint sectors, for the same underlying reason.

Physics Connection: this is an ordinary superselection rule

“Disjoint GNS sectors that no operator connects” already has a name in elementary quantum mechanics: **superselection**. You already know one example. No physical operator connects a state of one total electric charge to a state of a different total charge. So a superposition such as $|\psi\rangle = \alpha |Q=0\rangle + \beta |Q=1\rangle$ can never be distinguished from the corresponding mixture by any observable. The reason is that every physical operator O must conserve charge, $[O, \hat{Q}] = 0$. This is exactly a block-diagonal condition. Check it on a toy two-level charge operator $\hat{Q} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Write a general operator as $O = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$[O, \hat{Q}] = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & b \\ -c & 0 \end{pmatrix}.$$

Requiring this to vanish gives $b = c = 0$. So O must be block-diagonal in the charge sectors, and no charge-conserving operator can have a nonzero matrix element between them.

Different semiclassical vacua behave in the same way. The role of the conserved charge is played by an energy or entanglement difference that diverges as $N \rightarrow \infty$ (Chapter 1). It forces every operator that survives the large- N limit to have vanishing matrix elements between sectors. The mechanism that produces the split is different: a conserved quantity in one case, an infinite barrier in the other. But the resulting structure is the same superselection structure you already know: disjoint sectors, no operator that crosses between them, and a separate Hilbert space for each sector.

Why boundary time slices carry independent algebras

One more structural fact explains something that would otherwise look paradoxical. At finite N , the algebra of operators on one Cauchy slice determines the algebra everywhere. This is the time-slice axiom of Chapter 4: ordinary time evolution lets you reconstruct operators at any other time from data on one slice. In the strict $N \rightarrow \infty$ limit this is no longer true. The boundary theory becomes a **generalized free field**. Such a field is Gaussian and is specified completely by its two-point function. It has *no* equation of motion that governs its time evolution. As a result, algebras built on different Cauchy slices become inequivalent.

The reason lies in how the boundary Hamiltonian scales with N . The stress tensor has the schematic form $T^{\mu\nu} = N \text{Tr}(\dots)$. It has an overall factor of N relative to the normalized single-trace operators. So the ordinary boundary Hamiltonian $H = \int d^{d-1}x T^{00}$ has no finite $N \rightarrow \infty$ limit in any sector: it diverges. The *rescaled* versions $\hat{T}^{\mu\nu} \equiv T^{\mu\nu}/N$ and $\hat{H} \equiv H/N$ do survive the limit. But they generate only an infinitesimally slow flow, $i[\hat{H}, O(x)] = \frac{1}{N} \partial_t O(x)$. A full time translation of single-trace operators needs the unrescaled H , and that operator does not exist in the limit. **Ordinary time translation, generated by the integral of a local density over one Cauchy slice, does not survive the large- N limit.**

This does not contradict the fact that semiclassical states such as the vacuum clearly have time-translation symmetry. Within a given sector there is a genuine time-translation operator $\hat{h}_\Psi$ satisfying

$$i[\hat{h}_\Psi, O(x)] = \partial_t O(x). \quad (6.4)$$

But $\hat{h}_\Psi$ cannot be written as the integral of a local operator over a single time slice, the way H was. The next section, on the vacuum sector, makes this explicit and constructs $\hat{h}_\Omega$. The same thing happens for any global symmetry, not only time translation. We return to it in the section on $1/N$ corrections, for an internal $U(1)$ charge.

6.2 The vacuum sector

In the vacuum sector every part of the general story can be made explicit and checked. The reason is that here both sides of the duality are known in closed form. The vacuum $|\Omega\rangle$ is dual to empty global AdS, with bulk vacuum $|0\rangle_{\text{AdS}} \in \mathcal{H}_{\text{AdS}}^{\text{Fock}}$. A bulk field has the ordinary mode expansion

$$\phi(X) = \sum_k (u_k(X)a_k + u_k^*(X)a_k^\dagger), \quad a_k |0\rangle_{\text{AdS}} = 0, \quad (6.5)$$

and $\mathcal{H}_{\text{AdS}}^{\text{Fock}}$ is built by acting with creation operators $a_k^\dagger$ on the vacuum, just as for any ordinary Fock space.

On the boundary, vacuum correlators of single-trace operators have the standard large- N scaling: $\langle O \rangle = 0$, $\langle O_1 O_2 \rangle_c \sim O(N^0)$, and $\langle O_1 \cdots O_n \rangle_c \sim N^{2-n}$. So at leading order every higher correlator factorizes into a sum of products of two-point functions. This is exactly the Gaussian, generalized-free-field structure that (6.3) requires. Each single-trace operator behaves like a generalized free field, so $\mathcal{H}_\Omega^{\text{GNS}}$ has the required Fock-space structure. In fact

$$B(\mathcal{H}_\Omega^{\text{GNS}}) = (\pi_\Omega(\mathcal{S}))''.$$

This means that ω_Ω is a *pure* state with respect to $\mathcal{S}$, and $\mathcal{S} = \mathcal{A}_\Omega$. In the vacuum sector, single-trace operators are all there is. Nothing extra survives the large- N limit.

Worked calculation: matrix Wick contractions and large- N factorization

To see how large- N factorization produces a free Fock algebra from a matrix theory, work out the index contractions for an $N \times N$ hermitian matrix field $M_{ij}(x)$ with Gaussian propagator

$$\langle M_{ij}(x) M_{kl}(y) \rangle_0 = \delta_{il} \delta_{jk} G(x, y).$$

Here $G(x, y)$ is a scalar two-point function. Define the gauge-invariant, normalized single-trace operator

$$\mathcal{O}(x) \equiv \frac{1}{N} \text{Tr} (M(x)^2) = \frac{1}{N} \sum_{i,j=1}^N M_{ij}(x) M_{ji}(x),$$

understood as normal-ordered. This means contractions between the two M 's inside the same $\mathcal{O}$ are left out. Equivalently, the vacuum value $\langle \mathcal{O} \rangle_0 = N G(x, x)$ has been subtracted. The two-point correlator is

$$\langle \mathcal{O}(x) \mathcal{O}(y) \rangle_0 = \frac{1}{N^2} \sum_{i,j,k,l=1}^N \langle M_{ij}(x) M_{ji}(x) M_{kl}(y) M_{lk}(y) \rangle_0.$$

By Wick's theorem, there are two ways to contract each $M(x)$ with an $M(y)$:

1. Contract $M_{ij}(x)$ with $M_{kl}(y)$, and $M_{ji}(x)$ with $M_{lk}(y)$:

$$\langle M_{ij}(x) M_{kl}(y) \rangle_0 \langle M_{ji}(x) M_{lk}(y) \rangle_0 \stackrel{(1)}{=} (\delta_{il} \delta_{jk} G(x, y)) (\delta_{jk} \delta_{il} G(x, y)) \stackrel{(2)}{=} \delta_{il} \delta_{jk} G(x, y)^2.$$

(1) the Gaussian propagator, applied to each contracted pair. (2) a Kronecker delta squared equals itself, $\delta_{il} \delta_{il} = \delta_{il}$.

Summing over all four indices gives

$$\sum_{i,j,k,l} \delta_{il} \delta_{jk} = \sum_{i,j=1}^N 1 = N^2.$$

2. Contract $M_{ij}(x)$ with $M_{lk}(y)$, and $M_{ji}(x)$ with $M_{kl}(y)$:

$$\langle M_{ij}(x) M_{lk}(y) \rangle_0 \langle M_{ji}(x) M_{kl}(y) \rangle_0 \stackrel{(3)}{=} (\delta_{ik} \delta_{jl} G(x, y)) (\delta_{jl} \delta_{ik} G(x, y)) \stackrel{(4)}{=} \delta_{ik} \delta_{jl} G(x, y)^2.$$

(3) the Gaussian propagator again, with the indices of the second contraction pattern. (4) a Kronecker delta squared equals itself.

Summing over the indices again gives $\sum_{i,j} 1 = N^2$.

Now multiply by the normalization $\frac{1}{N^2}$:

$$\langle \mathcal{O}(x) \mathcal{O}(y) \rangle_0 = \frac{1}{N^2} (N^2 G(x, y)^2 + N^2 G(x, y)^2) = 2 G(x, y)^2 \sim O(N^0).$$

Next consider the four-point function $\langle \mathcal{O}(x_1) \mathcal{O}(x_2) \mathcal{O}(x_3) \mathcal{O}(x_4) \rangle_0$. Its contractions fall into two classes.

- **Disconnected contractions.** These contract the single traces in pairs, for example $\mathcal{O}_1$ with $\mathcal{O}_2$ and $\mathcal{O}_3$ with $\mathcal{O}_4$:

$$\langle \mathcal{O}(x_1) \mathcal{O}(x_2) \rangle_0 \langle \mathcal{O}(x_3) \mathcal{O}(x_4) \rangle_0 = (2G(x_1, x_2)^2) (2G(x_3, x_4)^2) \sim O(N^0).$$

There are three such pairings: (12)(34), (13)(24), and (14)(23).

- **Connected contractions.** These link all four operators in a single ring, for example $M(x_1) \rightarrow M(x_2) \rightarrow M(x_3) \rightarrow M(x_4) \rightarrow M(x_1)$. In double-line notation the ring has two closed index loops, one running around its inside and one around its outside. That gives two free index sums and a factor N^2 . With four normalization factors of $1/N$, the connected

four-point function scales as

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4) \rangle_c \sim \frac{1}{N^4} \times N^2 = \frac{1}{N^2} \rightarrow 0.$$

This agrees with the general planar scaling $\langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle_c \sim N^{2-n}$ at $n = 4$. The same scaling holds when planar gauge interactions are switched on, at fixed 't Hooft coupling $\lambda = g_{\text{YM}}^2 N$.

As a check at coincident points, with $G = 1$: here $\text{Tr } M^2$ is a sum of independent squared Gaussian entries, and adding up their cumulants gives exactly $\langle \mathcal{O}^2 \rangle_0 = 2$ and a fourth cumulant $48/N^2$. A Monte Carlo sample of random Hermitian matrices reproduces the variance 2.

So at strictly $N \rightarrow \infty$,

$$\begin{aligned} \langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4) \rangle_0 &= \langle \mathcal{O}_1\mathcal{O}_2 \rangle_0 \langle \mathcal{O}_3\mathcal{O}_4 \rangle_0 + \langle \mathcal{O}_1\mathcal{O}_3 \rangle_0 \langle \mathcal{O}_2\mathcal{O}_4 \rangle_0 \\ &\quad + \langle \mathcal{O}_1\mathcal{O}_4 \rangle_0 \langle \mathcal{O}_2\mathcal{O}_3 \rangle_0 + O(1/N^2). \end{aligned}$$

Connected n -point correlators vanish for all $n \geq 3$. So the single-trace operators obey Wick's theorem exactly in the limit, and they generate a generalized-free-field Fock space $\mathcal{H}_\Omega^{\text{GNS}}$.

The match between bulk and boundary mode expansions can be made explicit. Expand a single-trace operator directly on the GNS Hilbert space:

$$\pi_\Omega(O(x)) = \sum_k (v_k(x)b_k + v_k^*(x)b_k^\dagger), \quad b_k |1\rangle_\Omega = 0.$$

Match the boundary mode functions to the boundary limits of the bulk ones:

$$v_k(x) = \lim_{r \rightarrow \infty} r^\Delta u_k(X).$$

The extrapolate dictionary (6.1) then forces the mode operators to be the same, $a_k = b_k$. This establishes (6.3) explicitly in this one solvable case. Written in position space, the identification becomes

$$\phi(X) = \int d^d x K(X; x) \pi_\Omega(O(x)). \quad (6.6)$$

This is the **global HKLL construction**, named after Hamilton, Kabat, Lifschytz and Lowe. The kernel K is explicit, and it reconstructs the bulk field everywhere in AdS from boundary single-trace operators.

Worked derivation: the HKLL smearing kernel in AdS_3

To make the HKLL kernel $K(X; x)$ concrete, consider a free massless scalar field $\phi(z, t, x)$ in Poincaré AdS_3 with metric

$$ds^2 = \frac{R^2}{z^2} (dz^2 - dt^2 + dx^2), \quad z > 0.$$

The boundary is at $z = 0$. For $m^2 = 0$ the conformal dimension is $\Delta = 2$. The bulk Klein–Gordon equation $(\square - m^2)\phi = 0$, multiplied through by R^2/z^2 , reads

$$z \partial_z \left(\frac{1}{z} \partial_z \phi \right) - \partial_t^2 \phi + \partial_x^2 \phi = \partial_z^2 \phi - \frac{1}{z} \partial_z \phi - \partial_t^2 \phi + \partial_x^2 \phi = 0.$$

The second form follows from the product rule applied to $\partial_z(z^{-1}\partial_z\phi)$.

Go to Fourier space, with boundary momentum (ω, k) chosen timelike, $\omega^2 - k^2 \equiv q^2 > 0$. For $\phi(z, t, x) = f(z)e^{-i\omega t + ikx}$ the mode equation is

$$f''(z) - \frac{1}{z}f'(z) + q^2 f(z) = 0.$$

Setting $f(z) = z g(z)$ turns this into Bessel's equation of order 1 for $g(z)$. The solution that is regular in the interior is J_1 :

$$g''(z) + \frac{1}{z}g'(z) + \left(q^2 - \frac{1}{z^2}\right)g(z) = 0 \implies f(z) = C z J_1(qz).$$

Near the boundary, $z \rightarrow 0$, we have $J_1(qz) \approx \frac{qz}{2}$, so $f(z) \approx C \frac{q}{2} z^2$. The extrapolate dictionary (in Poincaré coordinates, $z^{-\Delta}$ plays the role of r^Δ) requires $\lim_{z \rightarrow 0} z^{-2} \phi(z, t, x) = \mathcal{O}(t, x) = e^{-i\omega t + ikx}$. This fixes $C = \frac{2}{q}$. Hence

$$\phi(z, t, x) = \frac{2z}{q} J_1(qz) \mathcal{O}(t, x).$$

Now go back to position space. The result is a smearing of $\mathcal{O}$ over a disk of radius z , with one of the boundary coordinates continued to imaginary values:

$$\phi(z, t, x) = \frac{1}{\pi} \int_{t'^2 + y'^2 < z^2} dt' dy' \mathcal{O}(t + t', x + iy').$$

To check it, insert $\mathcal{O}(t, x) = e^{-i\omega t + ikx}$:

$$\begin{aligned} & \frac{1}{\pi} \int_{t'^2 + y'^2 < z^2} dt' dy' e^{-i\omega(t+t') + ik(x+iy')} \\ & \stackrel{(1)}{=} \frac{e^{-i\omega t + ikx}}{\pi} \int_{t'^2 + y'^2 < z^2} dt' dy' e^{-i\omega t' - ky'} \\ & \stackrel{(2)}{=} \frac{e^{-i\omega t + ikx}}{\pi} \cdot \frac{2\pi z J_1(qz)}{q} \stackrel{(3)}{=} \frac{2z}{q} J_1(qz) \mathcal{O}(t, x). \end{aligned}$$

(1) pull the factor that does not depend on t', y' out of the integral. **(2)** the disk integral $\int_{|u| < z} d^2 u e^{ia \cdot u} = 2\pi z J_1(|a|z)/|a|$, continued to $a = (-\omega, ik)$, for which $a \cdot a = \omega^2 - k^2 = q^2$. **(3)** $e^{-i\omega t + ikx} = \mathcal{O}(t, x)$.

We also checked step (2) numerically for sample values of z, ω, k . The contour can be rotated back to real boundary coordinates. The smearing region then becomes the set of boundary points spacelike-separated from the bulk point, which is not compact, and the real-space kernel needs more care; we do not need its explicit form.

The disk formula has a clear physical meaning. To evaluate the local bulk operator $\phi(z, t, x)$ at radial depth z , one must integrate the boundary operator $\mathcal{O}$ over a region of size z . The deeper the operator sits in the bulk (the larger z), the larger the boundary region needed to reconstruct it. This is a direct example of the holographic UV/IR relation.

A further fact follows from the representation theory of the boundary conformal group. In global AdS the single-particle energy levels are evenly spaced: in units of $1/R$ they are $\Delta + 2n + l$, so successive radial excitations are spaced by 2. From this one can show the following result, which we state without proof:

$$B(\mathcal{H}_\Omega^{\text{GNS}}) = Y_{I_w}, \quad w \geq \pi R. \quad (6.7)$$

Here I_w is a boundary time band of width w . **The entire operator content of the bulk vacuum sector, meaning every bulk field everywhere in AdS, is already generated by boundary operators smeared over a time band of width πR .** You do not need the whole boundary history, or an infinite band. A band of width πR is already enough to reconstruct the entire bulk. This width is half of $2\pi R$, the time after which every single-particle mode returns to itself up to an overall phase. This is the clearest and most concrete preview of subregion-subalgebra duality, which Chapter 7 develops as a general principle.

Finally, the vacuum-sector time-translation operator $\hat{h}_\Omega$ from the previous section can be written down explicitly. It is the bulk energy integral over a Cauchy slice, schematically

$$\hat{h}_\Omega = \int d\rho d^{d-1}\Omega \rho^{d-1} T_{tt},$$

with the appropriate metric factors. Here T_{tt} is the bulk stress tensor, which is quadratic in ϕ at this order. Substitute the mode expansion (6.5) and use $a_k = b_k$. This turns $\hat{h}_\Omega$ into a specific quadratic expression in boundary single-trace operators, which is an operator satisfying (6.4). Consistent with (6.7), this expression involves integrating boundary operators over a time band of width πR , not a single instant.

6.3 The thermofield double state

Setup, and the Hawking–Page transition

Take two copies of the boundary CFT, CFT_R and CFT_L . At finite N , build the **thermofield double** (TFD) state

$$|\Psi_\beta\rangle = \frac{1}{\sqrt{Z_\beta}} \sum_n e^{-\beta E_n/2} |n\rangle_R |\Theta n\rangle_L, \quad Z_\beta = \sum_n e^{-\beta E_n}. \quad (6.8)$$

Here Θ is the CRT operator (charge conjugation, reflection, and time reversal). It is needed to match energy eigenstates correctly across the two copies. Tracing out L gives the ordinary thermal density matrix $\rho_\beta = \frac{1}{Z_\beta} e^{-\beta H_R}$ on R .

This is the finite-dimensional worked example of Chapter 4, now applied to a field theory. The

algebra $B(\mathcal{H}_R)$ is type I, $|\Psi_\beta\rangle$ is cyclic and separating for it, and the modular operator is

$$-\log \Delta_\beta = \beta(H_R - H_L). \quad (6.9)$$

This is the formula $\Delta_\Psi = \rho_R \otimes \rho_L^{-1}$ from Chapter 4, with $\rho_R = e^{-\beta H_R}/Z_\beta$ and $\rho_L = e^{-\beta H_L}/Z_\beta$. The factors of Z_β cancel in the logarithm.

In the large- N limit this system has a first-order phase transition at the **Hawking–Page temperature** T_{HP} . The free energy jumps in its scaling with N : it is of order N^0 below T_{HP} and of order N^2 above it. The dual bulk geometry jumps at the same point. Below T_{HP} it is *thermal AdS*, a thermal gas of particles in ordinary global AdS with no black hole. Above T_{HP} it is an *eternal black hole*.

Below T_{HP} . The energies that contribute to the sum in (6.8) stay of order N^0 , because higher energies are suppressed by the Boltzmann factor. So in the large- N limit these states lie inside the vacuum-sector GNS Hilbert spaces $\mathcal{H}_\Omega^R$ and $\mathcal{H}_\Omega^L$ built in the previous section. The GNS Hilbert space built from $|\Psi_\beta\rangle$ factorizes, $\mathcal{H}_{\Psi_\beta}^{\text{GNS}} = \mathcal{H}_\Omega^R \otimes \mathcal{H}_\Omega^L$. Correspondingly $Y_R = B(\mathcal{H}_\Omega^R)$ and $Y_L = B(\mathcal{H}_\Omega^L)$. **So Y_R remains type I.** The bulk dual is two separate copies of global AdS, with no geometric connection between them. The only link between them is the entanglement carried by the TFD state. The bulk geometry is disconnected.

Above T_{HP} . Now the thermal ensemble is dominated by states of energy of order N^2 . The finite- N sum (6.8) has no sensible limit as $N \rightarrow \infty$: neither the energies nor the eigenstates converge. In fact the one-point function of a single-trace operator in this state *diverges*, $\langle \Psi_\beta | O | \Psi_\beta \rangle \sim O(N)$. The fix is to use “renormalized,” mean-subtracted operators $\hat{O} \equiv O - \langle \Psi_\beta | O | \Psi_\beta \rangle$. Their connected correlators have sensible large- N limits of order N^0 , and they again obey large- N factorization. This produces a good GNS Hilbert space with a Fock-space structure. The resulting algebras $Y_R \equiv (\pi_{\Psi_\beta}(\mathcal{S}_\beta^{(R)}))''$ satisfy $Y'_R = Y_L$.

But now the entanglement between CFT_R and CFT_L has jumped to order N^2 . Because of this, it has been argued that Y_R **becomes type III₁**, and $\mathcal{H}_{\Psi_\beta}^{\text{GNS}}$ no longer factorizes into separate R and L pieces. On the gravity side this matches the eternal black-hole geometry (Fig. 6.1). The algebras Y_R, Y_L are identified with the bulk exterior algebras $\widetilde{\mathcal{M}}_R, \widetilde{\mathcal{M}}_L$. These are type III₁ for a simple reason: they are algebras of subregions in an ordinary continuum quantum field theory, which is the local-algebra story of Chapter 4.

Two puzzles, and how algebra resolves the first of them

The $T > T_{\text{HP}}$ phase is identified with an eternal black hole, and this identification passes every quantitative check. Even so, before the algebraic reformulation it carried two conceptual puzzles. Both are long-standing and well known in holography.

The factorization puzzle. The boundary theory clearly factorizes as $\text{CFT}_R \otimes \text{CFT}_L$. It is just two separate, non-interacting copies of the same CFT. But the bulk black hole is a single, *connected* spacetime. There are bulk operators, such as a Wilson line stretching from the left boundary through the interior to the right boundary (Fig. 6.1), that do not seem to split into an operator in $B(\mathcal{H}_R)$ times an operator in $B(\mathcal{H}_L)$. How can a connected bulk object correspond to anything in a boundary tensor product that is clearly disconnected?

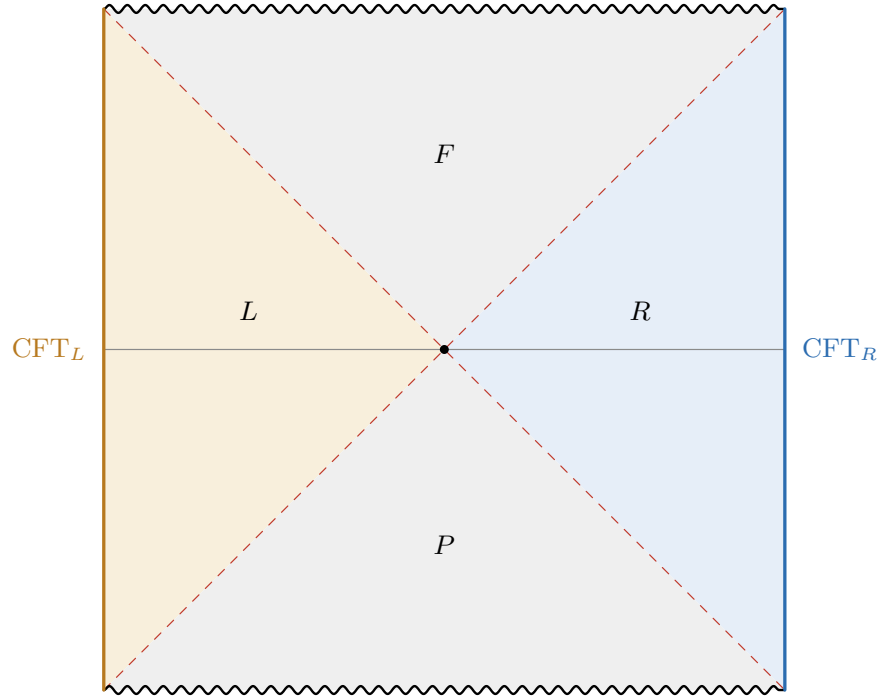


Figure 6.1: Penrose diagram of the two-sided eternal AdS black hole, dual to the thermofield double state $|\Psi_\beta\rangle$. The exterior regions R (blue) and L (gold) end on the AdS boundaries where CFT_R and CFT_L live, and are cut off from the interior regions F and P by the event horizons (dashed red), which cross at the bifurcation point (black dot). The wavy lines are the future and past singularities, and the gray horizontal line is the $t = 0$ slice, which runs from one boundary to the other through the Einstein–Rosen bridge.

The meeting-behind-the-horizon puzzle. There is no interaction at all between CFT_R and CFT_L : the total Hamiltonian is simply $H_R + H_L$, with no cross term. Yet bulk degrees of freedom that start separately in the R and L exterior regions can cross the horizon and interact with each other in the interior. How can two boundary theories that are disconnected, both causally and dynamically, produce bulk physics that is connected once you look behind the horizon?

The algebraic picture resolves the *first* puzzle directly. Once $N \rightarrow \infty$ is taken seriously, “ $B(\mathcal{H}_R)$ ” is the wrong object to compare with the bulk exterior algebra. The actual large- N limit of the algebra is Y_R . It is type III₁, not type I, precisely because of the order- N^2 entanglement between R and L . For a type III₁ algebra and its commutant, there is no tensor product structure $Y_R \otimes Y_L$ inside $B(\mathcal{H}^{\text{GNS}})$ of the kind an ordinary tensor product would give. So nothing prevents the presence of operators (such as the boundary counterpart of that Wilson line) that are not products of an R -piece and an L -piece. The apparent contradiction came from using type I tensor-product intuition in a regime where the algebra has already become type III₁.

The second puzzle, meeting behind the horizon, needs more tools. It needs the emergent commutant and the half-sided modular inclusion structure of Chapter 4. It is taken up in Chapter 8, where the bulk causal structure of the eternal black hole is built directly from boundary algebra data.

Three further remarks prepare the ground for algebraic ER=EPR in Chapter 8.

First, the whole qualitative conclusion here (disconnected bulk geometry below T_{HP} , connected black hole above it) could in principle have been *deduced* just from watching Y_R jump from type I to type III₁. No independent knowledge of the bulk geometry is needed.

Second, for $T > T_{\text{HP}}$ different temperatures β lie in different, non-overlapping GNS sectors. Their entanglement entropies differ by an amount of order N^2 , which is an infinite barrier in the strict limit (the mechanism of Chapter 1 again). Below T_{HP} , by contrast, every β shares one common GNS Hilbert space, and the different TFD states are just different vectors inside it.

Third, the ordinary Hamiltonians H_R, H_L do not survive the large- N limit as elements of the algebra. So (6.9) stops holding literally. But the modular operator Δ_β itself *does* survive, and it still generates boundary time translation in opposite directions on the two sides. Below T_{HP} it splits into an R part and an L part, $-\log \Delta_\beta = \beta(\hat{h}_R - \hat{h}_L)$, where $\hat{h}$ is the vacuum-sector time-translation operator of the previous section. Above T_{HP} it does not split at all. This tracks the transition from type I to type III₁ exactly.

6.4 General black holes

The thermofield double is special. It has an exact bifurcate horizon: the future and past horizons meet at a single surface in a perfectly symmetric way. A more generic two-sided semiclassical state $|\Psi\rangle$ is dual to a “long” black hole. Such a black hole has a genuine interior region I that separates R and L , and its horizons do not meet at a single bifurcation surface.

This case shows the subtlety noted at the start of this chapter: the inclusion $\mathcal{S} \subseteq \mathcal{A}_\Psi$ can now be *strict*. The renormalized single-trace algebras still reproduce the bulk exterior algebras, $Y_R = \widetilde{\mathcal{M}}_R$ and $Y_L = \widetilde{\mathcal{M}}_L$. But R and L together no longer cover a full Cauchy slice. The interior region I is missing. Bulk operators in I must still correspond to *some* boundary operators that survive

the large- N limit, since they belong to the same physical bulk field as the exterior operators, only evaluated at a different place. But they are not single-trace operators. They must belong to $\mathcal{A}_\Psi \setminus \mathcal{S}$.

Chapter 7 identifies exactly what these extra operators are. They are generated from single-trace operators by *modular flow*. This mechanism was previewed by the ergodic lemma of Chapter 4: a modular flow, run long enough, generates a whole larger algebra starting from a smaller subalgebra.

The same phenomenon (extra, non-single-trace operators needed for a black-hole interior) also occurs for a “long” one-sided black hole. A one-sided black hole formed by ordinary gravitational collapse, with matter falling in from empty space, is different. This contrast matters in Chapter 7. There, $\mathcal{A}_\Psi = \mathcal{S}$ exactly: single-trace operators are already everything, and nothing extra is needed.

6.5 A diagnostic of firewalls in generic highly excited states

Now consider a completely generic, highly excited state $|\Psi\rangle$ with energy $E_\Psi \sim O(N^2)$. We do not assume that it was prepared to have a nice bulk dual. It is just an arbitrary state of that energy. The boundary theory at such energies is expected to be chaotic. A typical state of that energy should then look thermal to any probe made of single-trace operators, to leading order in $1/N$:

$$\langle \Psi | O | \Psi \rangle \approx \langle O \rangle_\beta, \quad \langle \Psi | O_1 \cdots O_n | \Psi \rangle \approx \langle O_1 \cdots O_n \rangle_\beta, \quad (6.10)$$

with β fixed by matching the energy, $E_\Psi = E_\beta$. This is the familiar statement, in the spirit of the eigenstate thermalization hypothesis, that a single generic high-energy state reproduces thermal correlators. It says that, as far as single-trace operators can tell, Ψ has a smooth black-hole exterior. But does it also have a smooth *interior*, a genuine continuation of the geometry behind the horizon? Or does the interior simply fail to exist? The second possibility is called a **firewall**.

The previous section already gave the criterion needed to answer this. A smooth interior requires operators that survive the large- N limit *beyond* single-trace operators, that is, $\mathcal{S} \subsetneq \mathcal{A}_\Psi$. This gives a clean, checkable diagnostic. **Suppose a generic highly excited state satisfies the thermal condition (6.10) and also has $\mathcal{A}_\Psi = \mathcal{S}$, so that no operators beyond single-trace ones survive the limit. Then its bulk dual has a firewall.** In this case ω_Ψ is a *mixed* state on $\mathcal{A}_\Psi$. This violates the purity condition (6.2) assumed earlier for the “nice” semiclassical case.

For two-sided generic states there is an analogous statement with one more condition. The correlations between the two sides must vanish, $\langle \Psi | O_R O_L | \Psi \rangle_c \rightarrow 0$ as $N \rightarrow \infty$. This is consistent with there being no smooth, connected bridge between the two boundaries in that case.

6.6 Perturbative $1/N$ corrections

Corrections to bulk reconstruction

Everything so far worked at strictly leading order in G_N , or equivalently in $1/N$. Including subleading orders makes the bulk field theory interacting: the terms $\kappa S_3 + \kappa^2 S_4 + \cdots$ in the bulk action switch back on. Correspondingly, the boundary theory develops nonzero connected three-point and higher

functions, suppressed by powers of $1/N$.

One direct consequence is that the HKLL formula (6.6), exact at leading order, gets corrected. Take the simplest case of a cubic self-interaction ϕ^3 and solve the interacting bulk equation of motion perturbatively in κ . (We do not reproduce this computation.) The result is a series of extra terms in the bulk-to-boundary map:

$$\begin{aligned}\phi(X) = & \int d^d x K(X; x) O^{(0)}(x) \\ & + \kappa \int d^d x_1 d^d x_2 K(X; x_1, x_2) O^{(0)}(x_1) O^{(0)}(x_2) + \cdots\end{aligned}$$

The leading-order reconstruction picks up multi-trace corrections, with one extra factor of a single-trace operator for each order in $1/N$. These corrections underlie the topic of “ $1/N$ corrections to bulk locality,” a substantial research area of its own. We do not need to track every term here.

The structural point that matters for everything downstream is this. To any finite order in the $1/N$ expansion, the *type* of each algebra discussed above is expected to stay the same as at leading order. The heuristic reason is that the spectrum of the modular operator is controlled by its zeroth-order piece. On this picture, perturbation theory in $1/N$ does not change an algebra’s type; only the strict $N \rightarrow \infty$ limit can do that.

Conserved charges, and where gravitational dressing first appears

The second new feature matters more conceptually, for everything from Chapter 9 onward. Suppose the boundary CFT has a global $U(1)$ symmetry. Its conserved charge must be rescaled, $\hat{Q} \equiv Q/N$, in exactly the way $\hat{H} = H/N$ was, because the current itself scales as $N \text{Tr}(\cdots)$. For a single-trace operator of unit charge, with the convention $[Q, O(x)] = O(x)$,

$$[\hat{Q}, O(x)] = \frac{1}{N} O(x). \quad (6.11)$$

So the action of the rescaled charge is suppressed by $1/N$, exactly as in the time-translation story earlier in this chapter. Within a given sector there is again a genuine large- N operator $\hat{q}$ that generates the full rotation, $[\hat{q}, O(x)] = O(x)$. Like $\hat{h}_\Omega$ before it, $\hat{q}$ cannot be written as the integral of a local density over a single time slice.

The bulk dual of this boundary global $U(1)$ symmetry is a bulk $U(1)$ *gauge* symmetry. Here something new and physically important appears once $1/N$ corrections are included. Translate the suppressed commutator (6.11) to the bulk with the extrapolate dictionary. It forces the bulk gauge field strength and a charged bulk scalar to have a *nonzero commutator at spacelike separation*. This is an explicit violation of naive bulk microcausality, order by order in $1/N$.

The violation comes directly from the bulk Gauss-law constraint. The equation of motion $\nabla_M F^{MN} = \kappa J^N$ ties the electric field on a Cauchy slice to the charges on that same slice, because Gauss’s law is intrinsically nonlocal. The resolution comes back as the central mechanism of Chapter 9. A bulk charged field $\phi(z, x)$ is not gauge invariant on its own. To build a gauge-invariant operator, one

attaches a **Wilson line** that runs from the boundary to the point:

$$\Phi(z, x) = e^{-i\kappa V} \phi(z, x), \quad V = \int_0^z dz' A_z(z', x). \quad (6.12)$$

This is a **dressed observable**. The Wilson line is nonlocal: it depends on the gauge field along a whole path, not just at one point. This nonlocality produces the nonlocal commutation relations directly, and nothing mysterious is left over.

Worked calculation: Gauss's law and the dressed operator commutator

To see this mechanism explicitly in canonical quantization, consider the bulk gauge field $A_M = (A_t, A_z, A_x)$ on a constant-time Cauchy slice, in the gauge $A_t = 0$. We suppress the factors of the AdS metric; they do not affect the structure of the argument. The momentum conjugate to $A_z(z, x)$ is the electric field $E^z(z, x) = F^{zt}(z, x) = \partial_t A_z - \partial_z A_t$. The canonical commutation relation is

$$[A_z(z', x'), F^{zt}(z'', x'')] = i \delta(z' - z'') \delta(x' - x'').$$

The field F^{xt} is conjugate to A_x , so it commutes with A_z .

In the quantum theory, Gauss's law is a constraint on physical operators. Define the Gauss operator

$$G(z'', x'') \equiv \partial_{z''} F^{zt}(z'', x'') + \partial_{x''} F^{xt}(z'', x'') - \kappa J^t(z'', x'').$$

A physical, gauge-invariant operator must commute with G at every point. Take $\phi(z, x)$ to create one unit of charge at (z, x) , so that $[J^t(z'', x''), \phi(z, x)] = \delta(z'' - z) \delta(x'' - x) \phi(z, x)$.

The naive, undressed field fails this test. It commutes with the gauge field, $[\phi(z, x), F^{Mt}] = 0$, so only the charge term contributes: $[G(z'', x''), \phi(z, x)] = -\kappa \delta(z'' - z) \delta(x'' - x) \phi(z, x) \neq 0$. So ϕ is not gauge invariant.

Now compute with the dressed operator $\Phi(z, x) = e^{-i\kappa V(z, x)} \phi(z, x)$, where $V(z, x) = \int_0^z dz' A_z(z', x)$. First, the commutator with the electric field:

$$\begin{aligned} [\Phi(z, x), F^{zt}(z'', x'')] &\stackrel{(1)}{=} \left[e^{-i\kappa \int_0^z dz' A_z(z', x)}, F^{zt}(z'', x'') \right] \phi(z, x) \\ &\stackrel{(2)}{=} -i\kappa \left(\int_0^z dz' [A_z(z', x), F^{zt}(z'', x'')] \right) \Phi(z, x) \\ &\stackrel{(3)}{=} -i\kappa \left(\int_0^z dz' i \delta(z' - z'') \delta(x - x'') \right) \Phi(z, x) \\ &\stackrel{(4)}{=} \kappa \theta(z - z'') \delta(x - x'') \Phi(z, x). \end{aligned}$$

(1) move $\phi(z, x)$ out of the commutator. It commutes with F^{zt} , so only the Wilson-line factor has to be commuted. **(2)** $[V, F^{zt}]$ is a c -number, so $[e^{-i\kappa V}, F^{zt}] = -i\kappa [V, F^{zt}] e^{-i\kappa V}$ exactly. Then recombine $e^{-i\kappa V} \phi = \Phi$. **(3)** substitute the canonical commutator $[A_z(z', x), F^{zt}(z'', x'')] = i\delta(z' - z'')\delta(x - x'')$. **(4)** $(-i)(i) = 1$. The $\delta(z' - z'')$ collapses the z' integral. The result is 1 if $0 < z'' < z$ and 0 if $z'' > z$, which is the step function $\theta(z - z'')$.

Next, differentiate with respect to z'' :

$$\begin{aligned}\partial_{z''} [\Phi(z, x), F^{zt}(z'', x'')] &\stackrel{(5)}{=} \kappa \frac{d}{dz''} \theta(z - z'') \delta(x - x'') \Phi(z, x) \\ &\stackrel{(6)}{=} -\kappa \delta(z'' - z) \delta(x'' - x) \Phi(z, x).\end{aligned}$$

(5) differentiate the result of step (4). Only the step function depends on z'' . (6) $\frac{d}{dz''} \theta(z - z'') = -\delta(z'' - z)$, the derivative of a step function.

Finally, assemble the Gauss operator:

$$\begin{aligned}[G(z'', x''), \Phi(z, x)] &\stackrel{(7)}{=} -\partial_{z''} [\Phi(z, x), F^{zt}(z'', x'')] - \kappa [J^t(z'', x''), \Phi(z, x)] \\ &\stackrel{(8)}{=} \kappa \delta(z'' - z) \delta(x'' - x) \Phi(z, x) \\ &\quad - \kappa \delta(z'' - z) \delta(x'' - x) \Phi(z, x) = 0.\end{aligned}$$

(7) $[\partial F, \Phi] = -\partial[\Phi, F]$. The F^{xt} term drops out because Φ contains only A_z and ϕ , both of which commute with F^{xt} . (8) use step (6) for the first term. For the second, J^t commutes with the gauge field, so $[J^t, \Phi] = e^{-i\kappa V} [J^t, \phi] = \delta \Phi$.

So Gauss's law holds identically as an operator statement for the dressed field. The price is visible in step (4). For every point $z'' < z$ along the string that runs from the boundary to the insertion point, the commutator with the electric field is nonzero, even though that point is spacelike-separated from (z, x) . The nonlocal string of the Wilson line is the physical cost of gauge invariance.

The same story holds for spacetime symmetries in place of an internal $U(1)$. To make a bulk operator diffeomorphism-invariant, one must dress it to the boundary with a *gravitational* Wilson line. Equivalently, one can work in a specific gauge and solve the constraint directly, as in the Gauss-law calculation above. This dressing produces an analogous $1/N$ correction to the rescaled Hamiltonian, $\hat{H} = \hat{H}^{(0)} + \frac{1}{N} \hat{h}_\Omega + \dots$. **This is the first appearance, in perturbation theory, of the mechanism that Chapter 9 turns into an exact, nonperturbative construction.** Dressing an operator gravitationally to a physical reference is what makes it a genuine, gauge-invariant, physical observable. Here the reference is a Wilson line; in Chapter 9 it is an observer's own clock, through the crossed product of Chapter 5. The dressing is not optional decoration. It is forced by the bulk Gauss-law constraint that any theory with gravity or gauge fields must satisfy.

The large- N algebraic structure of AdS/CFT is now in place: semiclassical states as GNS sectors, single-trace algebras Y_O , a change of algebra type that tracks the Hawking–Page transition, and gravitational dressing already visible in perturbation theory. Chapter 7 assembles all of this into the central physical claim of these notes, **subregion-subalgebra duality**. This is the statement that a bulk causal region is identical, not merely related, to a specific emergent boundary operator subalgebra.

CHAPTER 7

Subregion-subalgebra duality

Chapters 2–6 built the tools for the idea of this chapter. The claim is simple to state, and its consequences are large: **in the strict large- N limit, an arbitrary bulk spacetime region is not merely *related to* a boundary operator subalgebra; it *is* one.** This is not an approximation, and there is no correction term to add at this order. The algebra of the bulk region and the boundary algebra are the same mathematical object, described in two different languages.

7.1 General formulation

Where the identification comes from

Chapter 6 identified two Hilbert spaces. The first is the bulk Fock space. It is built by ordinary quantization of small fluctuations around a classical background. The second is the boundary GNS Hilbert space. It is built, through the GNS construction, from the algebra of boundary operators that survive the large- N limit. Since these are the same Hilbert space, the algebras of all bounded operators on them are also the same:

$$B(\mathcal{H}_{\Psi}^{\text{Fock}}) = B(\mathcal{H}_{\Psi}^{\text{GNS}}). \quad (7.1)$$

This equation hides a small surprise. $B(\mathcal{H}_{\Psi}^{\text{Fock}})$ is naturally built from data on a single bulk Cauchy slice, because canonical quantization only needs one moment of time. $B(\mathcal{H}_{\Psi}^{\text{GNS}})$ is built from operators on the entire boundary spacetime, or, by the time-band result of Chapter 6, at least on a finite time band. These look like objects of different sizes. They are reconciled by the fact that the bulk has one more spatial dimension than the boundary. The data of this extra dimension, spread over one bulk Cauchy slice, is matched by a stretch of boundary *time*, not boundary space. This is a first hint of a general phenomenon that Chapter 8 develops in full: boundary time and bulk radial position are closely linked.

At the free (quadratic) level, different bulk fields do not interact. So both the Hilbert space and its operator algebra split into independent tensor factors, one for each field species i : $\mathcal{H}_{\Psi}^{\text{GNS}} = \bigotimes_i \mathcal{H}_{\Psi,i}^{\text{GNS}}$, with $\mathcal{H}_{\Psi,i}^{\text{Fock}} = \mathcal{H}_{\Psi,i}^{\text{GNS}}$ for each field separately. Nothing conceptually new happens here. It is bookkeeping: the identification holds field by field, not just for the whole theory at once.

The duality, stated precisely

Equation (7.1) is an identity between entire operator algebras. So it must also hold subalgebra by subalgebra. Take any open bulk subregion b of a Cauchy slice. It has a bulk operator algebra

$\widetilde{\mathcal{M}}_b \subset B(\mathcal{H}_\Psi^{\text{Fock}})$. (Throughout this chapter, bulk algebras carry a tilde, to keep them visually distinct from boundary algebras.) By (7.1), this is also a subalgebra of $B(\mathcal{H}_\Psi^{\text{GNS}})$. When we view it as a boundary algebra we call it $\mathcal{M}_b$:

$$\widetilde{\mathcal{M}}_b = \mathcal{M}_b. \quad (7.2)$$

Read this as a literal equality, not as an analogy or a matching of some properties. There is one algebra, with two descriptions.

Chapter 4 explained how an algebra, together with a state, encodes causal structure, entanglement, and modular flow. Since $\mathcal{M}_b = \widetilde{\mathcal{M}}_b$, **the boundary algebra $\mathcal{M}_b$ already encodes the corresponding structure of the bulk region b** . For example, the modular operator of $\widetilde{\mathcal{M}}_b$ describes how b is entangled with its complement. It is exactly the modular operator of $\mathcal{M}_b$, and that can be computed from boundary data alone. **This is why a bulk region can be fully “reconstructed” from the corresponding boundary algebra.** This statement is subregion-subalgebra duality. You have already met two special cases of it in Chapter 6. The identifications $\widetilde{\mathcal{M}}_R = Y_R$ and $\widetilde{\mathcal{M}}_L = Y_L$ of the two black-hole exteriors in the thermofield double are exactly (7.2), with b taken to be the whole right or the whole left exterior region.

At this order the bulk theory is effectively a free quantum field theory, and $\widetilde{\mathcal{M}}_b$ is one of its local algebras. So $\widetilde{\mathcal{M}}_b$ has every structural property of such algebras found in Chapter 4. By (7.2), the boundary algebra $\mathcal{M}_b$ has the same properties:

1. **Reeh–Schlieder:** the bulk “vacuum” $|0\rangle_{\phi_c}$ is cyclic and separating for $\widetilde{\mathcal{M}}_b$. Chapter 6 identified this vacuum with the boundary GNS vacuum, $|0\rangle_{\phi_c} = |1\rangle_\Psi$. So $|1\rangle_\Psi$ is cyclic and separating for $\mathcal{M}_b$ too.
2. **Causality:** $\widetilde{\mathcal{M}}_b = \widetilde{\mathcal{M}}_{\widehat{b}}$, where $\widehat{b}$ is the domain of dependence of b . In words, the algebra of a region equals the algebra of its full domain of dependence (the time-slice axiom of Chapter 4). So $\mathcal{M}_b$ reconstructs the whole of $\widehat{b}$, not only b itself.
3. **Type III₁:** every local algebra of a continuum quantum field theory is type III₁ (Chapter 4). So $\widetilde{\mathcal{M}}_b$ is type III₁, and $\mathcal{M}_b$ must be too.
4. **Additivity** for topologically trivial regions: $\widetilde{\mathcal{M}}_{b_1 \cup b_2} = \widetilde{\mathcal{M}}_{b_1} \vee \widetilde{\mathcal{M}}_{b_2}$.
5. **Haag duality:** $\widetilde{\mathcal{M}}_{b'} = \widetilde{\mathcal{M}}'_b$, where b' is the bulk causal complement of b on its Cauchy slice.

Each of these bulk statements becomes a checkable statement about boundary algebras once it is translated through (7.2). The worked examples in the rest of this chapter all come from such translations.

7.2 Entanglement wedge reconstruction, algebraically

Defining the boundary algebra X_A properly

Take a boundary spatial region A on a constant-time Cauchy slice. Its **Ryu–Takayanagi (RT) surface** γ_A is the bulk surface of minimal area anchored on ∂A . The bulk region b_A lying between

A and γ_A is the bulk region dual to A . The **entanglement wedge** $\hat{b}_A$ is the bulk domain of dependence of b_A (Figure 7.1). We write X_A for the boundary algebra that subregion-subalgebra duality assigns to b_A :

$$X_A \equiv \mathcal{M}_{b_A} \stackrel{(1)}{=} \widetilde{\mathcal{M}}_{b_A} \stackrel{(2)}{=} \widetilde{\mathcal{M}}_{\hat{b}_A}, \quad (7.3)$$

(1) subregion-subalgebra duality (7.2) for the region $b = b_A$. (2) the causality property: a bulk region's algebra equals that of its domain of dependence.

So $X_A \subset B(\mathcal{H}_\Psi^{\text{GNS}})$ is the boundary algebra dual to the entanglement wedge of A . Entanglement wedge reconstruction is usually stated as: physics inside $\hat{b}_A$ can be fully recovered from A alone. Equation (7.3) restates this as a literal identity between algebras.

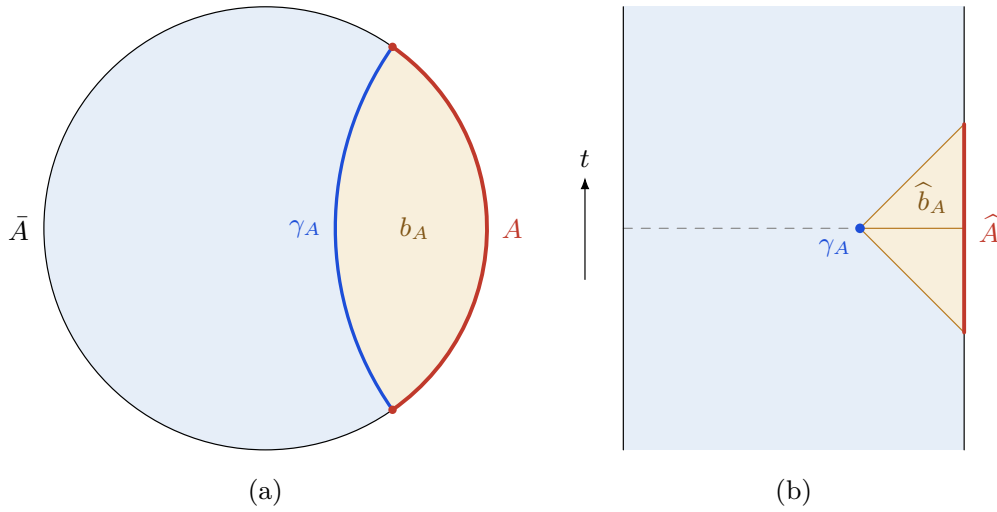


Figure 7.1: (a) A constant-time slice of AdS drawn as a disk, whose edge is the boundary: the Ryu–Takayanagi surface γ_A (blue) is the minimal curve anchored at the endpoints of the boundary region A (red), b_A (gold) is the bulk region between them, and $\bar{A}$ is the rest of the boundary. The entropy of A is $S(A) = \text{Area}(\gamma_A)/4G_N + S_{\text{bulk}}(b_A) = S_{\text{gen}}(b_A)$. (b) A side view, with time running upward, cut through the middle of A : the entanglement wedge $\hat{b}_A$ (gold), bounded by light rays, is the domain of dependence of b_A ; it meets the boundary along $\hat{A}$ (red), and its tip on the $t = 0$ slice (dashed) is γ_A (blue dot).

The boundary construction of X_A needs some care. It is used many times below, so here it is in detail. At finite N there is an ordinary von Neumann algebra $B_A^{(N)}$ of operators localized in A . It satisfies $B_A^{(N)} = B_{\hat{A}}^{(N)}$, where $\hat{A}$ is the boundary domain of dependence of A . This is the ordinary time-slice axiom. As $N \rightarrow \infty$, many operators in $B_A^{(N)}$ have no sensible limit and drop out. The right object is built from the operators that do survive the limit in the state Ψ :

$$X_A \equiv \pi_\Psi \left(\lim_{N \rightarrow \infty, \Psi} B_A^{(N)} \right)'' = X_{\hat{A}}.$$

Here π_Ψ is the GNS representation. The double commutant plays the same role as in the GNS discussion of Chapter 2: it makes the result a genuine von Neumann algebra. The equality with $X_{\hat{A}}$

holds because the finite- N time-slice axiom survives the limit. The content of entanglement wedge reconstruction is that this boundary-defined algebra equals the bulk algebra in (7.3).

A second algebra is easier to construct. The algebra $Y_{\hat{A}}$ is generated by single-trace operators restricted to $\hat{A}$. It is the single-trace algebra Y_O of Chapter 6, taken for the region $\hat{A}$. Single-trace operators in any region survive the large- N limit by construction, so

$$Y_{\hat{A}} \subseteq X_A. \quad (7.4)$$

The rest of this section studies this inclusion: what $Y_{\hat{A}}$ describes in the bulk, and what else X_A contains.

Causal wedge reconstruction: the “easy” part of X_A

Causal wedge reconstruction is also known as HKLL reconstruction (after Hamilton, Kabat, Lifschytz, and Lowe). It concerns the **causal wedge**

$$C_{\hat{A}} \equiv \tilde{J}^+(\hat{A}) \cap \tilde{J}^-(\hat{A}),$$

where $\tilde{J}^\pm$ denote the bulk causal future and past. So $C_{\hat{A}}$ is the set of bulk points that can both receive signals from $\hat{A}$ and send signals to $\hat{A}$. Causal wedge reconstruction states that bulk fields in $C_{\hat{A}}$ can be written directly as single-trace operators smeared over $\hat{A}$:

$$\Phi(X) = \int d^d x K_A^{(C)}(X; x) \pi_\Psi(O(x)), \quad X \in C_{\hat{A}}, \quad x \in \hat{A}. \quad (7.5)$$

Here $K_A^{(C)}$ is an explicit kernel. In general it is a distribution rather than a smooth function. This is the HKLL construction of Chapter 6, now applied to a general region instead of the whole boundary. Algebraically, it says

$$\tilde{\mathcal{M}}_{C_{\hat{A}}} = Y_{\hat{A}}.$$

For A equal to a full boundary, this is the black-hole exterior identification of Chapter 6.

The causal wedge always lies inside the entanglement wedge, $C_{\hat{A}} \subseteq \hat{b}_A$. This is a standard geometric fact. It says that causal reconstruction is always more conservative than entanglement-based reconstruction. Combine this fact with the definition (7.3) and with $\tilde{\mathcal{M}}_{C_{\hat{A}}} = Y_{\hat{A}}$. The inclusion (7.4) then follows at once: it is the algebraic image of the geometric inclusion $C_{\hat{A}} \subseteq \hat{b}_A$.

Where the rest of X_A comes from: modular flow, made explicit

The double-commutant definition fixes X_A abstractly. It does not say which operators X_A contains beyond $Y_{\hat{A}}$. On the gravity side, the entanglement wedge is generically strictly larger than the causal wedge. Equality holds only in special, highly symmetric cases. So generically $Y_{\hat{A}} \subsetneq X_A$.

Here the ergodic lemma of Chapter 4 does real work. The lemma concerns two von Neumann algebras $\mathcal{N} \subset \mathcal{X}$ and a vector that is cyclic and separating for both. It says that the modular flow of the larger algebra $\mathcal{X}$, applied to the smaller algebra $\mathcal{N}$, generates all of $\mathcal{X}$.

We apply it with $\mathcal{X} = X_A$. Since b_A is a genuine bulk local region, $|1\rangle_\Psi$ is cyclic and separating for $X_A = \widetilde{\mathcal{M}}_{b_A}$. Call the corresponding modular operator Δ_{X_A} . By (7.2), it equals the bulk modular operator $\widetilde{\Delta}_{b_A}$. For the smaller algebra we take Y_A^ϵ . This is the algebra of single-trace operators smeared within a thin time band of width ϵ around A . Picture a thin strip of boundary spacetime hugging the region A . Its causal wedge is a thin open bulk region next to A , so by the Reeh–Schlieder property $|1\rangle_\Psi$ is still cyclic and separating for Y_A^ϵ . The lemma then says that modular flow of the *full* algebra X_A regenerates all of X_A from this thin sliver:

$$X_A = \left\{ \Delta_{X_A}^{-is} O_\epsilon(\vec{x}) \Delta_{X_A}^{is} : \vec{x} \in A, s \in \mathbb{R} \right\}''.$$

Here $O_\epsilon(\vec{x})$ is a single-trace operator smeared over the thin band near the point $\vec{x}$. **The extra operators in X_A , beyond single-trace operators smeared over $\widehat{A}$, are single-trace operators carried along by modular flow.**

Write $O(s; \vec{x}) \equiv \Delta_{X_A}^{-is} O_\epsilon(\vec{x}) \Delta_{X_A}^{is}$ for these modular-flowed operators. At the free-field level of the large- N limit the state is Gaussian. The modular flow of a Gaussian state maps each smeared field to another field that is again linear in the original fields. A bulk field is itself linear in the fields. So a bulk field anywhere in the entanglement wedge, not only in the causal wedge, can be written as

$$\Phi(X) = \int_{-\infty}^{\infty} ds \int d\vec{x} K_A^{(E)}(X; s, \vec{x}) O(s; \vec{x}), \quad X \in \widehat{b}_A.$$

This is the entanglement-wedge generalization of the HKLL formula (7.5). Formulas of this type were first proposed on other grounds. Here their existence follows directly from modular theory. The kernel $K_A^{(E)}$ is in general not known in closed form.

A related result, the **JLMS relation** (Jafferis, Lewkowycz, Maldacena, and Suh, 2015), connects boundary and bulk modular flow at large but finite N . In the strict large- N limit it reduces to the statement $\Delta_{X_A} = \widetilde{\Delta}_{b_A}$ used above. The JLMS relation is often quoted without derivation. The box below shows how it fits together with two other holographic results.

Key Result / Core Concept: The JLMS relation and modular flow

Statement (JLMS). Let A be a boundary spatial region, with entanglement wedge b_A bounded by the RT surface γ_A . Work on the code subspace, the space of semiclassical states around a fixed background. Let σ be a reference state in it, and write the modular Hamiltonians with density matrices (at large but finite N , with a short-distance cutoff where needed):

$$H_A \equiv -\log \sigma_A, \quad H_{\text{bulk}} \equiv -\log \sigma_{b_A}.$$

1. The boundary and bulk modular Hamiltonians obey an operator identity on the code subspace,

$$H_A = \frac{\hat{A}[\gamma_A]}{4G_N} + H_{\text{bulk}} + \mathcal{O}(G_N),$$

where $\hat{A}[\gamma_A]$ is the area operator of the RT surface. (The area term is understood to include the local terms on γ_A , such as counterterms, that also appear in the FLM

formula below.)

2. For any bulk operator $\Phi \in \widetilde{\mathcal{M}}_{b_A}$ in the entanglement wedge,

$$\Delta_A^{-is} \Phi \Delta_A^{is} = \widetilde{\Delta}_{b_A}^{-is} \Phi \widetilde{\Delta}_{b_A}^{is}.$$

In words, boundary modular flow acts on every entanglement-wedge operator exactly as bulk modular flow does.

How the pieces fit. The argument below takes two inputs: the equality of bulk and boundary relative entropies (step 1), and the FLM formula (step 3). JLMS originally argued in the other direction. They started from FLM, derived the modular Hamiltonian identity, and then obtained the equality of relative entropies. Given FLM, the two statements are equivalent at this order. So the argument shows how the pieces fit together. It is not a derivation from first principles.

1. **Relative entropy equality.** Let σ be a reference background state, such as the global vacuum or the thermofield double. Let ρ be any nearby excited state in the code subspace. The input is that the relative entropy of the boundary region A equals that of the bulk entanglement wedge b_A , up to small corrections:

$$D(\rho_A \| \sigma_A) = D(\rho_{\text{bulk}} \| \sigma_{\text{bulk}}) + \mathcal{O}(G_N).$$

2. **Relative entropy in terms of modular Hamiltonians.** For any two density matrices, with $H^\sigma \equiv -\log \sigma$,

$$\begin{aligned} D(\rho \| \sigma) &\stackrel{(1)}{=} \text{Tr}(\rho \log \rho) - \text{Tr}(\rho \log \sigma) \\ &\stackrel{(2)}{=} -S(\rho) + \langle H^\sigma \rangle_\rho. \end{aligned}$$

(1) the definition of relative entropy. (2) $S(\rho) = -\text{Tr}(\rho \log \rho)$, and $-\log \sigma = H^\sigma$.

Applied to the boundary state and to the bulk state in the entanglement wedge, this gives

$$\begin{aligned} D(\rho_A \| \sigma_A) &= -S(\rho_A) + \langle H_A \rangle_\rho, \\ D(\rho_{\text{bulk}} \| \sigma_{\text{bulk}}) &= -S(\rho_{\text{bulk}}) + \langle H_{\text{bulk}} \rangle_\rho. \end{aligned}$$

3. **The FLM formula.** The Faulkner–Lewkowycz–Maldacena (FLM) formula is the quantum-corrected RT formula. It says that the boundary entanglement entropy equals the bulk generalized entropy, at leading and first subleading order:

$$S(\rho_A) = \left\langle \frac{\hat{A}[\gamma_A]}{4G_N} \right\rangle_\rho + S(\rho_{\text{bulk}}) + \mathcal{O}(G_N).$$

4. **Cancellation of the bulk entropy.** Start from the boundary relative entropy and use

steps 2, 3, and 1 in turn:

$$\begin{aligned}
D(\rho_A \| \sigma_A) &\stackrel{(1)}{=} -S(\rho_A) + \langle H_A \rangle_\rho \\
&\stackrel{(2)}{=} - \left\langle \frac{\hat{A}[\gamma_A]}{4G_N} \right\rangle_\rho - S(\rho_{\text{bulk}}) + \langle H_A \rangle_\rho + \mathcal{O}(G_N) \\
&\stackrel{(3)}{=} -S(\rho_{\text{bulk}}) + \langle H_{\text{bulk}} \rangle_\rho + \mathcal{O}(G_N).
\end{aligned}$$

(1) step 2 applied to the boundary state. (2) insert the FLM formula of step 3 for $S(\rho_A)$. (3) the relative entropy equality of step 1, with the bulk relative entropy written out using step 2.

The bulk entropy $S(\rho_{\text{bulk}})$ appears on both sides of the last equality, so it cancels. Rearranging what is left gives

$$\langle H_A \rangle_\rho = \left\langle \frac{\hat{A}[\gamma_A]}{4G_N} + H_{\text{bulk}} \right\rangle_\rho + \mathcal{O}(G_N).$$

5. **From expectation values to an operator identity.** This equality holds for *every* state ρ in the code subspace. The expectation values of a Hermitian operator in all density matrices supported on a subspace determine that operator's restriction to the subspace. So the two operators agree on the code subspace:

$$H_A = \frac{\hat{A}[\gamma_A]}{4G_N} + H_{\text{bulk}} + \mathcal{O}(G_N).$$

6. **Action on bulk operators.** Let $\Phi \in \widetilde{\mathcal{M}}_{b_A}$ be an operator localized in the interior of the entanglement wedge. The RT surface $\gamma_A = \partial b_A$ is the edge of the wedge, so it is spacelike separated from Φ . At leading order the area operator therefore commutes with Φ :

$$\left[\frac{\hat{A}[\gamma_A]}{4G_N}, \Phi \right] = 0.$$

Now compute the commutator of Φ with the boundary modular Hamiltonian:

$$[H_A, \Phi] \stackrel{(1)}{=} \left[\frac{\hat{A}[\gamma_A]}{4G_N} + H_{\text{bulk}}, \Phi \right] \stackrel{(2)}{=} [H_{\text{bulk}}, \Phi].$$

(1) the operator identity of step 5, at leading order. (2) the commutator is linear, and the area term commutes with Φ .

The operator $[H_{\text{bulk}}, \Phi]$ again lies in $\widetilde{\mathcal{M}}_{b_A}$, so the same step applies to it. Hence all nested commutators agree, $[H_A, [H_A, \dots [H_A, \Phi]]] = [H_{\text{bulk}}, [H_{\text{bulk}}, \dots [H_{\text{bulk}}, \Phi]]]$. Summing the exponential series of nested commutators then gives, formally,

$$e^{isH_A} \Phi e^{-isH_A} = e^{isH_{\text{bulk}}} \Phi e^{-isH_{\text{bulk}}}.$$

Finally, the full modular operator is $\Delta_A = \sigma_A \otimes \sigma_{\bar{A}}^{-1}$. Its $\sigma_{\bar{A}}$ factor commutes with Φ , so $\Delta_A^{-is} \Phi \Delta_A^{is} = \sigma_A^{-is} \Phi \sigma_A^{is} = e^{isH_A} \Phi e^{-isH_A}$. The same holds in the bulk. Therefore

$$\Delta_A^{-is} \Phi \Delta_A^{is} = \tilde{\Delta}_{b_A}^{-is} \Phi \tilde{\Delta}_{b_A}^{is}. \quad \blacksquare$$

When is the causal-wedge piece already everything, so that $X_A = Y_{\hat{A}}$? This happens exactly when the bulk modular flow of b_A acts *geometrically*. That means it acts as a pointwise coordinate transformation, like the Rindler-wedge boost of Chapter 4. A flow of this kind maps operators localized in $\hat{A}$ to operators that are still localized in $\hat{A}$, so it generates nothing new. This happens for a half-space or a spherical region in the vacuum, and for the full boundary in the thermofield double. These are exactly the cases already worked out explicitly. In them $X_A = Y_{\hat{A}}$ and $\hat{b}_A = C_{\hat{A}}$ hold exactly. In general, though, modular flow is *not* geometric, and the inclusion (7.4) is strict.

Physics Connection: Modular Flow and Operator Generation

You have already met a finite-dimensional version of this object in Chapter 4. There, the modular flow of a single entangled pair, $\sigma_s(A) = \rho_R^{-is} A \rho_R^{is}$, was an ordinary unitary conjugation. You worked it out explicitly on a 4×4 matrix. Here the same object, the modular flow of a boundary algebra, does something with no finite-dimensional analogue. Starting from a small algebra, it generates operators that were not in that algebra, and in fact the whole larger algebra. This is possible only because the algebras involved are infinite-dimensional (here, type III₁).

In finite dimensions the ergodic lemma has no content. Suppose $\mathcal{N} \subset \mathcal{M}$ act on a finite-dimensional Hilbert space $\mathcal{H}$, and a vector $|\Omega\rangle$ is cyclic for $\mathcal{N}$ and separating for $\mathcal{M}$. Because $|\Omega\rangle$ is separating for $\mathcal{M}$, the map $x \mapsto x|\Omega\rangle$ is injective on $\mathcal{M}$, so $\dim \mathcal{M} \leq \dim \mathcal{H}$. Because $|\Omega\rangle$ is cyclic for $\mathcal{N}$, the vectors $x|\Omega\rangle$ with $x \in \mathcal{N}$ fill all of $\mathcal{H}$, so $\dim \mathcal{N} \geq \dim \mathcal{H}$. Together these give $\dim \mathcal{N} \geq \dim \mathcal{M}$, and so $\mathcal{N} = \mathcal{M}$. In finite dimensions a proper subalgebra can never share such a vector with the larger algebra. There is nothing left for modular flow to fill in.

In infinite dimensions the situation is different. By the Reeh–Schlieder property, a small subalgebra, such as the thin sliver above, can have the same cyclic and separating vector as the larger algebra. Modular flow can then sweep out the entire larger algebra from the sliver. This is the “ergodic” lemma of Chapter 4, now doing physical work: it builds an entire entanglement wedge, or a black-hole interior, out of a thin sliver of boundary time. The definition of modular flow has not changed. What changed is that infinite dimensions let it do something a finite matrix conjugation cannot.

Two logically distinct sources of type III₁

Both X_A and $Y_{\hat{A}}$ are type III₁. This property can have two different physical origins, and it helps to keep them apart.

The first origin is the one met in Chapter 4. For a continuum quantum field theory at finite N , the

local algebra $B_A^{(N)}$ is already type III₁. The cause is the infinite amount of *short*-range entanglement across the boundary ∂A . In the bulk, this matches the fact that the RT surface reaches the AdS boundary, which is at infinite proper distance.

To isolate the second origin, put the boundary theory on a lattice. Then at any finite N the algebra $B_A^{(N)}$ is type I: it is an algebra of finite matrices, with minimal projections. Yet X_A and $Y_{\widehat{A}}$, defined through the large- N limit, are *still* type III₁. So their type III₁ property does not come from the type of $B_A^{(N)}$ before the limit. It comes from an infinite amount of *long*-range entanglement between A and its complement, which appears only in the strict $N \rightarrow \infty$ limit. In the bulk, this matches the infinite short-range entanglement of bulk fields across the RT surface γ_A (or, for $Y_{\widehat{A}}$, across the edge $\chi_{\widehat{A}}$ of the causal wedge).

So two independent mechanisms produce the same type of algebra, at two different stages of the same construction: finite N , and the strict $N \rightarrow \infty$ limit.

The same picture gives an algebraic way to locate the RT surface. Near γ_A , bulk modular flow acts as a local boost. This is the local-Rindler argument of Chapter 4, applied at the RT surface. The surface γ_A is the fixed (“invariant”) set of that boost. By the JLMS relation, boundary modular flow acts on wedge operators as bulk modular flow does. So **the RT surface can be characterized, without assuming a bulk metric, as the asymptotic fixed-point set of the boundary modular flow.** This gives an algebraic restatement of the procedure for finding the RT surface.

Extended gravitational systems: algebra without any geometry to point to

The entanglement-wedge and causal-wedge story extends beyond geometric boundary regions. Replace A by *any* von Neumann subalgebra $\mathcal{M}$ of the boundary theory. Then define $X_{\mathcal{M}}$ by the direct analogue of the double-commutant definition of X_A . This definition makes sense even when there is no bulk geometric picture at all.

One can go further. Couple a gravitational sector B (which has its own boundary dual) to an ordinary non-gravitational sector R . For example, R could be radiation that has already escaped to infinity. For any subsystem Q of the combined system $B \cup R$, define two algebras:

- the entanglement wedge algebra $X_Q \equiv \lim_{G_N \rightarrow 0, \Psi} B_Q$, whether or not it has a geometric meaning;
- the causal wedge algebra Y_Q , the algebra of Q ’s ordinary low-energy effective description.

This abstraction is exactly what the evaporating-black-hole discussion later in this chapter needs. There, Q will be the emitted Hawking radiation.

7.3 Quantum informational aspects of entanglement wedge reconstruction

Superadditivity, and its boundary origin

Entanglement wedge nesting is a standard geometric fact, which follows from the extremality of RT surfaces. For boundary regions $A_1 \subseteq A_2$, the entanglement wedges satisfy $\widehat{b}_{A_1} \subseteq \widehat{b}_{A_2}$. Algebraically this is just $X_{A_1} \subseteq X_{A_2}$, which follows immediately from $A_1 \subseteq A_2$ and the definition of X_A .

Nesting has a direct geometric consequence. For two regions A_1, A_2 on one Cauchy slice,

$$b_{A_1} \cup b_{A_2} \subseteq b_{A_1 \cup A_2}, \quad b_{A_1 \cap A_2} \subseteq b_{A_1} \cap b_{A_2}.$$

This is a consequence of nesting, not an extra assumption. In words, the entanglement wedge of a union is at least as big as the union of the entanglement wedges, and generically it is strictly bigger. This is **superadditivity of entanglement wedges**. Two AdS₃ examples make it concrete: two overlapping boundary intervals, and two disjoint intervals placed close together. In both cases the RT surface of the union “jumps” and encloses visibly more bulk than the two pieces do separately. Translated through the definition (7.3), superadditivity becomes a statement purely about boundary algebras:

$$X_{A_1} \vee X_{A_2} \subseteq X_{A_1 \cup A_2}, \quad X_{A_1 \cap A_2} \subseteq X_{A_1} \wedge X_{A_2}. \quad (7.6)$$

Compare this with the additivity axiom of Chapter 4 for an ordinary relativistic quantum field theory at finite N . For topologically trivial regions, that axiom says $B_{A_1}^{(N)} \vee B_{A_2}^{(N)} = B_{A_1 \cup A_2}^{(N)}$. **For the inclusions (7.6) to be strict, ordinary additivity has to fail in the strict large- N limit.** One can show that it does fail in the two AdS₃ examples above. The argument uses a classic result of Araki. It identifies $X_{A_1} \vee X_{A_2}$ concretely as the algebra of single-trace operators in a specific larger causal region. That algebra is strictly smaller than $X_{A_1 \cup A_2}$. (We quote this computation without reproducing it.) **Superadditivity of entanglement wedges is the bulk geometric image of the failure of additivity for boundary algebras in the strict large- N limit.** Algebras of local regions are then no longer “locally generated” in the way the algebras of an ordinary finite- N quantum field theory are.

Haag duality, by contrast, survives. Assume it holds for the bulk algebras, $\widetilde{\mathcal{M}}_{b'} = \widetilde{\mathcal{M}}'_b$. For a pure global state, A and its complement $\bar{A}$ share the same RT surface. So $b_{\bar{A}}$ is the bulk complement b'_A . Then Haag duality holds in the large- N limit:

$$X'_A = X_{\bar{A}}.$$

The single-trace algebra $Y_{\widehat{A}}$ behaves in the opposite way. It is additive by construction, but in general it does *not* satisfy Haag duality.

Quantum error correction, made precise

Entanglement wedge reconstruction has long been interpreted in the language of quantum error correction. All the information in b_A can be recovered from boundary data on A alone. So this information is protected against “erasing” the complementary region $\bar{A}$ entirely. Superadditivity makes this more precise and checkable.

Take an operator $\Phi(X)$ at a point X in a bulk region b . Suppose b lies in the *intersection* of the entanglement wedges of two different boundary regions A and $\tilde{A}$, but outside the entanglement wedge of $A \cap \tilde{A}$. Picture two overlapping boundary intervals. Their wedges overlap in a bulk region that reaches deeper than the wedge of their small common piece. Now use the HKLL-type formula (7.5). For concreteness, take a case in the vacuum sector (GNS representation π_Ω) where the causal and entanglement wedges coincide for both A and $\tilde{A}$. Then

$$\Phi(X) = \pi_\Omega(O_A(X)) = \pi_\Omega(O_{\tilde{A}}(X)). \quad (7.7)$$

Here $O_A(X)$ and $O_{\tilde{A}}(X)$ are two *different* boundary operators. They are built from different single-trace data, smeared over A and over $\tilde{A}$ respectively. Yet they coincide once represented on the GNS Hilbert space. In fact there is an infinite family of such reconstructions, one for every boundary region whose entanglement wedge contains b .

So a bulk degree of freedom $\Phi(X)$ in b cannot be assigned to any one boundary region. It is encoded collectively and redundantly across the boundary system. This is the defining feature of a quantum error-correcting code. This redundancy follows from superadditivity (7.6). It is not an extra postulate added to the holographic dictionary.

Worked calculation: a three-qubit toy code

To see this redundancy in elementary matrix algebra, we work through a small toy code with three qubits. It is a simplified cousin of the three-qutrit code of Almheiri, Dong, and Harlow, which is the standard example. The qubit version is not perfect, as we will see, but it already shows the key mechanism.

Let the bulk logical state be a single qubit at the center of the disk:

$$|\psi\rangle_L = \alpha |0\rangle_L + \beta |1\rangle_L, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The boundary consists of three physical qubits A, B, C . The encoding isometry $V : \mathbb{C}^2 \rightarrow (\mathbb{C}^2)^{\otimes 3}$ is defined by

$$|0\rangle_L \mapsto \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle), \quad |1\rangle_L \mapsto \frac{1}{\sqrt{2}}(|100\rangle + |011\rangle).$$

The four basis states that appear are distinct, so the two code states are orthonormal and V is an isometry. The encoded boundary state is

$$\begin{aligned} |\Psi(\alpha, \beta)\rangle &= \frac{\alpha}{\sqrt{2}}(|000\rangle_{ABC} + |111\rangle_{ABC}) \\ &\quad + \frac{\beta}{\sqrt{2}}(|100\rangle_{ABC} + |011\rangle_{ABC}). \end{aligned}$$

A single qubit. Compute the reduced density matrix of qubit C by tracing out qubits A and B :

$$\rho_C = \text{Tr}_{AB}(|\Psi\rangle\langle\Psi|) = \sum_{a,b \in \{0,1\}} \langle ab|\Psi\rangle \langle\Psi|ab\rangle.$$

Here $\langle ab|\Psi\rangle$ is a (unnormalized) state of qubit C . The four overlaps with the basis states of AB are

$$\begin{aligned}\langle 00|\Psi\rangle &= \frac{\alpha}{\sqrt{2}} |0\rangle_C, & \langle 00|\Psi\rangle \langle \Psi|00\rangle &= \frac{|\alpha|^2}{2} |0\rangle \langle 0|_C, \\ \langle 10|\Psi\rangle &= \frac{\beta}{\sqrt{2}} |0\rangle_C, & \langle 10|\Psi\rangle \langle \Psi|10\rangle &= \frac{|\beta|^2}{2} |0\rangle \langle 0|_C, \\ \langle 11|\Psi\rangle &= \frac{\alpha}{\sqrt{2}} |1\rangle_C, & \langle 11|\Psi\rangle \langle \Psi|11\rangle &= \frac{|\alpha|^2}{2} |1\rangle \langle 1|_C, \\ \langle 01|\Psi\rangle &= \frac{\beta}{\sqrt{2}} |1\rangle_C, & \langle 01|\Psi\rangle \langle \Psi|01\rangle &= \frac{|\beta|^2}{2} |1\rangle \langle 1|_C.\end{aligned}$$

Each AB basis state picks out exactly one of the four terms of $|\Psi\rangle$. Summing:

$$\begin{aligned}\rho_C &\stackrel{(1)}{=} \left(\frac{|\alpha|^2 + |\beta|^2}{2} \right) |0\rangle \langle 0|_C + \left(\frac{|\alpha|^2 + |\beta|^2}{2} \right) |1\rangle \langle 1|_C \\ &\stackrel{(2)}{=} \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \mathbf{1}_2.\end{aligned}$$

(1) add the four terms of the table, grouping the two that multiply $|0\rangle \langle 0|_C$ and the two that multiply $|1\rangle \langle 1|_C$. **(2)** normalization of the logical state, $|\alpha|^2 + |\beta|^2 = 1$.

So ρ_C is proportional to the identity matrix. It does not depend on the logical amplitudes α and β at all. The same four-overlap computation for qubit B (now tracing out A and C) gives the same answer, $\rho_B = \frac{1}{2} \mathbf{1}_2$. So qubit B alone and qubit C alone each contain **no** information about the central bulk qubit. Qubit A is different. The same computation gives $\rho_A = \frac{1}{2} \mathbf{1}_2 + \text{Re}(\alpha\beta^*) X_A$, which does depend on the logical state. (These three reduced density matrices were also checked numerically.) Three qubits are too few for a code in which *every* single qubit is blind to the logical information. The perfect version of this toy code uses three qutrits (three-level systems) instead.

Two qubits. Now consider reconstructing logical operators on *two* qubits, a boundary subregion of size 2.

- Logical** $Z_L = |0\rangle_L \langle 0| - |1\rangle_L \langle 1|$. Define the boundary operator $Z_{AB} \equiv Z_A \otimes Z_B$ acting on region AB . Then

$$\begin{aligned}(Z_A \otimes Z_B \otimes \mathbf{1}_C) |0\rangle_L &\stackrel{(1)}{=} \frac{1}{\sqrt{2}} ((+1)(+1) |000\rangle + (-1)(-1) |111\rangle) \\ &\stackrel{(2)}{=} |0\rangle_L, \\ (Z_A \otimes Z_B \otimes \mathbf{1}_C) |1\rangle_L &\stackrel{(1)}{=} \frac{1}{\sqrt{2}} ((-1)(+1) |100\rangle + (+1)(-1) |011\rangle) \\ &\stackrel{(2)}{=} -|1\rangle_L.\end{aligned}$$

(1) Z gives +1 on $|0\rangle$ and -1 on $|1\rangle$, applied to qubits A and B ; the identity leaves qubit C alone. **(2)** multiply the signs and compare with the encoding of $|0\rangle_L$ and $|1\rangle_L$.

Hence $(Z_{AB} \otimes \mathbf{1}_C) |\psi\rangle_L = Z_L |\psi\rangle_L$.

- Logical** $X_L = |0\rangle_L \langle 1| + |1\rangle_L \langle 0|$. Define the boundary operator $X_{AB} \equiv X_A \otimes \mathbf{1}_B$ on region

AB . (It acts nontrivially only on qubit A .) Then

$$\begin{aligned}(X_A \otimes \mathbf{1}_B \otimes \mathbf{1}_C) |0\rangle_L &\stackrel{(1)}{=} \frac{1}{\sqrt{2}}(|100\rangle + |011\rangle) \stackrel{(2)}{=} |1\rangle_L, \\(X_A \otimes \mathbf{1}_B \otimes \mathbf{1}_C) |1\rangle_L &\stackrel{(1)}{=} \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \stackrel{(2)}{=} |0\rangle_L.\end{aligned}$$

(1) X flips qubit A , $|0\rangle \leftrightarrow |1\rangle$, and leaves B and C alone. (2) compare with the encoding of $|1\rangle_L$ and $|0\rangle_L$.

Hence $(X_{AB} \otimes \mathbf{1}_C) |\psi\rangle_L = X_L |\psi\rangle_L$.

The logical X_L can also be reconstructed on the complementary region BC , as $X_{BC} \equiv \mathbf{1}_A \otimes X_B \otimes X_C$:

$$\begin{aligned}(\mathbf{1}_A \otimes X_B \otimes X_C) |0\rangle_L &\stackrel{(1)}{=} \frac{1}{\sqrt{2}}(|011\rangle + |100\rangle) \stackrel{(2)}{=} |1\rangle_L, \\(\mathbf{1}_A \otimes X_B \otimes X_C) |1\rangle_L &\stackrel{(1)}{=} \frac{1}{\sqrt{2}}(|111\rangle + |000\rangle) \stackrel{(2)}{=} |0\rangle_L.\end{aligned}$$

(1) flip qubits B and C in each basis state: $|000\rangle \rightarrow |011\rangle$, $|111\rangle \rightarrow |100\rangle$, $|100\rangle \rightarrow |111\rangle$, $|011\rangle \rightarrow |000\rangle$. (2) compare with the encoding.

Flipping only qubit B would not work: $X_B |0\rangle_L = \frac{1}{\sqrt{2}}(|010\rangle + |101\rangle)$ lies outside the code subspace. Both $X_A \otimes \mathbf{1}_B \otimes \mathbf{1}_C$ and $\mathbf{1}_A \otimes X_B \otimes X_C$ act on the code subspace as the central bulk operator $\Phi(0) = X_L$. Yet they are supported on disjoint, complementary boundary regions (A versus BC). The same kind of redundancy holds for Z_L . Repeating the sign computation above with Z on qubits A and C shows that $Z_A \otimes \mathbf{1}_B \otimes Z_C$ also acts as Z_L . So Z_L can be built on AB or on AC .

To summarize: each of the regions AB and AC reconstructs the full logical algebra (both X_L and Z_L). The region BC does not. It reconstructs X_L but not Z_L . The reason is that every operator on BC commutes with X_A , which acts as X_L , so the logical action of any operator on BC must commute with X_L . (A search over all three-qubit Pauli strings confirms this list of reconstructions numerically.) In the three-qutrit code, by contrast, any two of the three qutrits reconstruct every logical operator. Even in this imperfect qubit version, the same logical operator lives on several different boundary regions. This is the algebraic mechanism of holographic error correction.

One caveat about this picture. The “bulk Hilbert space as a code subspace” picture from quantum error correction is literally meaningful only at finite N . The bulk semiclassical Hilbert space, on the other hand, is only precisely defined at $N = \infty$. So holographic error correction is better understood as a framework connecting the two regimes. Any finite- N code is necessarily only *approximately* isometric, and should not be read as a literal, exact implementation of the holographic dictionary.

Physics Connection: Redundant Purifications and Code Subspaces

The operator $\Phi(X)$ can be written using completely different boundary data (region A or region $\tilde{A}$), while remaining, in every observable sense, the same operator. This is a large-scale version of a fact you already know from ordinary quantum information: **purification is never unique**.

Take a single mixed qubit, $\rho = \text{diag}(0.7, 0.3)$. One purification uses a second qubit as the

ancilla, $|\Psi\rangle_1 = \sqrt{0.7}|00\rangle + \sqrt{0.3}|11\rangle$. A completely different one uses, say, a 3-level ancilla, $|\Psi\rangle_2 = \sqrt{0.7}|0\rangle|a\rangle + \sqrt{0.3}|1\rangle|b\rangle$, for any orthonormal $|a\rangle, |b\rangle$ in the bigger ancilla space. Trace out the ancilla in either case and you get back the same ρ . You can check this directly from the definition of the partial trace: the result depends only on the Schmidt coefficients $\sqrt{0.7}, \sqrt{0.3}$, not on which ancilla states carry them. **No measurement confined to the original qubit can tell you which purification, meaning which ancilla entangled in which way, is “really” on the other side.** The information is present, but it is not attached to any single, canonical description of the environment. It is attached only to the reduced state itself. The double reconstruction (7.7) works the same way, at the scale of an entire holographic boundary instead of one ancilla qubit. The observable content of $\Phi(X)$ on the GNS Hilbert space is fixed. But the choice of boundary data used to build it (region A or region $\tilde{A}$) is as non-unique as the choice of ancilla above. As in the qubit example, no measurement of the reconstructed operator itself can tell you which region did the reconstructing. What is new here, with no counterpart in the single-qubit example, is the *physical* content of that redundancy. It is not just a bookkeeping fact about how ρ can be written as a partial trace. Superadditivity (7.6) ties it to real bulk geometry. It turns “purification is not unique” into “a bulk region is stored redundantly enough to survive the erasure of a boundary subregion.” That is the operational content of a quantum error-correcting code.

7.4 An algebraic formulation of entanglement islands

The island phenomenon, restated algebraically

The **entanglement island** phenomenon is central to modern derivations of the Page curve for an evaporating black hole. It says the following. After the Page time t_P , the black-hole interior stops being part of the black hole’s own entanglement wedge. Instead it becomes part of the entanglement wedge of the emitted radiation R .

Split the full system into the black-hole boundary theory B and the radiation R (Figure 7.2(b)). Before t_P , the minimal quantum extremal surface for B is empty. So B ’s entanglement wedge is the whole Cauchy slice: the interior I and the exterior O together. After t_P , a new, nontrivial quantum extremal surface α takes over, and B ’s entanglement wedge shrinks to the exterior O alone. Algebraically,

$$X_B = \begin{cases} \widetilde{\mathcal{M}}_O \vee \widetilde{\mathcal{M}}_I & t < t_P, \\ \widetilde{\mathcal{M}}_O & t > t_P. \end{cases}$$

Since $B \cup R$ is the whole system, once $\widetilde{\mathcal{M}}_I$ drops out of X_B it must appear in R ’s description instead. We use the entanglement-wedge algebra X_R and the causal-wedge algebra Y_R from the discussion of extended gravitational systems earlier in this chapter. The transfer is then stated precisely as

$$X_R = \begin{cases} Y_R & t < t_P, \\ Y_R \vee \widetilde{\mathcal{M}}_I & t > t_P. \end{cases}$$

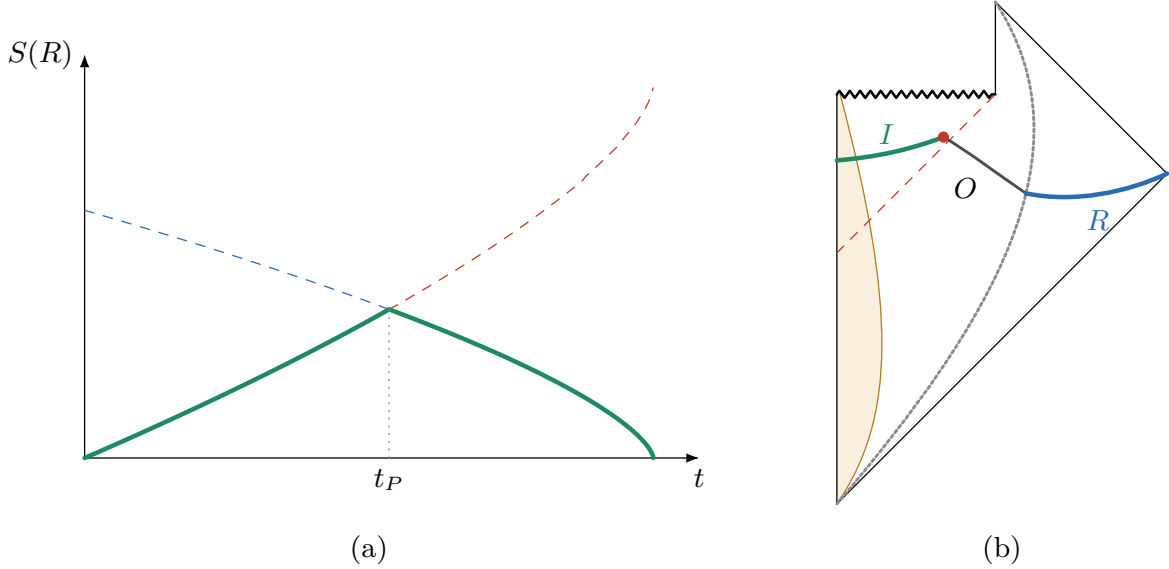


Figure 7.2: (a) Entropy of the radiation R against time. Hawking’s no-island answer (dashed red after t_P) keeps growing and the black hole’s area term $A_{\text{BH}}/4G_N$ (dashed blue before t_P) keeps shrinking; the true entropy (green) follows the smaller of the two, turning over at the Page time t_P . (b) Penrose diagram of a black hole that forms from a collapsing star (gold) and evaporates completely, drawn after [9]. The dashed red line is the event horizon and the zigzag the singularity; after the evaporation ends, the centre $r = 0$ continues straight up. The dotted curve separates the region near the black hole, described by B , from the far region where the radiation collects (for an AdS black hole this is the boundary to which the bath is attached). On a Cauchy slice after t_P , the quantum extremal surface (red dot) lies just inside the horizon. The island I (green) behind it joins the radiation region R (blue) in the radiation’s entanglement wedge, while the exterior O between the red dot and the dotted curve belongs to B .

This lets you *detect* an island using only data intrinsic to the radiation system, with no reference to B at all. Define

$$I_R \equiv Y'_R \cap X_R.$$

These are the operators in X_R that commute with every operator in Y_R . For $t > t_P$, this intersection is exactly $\widetilde{\mathcal{M}}_I$. An island exists exactly when the intersection is nontrivial, that is, larger than the multiples of the identity. In that case there are operators that survive the semiclassical limit but lie outside R ’s ordinary low-energy effective description. As with X_A versus $Y_{\widehat{A}}$ above, these extra operators are generated from Y_R by the modular flow of X_R . It is the same mechanism once more. The same definition, $I_Q \equiv Y'_Q \cap X_Q$, applies to any subsystem Q of an extended gravitational system, not just to radiation. An island for Q is whatever survives the semiclassical limit but sits outside Q ’s own low-energy description.

Now return to an ordinary boundary region A , so that $Y_Q = Y_{\widehat{A}}$. Then the island algebra I_A consists of the operators in X_A that commute with all of $Y_{\widehat{A}}$. These are the operators that modular flow adds to $Y_{\widehat{A}}$. Geometrically, they belong to the part of the entanglement wedge b_A lying outside the causal wedge. Write $c_{\widehat{A}}$ for the part of b_A inside the causal wedge. Then $b_A = c_{\widehat{A}} \cup i_A$. This is an ordinary geometric decomposition: a causal-wedge part next to the boundary, plus a deeper leftover

piece i_A between the causal wedge and the RT surface. It gives

$$X_A = Y_{\hat{A}} \vee \widetilde{\mathcal{M}}_{i_A}, \quad I_A = \widetilde{\mathcal{M}}_{i_A}.$$

So, in this language, an island is exactly the piece of an entanglement wedge that causal-wedge (HKLL) reconstruction alone can never reach.

Key Result / Core Concept: Derivation of the Page Curve from the Quantum Extremal Island Rule

The island rule for radiation. Consider an evaporating black hole coupled to a non-gravitational radiation reservoir. The entropy of the radiation R at boundary time t is found by extremizing the generalized entropy over candidate islands I , whose boundaries ∂I are quantum extremal surfaces (QES), and then taking the minimum over the extrema:

$$S(R) = \min_{\text{QES } I} \text{ext}_I \left[\frac{\text{Area}(\partial I)}{4G_N} + S_{\text{bulk}}(R \cup I) \right].$$

The Page curve in a toy model. The model below keeps only the features needed to see the shape of the Page curve.

1. **Evaporating black-hole geometry.** Let the black hole form at $t = 0$ with initial horizon area A_0 and Bekenstein–Hawking entropy $S_0 = A_0/4G_N$. As it radiates into the reservoir, its horizon area decreases. Let Γ be the rate at which its Bekenstein–Hawking entropy decreases, and take Γ constant. (In a real evaporating black hole the rate changes with time; a constant rate is a simplification.) Then

$$S_{\text{BH}}(t) \equiv \frac{A_{\text{BH}}(t)}{4G_N} = S_0 - \Gamma t, \quad t \in [0, t_{\text{evap}}],$$

where $t_{\text{evap}} = S_0/\Gamma$ is the total evaporation time.

2. **Saddle 1: no island ($I = \emptyset$).** For $I = \emptyset$ there is no boundary, $\partial I = \emptyset$, so $\text{Area}(\partial I) = 0$. The generalized entropy reduces to the bulk field-theory entropy of the radiation:

$$S_{\text{no-island}}(R) = S_{\text{bulk}}(R).$$

In Hawking’s semiclassical calculation, each outgoing quantum c_k in the radiation is entangled with an infalling partner mode b_k behind the horizon:

$$|\text{Hawking pair}\rangle_k \approx \frac{1}{\sqrt{2}} (|0\rangle_{c_k} |0\rangle_{b_k} + |1\rangle_{c_k} |1\rangle_{b_k}).$$

The reservoir R collects only the outgoing quanta c_k . The partners b_k stay behind the horizon. So tracing them out leaves R in a mixed state, and each new pair adds to its entropy. In the toy model we assume that the radiation entropy grows at the same rate Γ at which the black-hole entropy falls. (For a real black hole the two rates differ by a factor

of order one. This moves the crossing point below, but not the shape of the curve.) Then

$$S_{\text{no-island}}(R) = \int_0^t dt' \frac{dS_{\text{rad}}}{dt'} = \Gamma t.$$

This grows without bound as t increases. Eventually it exceeds the Bekenstein–Hawking entropy of the remaining black hole. But a black hole with entropy S_{BH} can be entangled with the radiation by at most S_{BH} . This conflict is the **Hawking information paradox**.

3. **Saddle 2: an island** ($I \neq \emptyset$). A second, non-empty extremum appears. Its boundary ∂I is a quantum extremal surface located just inside the event horizon. Its displacement from the horizon vanishes as $G_N \rightarrow 0$. The region I covers the black-hole interior behind the horizon. The generalized entropy of this saddle is

$$S_{\text{island}}(R) = \frac{\text{Area}(\partial I)}{4G_N} + S_{\text{bulk}}(R \cup I).$$

Now evaluate the bulk matter entropy $S_{\text{bulk}}(R \cup I)$. The union $R \cup I$ contains *both* the outgoing Hawking quanta $c_k \in R$ and their infalling partners $b_k \in I$. Each pair $|\text{Hawking pair}\rangle_k$ is a pure state, so a pair lying entirely inside $R \cup I$ contributes no entropy. The infalling and outgoing modes **purify each other**. What remains is the entropy of modes near ∂I . It is of order G_N^0 and does not grow like Γt :

$$S_{\text{bulk}}(R \cup I) = \mathcal{O}(G_N^0).$$

So the large, steadily growing contribution is absent, and the area term dominates:

$$S_{\text{island}}(R) \stackrel{(1)}{=} \frac{\text{Area}(\partial I)}{4G_N} + \mathcal{O}(G_N^0) \stackrel{(2)}{\approx} \frac{A_{\text{BH}}(t)}{4G_N} \stackrel{(3)}{=} S_0 - \Gamma t.$$

- (1) the bulk term $S_{\text{bulk}}(R \cup I)$ is only $\mathcal{O}(G_N^0)$, because the Hawking pairs purify each other.
- (2) the quantum extremal surface sits just inside the horizon, so its area is the horizon area up to tiny corrections, and the $\mathcal{O}(G_N^0)$ term is negligible next to terms of order $1/G_N$.
- (3) the linear decrease of the horizon area from step 1.

4. **The Page time and the phase transition.** By the island rule, the entropy is the smaller of the two saddle values:

$$S(R) \stackrel{(1)}{=} \min \{S_{\text{no-island}}(R), S_{\text{island}}(R)\} \stackrel{(2)}{=} \min \{\Gamma t, S_0 - \Gamma t\}.$$

- (1) the island rule: take the smallest of the extremal values.
- (2) substitute the two saddle values found in steps 2 and 3.

Setting the two branches equal gives the **Page time** t_P :

$$\Gamma t_P = S_0 - \Gamma t_P \implies t_P = \frac{S_0}{2\Gamma} = \frac{1}{2} t_{\text{evap}}.$$

Over the lifetime of the black hole, the resulting entropy is

$$S(R) = \begin{cases} \Gamma t & t < t_P \quad (\text{no island}), \\ S_0 - \Gamma t = \frac{A_{\text{BH}}(t)}{4G_N} & t > t_P \quad (\text{island } I). \end{cases}$$

The early phase, with no island, is the Hawking phase. The late phase, with the interior island I , is the unitary phase. At $t = t_{\text{evap}}$, $S(R) \rightarrow 0$ as the black hole disappears. This is the value unitary evolution requires: the final radiation is in a pure state.

5. **Algebraic translation.** For $t < t_P$, the radiation algebra is simply $X_R = Y_R$. For $t > t_P$, the island forms, and $X_R = Y_R \vee \widetilde{\mathcal{M}}_I$. The modular flow of X_R then generates the entire interior algebra $\widetilde{\mathcal{M}}_I$ from the radiation data alone. In this language, the resolution of the paradox needs no change to semiclassical effective field theory. What changes at t_P is which algebra the interior operators belong to. ■

7.5 Boundary description of a bulk causal diamond

This section works through several concrete examples of the duality applied to regions that *do not touch the boundary at all*. These are purely interior bulk regions, described entirely through the commutant structure of boundary algebras.

A diamond in the center of AdS, defined purely algebraically

In the vacuum sector, take a boundary time band I_w of width $w < \pi R$, where R is the AdS radius. Its causal wedge is a “spherical Rindler region” W_{ρ_w} . This is the part of global AdS outside a sphere of radius $\rho_w = R \tan(\frac{\pi}{2} - \frac{w}{2R})$ around the center. It is identified with the single-trace time-band algebra: $\widetilde{\mathcal{M}}_{W_{\rho_w}} = Y_{I_w}$. (For $w \geq \pi R$, W_{ρ_w} contains an entire Cauchy slice, and this reduces to the full-boundary statement of Chapter 6.) Now take the *commutant* of both sides, and use Haag duality, $\widetilde{\mathcal{M}}_{b'} = \widetilde{\mathcal{M}}'_b$:

$$\widetilde{\mathcal{M}}_{D_{\rho_w}} = Y'_{I_w}.$$

Here $D_{\rho_w} = W'_{\rho_w}$ is a spherical causal diamond centered in global AdS, as far from the boundary as possible. **The boundary description of this diamond is not geometric at all. It is defined purely algebraically, as the commutant of a boundary time-band algebra**, with no boundary region of its own to point to.

As $w \rightarrow \pi R$, the time band covers nearly the whole boundary, and the causal wedge W_{ρ_w} covers nearly the whole bulk. Correspondingly, the diamond D_{ρ_w} shrinks to an arbitrarily small, local patch of nearly flat spacetime. At every stage it is described by the commutant of an ever-larger boundary time-band algebra. This is a precise, operator-algebraic version of the holographic **IR/UV relation**: probing longer boundary time scales (a bigger Y_{I_w}) is dual to probing shorter bulk distance scales (a smaller diamond D_{ρ_w}).

Detecting a horizon from commutant structure alone

Now repeat this in the thermofield-double black hole above the Hawking–Page temperature T_{HP} . Take $Y_{I_w}^{(R)}$, the algebra of a time band of width w on the right boundary. Its causal wedge is a spherical wedge region W_{ρ_w} lying strictly inside the right black-hole exterior. *No matter how large w is taken*, W_{ρ_w} never covers the entire $t = 0$ slice of the exterior region. The horizon is what prevents this. Algebraically,

$$(Y_{I_w}^{(R)})' \cap Y_R \neq \mathbb{C} \mathbf{1}, \quad \text{for every } w. \quad (7.8)$$

Here $Y_R = \widetilde{\mathcal{M}}_R$ is the algebra of the whole right exterior, and $\mathbb{C} \mathbf{1}$ denotes the multiples of the identity. **The existence of a bulk horizon is detected on the boundary by the following fact: no matter how wide a time band you take, its commutant inside the R algebra never becomes trivial.**

Compare this with the vacuum case just above. There, growing w up to πR eventually made the causal wedge cover an entire Cauchy slice. There was no horizon, and correspondingly the commutant of the time band became trivial once $w \geq \pi R$. Here, above the Hawking–Page transition, this never happens, for any w . This gives a clean diagnostic, intrinsic to the boundary, that distinguishes a horizon-free geometry (such as empty AdS) from a black hole. It uses nothing but the way commutants of nested time bands behave as the bands grow.

A single-sided collapsing black hole, and the emergence of a horizon in real time

The richest example is a black hole formed by collapse. Take $|\Psi\rangle$ dual to a single-sided black hole formed by ordinary gravitational collapse: a shell of matter falls inward from the boundary and forms a horizon at late times. On the boundary side, this corresponds to $|\Psi\rangle$ thermalizing over time.

Take first an early time band I_0 , wide enough that its causal wedge already covers a full Cauchy slice. It gives $Y_{I_0} = B(\mathcal{H}_{\text{bulk}})$, an ordinary type I algebra. No horizon has formed yet, so nothing blocks full reconstruction. Next take, at late times well after the collapse, a semi-infinite time band I_1 . It reconstructs only the black-hole exterior, $\widetilde{\mathcal{M}}_R = Y_{I_1}$. Its commutant Y_{I_1}' gives an emergent “mirror” interior algebra $\widetilde{\mathcal{M}}_L$. So a *single-sided* collapse geometry reproduces the same split structure as the two-sided eternal black hole of the thermofield double:

$$Y_{I_0} = Y_{I_1} \vee Y_{I_1}' = \widetilde{\mathcal{M}}_R \vee \widetilde{\mathcal{M}}_L.$$

The type of the time-band algebra changes qualitatively as the system evolves. It is type I at early times, before a horizon exists. Once the system has thermalized, it is type III_1 for a time band of *any* width, however large, as long as the band’s earliest endpoint stays fixed at some late reference time. **This change of algebra type, tracked purely from boundary data, can serve as an operator-algebraic definition of horizon formation and thermalization in real time.** This is the idea behind the “causal depth” diagnostic that Chapter 8 makes precise.

The general theorem, and its limits

All the examples above shared a convenient special feature. The boundary region considered was exactly the intersection of its own causal wedge with the boundary. This fails in general: the causal wedge of a boundary region can meet the boundary in a strictly larger region. Correspondingly, for a generic boundary region Y , the naive single-trace algebra Y_Y is not a von Neumann algebra on its own. Its double commutant can contain single-trace operators supported on a region strictly larger than Y . This is the same phenomenon that lies behind the failure of additivity in the superadditivity discussion earlier in this chapter.

A precise theorem, which we quote without proof, says exactly when the naive causal-wedge story works. Write

$$C_Y \equiv (\tilde{J}^+[Y] \cap \tilde{J}^-[Y])''$$

for the generalized causal wedge. Here the double prime on a set of points means its bulk causal completion (the causal complement of the causal complement). Let B denote the boundary manifold. The theorem states: Y_Y admits standard causal-wedge reconstruction, and is already a von Neumann algebra, if and only if Y is **causally convex** and $C_Y \cap B = Y$. In that case $Y_Y = \widetilde{\mathcal{M}}_{C_Y}$ exactly, with commutant $Y'_Y = \widetilde{\mathcal{M}}_{C'_Y}$. If Y fails this condition, Y''_Y instead reconstructs the algebra of the larger region $Y_{\max} \equiv C_Y \cap B$. This gives a clean, general answer to how much bigger the double commutant can be.

7.6 Generalized entropy and subregion-subalgebra duality at finite N

Everything above was stated strictly at $N = \infty$. This closing section asks what survives at finite but large N . There, a bulk subregion can no longer be sharply defined, because spacetime itself fluctuates. The section gives a significant piece of indirect evidence that the framework nonetheless extends.

Take the thermofield double above T_{HP} . At finite N , $B(\mathcal{H}_R)$ is an ordinary type I algebra, with a well-defined entanglement entropy S_R . At large N , S_R should be given by the **generalized entropy** of the dual black hole,

$$S_{\text{gen}} \equiv \frac{A_{\text{hor}}}{4G_N(\epsilon)} + S_{\text{bulk}}(\epsilon). \quad (7.9)$$

Here ϵ is a bulk short-distance cutoff, and $G_N(\epsilon)$ is the corresponding bare Newton constant.

At strict $G_N \rightarrow 0$, $\widetilde{\mathcal{M}}_R$ is type III₁. So S_{bulk} is not even defined until $\widetilde{\mathcal{M}}_R$ is regularized into a type I algebra $\widetilde{\mathcal{M}}_R^\epsilon$ using the cutoff ϵ . The entropy of that regularized algebra has a leading UV divergence, $S_{\text{bulk}}(\epsilon) = a A_{\text{hor}}/\epsilon^{d-1} + \dots$, where a is a constant that depends on the field content and on the regulator. This is the familiar area-law divergence of Chapter 4 again. It is matched by a divergence in the bare coupling $G_N(\epsilon)$ in the first term. Neither term in (7.9) is separately finite as $\epsilon \rightarrow 0$. But there are strong indications, from independent gravitational calculations, that the two divergences cancel exactly, leaving a finite $S_{\text{gen}} = \lim_{\epsilon \rightarrow 0} (\frac{A_{\text{hor}}}{4G_N(\epsilon)} + S_{\text{bulk}}(\epsilon))$.

Worked calculation: cancellation of UV divergences in generalized entropy

To see this cancellation explicitly, consider a free, minimally coupled scalar field in $d + 1$ spacetime dimensions. Close to a non-extremal horizon, the metric looks like Rindler space:

$$ds^2 = -\kappa^2 \rho^2 dt^2 + d\rho^2 + dx_\perp^2.$$

Here ρ is the proper distance to the horizon at $\rho = 0$, and $\kappa = 2\pi/\beta$ is the surface gravity. The coordinates $x_\perp \in \mathbb{R}^{d-1}$ run along the horizon, whose total area is $A_{\text{hor}} = \int d^{d-1}x_\perp$.

Choice of regulator. The coefficient of a power-law divergence depends on how the theory is regulated. For example, a “brick wall” at proper distance ϵ from the horizon gives, for $d = 3$, a thermal entropy $A_{\text{hor}}/(360\pi\epsilon^2)$, while the regulator used below gives $A_{\text{hor}}/(48\pi\epsilon^2)$. A cancellation can only be tested if both terms of (7.9) are regulated in the same way. We use a proper-time (heat-kernel) cutoff throughout: every heat-kernel integral $\int ds$ is restricted to proper times $s \geq \epsilon^2$. This removes the contribution of distances shorter than about ϵ .

Bulk entropy. The entanglement entropy of the region outside the horizon can be computed by the replica method. One puts the field on a cone of opening angle $2\pi\alpha$ in the (t, ρ) plane (in Euclidean signature) and uses $S = (1 - \alpha\partial_\alpha) \log Z_\alpha$ at $\alpha = 1$. The heat kernel on such a two-dimensional cone contains an s -independent term $(1 - \alpha^2)/(12\alpha)$; this is a standard result, quoted here. The operation $(1 - \alpha\partial_\alpha)$ at $\alpha = 1$ turns this term into $\frac{1}{6}$. The transverse directions contribute a factor $A_{\text{hor}}(4\pi s)^{-(d-1)/2}$. So, with $\log Z_\alpha = \frac{1}{2} \int_{\epsilon^2}^\infty \frac{ds}{s} \text{Tr } e^{s\Box}$,

$$\begin{aligned} S_{\text{bulk}}(\epsilon) &\stackrel{(1)}{=} \frac{1}{2} \cdot \frac{1}{6} A_{\text{hor}} \int_{\epsilon^2}^\infty \frac{ds}{s} (4\pi s)^{-(d-1)/2} + S_{\text{bulk}}^{\text{finite}} \\ &\stackrel{(2)}{=} \frac{c_{d-1} A_{\text{hor}}}{\epsilon^{d-1}} + S_{\text{bulk}}^{\text{finite}}, \quad c_{d-1} = \frac{1}{6(d-1)(4\pi)^{(d-1)/2}}. \end{aligned}$$

(1) the replica formula applied to the cone term of the heat kernel, times the transverse factor; the remaining, cutoff-independent part is collected in $S_{\text{bulk}}^{\text{finite}}$. (2) $\int_{\epsilon^2}^\infty ds s^{-(d+1)/2} = \frac{2}{d-1} \epsilon^{-(d-1)}$.

For $d = 3$ this gives $c_2 = 1/(48\pi)$. On its own, $S_{\text{bulk}}(\epsilon) \rightarrow +\infty$ as $\epsilon \rightarrow 0$, as it must for a type III₁ local algebra.

Renormalization of Newton’s constant. Quantum fluctuations of the same scalar also correct the gravitational effective action. In Euclidean signature, integrating out the scalar with the same cutoff gives

$$I_{\text{eff}}[g] = -\frac{1}{16\pi G_N(\epsilon)} \int d^{d+1}x \sqrt{g} R + \frac{1}{2} \text{Tr} \log(-\Box) + \dots$$

For a minimally coupled scalar, the heat-kernel expansion is $\text{Tr } e^{s\Box} = (4\pi s)^{-(d+1)/2} \int d^{d+1}x \sqrt{g} (1 + \frac{s}{6} R + \mathcal{O}(s^2))$, and $\frac{1}{2} \text{Tr} \log(-\Box) = -\frac{1}{2} \int_{\epsilon^2}^\infty \frac{ds}{s} \text{Tr } e^{s\Box}$. The term proportional to R has the same form as the Einstein–Hilbert term, so it shifts the coefficient of $\int \sqrt{g} R$:

$$\begin{aligned} \frac{1}{16\pi G_{N,\text{ren}}} &\stackrel{(1)}{=} \frac{1}{16\pi G_N(\epsilon)} + \frac{1}{12} (4\pi)^{-(d+1)/2} \int_{\epsilon^2}^\infty ds s^{-(d+1)/2} \\ &\stackrel{(2)}{=} \frac{1}{16\pi G_N(\epsilon)} + \frac{c_{d-1}}{4\pi \epsilon^{d-1}}. \end{aligned}$$

(1) collect the coefficient of $-\int \sqrt{g} R$: the bare term, plus $\frac{1}{2} \cdot \frac{1}{6}$ times the proper-time integral from the R term of the heat kernel. (2) do the integral as before, and use the definition of c_{d-1} .

Multiplying by 4π gives

$$\frac{1}{4G_N(\epsilon)} = \frac{1}{4G_{N,\text{ren}}} - \frac{c_{d-1}}{\epsilon^{d-1}}.$$

For $d = 3$ this is the familiar one-loop result $1/G_{N,\text{ren}} = 1/G_N(\epsilon) + 1/(12\pi\epsilon^2)$ for a single scalar. (The coefficients c_{d-1} in the entropy and in the coupling were also checked to agree with a computer-algebra evaluation of the integrals, for several values of d .)

The cancellation. Now substitute this coupling into the generalized entropy $S_{\text{gen}}(\epsilon)$:

$$\begin{aligned} S_{\text{gen}}(\epsilon) &\stackrel{(1)}{=} \frac{A_{\text{hor}}}{4G_N(\epsilon)} + S_{\text{bulk}}(\epsilon) \\ &\stackrel{(2)}{=} \left(\frac{A_{\text{hor}}}{4G_{N,\text{ren}}} - \frac{c_{d-1} A_{\text{hor}}}{\epsilon^{d-1}} \right) + \left(\frac{c_{d-1} A_{\text{hor}}}{\epsilon^{d-1}} + S_{\text{bulk}}^{\text{finite}} \right) \\ &\stackrel{(3)}{=} \frac{A_{\text{hor}}}{4G_{N,\text{ren}}} + S_{\text{bulk}}^{\text{finite}}. \end{aligned}$$

(1) the definition (7.9) of generalized entropy with cutoff ϵ . (2) insert the renormalized coupling $1/4G_N(\epsilon) = 1/4G_{N,\text{ren}} - c_{d-1}/\epsilon^{d-1}$ in the first term and the replica result for $S_{\text{bulk}}(\epsilon)$ in the second. (3) the two $c_{d-1} A_{\text{hor}}/\epsilon^{d-1}$ terms have opposite signs and cancel.

The cutoff-dependent term $\epsilon^{-(d-1)}$ cancels exactly. Both coefficients come from the same heat-kernel coefficient $\frac{1}{6}$, which is why they match. So, at one loop and for this field, the generalized entropy S_{gen} does not depend on the cutoff, even though its geometric part and its field-theory part are separately ill-defined without a regulator. For fields with non-minimal curvature couplings, or for gauge fields, extra contributions localized on the horizon appear, and the matching needs more care. The crossed product of Chapter 5 gives an operator-algebraic version of the same idea: it produces a type II algebra whose entropy is finite and agrees with the generalized entropy up to a state-independent constant.

This finiteness is real, if indirect, evidence for the extension this chapter has been building toward. There should exist a boundary regularization at finite N that produces a type I algebra Y_R^ϵ with $\widetilde{\mathcal{M}}_R^\epsilon = Y_R^\epsilon$. Nobody currently knows how to write this regularization down explicitly for a general boundary theory. Now push ϵ down to the Planck length ℓ_p . There, the clean separation between the two terms of (7.9) breaks down, since $1/G_N \sim 1/\epsilon^{d-1}$. The natural expectation is

$$\widetilde{\mathcal{M}}_R^\epsilon = Y_R^\epsilon = B(\mathcal{H}_R), \quad \epsilon \sim \ell_p.$$

The conjecture, then, is the following. The emergent, large- N , type III₁ algebra Y_R is the strict $\epsilon \rightarrow 0$ endpoint of a continuous family of type I algebras at finite N . This family reaches all the way down to the ordinary finite- N algebra $B(\mathcal{H}_R)$ itself. In this sense subregion-subalgebra duality, suitably reinterpreted, would survive at finite N , not only as a strict $N = \infty$ statement.

Subregion-subalgebra duality is now established as a general principle. We have worked it through for the vacuum sector, the eternal and evaporating black holes, and purely algebraic bulk diamonds. Chapter 8 turns to a further consequence: reading bulk *causal structure itself* directly off the type

and commutant structure of boundary algebras. This includes the formation of a horizon and the connectivity of spacetime, and no bulk metric is assumed anywhere in the argument.

CHAPTER 8

Emergence of bulk geometric concepts

Chapter 7 established that a bulk region *is* a boundary algebra. This chapter pushes that identification to its limit: causal structure, the formation of a horizon, and even whether two boundary theories are joined by a wormhole at all, can each be read directly off the type and commutant structure of boundary algebras — with no bulk metric assumed anywhere in the argument. This closes the two puzzles raised in Chapter 6 (the factorization puzzle was already resolved there; the meeting-behind-the-horizon puzzle is resolved here, in the section on Kruskal-like time).

8.1 Algebraic characterization of bulk causal structure

Why boundary commutants can encode more than boundary causality

Boundary operators separated in a spacelike way automatically commute — ordinary microcausality. But *timelike*-separated boundary operators are not required to commute at all; whether they do depends on the specific representation, which is fixed by the state’s two-point functions. This freedom is not a bug — it is exactly the room needed for boundary commutant structure to encode something richer than boundary causality alone: by subregion-subalgebra duality, it encodes bulk causal structure, in one higher dimension.

The causal depth parameter

Recall from Chapter 6 that in empty AdS, a boundary time band I_w generates the entire algebra, $Y_{I_w} = B(\mathcal{H}_\Omega^{\text{GNS}})$, once $w \geq \pi R$ — geometrically, πR is exactly the minimal width for which light rays from the band already cover an entire bulk Cauchy slice. This motivates a purely boundary-intrinsic definition:

Definition (causal depth $T(t)$). For a single-sided semiclassical state $|\Psi\rangle$, $T(t)$ is the largest w for which the time-band algebra $Y_{I_w(t)}$ (centered at boundary time t) still has a *nontrivial* commutant.

For empty AdS, $T(t) = \pi R$: finite and time-independent, matching the geometric statement that light rays from a band of exactly that width already reach everywhere. For a general horizon-free asymptotically-AdS geometry, $T(t)$ is expected to stay finite for all time. For a single-sided eternal black hole (Figure 8.1(b)), by contrast, Y_{I_w} has a *nontrivial* commutant for *every* finite w — no matter how wide a time band you take, you can never quite reach the full exterior algebra (only in the strict $w \rightarrow \infty$ limit does the commutant become trivial) — so $T(t) = \infty$, for all t : **a purely**

boundary-intrinsic signature of a horizon, requiring no bulk metric to state. For a black hole formed by collapse (the single-sided collapse example of Chapter 7), $T(t)$ starts finite and grows monotonically, diverging only as $t \rightarrow \infty$ — the boundary-side signature of a horizon actually *forming*, in real time, rather than always having been there.

A two-sided version, $T_R(t)$, is defined the same way but using the *relative* commutant of Y_{I_w} within Y_R (the right boundary’s own single-trace algebra) — for the thermofield double below T_{HP} , this reduces to ordinary empty AdS’s πR ; above T_{HP} , it diverges for all time, exactly reproducing the bifurcating-horizon diagnostic of Chapter 7, the statement that $(Y_{I_w}^{(R)})' \cap Y_R$ is nontrivial for every w , in this new language.

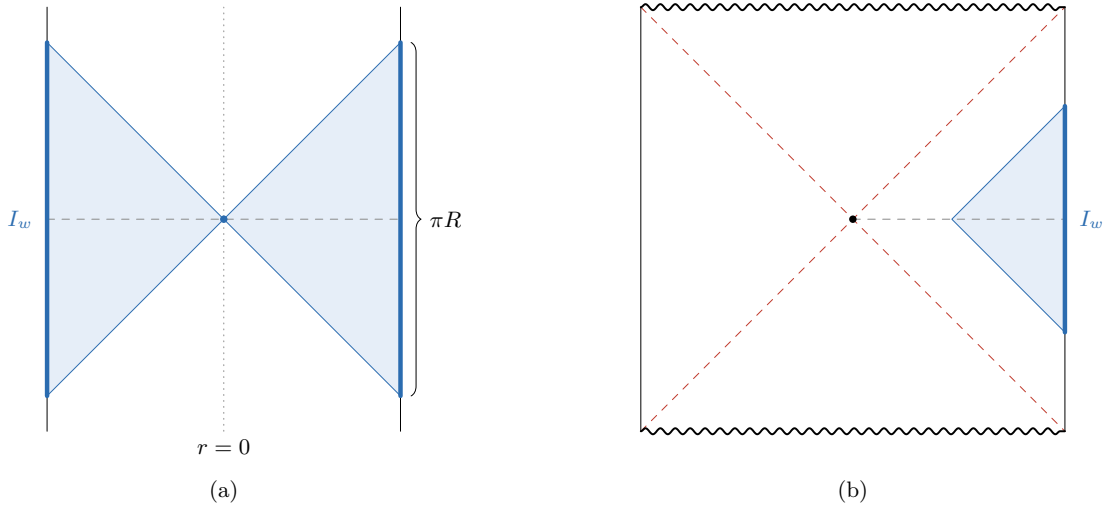


Figure 8.1: (a) Empty AdS drawn as a strip: the vertical edges are the boundary and the dotted line is the center $r = 0$. Light rays (blue lines) from the ends of a boundary time band I_w of width $w = \pi R$ meet at $r = 0$, so the region they enclose (shaded) contains the whole $t = 0$ slice (dashed), giving causal depth $T = \pi R$. (b) Penrose diagram of the eternal black hole: for a band I_w of any finite width on the right boundary, the region its light rays enclose (shaded) stops short of the horizons (dashed red) and their crossing point, so it never covers the whole exterior $t = 0$ slice and $T = \infty$.

T as a measure of lost determinism

T has a second meaning, and it explains why the whole construction is possible. At finite N , knowing the operator algebra on a single Cauchy slice determines the whole theory (ordinary causal time evolution, the time-slice axiom of Chapter 4). In the strict large- N limit, this stops being true — there are no equations of motion relating single-trace operators at different times (Chapter 6) — but the theory isn’t completely disconnected across time either: correlations (encoded in two-point functions) still relate a time band’s algebra to the rest of the theory, just not through equations of motion. T is exactly the minimal width of a band whose algebra is *still enough*, via these correlations rather than dynamics, to recover the whole system. Read this way, $T = 0$ is **full determinism (an honest equation of motion)**; T finite is a **partial, quantifiable loss of determinism**; $T = \infty$, as in a black hole, is a **complete loss of determinism** — no finite

window of boundary time, however wide, is ever enough. The emergence of the bulk radial direction is, in this precise sense, tied to the boundary theory progressively losing determinism as $N \rightarrow \infty$.

The precise version of this statement relates T to the *spectral function* $\rho(\omega)$, the Fourier transform of the commutator $\langle \Psi | [O(t), O(0)] | \Psi \rangle$. It connects to an old question of Kolmogorov's: how long must you watch a fluctuating signal before you can predict all of its future? The general theorem belongs to harmonic analysis and is quoted rather than proved here. What can be done with ordinary tools is to see, in two contrasting examples, why a discrete spectrum gives a finite T and a continuous spectrum does not.

Worked calculation: spectral functions and the divergence of causal depth

The spectral function is

$$\rho(\omega) \equiv \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \Psi | [O(t), O(0)] | \Psi \rangle.$$

It records which frequencies the operator O can put into the system, and with what weight.

1. **Vacuum state $|\Omega\rangle$ (empty AdS).** In global AdS_{d+1} of radius R , the modes created by O have the discrete energies $\omega_n = (\Delta + 2n)/R$, $n = 0, 1, 2, \dots$, exactly like the evenly spaced levels of a harmonic oscillator. The Wightman function is a sum over these levels,

$$\langle \Omega | O(t) O(0) | \Omega \rangle = \sum_{n=0}^{\infty} c_n e^{-i\omega_n t}, \quad c_n \geq 0,$$

and so

$$\begin{aligned} \rho(\omega) &\stackrel{(1)}{=} \int dt e^{i\omega t} \sum_n c_n (e^{-i\omega_n t} - e^{i\omega_n t}) \\ &\stackrel{(2)}{=} 2\pi \sum_n c_n (\delta(\omega - \omega_n) - \delta(\omega + \omega_n)). \end{aligned} \tag{8.1}$$

(1) the commutator is $\langle O(t) O(0) \rangle - \langle O(0) O(t) \rangle$, and the second term is the complex conjugate of the first. (2) $\int dt e^{i(\omega - \omega_n)t} = 2\pi \delta(\omega - \omega_n)$.

The spectrum is a comb of spikes with fixed spacing $2/R$. Now ask: which time bands are wide enough to separate the modes from one another? A smeared operator $\int dt f(t) O(t)$ with f supported in a band of width w picks up mode n with weight $\tilde{f}(\omega_n)$. On a window of width exactly

$$w = \frac{2\pi}{\text{spacing}} = \frac{2\pi}{2/R} = \pi R,$$

the functions $e^{-i\omega_n t}$ are mutually orthogonal, just as e^{int} are orthogonal on $[0, 2\pi]$ in an ordinary Fourier series. So by choosing f to be one of these exponentials one can switch on a single mode and nothing else. Every mode operator is then in the band algebra, and the band algebra is the whole algebra of the vacuum sector. On a shorter window the exponentials overlap and cannot be separated. A direct numerical check with the first 40 modes confirms this: the overlap (Gram) matrix of the $e^{-i\omega_n t}$ is exactly the identity on a window of width

πR , while on a window of width $0.9\pi R$ its smallest eigenvalue has already dropped to about 6×10^{-5} , and at $0.6\pi R$ it is zero to machine precision. The threshold width is $T = \pi R$, exactly the light-crossing value found geometrically.

2. **Thermal state $|\Psi_\beta\rangle$ (black hole).** Take the thermal two-point function of an operator of dimension Δ in a one-dimensional thermal system at inverse temperature β ,

$$\langle O(t)O(0) \rangle_\beta = \left(\frac{\pi}{i\beta \sinh(\frac{\pi}{\beta}(t - i\epsilon))} \right)^{2\Delta}.$$

Its Fourier transform and the resulting spectral function are

$$\begin{aligned} G_\beta(\omega) &\equiv \int dt e^{i\omega t} \langle O(t)O(0) \rangle_\beta = \frac{1}{\Gamma(2\Delta)} \left(\frac{2\pi}{\beta} \right)^{2\Delta-1} e^{\beta\omega/2} \left| \Gamma\left(\Delta + \frac{i\beta\omega}{2\pi}\right) \right|^2, \\ \rho(\omega) &\stackrel{(1)}{=} G_\beta(\omega)(1 - e^{-\beta\omega}) \stackrel{(2)}{=} \frac{2}{\Gamma(2\Delta)} \left(\frac{2\pi}{\beta} \right)^{2\Delta-1} \sinh\left(\frac{\beta\omega}{2}\right) \left| \Gamma\left(\Delta + \frac{i\beta\omega}{2\pi}\right) \right|^2. \end{aligned} \quad (8.2)$$

- (1) the KMS condition of Chapter 4: in a thermal state $\langle O(0)O(t) \rangle$ transforms to $e^{-\beta\omega} G_\beta(\omega)$.
 (2) $e^{\beta\omega/2}(1 - e^{-\beta\omega}) = 2 \sinh(\beta\omega/2)$.

The formula for $G_\beta(\omega)$ was checked by numerical integration for $\Delta = 1, 1.5, 2.3$ at several frequencies, agreeing to all twenty digits computed. For $\Delta = 1$ it reduces to $G_\beta(\omega) = 2\pi\omega/(1 - e^{-\beta\omega})$, which is $2\pi\omega$ times the Bose–Einstein factor, as it should be for a thermal gas of bosons, and then $\rho(\omega) = 2\pi\omega$ exactly.

The decisive difference from the vacuum is the *support* of ρ . At high frequency, Stirling’s formula gives $\rho(\omega) \rightarrow \frac{2\pi}{\Gamma(2\Delta)} |\omega|^{2\Delta-1} \text{sgn } \omega$, the same power law as in the vacuum (the ultraviolet does not know about the temperature; numerically the ratio to this power law is 1.0247 at $\beta\omega = 20$ and 1.0001 at $\beta\omega = 400$). But unlike the vacuum comb, $\rho(\omega)$ is a smooth function that is nonzero at *every* real frequency. There is no spacing between levels. Using the same reasoning as in the vacuum case, separating frequencies spaced by $\delta\omega$ needs a window of width $2\pi/\delta\omega$, and a continuous spectrum has $\delta\omega \rightarrow 0$. No finite band can single out the individual modes, and the causal depth is $T = \infty$.

The vacuum argument is complete as it stands. The thermal argument shows why no finite band could work by mode separation. The stronger statement, that the commutant of every finite band is genuinely nontrivial, is the harmonic-analysis theorem quoted above. Physically, the continuous spectrum is the boundary imprint of the horizon: a black hole absorbs perturbations at every frequency, and that damping destroys the sharp recurrences that let a finite window of the vacuum reconstruct everything.

8.2 Emergence of Kruskal-like time, and resolving the meeting-behind-the-horizon puzzle

Two kinds of bulk time

The eternal AdS black hole geometry has two qualitatively different notions of time. **Schwarzschild time** is an honest isometry — it maps the R (or L) exterior region to itself, never leaving it. This is exactly the boundary time of the thermofield double: Chapter 6 already established Schwarzschild time as the modular time of $\widetilde{\mathcal{M}}_R = Y_R$. **Kruskal time**, by contrast, is not an isometry at all. It is the time that actually carries a point from the exterior R region across the horizon into the interior F (future) or P (past) regions (see the Penrose diagram, Figure 6.1, in Chapter 6). The Kruskal null coordinates U, V are exactly the light-cone coordinates $x^\mp$ of the Rindler-wedge discussion in Chapter 4, since near the horizon the geometry looks locally like flat Rindler space. Schwarzschild time has an obvious boundary home (it's just ordinary boundary time evolution); Kruskal time, at first glance, has no boundary home at all — boundary time simply runs to $\pm\infty$ as you approach the horizon and has nothing left to say about what happens beyond it.

Kruskal time from half-sided modular inclusion

Here the half-sided modular inclusion machinery of Chapter 4, introduced there as a purely algebraic curiosity and worked out on an abstract Rindler-wedge light-cone example, turns out to be *exactly* the tool needed to manufacture Kruskal time out of nothing but boundary data. Take $\mathcal{M} = Y_R$ (the right boundary's full single-trace algebra, cyclic-separating for $|\Psi_\beta\rangle$), and $\mathcal{N} = Y_-$, the subalgebra generated by single-trace operators supported on the semi-infinite band $t < 0$. By the time-band duality of Chapter 7, Y_- is dual to a specific bulk wedge region W_- (the part of the right exterior that can send and receive signals from the half-band $t < 0$), itself type III₁, with $|\Psi_\beta\rangle$ cyclic and separating for it too. Because Y_R 's modular flow is literally boundary time translation, Y_- automatically satisfies exactly the half-sided modular inclusion condition of Chapter 4. So that machinery applies immediately, producing a genuine positive generator

$$G_+ \equiv \frac{1}{2\pi}(K_{Y_R} - K_{Y_-}), \quad (8.3)$$

generating a brand-new time flow on $Y_R = \widetilde{\mathcal{M}}_R$ that leaves $|\Psi_\beta\rangle$ fixed. Repeating with $\mathcal{N} = Y_+$ (the band $t > 0$) gives a second, independent generator G_- .

Near the horizon, where, exactly as in the abstract example of Chapter 4, the geometry reduces to flat Rindler space, these two generators act exactly as null translations in the two light-cone directions,

$$\begin{aligned} e^{iG_+s}\phi(X)e^{-iG_+s} &= \phi(X_s), \quad X_s = (U + s, V, x_\perp) \quad (V \ll 1), \\ e^{iG_-s}\phi(X)e^{-iG_-s} &= \phi(X_s), \quad X_s = (U, V + s, x_\perp) \quad (U \ll 1) \end{aligned}$$

$G_\pm$, **built purely from boundary modular data, literally translate a bulk operator across the horizon**, and the combinations $p \equiv G_- - G_+$, $h \equiv G_+ + G_-$ generate ordinary spatial and genuine Kruskal-time translation respectively, right in the near-horizon region.

This resolves the meeting-behind-the-horizon puzzle of Chapter 6 directly: the boundary Hamiltonian $H = H_R + H_L$ genuinely has no term coupling R to L — and yet the entanglement structure of $|\Psi_\beta\rangle$ itself, purely through the algebraic relationship between Y_R and its subalgebras $Y_\pm$, generates emergent operators $G_\pm$ that couple the two sides and translate operators from R and

L into causal contact behind the horizon. No interaction term was ever needed in H ; the coupling is already latent in how entangled the state is, made manifest only once you build the right algebraic objects from it.

For the BTZ black hole the full computation reproduces the *exact* causal structure of the black hole geometry, not just a qualitative picture. Flow an operator, $\Phi(X, s) \equiv e^{iG_+s}\phi(X)e^{-iG_+s}$, starting at $X = (U_0, V_0, x_\perp)$. For s below the threshold $s_0 = -U_0$, $\Phi(X; s)$ stays entirely within $\widetilde{\mathcal{M}}_R$. Once s passes s_0 , operators from $\widetilde{\mathcal{M}}_L$ suddenly appear in $\Phi(X; s)$: the flowed point has crossed the horizon. Commutators tell the same story. For two points $X_1 \in R$ and $X_2 \in L$, $[\Phi(X_1, s), \phi(X_2)]$ is exactly zero until s passes the threshold $s_{12} = U_2 - U_1$, and nonzero after it. **Sharp causal structure comes out of an evolution built entirely from algebraic data, with no bulk light cone put in by hand.** The near-horizon part of this computation can be done explicitly, and that is the content of the next derivation.

Worked derivation: Kruskal coordinates and horizon crossing in BTZ

The non-rotating BTZ black hole in AdS_3 has metric

$$ds^2 = -\frac{r^2 - r_+^2}{R^2} dt^2 + \frac{R^2}{r^2 - r_+^2} dr^2 + \frac{r^2}{R^2} d\phi^2.$$

The surface gravity at the horizon $r = r_+$ is $\kappa = r_+/R^2$, and the Hawking temperature is $1/\beta = \kappa/2\pi$. The tortoise coordinate is

$$r^*(r) \equiv \int \frac{R^2}{r^2 - r_+^2} dr = \frac{1}{2\kappa} \log \left(\frac{r - r_+}{r + r_+} \right) \in (-\infty, 0),$$

and in the right exterior R ($r > r_+$) the Kruskal null coordinates are

$$U \equiv -e^{-\kappa(t-r^*)}, \quad V \equiv e^{\kappa(t+r^*)}.$$

In region R one has $U < 0$ and $V > 0$. As $r \rightarrow r_+$, $r^* \rightarrow -\infty$ and $UV = -e^{2\kappa r^*} \rightarrow 0$. The future horizon $\mathcal{H}^+$ is the null surface $U = 0$, $V > 0$, and the black hole interior F is the region $U > 0$, $V > 0$.

First, the modular flow of Y_R in these coordinates. Shifting Schwarzschild time, $t \rightarrow t + a$, gives $U \rightarrow e^{-\kappa a}U$ and $V \rightarrow e^{\kappa a}V$, so

$$K_{Y_R} \stackrel{(1)}{=} \beta H_R \stackrel{(2)}{=} \beta \kappa (V \partial_V - U \partial_U) \stackrel{(3)}{=} 2\pi (V \partial_V - U \partial_U). \quad (8.4)$$

(1) Chapter 6: the modular Hamiltonian of Y_R in the thermofield double is β times the right boundary Hamiltonian. **(2)** $\partial_t = \kappa(V \partial_V - U \partial_U)$, read off from the transformation of U, V under $t \rightarrow t + a$ (checked by finite differences). **(3)** $\beta = 2\pi/\kappa$.

The modular flow is a Lorentz boost in the (U, V) plane: one unit of modular time sends $U \rightarrow e^{-2\pi s}U$, $V \rightarrow e^{2\pi s}V$, and it leaves the horizon $U = 0$ in place. This is exactly the Rindler boost of Chapter 4.

The half-sided modular generator $G_+ = \frac{1}{2\pi}(K_{Y_R} - K_{Y_-})$ acts differently. Near the horizon it generates a rigid translation along the null direction U , normalised so that

$$\Phi(s) \equiv e^{iG_+s} \phi(U_0, V_0) e^{-iG_+s} = \phi(U_0 + s, V_0).$$

Follow the coordinate $U(s) = U_0 + s$ of a point that starts in the right exterior, $U_0 < 0$:

1. For $0 \leq s < -U_0$: $U(s) < 0$. The point is still in the right exterior, and $\Phi(s) \in \widetilde{\mathcal{M}}_R = Y_R$.
2. At $s = s_0 \equiv -U_0$: $U(s_0) = 0$. The point sits on the future horizon $\mathcal{H}^+$.
3. For $s > -U_0$: $U(s) > 0$ and $V_0 > 0$, so the point is inside the black hole, in region F .

Now bring in an operator $\phi(X_L)$ at a point $X_L = (U_L, V_L)$ of the left exterior, where $U_L > 0$ and $V_L < 0$. Near the horizon the metric is $ds^2 \propto -dU dV$, so two points are causally connected exactly when $\Delta U \Delta V \geq 0$. Here $\Delta V = V_0 - V_L > 0$, so the condition is $\Delta U = U_0 + s - U_L \geq 0$. By microcausality the commutator $[\Phi(s), \phi(X_L)]$ vanishes for $s < U_L - U_0$ and is generically nonzero once $s \geq U_L - U_0$. This is the threshold s_{12} quoted above, now derived from the geometry.

Near the horizon, then, the generator G_+ , built from boundary modular data alone, carries a right-exterior operator into the interior, where it can meet an operator sent in from the left. That is the resolution of the meeting-behind-the-horizon puzzle, seen in coordinates.

In a suitable large-conformal-weight limit this flow even becomes a genuinely *local* bulk transformation, with trajectories running smoothly into the black hole singularity as s approaches a further critical value. Together with the boost K , the generators $G_\pm$ close into an $SL(2, \mathbb{R})$ algebra, the two-dimensional conformal structure already met abstractly in Chapter 4. These two statements are quoted without proof. A consistency check is immediate, however: below T_{HP} , Y_R is type I, and half-sided modular inclusion cannot occur for a type I algebra (Chapter 4 showed that it requires type III₁). This matches the bulk fact that the two boundaries are then disconnected, with no horizon and nothing to cross.

8.3 Emergent spacetime connectivity: algebraic ER=EPR

Why the naive slogan needs fixing

The naive ER=EPR slogan says that any two entangled gravitational systems are connected by some kind of Einstein–Rosen bridge. Examined carefully, it runs into two problems. **First**: below T_{HP} in the thermofield double, R and L are certainly entangled (by an amount of order G_N^0), but the bulk dual is two entirely *disconnected* copies of AdS (Chapter 6). So “entangled $\Rightarrow$ connected” is already false as stated, unless one calls this a “quantum wormhole” without any independent definition of what that means. **Second**, and sharper: even requiring a large, $O(1/G_N)$ amount of entanglement does not fix it. Consider an evaporating black hole at a time $t < t_P$ before the Page time. The black hole and the radiation it has already emitted share $O(1/G_N)$ entanglement, yet the two systems are *classically disconnected*. The amount of entanglement, at any threshold, is not the right diagnostic.

The fix: entanglement *structure*, not entanglement *amount*

Two facts established earlier point to the right diagnostic. Chapter 6 showed that a classical Einstein–Rosen bridge corresponds to Y_R and Y_L both being type III₁. The previous section showed that type III₁ is what makes half-sided modular inclusion, and hence a causal connection through the horizon, possible at all. **Algebraic ER=EPR** turns this observation into a three-way classification. Take two entangled systems R_1 and R_2 in a pure semiclassical state, with bulk dual $W_{R_1 R_2}$ (every part of which touches some boundary) and algebras $\mathcal{M}_{R_1}, \mathcal{M}_{R_2}$. Then:

- $W_{R_1 R_2}$ is **disconnected** $\iff \mathcal{M}_{R_1}$ and $\mathcal{M}_{R_2}$ are both type I.
- $W_{R_1 R_2}$ has a **classical** wormhole $\iff \mathcal{M}_{R_1}$ and $\mathcal{M}_{R_2}$ are both type III₁ *and* $W_{R_1 R_2}$ is classical.
- $W_{R_1 R_2}$ has a **quantum** wormhole $\iff W_{R_1 R_2}$ is **quantum volatile** and $\mathcal{M}_{R_1}, \mathcal{M}_{R_2}$ are not type I.

A bulk spacetime is **quantum volatile** when it fails to become classical in the $G_N \rightarrow 0$ limit even though that limit is being taken. Concretely, either diffeomorphism-invariant fluctuations do not die away as $G_N \rightarrow 0$, or some geometric quantity (a length, an area, a volume) grows like G_N^{-a} instead of staying finite. The standard example is the evaporating black hole before the Page time. The interior connecting the black hole to its radiation has a *proper length* of order $1/G_N$, which diverges as $G_N \rightarrow 0$. That is why it counts as quantum volatile rather than as an ordinary classical wormhole, even though the entanglement supporting it is large.

Physics Connection: why “structure,” not “amount,” already matters in ordinary quantum information

The classification uses the *type* of $\mathcal{M}_{R_1}$ and $\mathcal{M}_{R_2}$, not the amount of entanglement. The same distinction already appears for three qubits. Compare the GHZ state $|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ with the W state $|\text{W}\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$. Trace out qubit C from each. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the remaining two-qubit states are

$$\rho_{AB}^{\text{GHZ}} = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}, \quad \rho_{AB}^{\text{W}} = \begin{pmatrix} \frac{1}{3} & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Apply the Peres–Horodecki test: transpose the B index and look for a negative eigenvalue. For ρ_{AB}^{GHZ} the eigenvalues after partial transpose are $\{0, 0, \frac{1}{2}, \frac{1}{2}\}$, all non-negative, so **no entanglement is left** between A and B . For ρ_{AB}^{W} they are $\{\frac{1-\sqrt{5}}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1+\sqrt{5}}{6}\} \approx \{-0.206, 0.333, 0.333, 0.539\}$. One is negative, so **A and B are still entangled** after C is gone.

By the simplest measure, GHZ is the *more* entangled of the two. The entropy of one qubit against the other two is 1 bit for GHZ and 0.918 bits for W. Yet removing a qubit destroys all remaining entanglement in GHZ and not in W. The difference lies in how the entanglement is

arranged. GHZ puts all of its correlation into a three-way correlation with nothing left for any pair, while W spreads it out so that every pair keeps some. Algebraic ER=EPR applies the same idea to gravitational systems. Type I, type III₁, and quantum volatile describe how entanglement is organised between $\mathcal{M}_{R_1}$ and $\mathcal{M}_{R_2}$. Two systems with comparable entanglement entropy can land in different classes, just as ρ_{AB}^{GHZ} and ρ_{AB}^{W} do.

This resolves both problems. Below T_{HP} , both boundary algebras are type I, and algebraic ER=EPR correctly reports *no* wormhole of either kind. For the evaporating black hole before the Page time, the entanglement wedge connecting the black hole and the radiation is quantum volatile (its interior length is of order $1/G_N$), so the connection is a *quantum* wormhole rather than either “no wormhole” or an ordinary classical one. After the Page time the entanglement wedge becomes classical, and it connects to the radiation through a classical wormhole anchored at the quantum extremal surface. A refinement of the causal depth of the first section, the *modular* depth, built from modular rather than ordinary time bands, sharpens the distinction further. The boundary algebra is type III₁ in both regimes, but the modular depth is finite before the Page time and infinite after it.

The whole classification assumes the strict $\alpha' \rightarrow 0$ (large 't Hooft coupling) limit, in which the bulk has ordinary geometry. In the stringy regime, taken up next, type III₁ alone is no longer enough to guarantee connectivity.

8.4 Stringy geometry and stringy black holes

Everything so far assumed the bulk is described by ordinary Einstein gravity coupled to matter, which is valid in the double limit $N \rightarrow \infty$, $\lambda \rightarrow \infty$ ('t Hooft coupling large, equivalently $\alpha' \rightarrow 0$). At finite λ , that is, at finite string tension, an infinite tower of massive string modes appears. The notion of a sharp bulk causal region, built throughout this chapter from field-theory causal wedges and RT surfaces, becomes delicate, because strings are extended objects and interact non-locally on the string scale. The dictionary between bulk causal structure and boundary commutants must therefore be modified.

Boundary operator algebras can still be used in this regime. They still probe causal structure, and they still define an analogue of a horizon. The half-sided modular inclusion mechanism behind Kruskal time also carries over to a stringy black hole. The main new lesson is negative: once stringy effects are included, type III₁ structure alone no longer guarantees connectivity, and the algebraic ER=EPR classification needs an extra ingredient. This part of the subject is still being developed, and the details are not needed for anything that follows. The central chain of ideas, from algebra to type of entanglement to emergent spacetime, stands without it.

Bulk causal structure, horizon formation, and spacetime connectivity have now all been read directly off boundary algebra data. Chapter 9 turns to a different kind of question. Rather than analysing an existing holographic theory, it builds simple, fully solvable *toy models* of quantum gravity directly out of operator algebras. The crossed product of Chapter 5 returns there, no longer as an abstract construction but as the mechanism by which an observer’s own clock produces the Bekenstein–Hawking area term and the de Sitter entropy.

CHAPTER 9

Algebraic approaches to quantum-gravity regimes

Chapters 6–8 analyzed an existing structure — AdS/CFT — using the machinery built in Chapters 2–5. This chapter does something different and, in a sense, more ambitious: it builds simple, completely solvable *toy models* of quantum gravity directly out of operator algebras, with the crossed product of Chapter 5 appearing not as an abstract mathematical device but as the literal, physical mechanism that turns a quantum field theory’s type III₁ algebra into the finite, well-defined entropy of a black hole or of de Sitter space. This is the payoff the whole crossed-product construction was built for.

9.1 A model of gravitational dressing from observers

Setting up the constraint

Start with an ordinary quantum field theory on Hilbert space $\mathcal{H}_Q$, and let $\mathcal{M}$ be the algebra of some region, $\mathcal{M}'$ its commutant, both type III₁, with a cyclic-separating reference $|\Psi\rangle$ and modular Hamiltonian K generating the usual internal time flow on $\mathcal{M}$ and $\mathcal{M}'$ (Chapter 4).

Now attach two observers, R and L , each with their own Hamiltonian $\hat{q}_R, \hat{q}_L$ and conjugate clock-reading operators $\hat{p}_R, \hat{p}_L$ (with clock-reading eigenstates $|\tau\rangle_R, |\tau\rangle_L$). Demand the *whole* system — field theory plus both observers — be invariant under translations generated by

$$H = K + \hat{q}_R - \hat{q}_L.$$

This is a toy **Hamiltonian constraint**, built to mimic the way general relativity’s own Hamiltonian constraint works: there is no external clock, only relative readings between subsystems, exactly the way diffeomorphism invariance in gravity means only relational data (not absolute time) is physical.

The naive way to build gauge-invariant states — projecting with $\Pi \equiv \int_{-\infty}^{\infty} dt e^{-iHt}$ — fails, because $\Pi^2 \propto \Pi \int dt$ diverges: the projector is not normalizable, the same obstruction met (and resolved) in Chapter 5. The fix is to define physical states as *equivalence classes* $[\psi]$ under $\Pi|\psi_1\rangle = \Pi|\psi_2\rangle$ (states related by the gauge flow e^{iHt} count as the same physical state), with inner product $(\psi|\phi) \equiv \langle\psi|\Pi|\phi\rangle$, which depends only on the equivalence classes, as can be checked directly.

Gauge-fixing, and two equivalent presentations

Any state can be gauge-fixed by pinning the L -observer’s clock to a specific reading τ : expanding a general state in the joint clock-reading $\times$ field basis, one shows explicitly that $|\psi\rangle \sim |\tau\rangle_L \otimes |\psi_\tau\rangle_{RQ}$ for

$|\psi_\tau\rangle_{RQ} \equiv {}_L\langle\tau|\Pi|\psi\rangle$, and this map from the physical (gauge-invariant) Hilbert space to $\mathcal{H}_R \otimes \mathcal{H}_Q$ is an honest isometry, checked directly from the definitions. In this gauge, $\hat{q}_L$ is no longer an independent dynamical variable — the constraint $H = 0$ solves it as a composite operator, $\hat{q}_L = \hat{q}_R + K$. Symmetrically, gauge-fixing the R -observer's clock instead gives $\hat{q}_R = \hat{q}_L - K$. These are two different, but physically equivalent, ways of describing exactly the same gauge-invariant physics.

Dressed operators, and the crossed product falls out automatically

A gauge-invariant (physical) operator must commute with H . Starting from $A \in \mathcal{M}$ (or $A' \in \mathcal{M}'$), the gauge-invariant, “dressed” version — literally the operator A , tagged with the R -observer's clock reading τ — is built by integrating over the gauge group,

$$\hat{A}_R(\tau) \equiv \int dt e^{iHt} (|\tau\rangle\langle\tau|_R \otimes A) e^{-iHt} = e^{iK\hat{p}_R} A(-\tau) e^{-iK\hat{p}_R},$$

with the mirror construction $\hat{A}'_L(\tau)$ for $A' \in \mathcal{M}'$, and by construction $[\hat{A}_R(\tau), H] = 0$. The clock reading τ appearing in $\hat{A}_R(\tau)$ is now a *quantum number*, not a classical coordinate: in a generic state (not an eigenstate of $\hat{p}_R$), it genuinely fluctuates. **The dressed operators live in a quantum spacetime.** The algebra generated by all such dressed operators, together with clock translations, is

$$\widehat{\mathcal{M}} \equiv \{ \hat{A}_R = e^{iK\hat{p}_R} A e^{-iK\hat{p}_R}, e^{i\hat{q}_R s} \mid A \in \mathcal{M}, s \in \mathbb{R} \}'' ,$$

and this is, symbol for symbol, exactly the crossed-product algebra $\widehat{\mathcal{M}}$ constructed abstractly in Chapter 5, with the clock $\hat{q}$ there identified with the R -observer's own Hamiltonian $\hat{q}_R$ here.

Worked derivation: solving the relational constraint and dressing operators

To see this constraint solving and dressing algebraically, work out the quantum mechanics of the relational Hamiltonian constraint:

$$H = K + \hat{q}_R - \hat{q}_L = 0.$$

Let the observers' clock variables satisfy the canonical commutation relations $[\hat{p}_L, \hat{q}_L] = -i$ and $[\hat{p}_R, \hat{q}_R] = -i$. In the clock-reading representation $|\tau_L\rangle$, the Hamiltonian operator acts as a differential operator $\hat{q}_L = i \frac{\partial}{\partial \tau_L}$.

A physical (gauge-invariant) wavefunction $|\Psi_{\text{phys}}\rangle$ must satisfy the Wheeler–DeWitt-like equation:

$$H |\Psi_{\text{phys}}\rangle = 0 \implies \left(K + \hat{q}_R - i \frac{\partial}{\partial \tau_L} \right) \psi(\tau_L) = 0.$$

Integrating this first-order differential equation immediately yields:

$$\psi(\tau_L) = e^{-i(K+\hat{q}_R)\tau_L} \psi(0).$$

Thus the entire τ_L -dependence is completely determined by the data on the initial slice $\tau_L = 0$. Choosing the gauge $\tau_L = 0$ provides a faithful, isometric embedding of the physical Hilbert space

into $\mathcal{H}_R \otimes \mathcal{H}_Q$:

$$|\Psi_{\text{phys}}\rangle \longleftrightarrow |\psi(0)\rangle_{RQ} \in \mathcal{H}_R \otimes \mathcal{H}_Q.$$

In this gauge, the L -clock momentum is eliminated via the constraint $\hat{q}_L = K + \hat{q}_R$.

Now consider dressing an operator $A \in \mathcal{M}$ to make it gauge-invariant under the flow e^{iHt} . The group-averaged operator is:

$$\hat{A}_R(\tau) \equiv \int_{-\infty}^{\infty} dt e^{iHt} (|\tau\rangle\langle\tau|_R \otimes A) e^{-iHt}.$$

Evaluate it on the physical gauge slice $\tau_L = 0$:

$$\begin{aligned} \hat{A}_R(\tau) &\stackrel{(1)}{=} \int_{-\infty}^{\infty} dt e^{i(K+\hat{q}_R)t} (|\tau\rangle\langle\tau|_R \otimes A) e^{-i(K+\hat{q}_R)t} \\ &\stackrel{(2)}{=} \int_{-\infty}^{\infty} dt (|\tau+t\rangle\langle\tau+t|_R) \otimes (e^{iKt} A e^{-iKt}). \end{aligned}$$

(1) on the slice $\tau_L = 0$ the constraint replaces $\hat{q}_L$ by $K + \hat{q}_R$, so $e^{iHt} = e^{i(K+\hat{q}_R)t} e^{-i\hat{q}_L t}$ reduces to $e^{i(K+\hat{q}_R)t}$ acting on $\mathcal{H}_R \otimes \mathcal{H}_Q$. (2) K and $\hat{q}_R$ act on different factors and commute, so the exponential splits; $e^{i\hat{q}_R t}$ shifts the clock state, $e^{i\hat{q}_R t} |\tau\rangle_R = |\tau+t\rangle_R$, while e^{iKt} flows A in modular time.

Since the clock momentum operator $\hat{p}_R$ is diagonal in the clock basis ($\hat{p}_R |\tau\rangle_R = \tau |\tau\rangle_R$), the t -integral collapses once the clock reading is identified with $\hat{p}_R$: setting $t = \hat{p}_R - \tau$ inside the integral gives the closed form

$$\hat{A}_R(\tau) = e^{iK\hat{p}_R} A(-\tau) e^{-iK\hat{p}_R}.$$

Check directly that this dressed operator commutes with the total generator $K + \hat{q}_R$:

$$\begin{aligned} [K + \hat{q}_R, \hat{A}_R(\tau)] &\stackrel{(3)}{=} [K, \hat{A}_R(\tau)] + [\hat{q}_R, e^{iK\hat{p}_R} A(-\tau) e^{-iK\hat{p}_R}] \\ &\stackrel{(4)}{=} i \frac{\partial}{\partial \tau} \hat{A}_R(\tau) - i \frac{\partial}{\partial \tau} \hat{A}_R(\tau) = 0. \end{aligned}$$

(3) bilinearity of the commutator over the sum $K + \hat{q}_R$. (4) $[K, \hat{A}_R(\tau)]$ is the generator of modular flow acting on $A(-\tau)$, which is $i\partial_\tau \hat{A}_R(\tau)$; the second commutator, evaluated with $[\hat{q}_R, e^{\pm iK\hat{p}_R}] = \mp K e^{\pm iK\hat{p}_R}$ (from $[\hat{q}_R, \hat{p}_R] = i$), gives the same term with the opposite sign.

The operator $\hat{A}_R(\tau)$ is genuinely gauge-invariant. The algebra generated by $\{\hat{A}_R, e^{i\hat{q}_R s}\}$ is identically the crossed-product algebra $\mathcal{M} \rtimes_{\sigma} \mathbb{R}$.

Gravitational dressing to a physical observer's clock is the crossed product by the modular group — not an analogy, a literal identity, checkable by comparing the two constructions gauge-fixing choice for gauge-fixing choice: fixing the L -observer's clock at $\tau = 0$ reproduces exactly the first presentation of $\widehat{\mathcal{M}}$ and $\widehat{\mathcal{M}}'$ in Chapter 5; fixing the R -observer's instead reproduces the second presentation, in which undressed elements of $\mathcal{M}$ sit directly inside $\widehat{\mathcal{M}}$. And since K here is a genuine modular Hamiltonian, the theorem of Chapter 5 applies immediately: $\widehat{\mathcal{M}}$ and $\widehat{\mathcal{M}}'$ are type II_{∞} , with honest density operators and entropies — no further argument needed, since it is literally the same construction already proven there.

9.2 Application I: quantum volatile black hole spacetime and its entropy

Apply the model directly to the eternal AdS black hole: $\mathcal{H}_Q$ is the bulk QFT in the black-hole geometry at $G_N \rightarrow 0$, $|\Psi\rangle$ is the Hartle–Hawking vacuum, $\mathcal{M} = \widetilde{\mathcal{M}}_R$, $\mathcal{M}' = \widetilde{\mathcal{M}}_L$, and K generates ordinary Schwarzschild time. With two genuine asymptotic boundaries available, it is natural to identify the R, L “observers” of the previous section with the two boundaries themselves, and $\hat{q}_R, \hat{q}_L$ with (mean-subtracted) perturbations of the boundary Hamiltonians H_R, H_L around their background values — concretely, in a microcanonical thermofield double peaked around energy $E_0 \sim O(N^2)$ with an $O(N^0)$ spread, $\hat{q}_R \equiv H_R - \langle H_R \rangle$.

The crossed-product algebras are type II_∞ — and this genuinely reflects a change in the physical spacetime structure. The gauge-invariant, but now quantum, quantity $\hat{p}_R + \hat{p}_L$ (the time separation between the R and L boundaries, represented geometrically by a geodesic running between them) has $O(1)$ fluctuations as $G_N \rightarrow 0$ — precisely an example of the **quantum volatile geometry** met in Chapter 8: **the two boundaries are connected by a genuine quantum wormhole**, not a classical one.

Now the entropy payoff, which closes the loop opened at the start of Chapter 5. For a semiclassical excited state $|\widehat{\Phi}\rangle = |\Phi\rangle \otimes |g\rangle$ (exactly the construction of Chapter 5), the crossed-product entropy formula of Chapter 5 gives

$$S_{\mathcal{M}_R}^{(\widehat{\Phi})} = -S(\Phi\|\Psi) + \text{const.}$$

Separately — a gravitational computation quoted here rather than re-derived — it has been shown that, assuming the system relaxes back to the equilibrium state $|\Psi\rangle$ as $t \rightarrow \infty$, this same relative entropy is exactly (minus, up to an additive constant) the ordinary generalized entropy of the excited state, $S(\Phi\|\Psi) = -S_{\text{gen}}^{(\Phi)} + \text{const.}$ Combining the two:

$$S_{\mathcal{M}_R}^{(\widehat{\Phi})} = S_{\text{gen}}^{(\Phi)} + \text{const.}$$

The type II entropy of the gravitationally dressed algebra literally *is* the generalized (Bekenstein–Hawking + bulk-matter) entropy of the black hole, up to a state-independent constant. Every algebraic step in this chain — crossed product, type II, its entropy formula — was proved in full generality in Chapter 5; the only new input is the quoted gravitational identity relating relative entropy to generalized entropy.

9.3 Application II: de Sitter entropy

The puzzle

De Sitter space has no boundary at all — quantum gravity there faces a much deeper “problem of time” than AdS does, with no asymptotic region to anchor observables to. A static observer (sitting at a pole of the spatial sphere) only ever accesses a **static patch**, bounded by a cosmological horizon at the de Sitter radius R (Fig. 9.1). The horizon carries the celebrated Gibbons–Hawking

entropy, $S_{\text{dS}} = A_{\text{hor}}^{(0)}/4G_N$ — but its physical meaning has been debated for decades (does its exponential count a genuine, finite-dimensional de Sitter Hilbert space, for instance?), with little concrete progress.

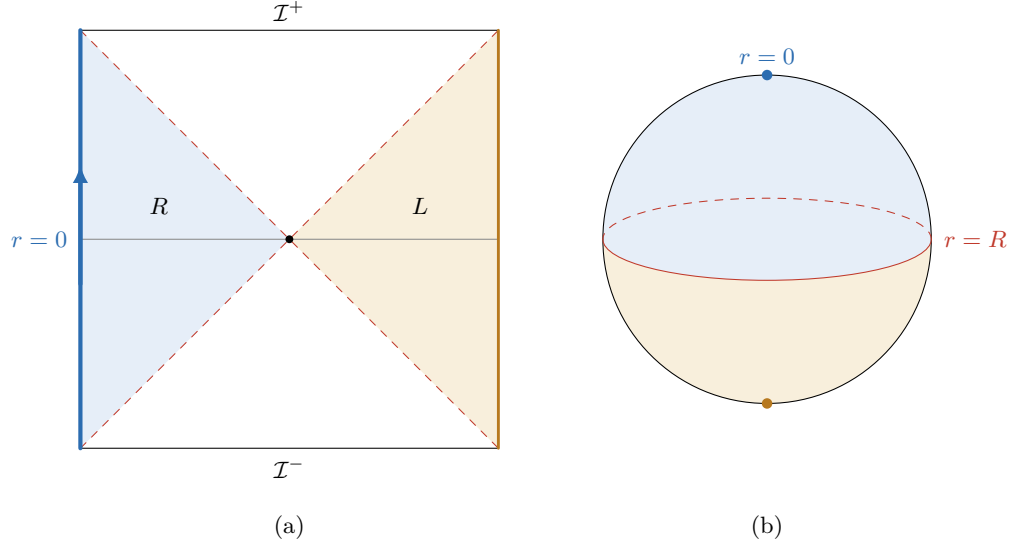


Figure 9.1: (a) Penrose diagram of global de Sitter space. The left edge is the worldline of a static observer at the north pole $r = 0$ (blue arrow); the cosmological horizons (dashed red diagonals) confine everything this observer can ever see to the static patch R (blue), while the patch L (gold) belongs to an observer at the south pole (right edge). The top and bottom edges are future and past infinity $\mathcal{I}^\pm$, and the gray line is the $t = 0$ slice. (b) That slice is a sphere: its northern half (blue) lies in R and ends on the horizon $r = R$ (red equator), which is the black dot in (a).

The *generalized* de Sitter entropy, $S_{\text{gen}} = A_{\text{hor}}/4G_N + \tilde{S}_R$, has a strange feature that is the opposite of a black hole's behavior: **exciting matter in the static patch decreases S_{gen}** , because it shrinks the cosmological horizon area faster than it grows the matter entanglement $\tilde{S}_R$. This can be checked completely explicitly: put a black hole of mass parameter μ inside the static patch (giving a black-hole horizon at r_b nested inside the cosmological horizon at $r_c > r_b$), represent $\tilde{S}_R$ by the black-hole horizon area itself, and the generalized entropy comes out to $S_{\text{gen}} = \omega_{d-2}(r_c^{d-2} + r_b^{d-2})/4G_N$, which is *always less* than S_{dS} — so S_{dS} is the maximum possible value of S_{gen} , achieved exactly by empty de Sitter.

Worked calculation: why matter excitations decrease de Sitter entropy

To see this thermodynamic behavior in detail, consider a four-dimensional Schwarzschild–de Sitter spacetime with metric:

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega_2^2, \quad f(r) = 1 - \frac{2G_N M}{r} - \frac{r^2}{R^2}.$$

1. **Empty de Sitter space** ($M = 0$): The horizon condition $f(r) = 1 - r^2/R^2 = 0$ gives a single cosmological horizon at $r_c^{(0)} = R$. The cosmological horizon area is $A_{\text{dS}} = 4\pi R^2$, giving

the Gibbons–Hawking entropy:

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G_N} = \frac{\pi R^2}{G_N}.$$

2. **Matter excitation: small black hole** ($0 < G_N M \ll R$): For a localized excitation of mass M , the horizon equation $f(r) = 0$ has two positive roots: a black hole horizon at r_b and a shifted cosmological horizon at $r_c < R$. Let $r_c = R(1 - \delta)$ with $\delta \ll 1$. Expanding $f(r_c) = 0$ to first order in δ and M :

$$1 - \frac{2G_N M}{R} - (1 - 2\delta) \approx 0 \implies \delta = \frac{G_N M}{R}.$$

Therefore the cosmological horizon shrinks to $r_c \approx R \left(1 - \frac{G_N M}{R}\right)$. Its shifted area is:

$$A_c(M) = 4\pi r_c^2 \approx 4\pi R^2 \left(1 - \frac{2G_N M}{R}\right) = A_{\text{dS}} - 8\pi G_N M R.$$

The black hole horizon is at $r_b \approx 2G_N M$, with area $A_b(M) = 4\pi r_b^2 \approx 16\pi G_N^2 M^2 \sim O(G_N^2 M^2)$. The total generalized entropy is the sum of both horizon areas:

$$\begin{aligned} S_{\text{gen}}(M) &= \frac{A_c(M)}{4G_N} + \frac{A_b(M)}{4G_N} \stackrel{(1)}{=} \frac{A_{\text{dS}} - 8\pi G_N M R}{4G_N} + O(G_N M^2) \\ &\stackrel{(2)}{=} S_{\text{dS}} - 2\pi M R + O(G_N M^2). \end{aligned}$$

(1) substitute the shifted cosmological area $A_c(M) = A_{\text{dS}} - 8\pi G_N M R$, and note that the black-hole term $A_b/4G_N \approx 4\pi G_N M^2$ is second order in M . (2) split the fraction: $A_{\text{dS}}/4G_N = S_{\text{dS}}$ and $8\pi G_N M R/4G_N = 2\pi M R$.

Here $\beta_{\text{dS}} = 2\pi R$ is the inverse Gibbons–Hawking temperature. Thus, to linear order in perturbation:

$$S_{\text{gen}}(M) - S_{\text{dS}} = -\beta_{\text{dS}} M < 0 \quad \text{for all } M > 0.$$

Any localized energy excitation inside the static patch strictly **lowers** the generalized entropy. Consequently, empty de Sitter ($M = 0$) is the unique state of **maximal entropy**.

Recognizing the type II₁ signature

This behavior — a reference state whose entropy can only ever decrease under perturbation — is exactly what appeared, worked out on an explicit example, in the type II₁ Bell-pair chain of Chapter 3, where perturbing away from the maximally entangled $|\Phi_{\pi/4}\rangle$ always gives $S_{\mathcal{M}}(\Psi) \leq 0$. **Empty de Sitter behaves exactly like the maximally entangled reference state of a type II₁ algebra, with R and L static patches playing the role of the two entangled halves.**

Making this precise uses exactly the observer model above, with one small but crucial modification. Take $\mathcal{H}_Q$ to be the QFT Hilbert space on global de Sitter, $|\Psi\rangle$ the Bunch–Davies vacuum, $\mathcal{M}, \mathcal{M}'$ the R - and L -patch algebras (modular Hamiltonian K = the boost generator of t -translations), and $\hat{q}_R, \hat{q}_L$ the Hamiltonians of static observers at the two poles. This reproduces $\widehat{\mathcal{M}}_R, \widehat{\mathcal{M}}_L$ exactly as before — *except* that now $\hat{q}_R, \hat{q}_L$ are required to be **bounded below** (non-negative eigenvalues —

physically, the sensible statement that a real observer's energy cannot be negative). By exactly the mechanism checked in Chapter 5 (restricting the clock spectrum to $q \geq 0$ turns a divergent trace into a normalized one), $\widehat{\mathcal{M}}_R, \widehat{\mathcal{M}}_L$ become type II_1 , not type II_∞ . The type II_1 entropy relative to the maximally entangled reference state $|\widehat{\Psi}_T\rangle = e^{-\hat{q}/2} \otimes |0\rangle_{\text{BD}}$, with trace $\text{tr}(\cdots) \equiv \langle \widehat{\Psi}_T | \cdots | \widehat{\Psi}_T \rangle$, is negative exactly the way the Chapter 3 example was, and for the identical reason: there is no unentangled reference to fall below zero from, only a maximally entangled one to fall below.

One more consistency check resolves an objection that looks fatal at first glance: if empty de Sitter is a maximally entangled state, should it not have infinite temperature (the way a maximally entangled Bell pair has infinite temperature in the entanglement-Hamiltonian sense)? The resolution: the state $|\widehat{\Psi}_T\rangle$ has density operator $\rho = e^{-K}$ on the crossed-product algebra — a thermal state at *dimensionless* temperature exactly 1, the universal modular temperature of Tomita–Takesaki theory (Chapter 4). Converting this dimensionless $\beta = 1$ into physical units using the static observer's own proper time t reproduces exactly the known, finite de Sitter temperature $T_{\text{dS}} = 1/2\pi R$. There is no contradiction: “maximally entangled” was always a dimensionless, modular-time statement, and the finite physical temperature only appears once you convert to an observer's physical clock.

Property	Black Hole Exterior (AdS/Minkowski)	de Sitter Static Patch
Spacetime Horizon	Event horizon (null boundary of black hole interior)	Cosmological horizon (null boundary of observer static patch)
Horizon Temperature	Hawking temperature $T_H = \frac{1}{8\pi G_N M}$	Gibbons–Hawking temperature $T_{\text{dS}} = \frac{H}{2\pi} = \frac{1}{2\pi R}$
Killing Vector $\chi = \partial_t$	Timelike outside; null on horizon; timelike again at spatial infinity	Timelike inside static patch; null on cosmological horizon; spacelike outside
Reference Vacuum State	Hartle–Hawking vacuum $ \Psi_{\text{HH}}\rangle$	Bunch–Davies vacuum $ 0_{\text{BD}}\rangle$
Clock Hamiltonian $\hat{q}$	$\hat{q} \in (-\infty, \infty)$ (unbounded real line)	$\hat{q} \geq 0$ (energy bounded below for physical observer)
Crossed Product Algebra	Type II_∞ factor ($\text{tr}(\mathbf{1}) = \int_{-\infty}^{\infty} dq e^{-q} = \infty$)	Type II_1 factor ($\text{tr}(\mathbf{1}) = \int_0^{\infty} dq e^{-q} = 1$)
Entropy Formula	$S_{\widehat{\mathcal{M}}} = -S(\Phi \parallel \Psi) + \text{const} = S_{\text{gen}} + \text{const}$	$S_{\widehat{\mathcal{M}}} \leq 0$ (relative to maximally entangled empty de Sitter)
Perturbation Response	$\Delta S_{\text{gen}} > 0$ (adding matter increases horizon area)	$\Delta S_{\text{gen}} = -\beta_{\text{dS}} M < 0$ (matter shrinks cosmological horizon)
Maximal Entropy State	None (entropy can grow unboundedly with mass M)	Empty de Sitter ($M = 0$, $S_{\text{max}} = S_{\text{dS}} = \frac{\pi R^2}{G_N}$)

Table 9.1: Comprehensive algebraic and thermodynamic comparison between the black hole exterior and the de Sitter static patch under the crossed-product construction.

9.4 Application III: generalized entropy for a local spacetime region

The black hole and de Sitter examples both had a special simplifying feature: a single reference state existed whose modular Hamiltonian coincided exactly with an honest geometric (isometric)

time flow. A more general region has no such flow. A standard example is Schwarzschild–de Sitter: a black hole sitting inside a de Sitter universe, so the observer’s accessible region is bounded by *two* horizons at different temperatures, $r_b < r_c$. With no common temperature, there is no single global geometric modular flow at all. The construction generalizes cleanly: decompose the region’s algebra as $\mathcal{M}_R = \mathcal{M}_c \otimes \mathcal{M}_b$ (cosmological- and black-hole-horizon factors), choose a state making *each* factor’s modular flow separately geometric, and introduce two independent pairs of observer Hamiltonians, one for each horizon, with a separate constraint for each. The resulting type is not what naive analogy would suggest: even with both sets of observer Hamiltonians bounded below, the algebra comes out type II_∞ , not type II_1 — and its entropy reproduces exactly the expected sum of both horizon areas plus bulk matter entropy,

$$S_{\text{gen}} = \frac{A_{\text{cos}}}{4G_N} + \frac{A_{\text{BH}}}{4G_N} + \tilde{S}_R.$$

This shows the crossed-product mechanism is robust and general, not a special property of the two simplest (black hole, de Sitter) examples.

9.5 A model of dynamical observers

Every model so far treated the observer’s own trajectory as fixed, non-dynamical — only its clock was quantum mechanical. A more honest model lets the observer itself be a genuine, finite-mass dynamical particle, worked out for two-dimensional de Sitter space: promoting the particle’s mass to a dynamical variable $Q(\tau)$ with conjugate $P(\tau)$ and gauging the full de Sitter isometry group (rather than just time translations) produces a richer physical Hilbert space, with gauge-invariant, “dressed” field operators ϕ_R built by projecting onto observer-frame matrix elements. A key structural result: the resulting observer algebra $\mathcal{A}_R$ turns out to be a direct integral of type I_∞ factors — because a genuinely dynamical observer can change its own trajectory, there is no fixed subregion its dressed operators stay confined to, so they end up smeared, state-dependently, over an entire Cauchy slice. The static-observer limit of the de Sitter discussion above is recovered by sending the observer’s mass $\Lambda \rightarrow \infty$ — but this limit does *not* commute with taking the double commutant: take $\Lambda \rightarrow \infty$ first and you recover the emergent type II algebra (and its Gibbons–Hawking entropy); take the double commutant first and then send $\Lambda \rightarrow \infty$, and you get type I instead. The type II structure — and the cosmological horizon’s very existence as an entropic object — is itself an emergent, order-of-limits-dependent phenomenon.

Physics Connection: bounded-below spectra are why a Gibbs state exists at all

The distinction running through Chapter 5 and the de Sitter discussion above — whether the observer’s clock spectrum is bounded below decides type II_1 versus type II_∞ — is exactly the same distinction you already navigate every time you write down a partition function. An ordinary Gibbs state $\rho = e^{-\beta H}/Z$ only exists as a normalizable density matrix if $Z = \text{Tr } e^{-\beta H}$ converges, and that convergence is exactly the statement that H is bounded below: for the

harmonic oscillator, $H = \omega(n + \frac{1}{2})$ with $n = 0, 1, 2, \dots$, so

$$Z = \sum_{n=0}^{\infty} e^{-\beta\omega(n+1/2)} = \frac{e^{-\beta\omega/2}}{1 - e^{-\beta\omega}}$$

converges precisely because the sum is bounded below at $n = 0$. This is the discrete, everyday version of the exact integral check carried out in Chapter 5, $\int_0^\infty dq e^{-q} = 1$ versus $\int_{-\infty}^\infty dq e^{-q} = \infty$. A harmonic oscillator or a hydrogen atom has a ground state, and hence a sensible thermal state at any temperature, *because* its spectrum is bounded below; a system whose energy were unbounded below would have no normalizable Gibbs state at any temperature. The observer models of this chapter are doing nothing more exotic: a physical observer’s clock Hamiltonian $\hat{q}$ is bounded below for the same reason a real physical system’s energy is — because a real system has a ground state — and it is exactly this ordinary fact, not a new gravitational input, that produces the finite trace (type II₁) and hence the finite Gibbons–Hawking entropy of the cosmological horizon.

With a large but finite observer mass, the observer’s position becomes increasingly uncertain over time: an initial position uncertainty $1/\Lambda$ grows to $\sim e^{2\pi\tau/\beta_{\text{dS}}}/\Lambda$ after proper time τ , from the exponential expansion of de Sitter. This defines a natural **scrambling time** $\tau_s \equiv \frac{\beta_{\text{dS}}}{2\pi} \log \Lambda$, after which the observer has an $O(1)$ chance of leaving its original static patch. This scrambling shows up concretely in an out-of-time-order correlator (a standard chaos diagnostic): even though the matter field itself is free and does not directly interact with the observer, the gravitational constraint couples them nontrivially — inserting an operator along the observer’s worldline physically “kicks” its trajectory, and this recoil produces genuine quantum chaotic behavior, with a decay rate matching a Lyapunov exponent of $4\pi/\beta_{\text{dS}}$ — twice the universal chaos bound, matching a mechanism identified independently in other contexts.

9.6 Operator algebras of JT gravity

The chapter closes with a remarkable exact result, in a setting simple enough to solve completely rather than only perturbatively: **Jackiw–Teitelboim (JT) gravity**, a two-dimensional gravity theory (with a scalar “dilaton” field Φ) coupled to matter, whose gravitational sector reduces entirely to boundary dynamics once the dilaton’s own equation of motion is solved, forcing the bulk geometry to be locally exact AdS_2 . The theory’s gauge freedom (the isometry group of AdS_2) locks the two boundaries’ Hamiltonians together, $H_L = H_R \equiv H$, reducing the classical phase space to just two variables — equivalently, the boundary geodesic separation ℓ and its conjugate momentum — which can be canonically quantized into a Hilbert space $L^2(\mathbb{R})$, with an (overcomplete) basis of states $|\beta\rangle$, $\beta \in \mathbb{R}_{>0}$, built from a Euclidean path integral. Including matter fields (which do not couple directly to the dilaton), the full quantum-gravitational Hilbert space becomes $\mathcal{H} = L^2(\mathbb{R}) \otimes \mathcal{H}_{\text{matt}}$, with a basis built from path integrals with matter insertions on the Euclidean boundary.

This is one of the very few places in the entire subject where “the operator algebra of quantum gravity” is known in exact, closed form: the full algebra of gauge-invariant JT-gravity observables

(built including the boundary Hamiltonian H_R itself, at genuinely finite G_N — with no crossed product, no auxiliary observer, and no perturbative expansion needed as an approximation) is **type II**. This result is stated here without proof. Unlike every earlier example in this chapter, it is not a statement about a semiclassical, large- N construction dressed onto an auxiliary clock; it is an exact statement about the full quantum-gravitational theory itself, at finite coupling. It stands as concrete proof of principle that “the algebra of quantum gravity,” at finite G_N , is a well-posed and answerable question — not merely a semiclassical approximation scheme — at least in this simplest of solvable models.

The concrete applications of this chapter — black hole, de Sitter, Schwarzschild–de Sitter, dynamical observers, and exact JT gravity — are all built from the same single mechanism: gravitational dressing to a physical observer is the crossed product by the modular group, and it is this, and nothing more exotic, that converts an intractable type III_1 algebra with no entropy into a genuine type II algebra whose entropy is exactly the gravitational entropy physicists already knew to expect. Chapter 10 closes with a summary and some more speculative remarks on what all of this might mean for the mathematical structure of quantum gravity at finite G_N .

Conclusions and outlook

10.1 Summary

The main results of these notes fit into five statements. Each one is listed below together with the chapter where it was built, so any one of them can be traced back to its full derivation.

- In the $G_N \rightarrow 0$ limit, a general bulk subregion is defined by a boundary operator algebra — typically one with no direct boundary geometric description at all — carrying an emergent type III₁ structure, sourced by the infinite long-range entanglement that only appears in the strict large- N limit (Chapters 6 and 7). This machinery is not restricted to the strict Einstein ($\lambda \rightarrow \infty$) limit; it extends, with modification, into the stringy regime (Chapter 8).
- Bulk causal structure is encoded in boundary *commutant* structure — not merely boundary causality (which only forces spacelike-separated operators to commute), but the richer, timelike commutant structure that subregion-subalgebra duality inherits from the bulk. Quantitative tools like the causal depth parameter $T(t)$ (Chapter 8) make this a checkable, numerical diagnostic of horizons and global causal structure, and these tools too extend into the stringy regime.
- Modular flow, and its refinement half-sided modular flow, characterize the emergence of time itself in the bulk — resolving the meeting-behind-the-horizon puzzle via emergent Kruskal time, and motivating a sharpened, algebraic version of the ER=EPR proposal (both in Chapter 8).
- Large- N boundary algebras split cleanly into two kinds. An **entanglement wedge algebra** X is, by definition, the large- N limit of a genuine finite- N algebra B — it always admits a finite- N ancestor. A **causal wedge algebra** Y , by contrast, is intrinsically a large- N , semiclassical construct, built directly from single-trace operators via the extrapolate dictionary, with no finite- N definition of its own. Neither is guaranteed to have a bulk geometric meaning once you leave the strict Einstein regime — and both constructions apply just as well to non-gravitational systems entangled with a holographic partner (Chapter 7), with no need for the non-gravitational side to have its own large- N parameter at all.
- Simple operator-algebraic toy models — static observers, dynamical observers, and exactly solvable JT gravity (all in Chapter 9) — turn this machinery into concrete physics: the physical origin of generalized gravitational entropy for black holes, de Sitter space, and general regions; the emergence of a cosmological horizon as an observer’s mass is taken to infinity; and, in JT gravity, a complete, finite- G_N example where the operator algebra of quantum gravity is known in exact closed form, even though the Hilbert space itself does not factorize.

10.2 The mathematical structure of quantum gravity

This final section is more speculative than the rest of the notes. It returns to the question raised in Chapter 1 — *what happens to the wavefunction, and to the Hilbert space itself, once you take the algebra seriously as the fundamental object?* — and answers it at the largest scale available.

Why quantum gravity cannot have one single Hilbert space

Ordinary quantum mechanics treats the Hilbert space as fundamental: states are density operators living in it, observables are operators acting on it. There is a concrete reason why this cannot be the right starting point for quantum gravity. It is widely believed that no physical process in quantum gravity can change a spacetime’s *asymptotic* structure — there is no physical process that takes a state of type IIB string theory on $\text{AdS}_5 \times S^5$ to a state on $\text{AdS}_3 \times S^3 \times K3$, or to either of these from ten-dimensional flat Minkowski space. Since a genuine physical process is exactly what it means for two states to live in the same Hilbert space (you can always in principle evolve from one to the other), **each of these asymptotic backgrounds must correspond to a genuinely separate Hilbert space**. This is not a conjecture; it is already visible directly in the existing dictionary: $\text{AdS}_5 \times S^5$ is dual to a four-dimensional CFT with its own Hilbert space, $\text{AdS}_3 \times S^3 \times K3$ to a two-dimensional CFT with a completely different one, and nothing maps a state of one to a state of the other. Yet all of these are supposed to be different vacua of the very same underlying theory — type IIB string theory. **Without one single, global Hilbert space to hold all of these together, what mathematical object is left to call “the theory”?**

The resolution you have already met, twice

This is exactly the same shape of question that opened Chapter 1, and the resolution is precisely the demotion of the Hilbert space worked through in detail in Chapter 2, applied now at the grandest possible scale. Look first at a much smaller-scale version of the same puzzle, already familiar from ordinary QFT in curved spacetime: a free scalar field in flat Minkowski space can be quantized using ordinary Minkowski time, *or* using Rindler time (Chapter 4). Both are perfectly sensible choices, describing different physical situations (an inertial versus an accelerated observer), and yet they produce genuinely, unitarily *inequivalent* Hilbert spaces. You could declare these to be two different theories, but that is clearly the wrong instinct — there is an obvious sense in which it is the same free scalar field theory, just quantized around two different reference states. **Algebraic QFT resolves this by defining the theory as a pair $(\mathcal{A}, \mathcal{S})$ — an algebra $\mathcal{A}$ of field operators, together with a space $\mathcal{S}$ of allowed states (in the abstract, linear-functional sense of Chapter 2, not vectors in any particular Hilbert space) — and letting the Hilbert space be a *derived*, state-dependent object, produced fresh by the GNS construction of Chapter 2 for whichever state $\omega \in \mathcal{S}$ you happen to pick.** Minkowski quantization and Rindler quantization are simply two different GNS representations of the very same underlying algebra $\mathcal{A}$ — genuinely different Hilbert spaces, but manifestly the same theory, because the algebra never changed.

The proposal is to do exactly this, one level up, for the whole of quantum gravity.

Postulate that a quantum-gravity theory at finite G_N is specified by a single abstract $*$ -algebra $\mathcal{A}$ (not tied to any particular asymptotic structure) together with some allowed collection of states on it. Different asymptotic structures — $\text{AdS}_5 \times S^5$, $\text{AdS}_3 \times S^3 \times K3$, flat space, de Sitter — simply correspond to *different states* ω on this one algebra, each producing its own GNS Hilbert space $\mathcal{H}_\omega$ via exactly the construction of Chapter 2, with the corresponding boundary CFT (where one is known) simply *being* that GNS representation. Concretely: $\omega_{\text{AdS}_5 \times S^5}$ is the state whose GNS Hilbert space is that of $\mathcal{N} = 4$ super-Yang-Mills; $\omega_{\text{AdS}_3 \times S^3 \times K3}$ is the state whose GNS space is the dual two-dimensional CFT's; and, in principle, flat space and de Sitter correspond to further states of the same underlying algebra A_{IIB} , even though an explicit boundary description for those cases is not yet known.

Physics Connection: levels of algebraic emergence

This is the same logical move made twice already in these notes. Lined up side by side, the three levels are each a strictly more drastic version of the same idea:

- **States (Chapter 2, Slavnov's statistical picture):** an ordinary quantum *state* ω (or ρ , or $|\psi\rangle$) is not fundamental — it is a derived bookkeeping device (an ensemble average, in Slavnov's language) built on top of the more primitive pairing of algebra and individual measurement outcome.
- **Hilbert spaces (Chapter 2's GNS construction, and the Minkowski–Rindler example just above):** the *Hilbert space itself* is not fundamental — a single algebra $\mathcal{A}$ can produce many inequivalent Hilbert spaces, one per choice of reference state, via GNS.
- **Spacetime backgrounds (this section):** even the question “which spacetime background am I in” is not fundamental data fed into the theory — it, too, is just a choice of state on one underlying algebra, with the entire geometric structure of a spacetime (AdS , flat space, de Sitter, and everything built on top of it in these notes — causal structure, horizons, entropy) emerging as a *consequence* of that choice, not an ingredient of it.

Nothing about the mathematics changed between these three instances — it is the identical GNS construction, applied at the level of an individual measurement, then at the level of an ordinary quantum system, then at the level of an entire spacetime background. What changed is only how much structure you are willing to regard as emergent rather than fundamental, and each chapter of these notes have pushed that boundary a little further out.

At present, there is no known background-independent definition of the algebra A_{IIB} itself, nor a full characterization of which states are allowed. String field theory is a plausible route toward one, since for any fixed background state it already supplies both the Hilbert space and the background-dependent algebra, with its equations of motion available to help identify further consistent states systematically. The algebraic program developed in these notes — subregion-subalgebra duality, modular time, the crossed product, generalized entropy from observer dressing — can be read as a first, concrete step toward exactly this larger goal: a formulation of quantum gravity in which asymptotic structure, the cosmological constant, the Hilbert space, and the very number of degrees

of freedom are not fixed ingredients handed to the theory in advance, but *derived, state-dependent consequences* of a single, more primitive algebraic structure.

Further reading

Two technical computations are not reproduced in these notes; both can be found in [1]. The first consists of further finite-dimensional examples of infinite entanglement coexisting with a factorizable Hilbert space — fine-tuned counterexamples to the general expectation, discussed in Chapter 2, that infinite entanglement generically prevents factorization. The second is the explicit solution of the KMS relation that gives $\hat{\Delta} = \Delta_\Psi$ for the crossed-product algebra, the result used without derivation in Chapter 5. Neither changes the conceptual picture developed here.

These notes began with a wavefunction and asked what it really is. The answer that emerged, one worked example at a time, is that the wavefunction, the Hilbert space it lives in, and finally the spacetime itself are each the most convenient representation of something more primitive: an algebra of observables, together with a choice of state on it. Which parts of the familiar picture are fundamental, and which are representations of something deeper, is the same question at every level — for a single spin, for a quantum field in a Rindler wedge, for a black hole, and for the universe as a whole.

Bibliography

- [1] H. Liu, *Lectures on entanglement, von Neumann algebras, and emergence of spacetime*, arXiv:2510.07017 [hep-th].
- [2] E. Witten, *Notes on some entanglement properties of quantum field theory*, Rev. Mod. Phys. **90**, 045003 (2018), arXiv:1803.04993 [hep-th].
- [3] D. A. Slavnov, *Statistical algebraic approach to quantum mechanics*, Theor. Math. Phys. **129**, 1385 (2001).
- [4] F. Strocchi, *An Introduction to the Mathematical Structure of Quantum Mechanics*, 2nd ed., World Scientific (2008).
- [5] M. Van Raamsdonk, *Building up spacetime with quantum entanglement*, Gen. Rel. Grav. **42**, 2323 (2010), arXiv:1005.3035.
- [6] J. Maldacena and L. Susskind, *Cool horizons for entangled black holes*, Fortsch. Phys. **61**, 781 (2013), arXiv:1306.0533.
- [7] S. Ryu and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*, Phys. Rev. Lett. **96**, 181602 (2006), hep-th/0603001.
- [8] V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, *An algebra of observables for de Sitter space*, JHEP **02** (2023) 082, arXiv:2206.10780.
- [9] A. Almheiri, T. Hartman, J. Maldacena, E. Shaghoulian, and A. Tajdini, *The entropy of Hawking radiation*, Rev. Mod. Phys. **93**, 035002 (2021), arXiv:2006.06872.
- [10] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed., Springer (1996).